BIO201 – MIDTERM MERGED PPT HANDOUTS
(Lecture 01 to 88)
Cell Biology

Prepared By:

All About Biotech

AARISH ALI & TEAM (+96569002641)

Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

Cell Biology

Introduction

Introduction

Course Outline

Atom→ Molecules→
Cell Structures→ Cells

Cell Functions

Propagation/Heredity

Expression Regulation

Cell Communication

Applied Cell Biology

Introduction

Course Objective

Understand the molecular logic of the "living state"

How a molecule's function arises from its structure,

How this information is conveyed to the rest of the cellular machinery

System Manipulation

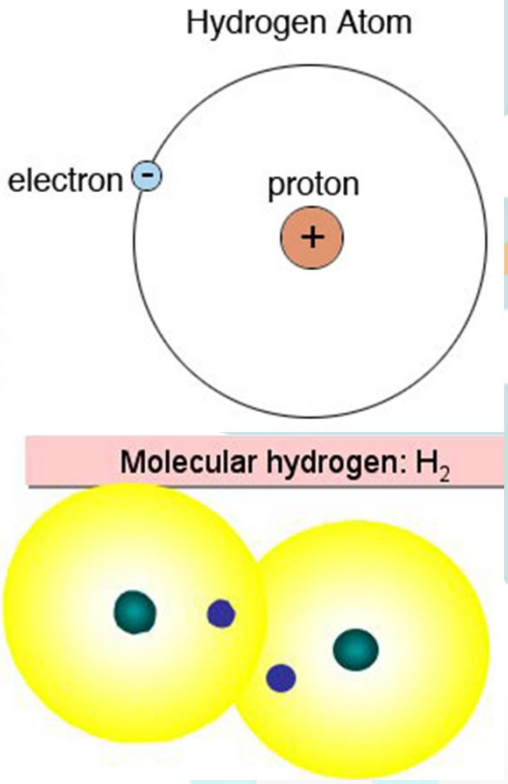
END

CELL BIOLOGY

Composition of Matter

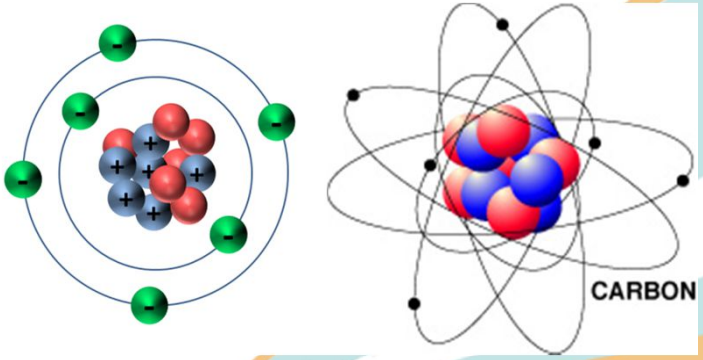
Composition of Matter

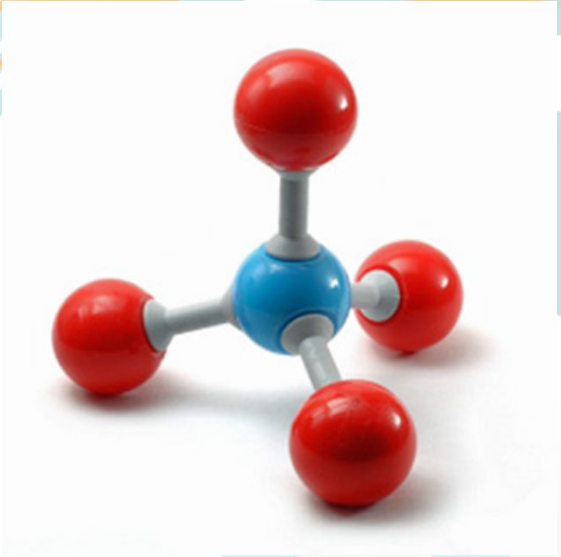
Atomic Structure



Composition of Matter

Carbon Atom

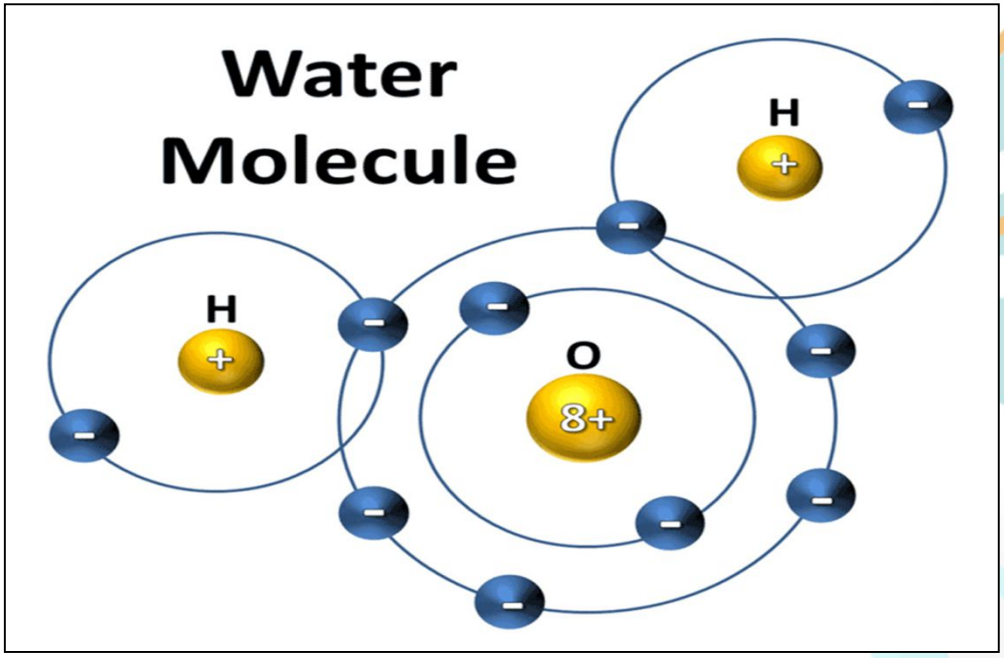




Composition of Matter

Electronegativity

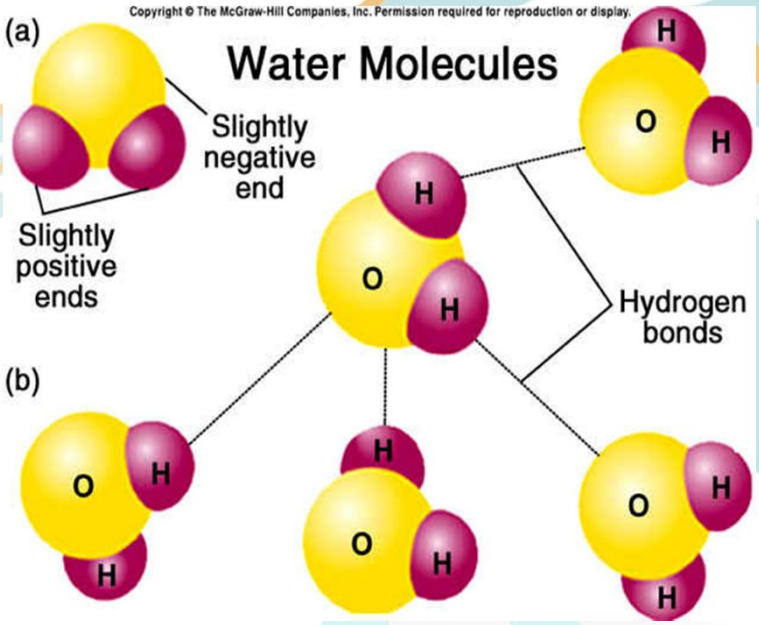
The tendency of an atom to attract electrons towards itself.



Composition of Matter

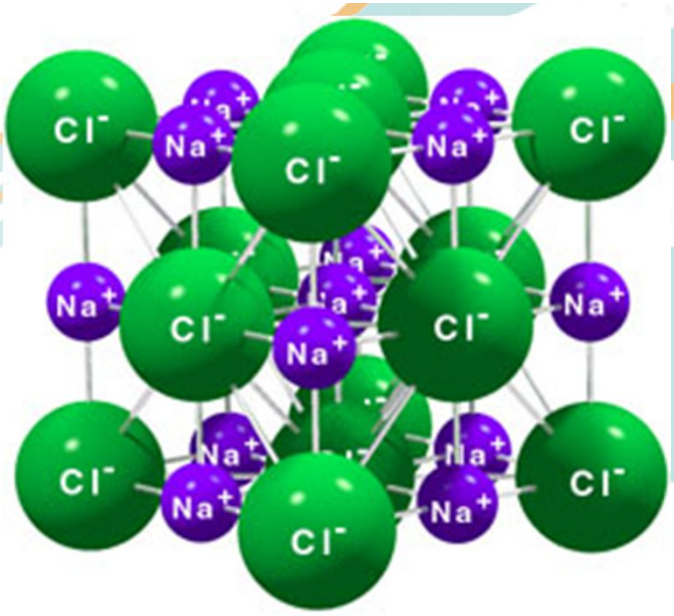
Polarity of H_2O

Results in Hydrogen Bonding



Composition of Matter

Ionic Bond

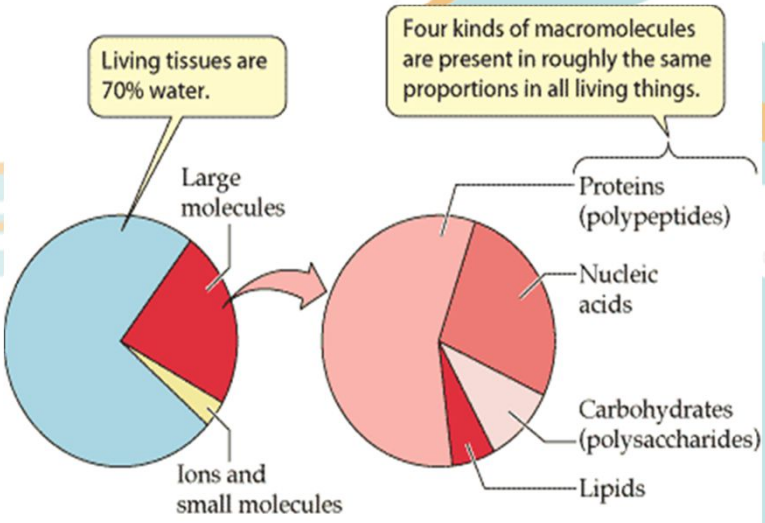


CELL BIOLOGY

Molecules of Life

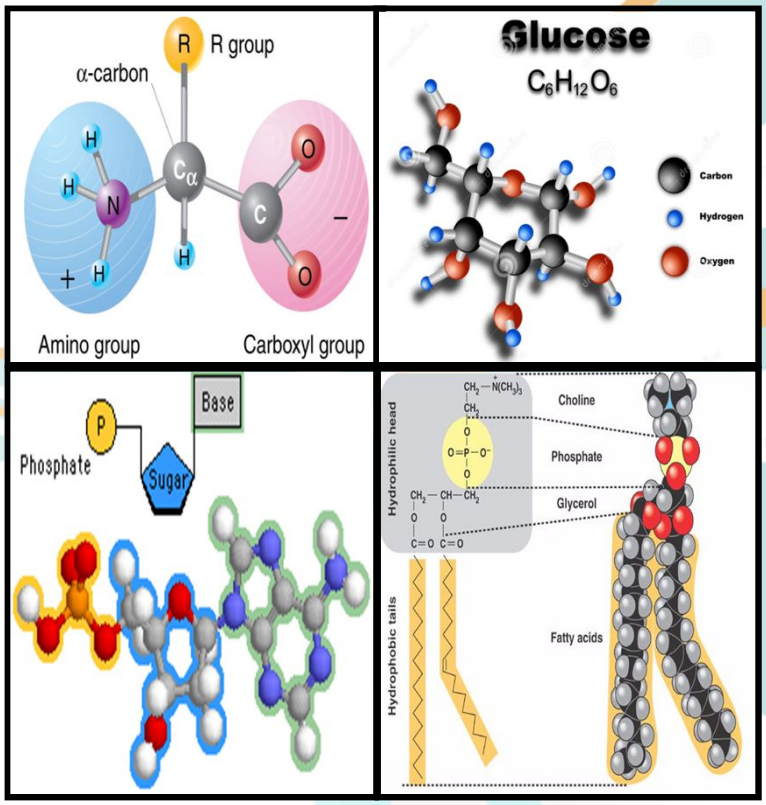
Molecules of Life

Composition of Life



Molecules of Life

Building Blocks of Life



Molecules of Life

Polymers & Monomers

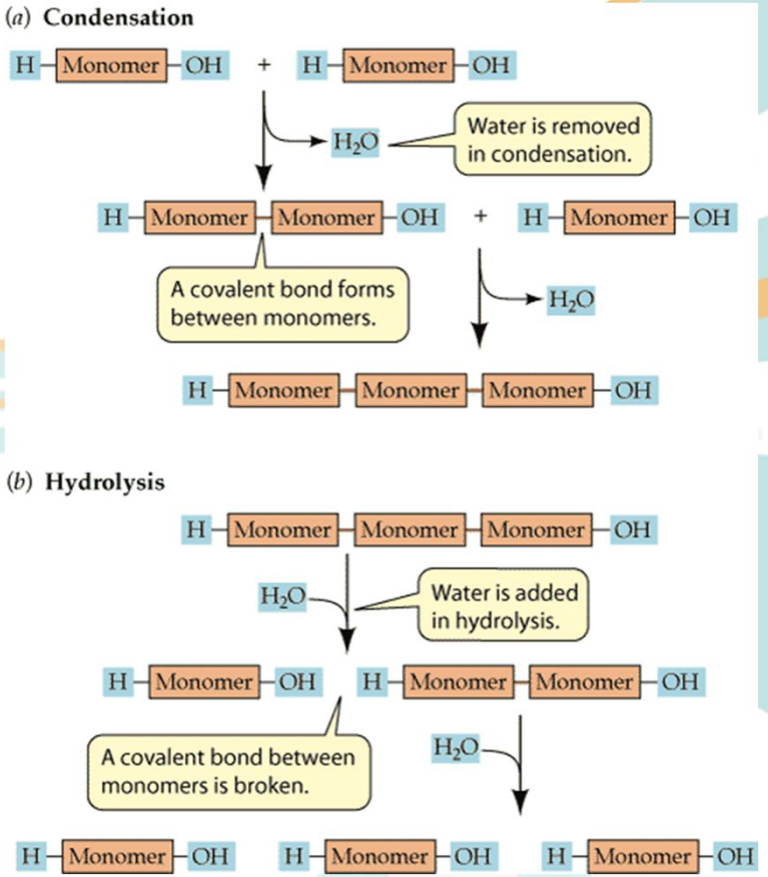
Macromolecules are polymers of smaller monomers molecules

Condensation reactions: Monomers join H_2O Removal

Hydrolysis reactions: Use water to break polymers into monomers.

Molecules of Life

Condensation & Hydrolysis



Proteins: Polymers of Amino Acids

Proteins: Polymers of Amino Acids

Protein Functions

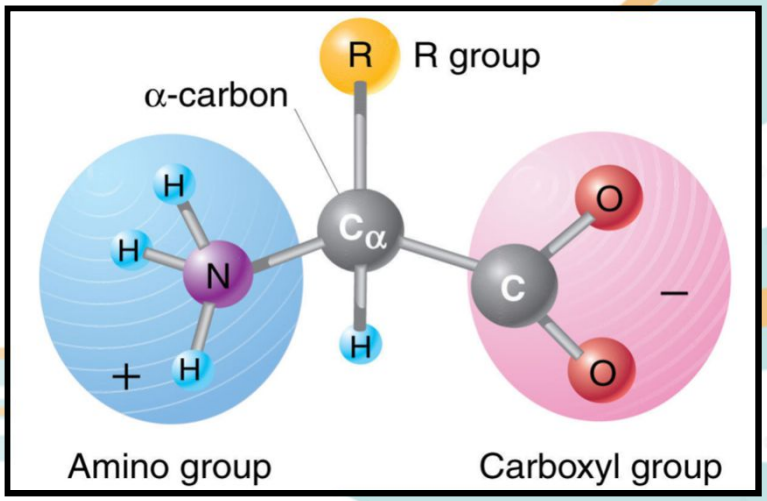
Most Abundant Polymer in the cell

Support, protection, catalysis, transport, defense, regulation & movement

Do not store genetic information

Proteins: Polymers of Amino Acids

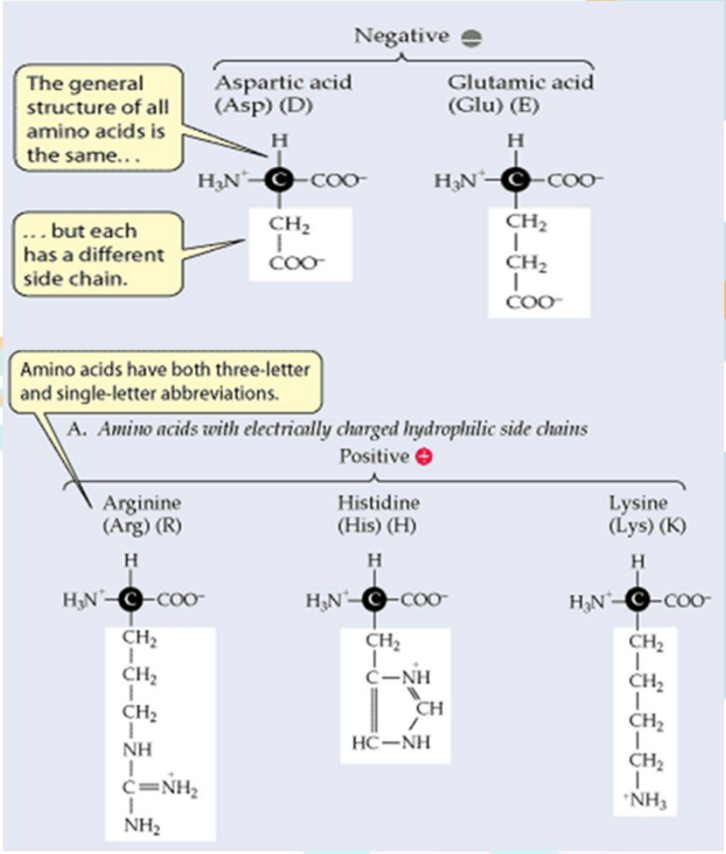
Amino Acid Structure



The side chains or R groups give different properties to each of the amino acids

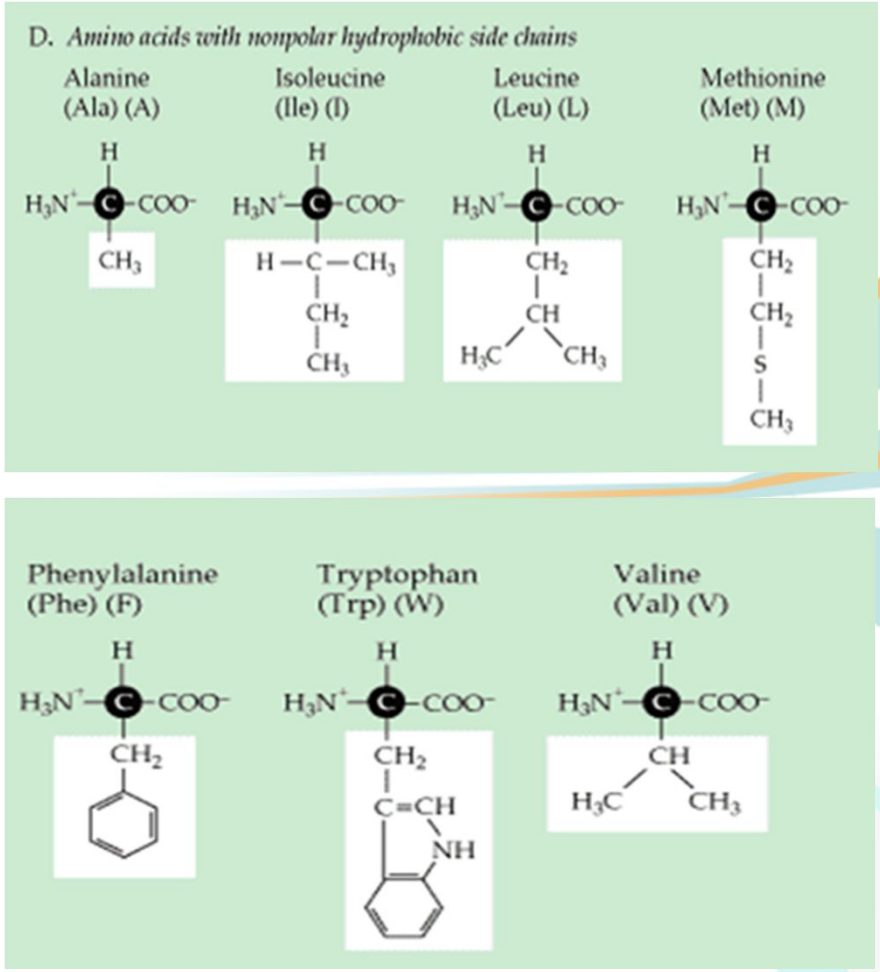
Proteins: Polymers of Amino Acids

Charged R Groups



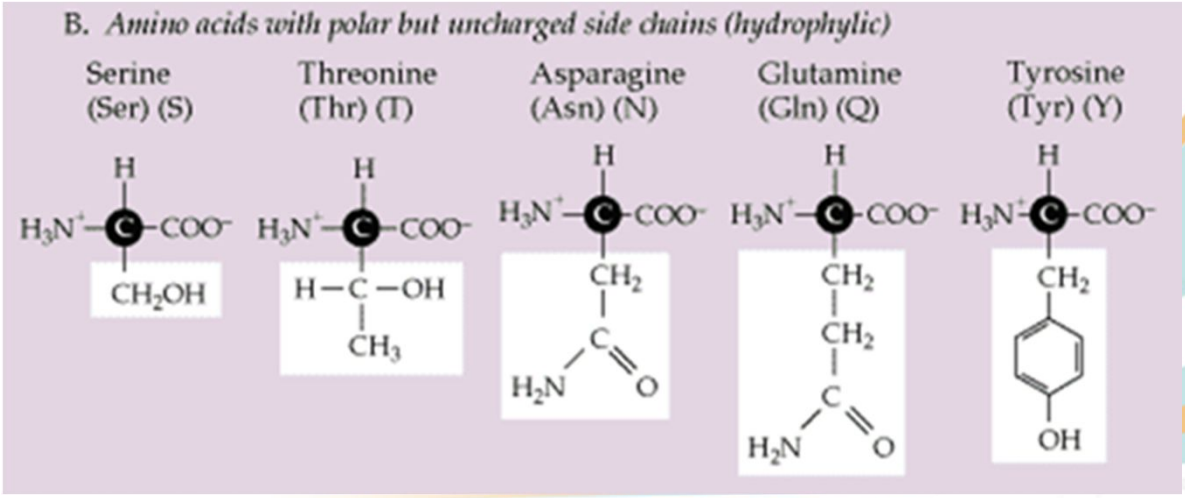
Proteins: Polymers of Amino Acids

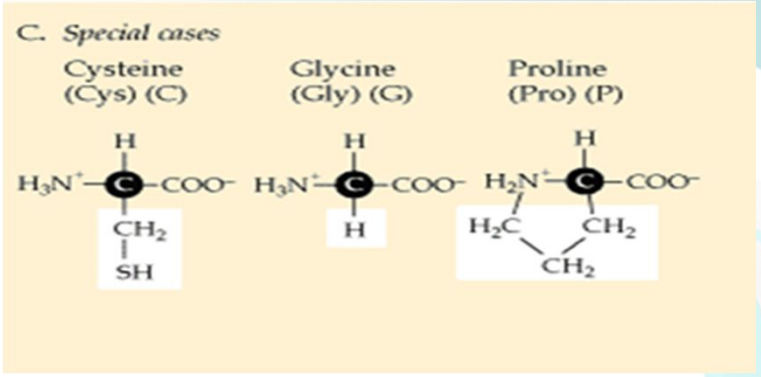
Nonpolar R Groups



Proteins: Polymers of Amino Acids

Polar & Special R Groups



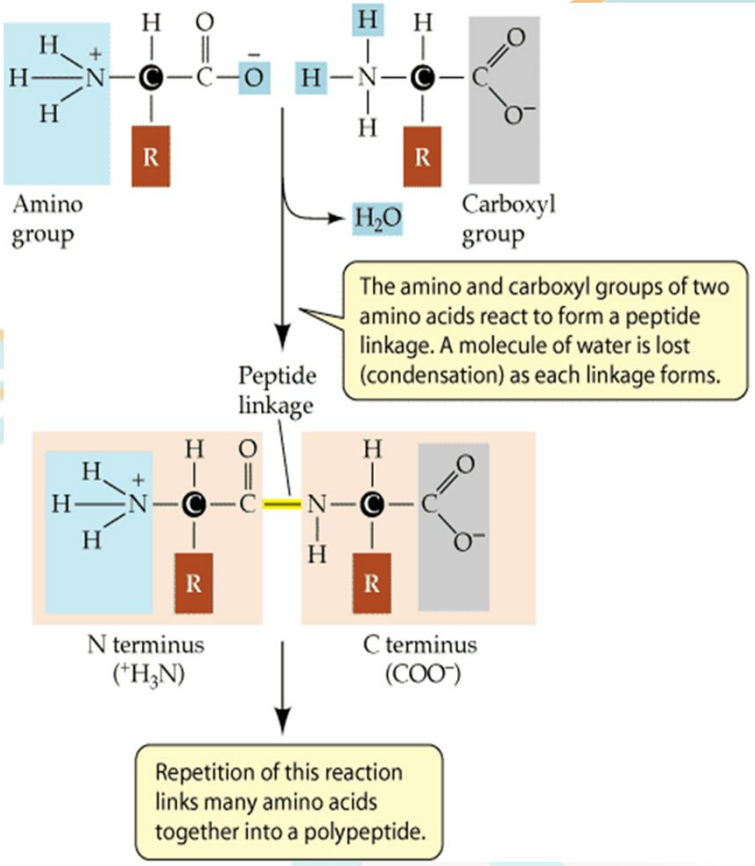


CELL BIOLOGY

Proteins: Structure

Proteins: Polymers of Amino Acids

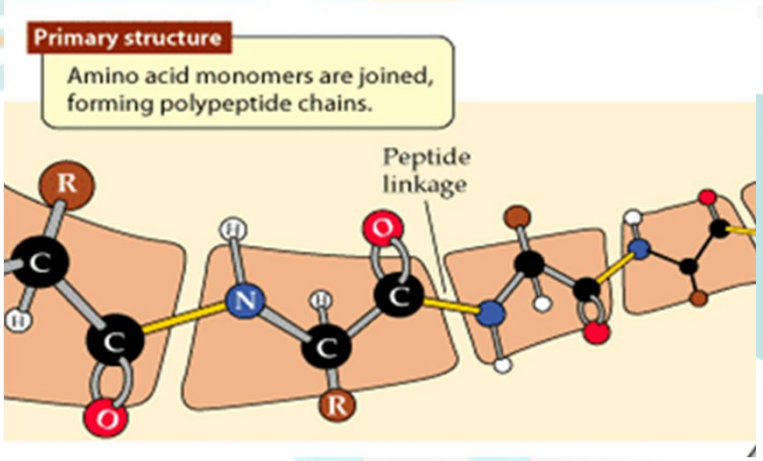
The Peptide Bond



Proteins: Structure

Primary Structure

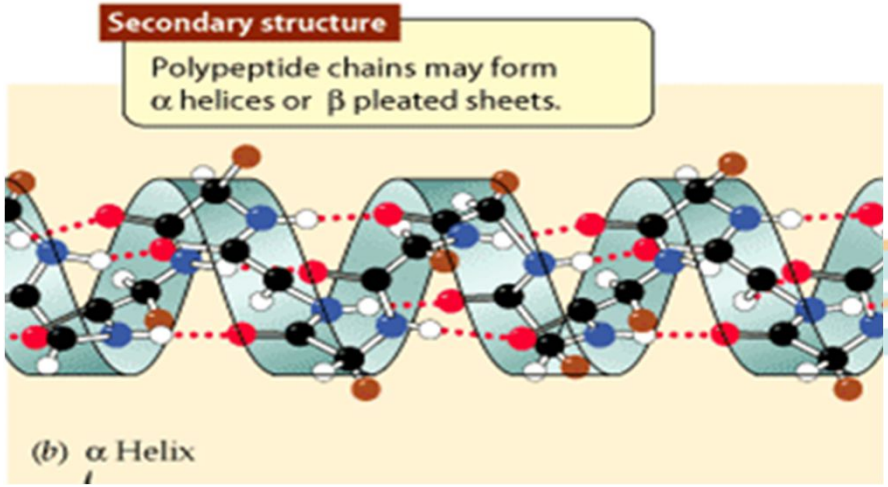
Sequence of amino acids bonded by peptide linkages (Diversity 20n)

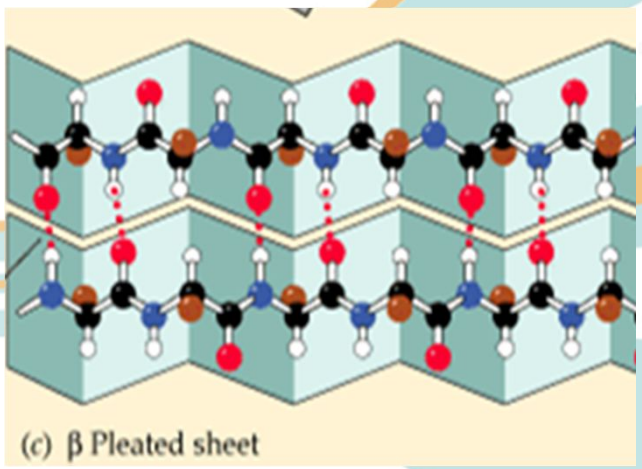


Proteins: Structure

Secondary Structure

α helices (hydrogen bonds between amino acid residues)

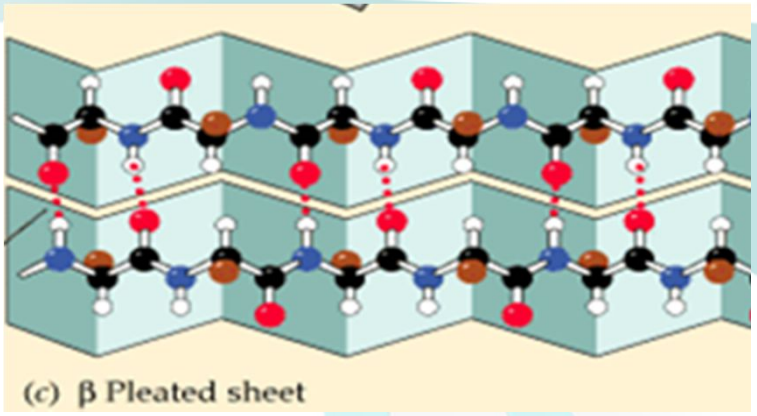




Proteins: Structure

Secondary Structure

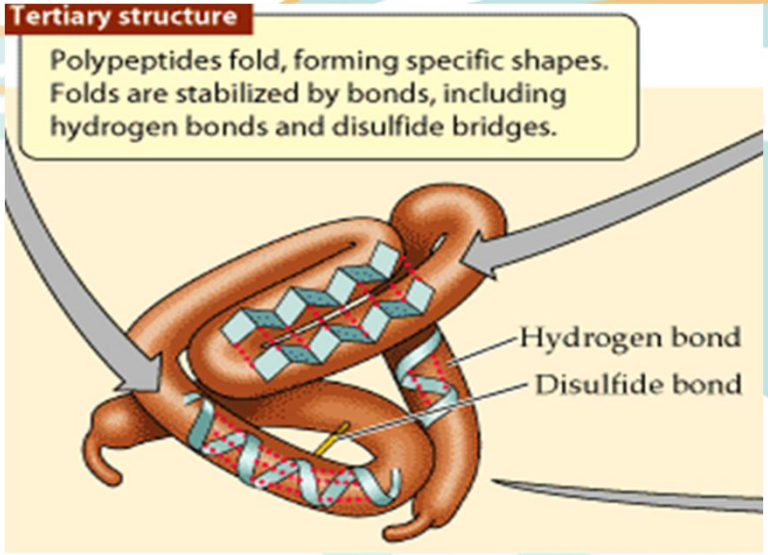
β pleated sheets (hydrogen bonds between amino acid residues)



Proteins: Structure

Tertiary Structure

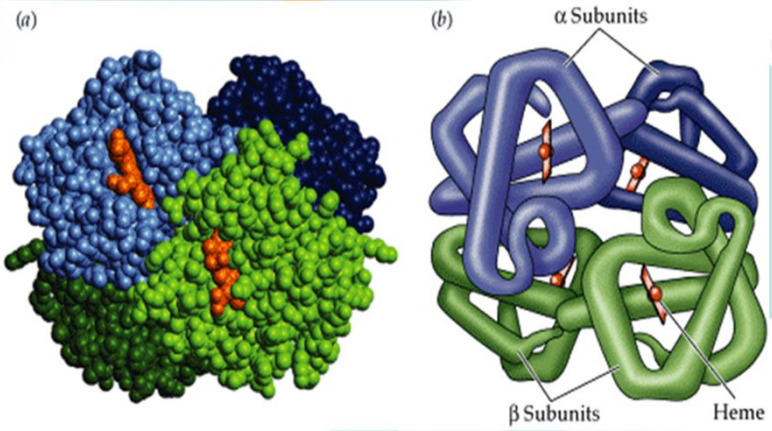
Generated by bending and folding of the polypeptide chain



Proteins: Structure

Quaternary structure

Arrangement of multiple folded protein molecules in a multi-subunit complex.



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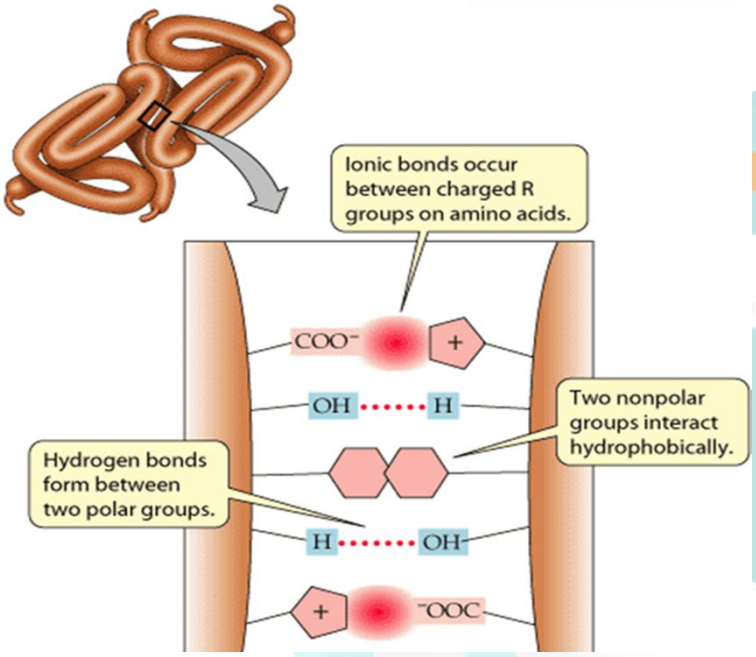
Protein Interactions

Protein Interactions

Weak interactions

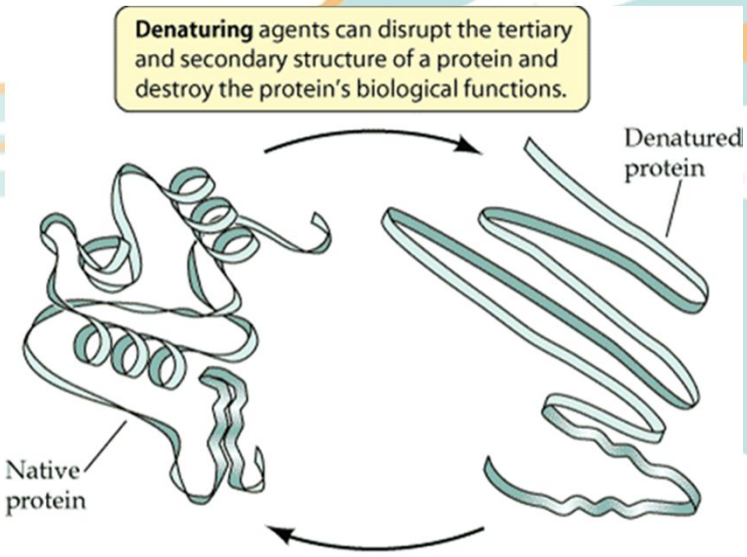
3D structure



Protein Interactions

Protein: Denaturation

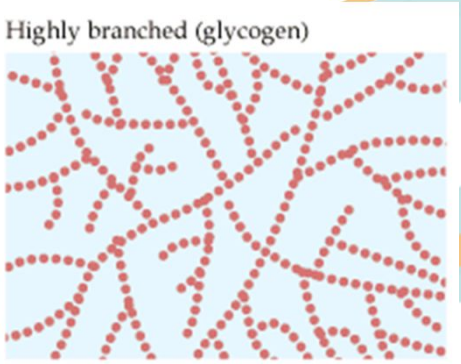
Heat, pH change: Tertiary and secondary structure as well as biological function.

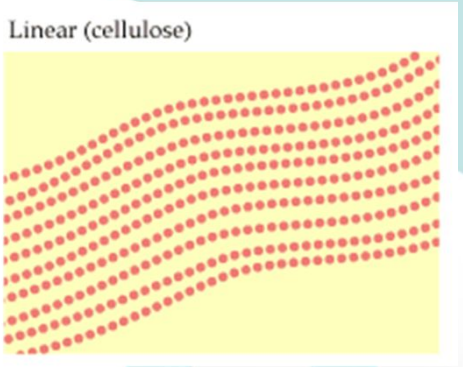


Glycosidic linkages

Carbohydrates

Disaccharide α or β orientation

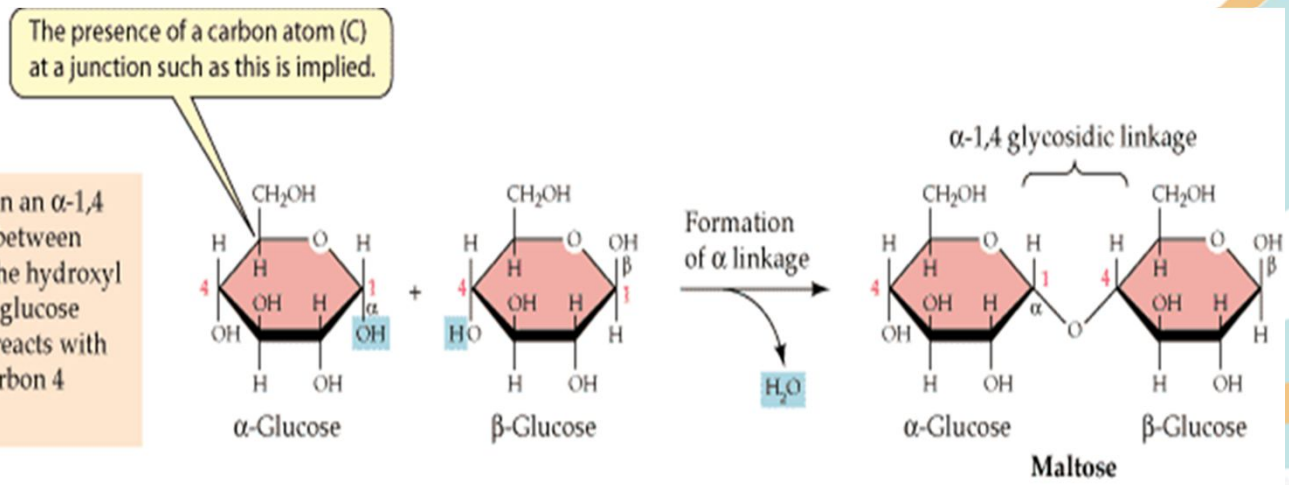




Glycosidic linkages

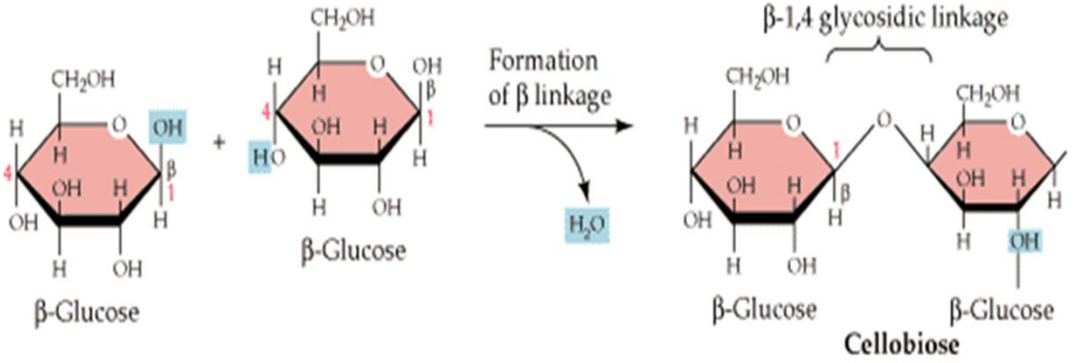
Disaccharide α or β orientation

Maltose is produced when an α-1,4 glycosidic linkage forms between two glucose molecules. The hydroxyl group on carbon 1 of one glucose in the α (down) position reacts with the hydroxyl group on carbon 4 of the other glucose.



Digestible

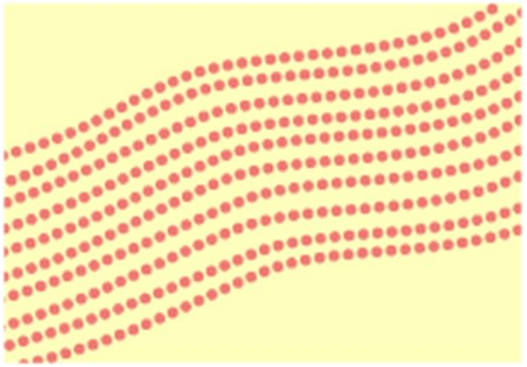
In **cellobiose**, two glucoses are linked by a β-1,4 glycosidic linkage.



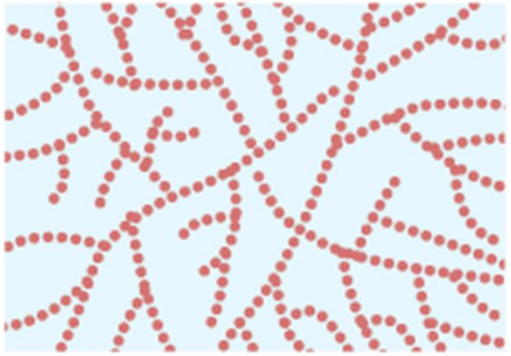
Glycosidic linkages

Cellulose & Glycogen

Linear (cellulose)

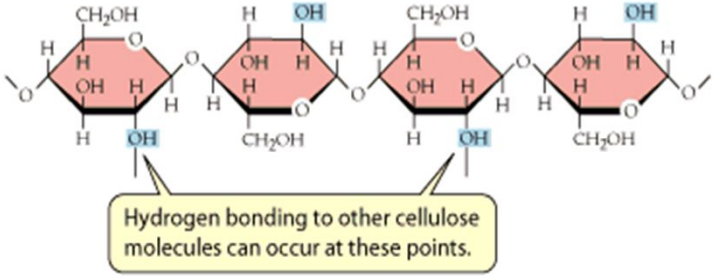


Highly branched (glycogen)



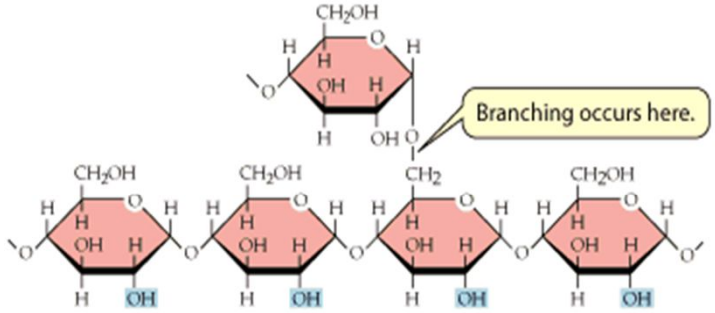
(a) Molecular structure

Cellulose



Cellulose is an unbranched polymer of glucose with β-1,4 glycosidic linkages that are chemically very stable.

Starch and glycogen



Glycogen and starch are polymers of glucose with α-1,4 glycosidic linkages. α-1,6 glycosidic linkages produce branching at carbon 6.

Glycosidic linkages

Cellulose & Starch

Lipids: Nonpolar Hydrocarbons

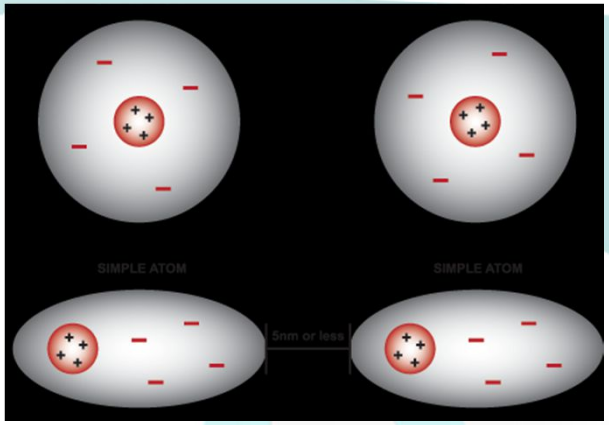
Lipids: Nonpolar Hydrocarbons

Week Bonds

Hydrophobic molecules

Water insoluble

Aggregate together (by hydrophobic and Van der Waals force)



Lipids: Nonpolar Hydrocarbons

Functions

Store energy

Phospholipids membranes

Carotenoids & Light

Hormones and vitamins

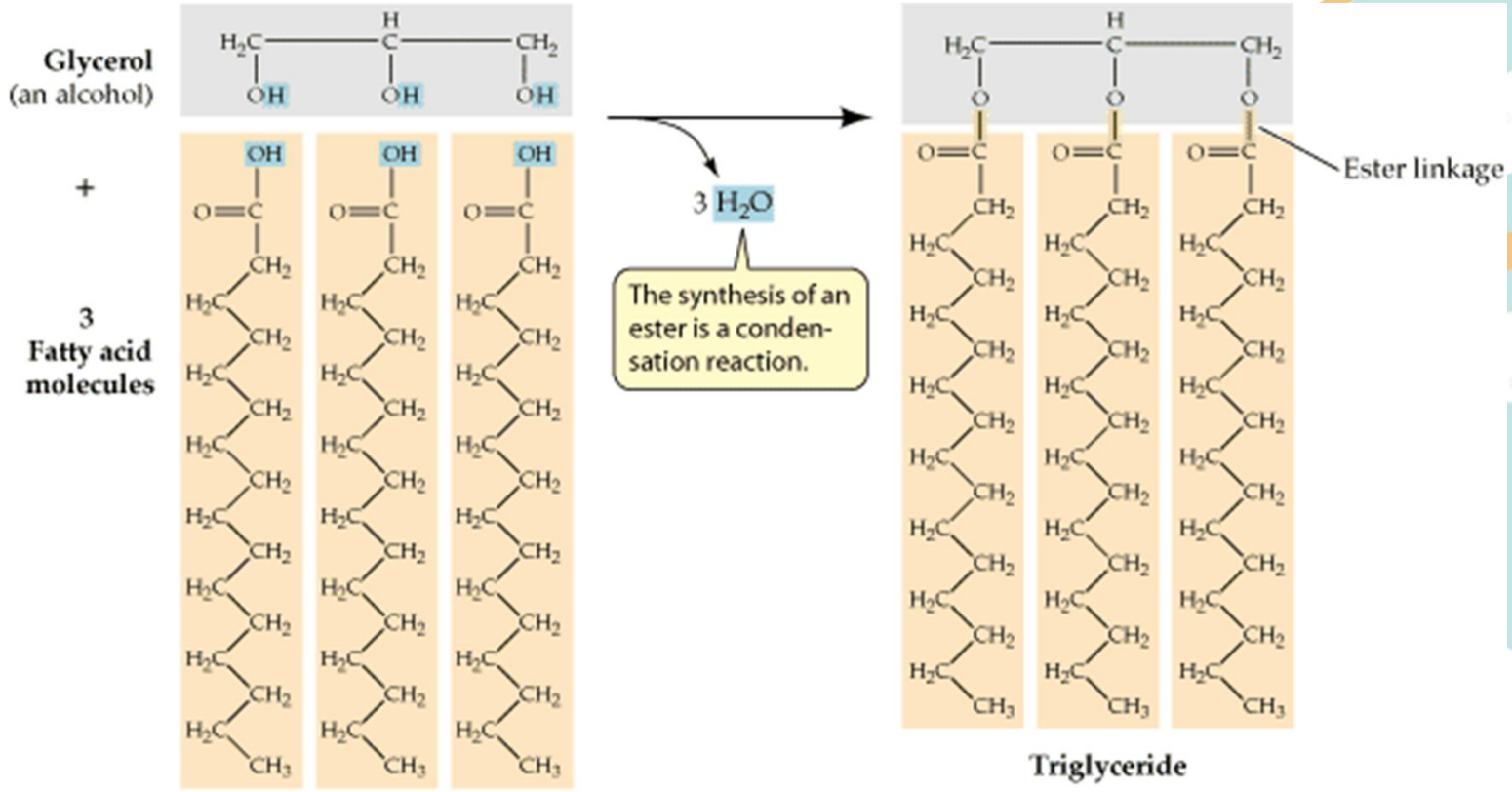
Animal fat is insulator

Insulation of nerves

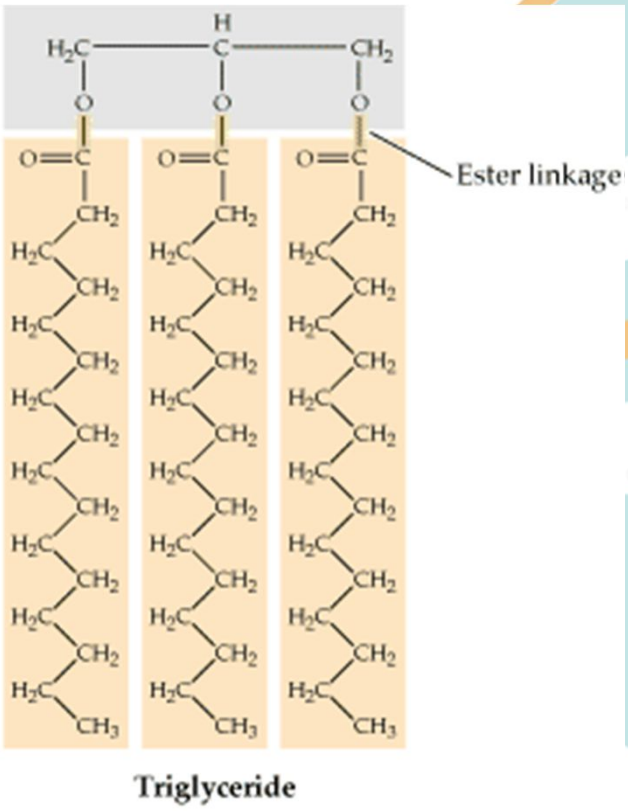
Water repellant for skin, fur and feathers

Lipids: Synthesis

Fats & oils: Triglycerides: (covalent ester linkage vs Hydrophobic bond)



Lipids: Synthesis

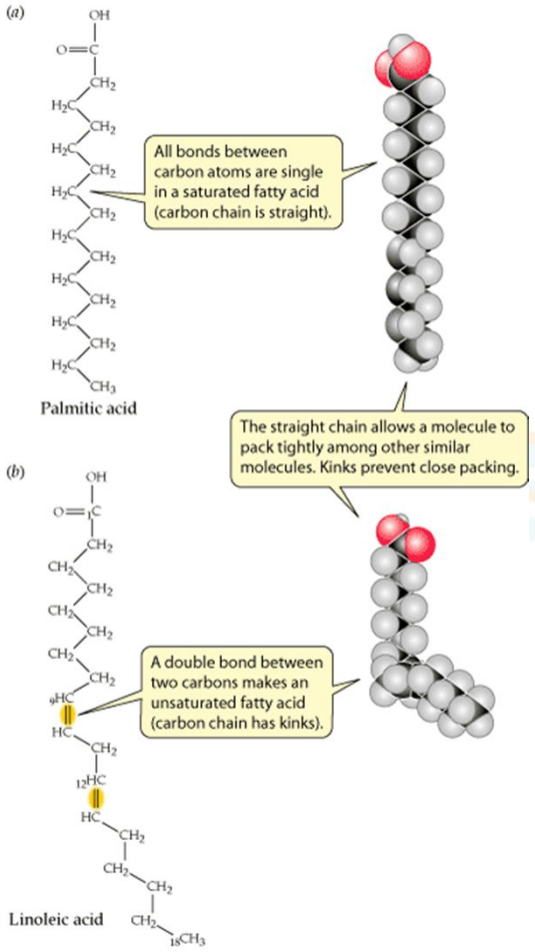


CELL BIOLOGY

Effort By: AARISH ALI

Phospholipids

Lipids: Saturated & unsaturated



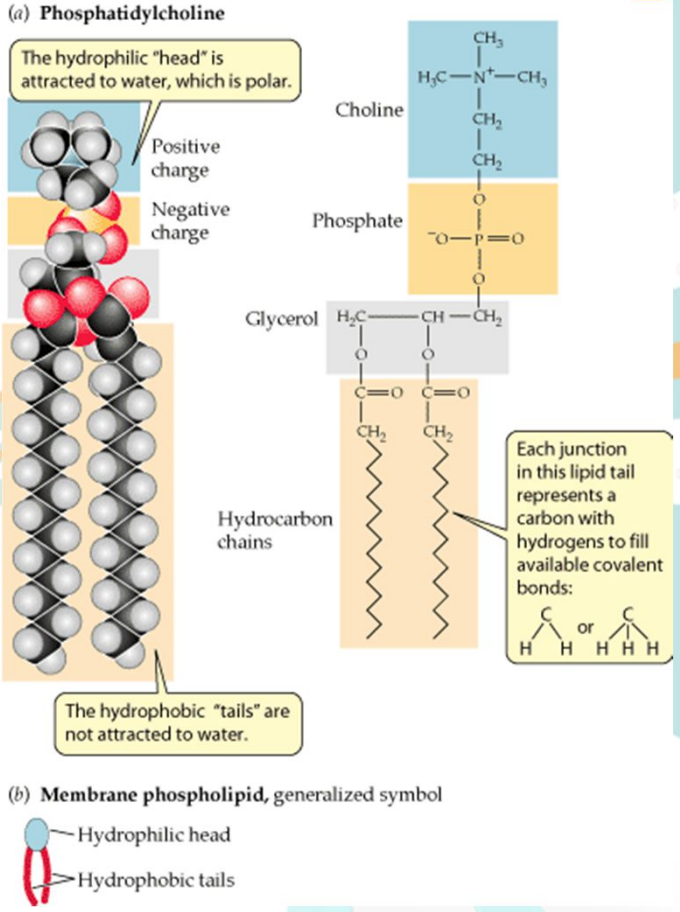
Fats (L) & Oils (S)

Saturated fatty acids have a hydrocarbon chain with no double bonds.

The hydrocarbon chains of unsaturated fatty acids have one or more double bonds that bend the chain, making close packing less possible.

Phospholipids

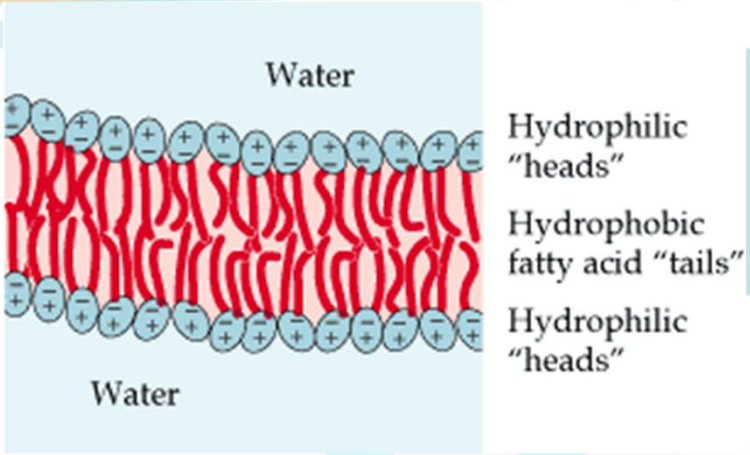
Amphiphatic Molecule



P-lipids: Aqueous Environment

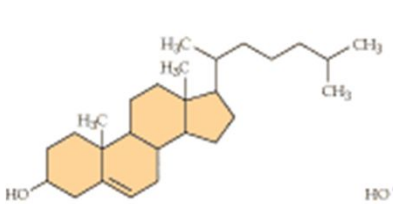
The Bilayer

Two molecules thick head groups directed outward, The tails are packed together in the interior of the bilayer.

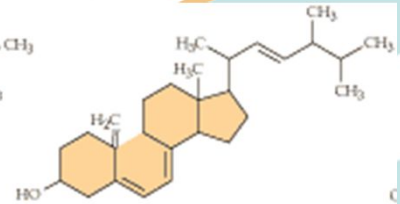


Lipids: Cholesterol

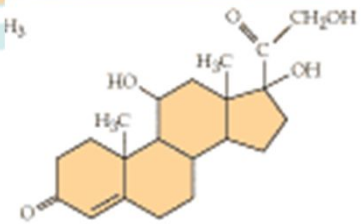
Vitamins & Signals



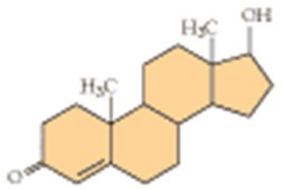
Cholesterol is a constituent of membranes and is the source of steroid hormones.



Vitamin D_2 can be produced in the skin by the action of light on a cholesterol derivative.



Cortisol is a hormone secreted by the adrenal glands.



Testosterone is a male sex hormone.

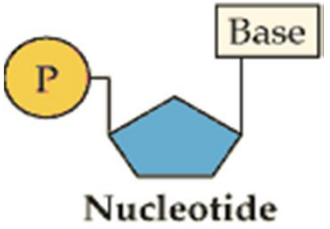
END

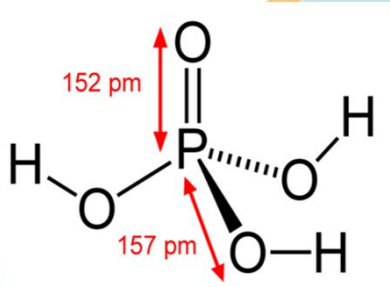
CELL BIOLOGY

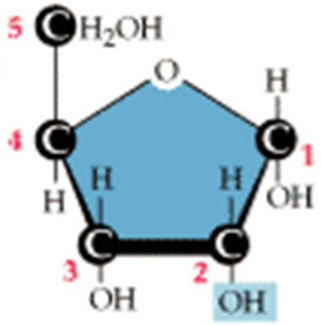
Nucleic Acids

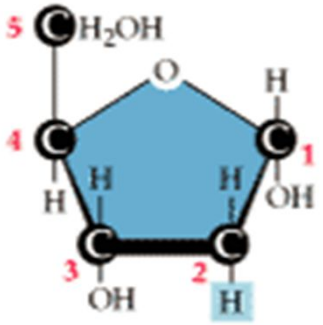
Nucleic Acids

DNA & RNA



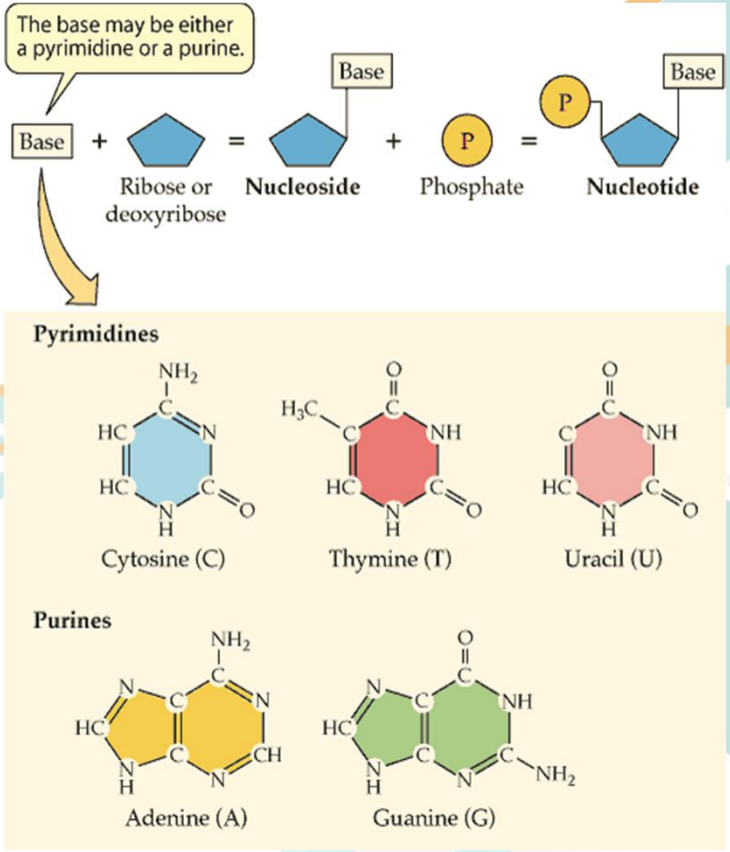






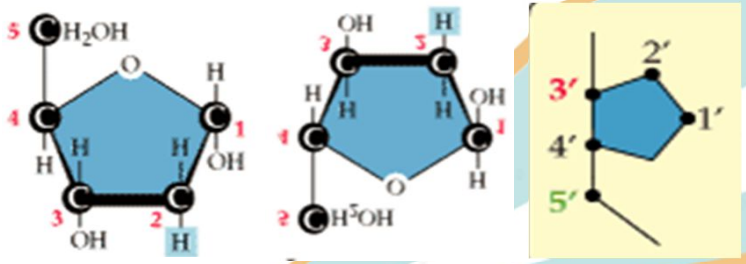
Nucleic Acids

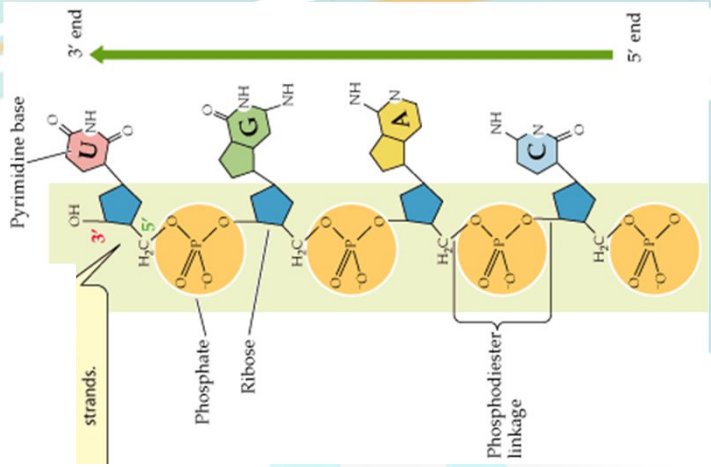
The Bases



Nucleic Acids

Prime Ends (3′& 5′)





END

CELL BIOLOGY

RNA & DNA

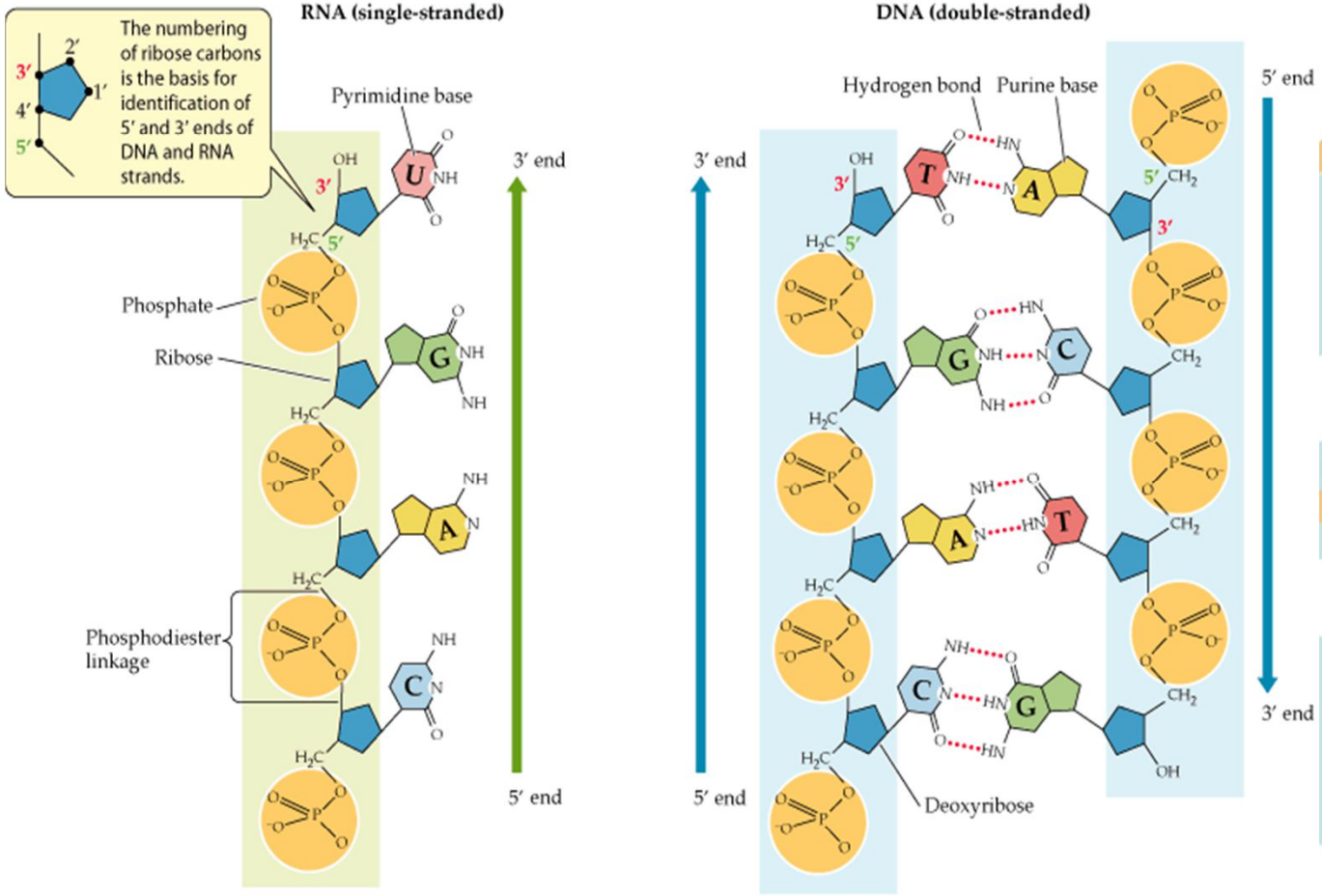
Nucleic Acids

RNA vs. DNA

3.3 Distinguishing RNA from DNA

NUCLEIC ACID	SUGAR	BASES
RNA	Ribose	Adenine Cytosine Guanine Uracil
DNA	Deoxyribose	Adenine Cytosine Guanine Thymine

Nucleic Acids



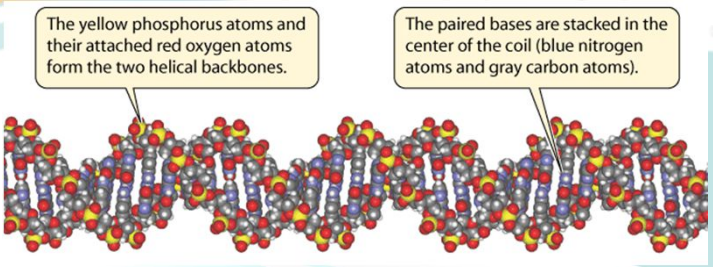
In RNA, the bases are attached to ribose. The bases in RNA are the purines adenine (A) and guanine (G) and the pyrimidines cytosine (C) and uracil (U).

In DNA, the bases are attached to deoxyribose, and the base thymine (T) is found instead of uracil. Hydrogen bonds between purines and pyrimidines hold the two strands of DNA together.

Nucleic Acids

DNA Uniform Width

Information in Sequence not Shape



Chicken vs. Egg: What Came 1st

Protein or DNA

Genetic information codes for enzymes

Enzymes are required to decipher genetic information and replicate it

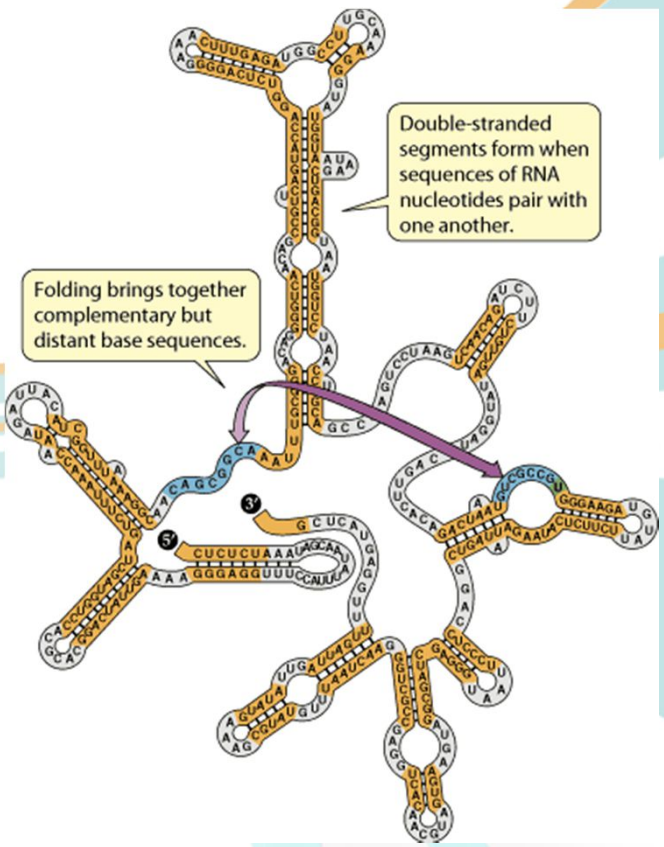
RNA Genetic Material & Enzyme

Many viruses use RNA as their hereditary material

RNAs can achieve chemical catalysis, like enzymes: ribosome the active site is composed entirely of RNA (peptide bond formation)

RNA: Information in Sequence and Shape

Why Then DNA?



END

CELL BIOLOGY

Unit of Life

Unit of Life

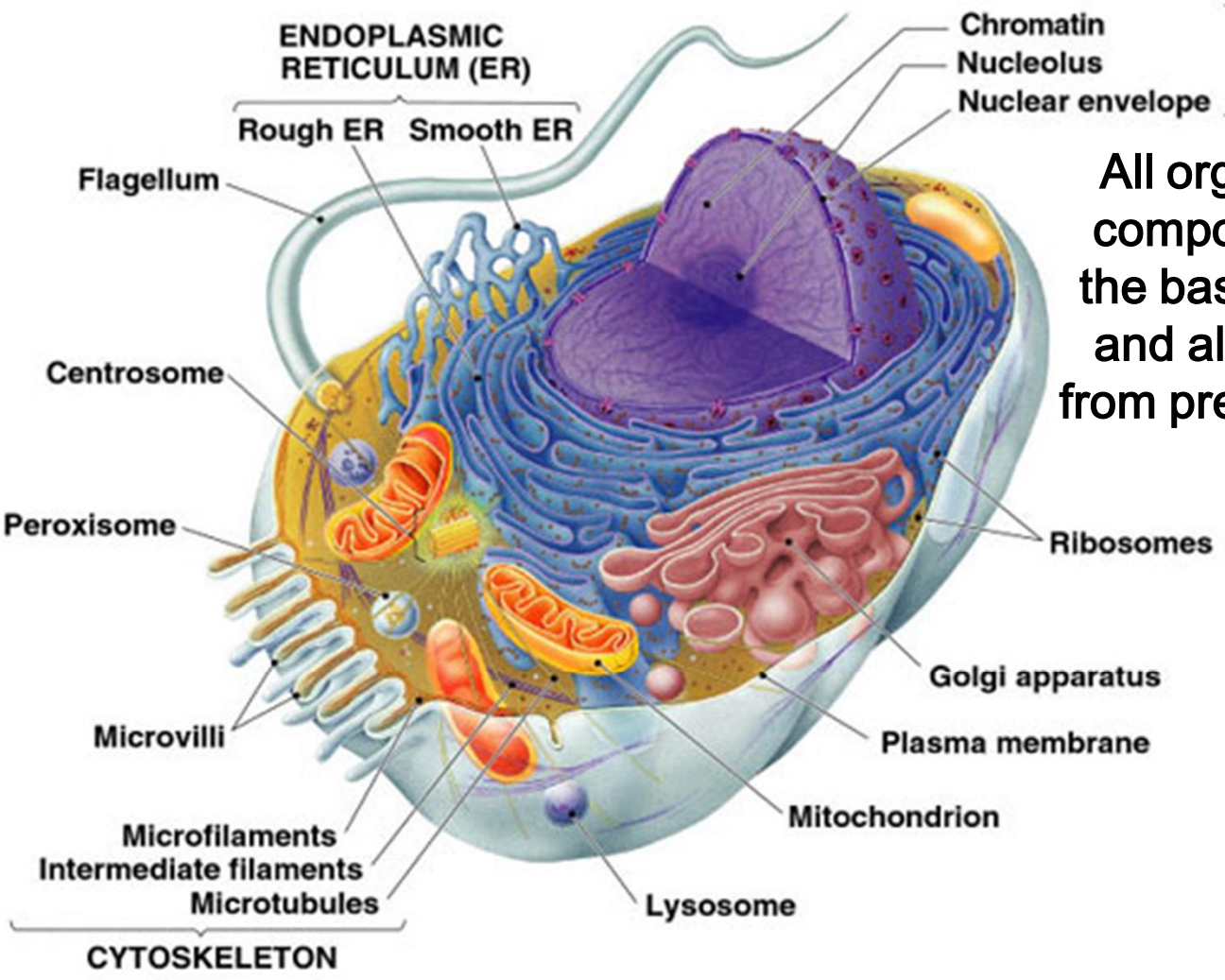
Cell Structure

Cellular compartments

Organelles

Structure & Function

Unit of Life



All organisms are composed of cells the basic unit of life and all cells come from preexisting cells

Not a random collection of molecules

Life

collection of macromolecules that can perform unique functions because the are enclosed in structural compartment that provides consistency (homeostasis).

CELL BIOLOGY

Journey into the Cell

Unit of Life

View animation

END

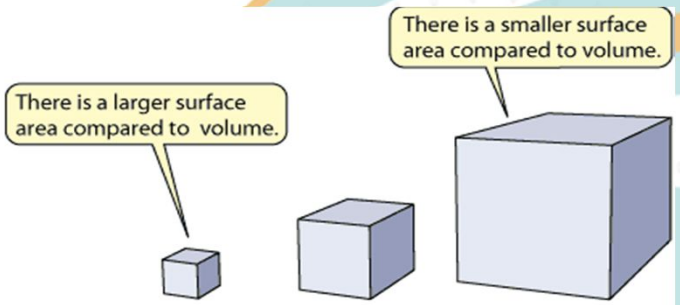
CELL BIOLOGY

Size Matters

Size Matters

Cell Structure

Cells are small to maintain large surface area to volume ratio



	1-mm cube	2-mm cube	4-mm cube
Surface area	6 sides × 1^2 = 6 mm^2	6 sides × 2^2 = 24 mm^2	6 sides × 4^2 = 96 mm^2
Volume	1^3 = 1 mm^3	2^3 = 8 mm^3	4^3 = 64 mm^3
Surface area-to-volume ratio	6/1	3/1	1.5/1

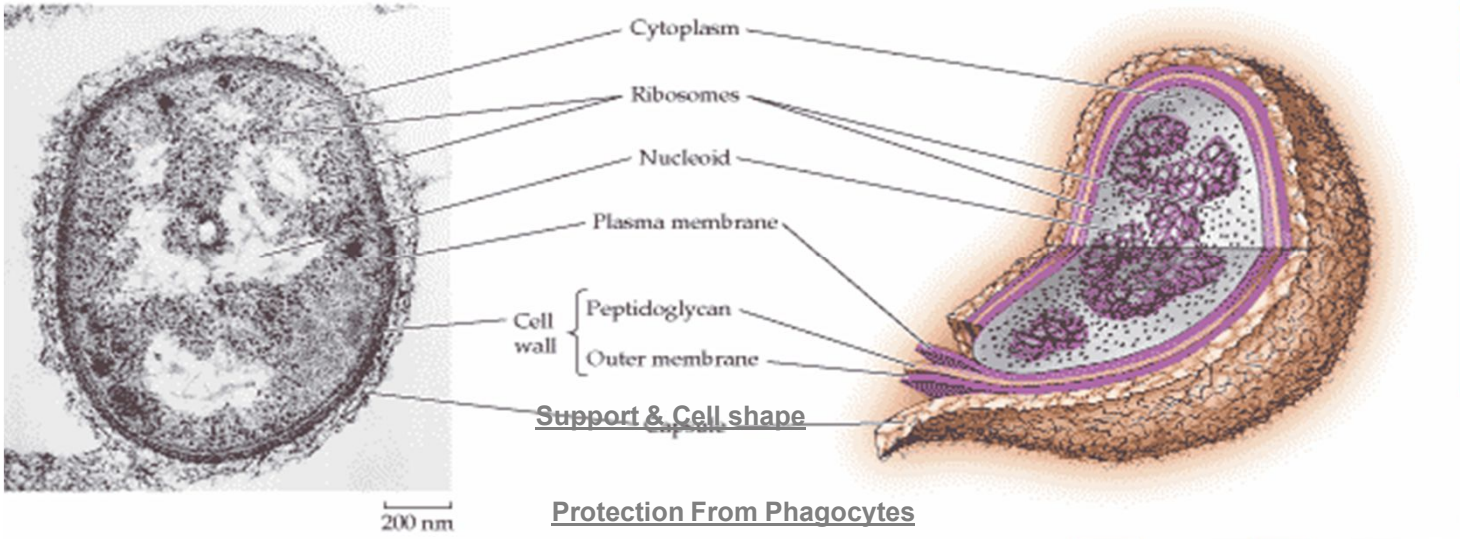
Prokaryotic Cells

No membrane enclosed internal compartments.

Plasma membrane regulates traffic (barrier).

Nucleoid region contains DNA.

Most have cell wall.



Support & Cell shape

Protection From Phagocytes

Prokaryotic Cells

Special Features

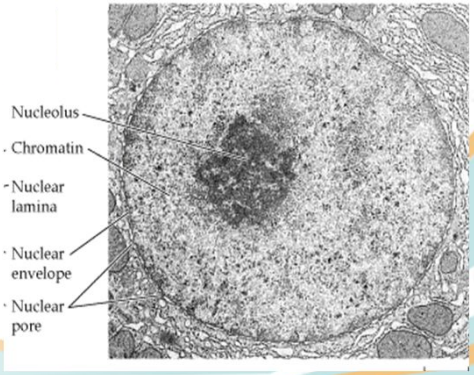
Cyanobacteria Chlorophyll containing have folds of plasma membrane, other have mesosomes (energy).

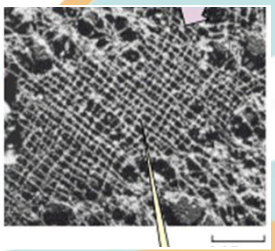
Some have actin like filaments and other have Flagella made-up of Flagellin.

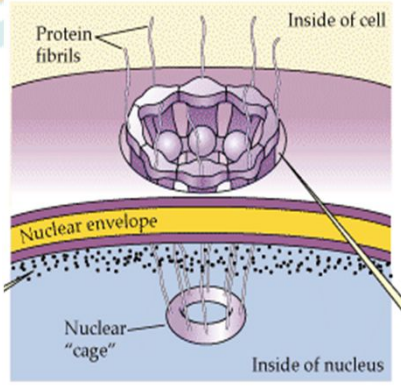
END

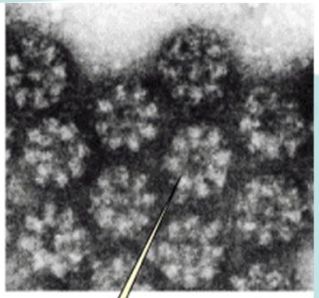
The Nucleus

Nuclear EMs









An octagon of protein complexes surrounds each **nuclear pore**. Protein fibrils on the nuclear side form a cagelike structure.

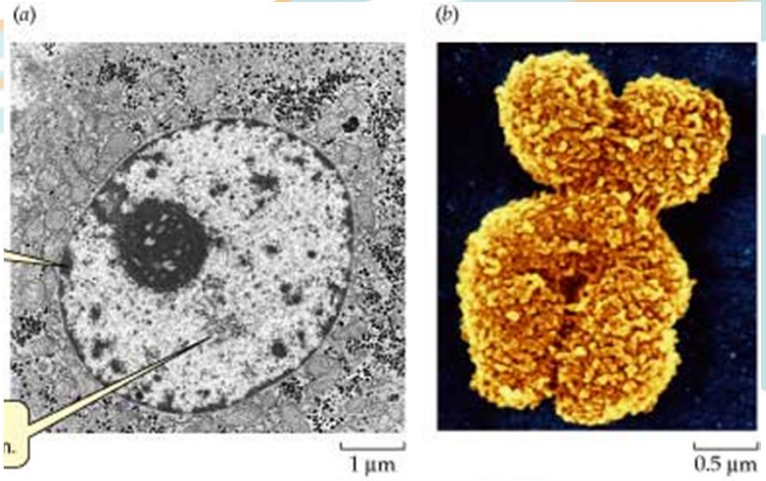
The Nucleus

Chromatin

Euchromatin

Hetrochromatin

Chromosome



END

CELL BIOLOGY

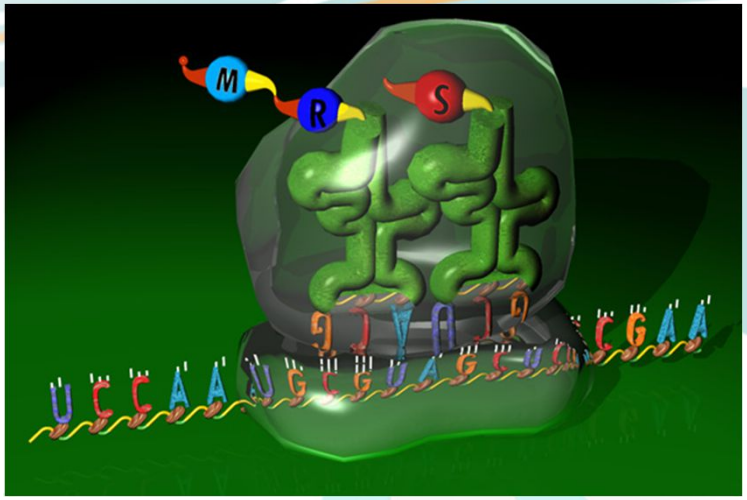
Ribosome

Ribosome

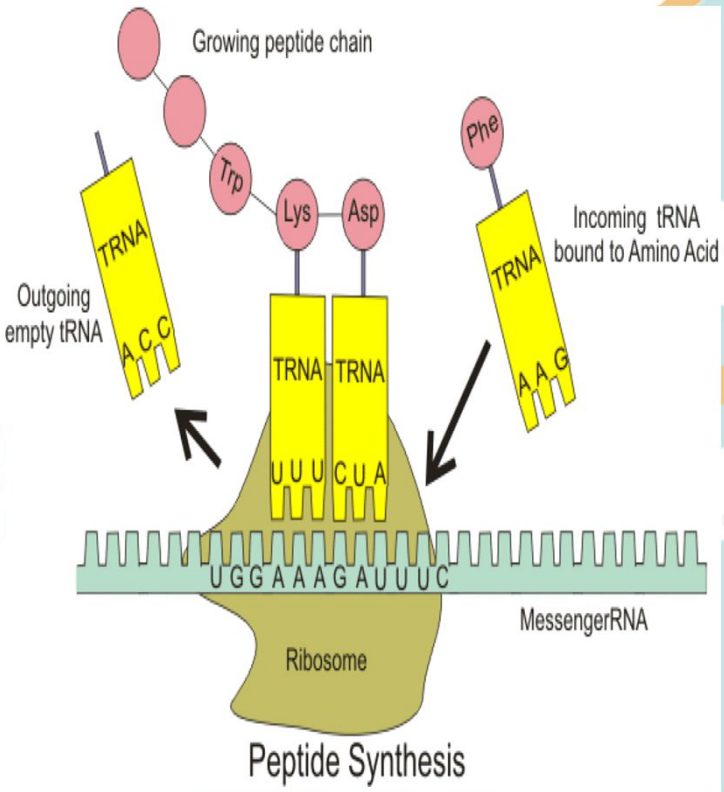
Protein & RNA

Free or attached to ER involved in protein synthesis.

Mitochondria & Chloroplasts.



Ribosome

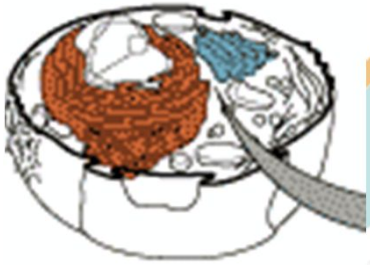


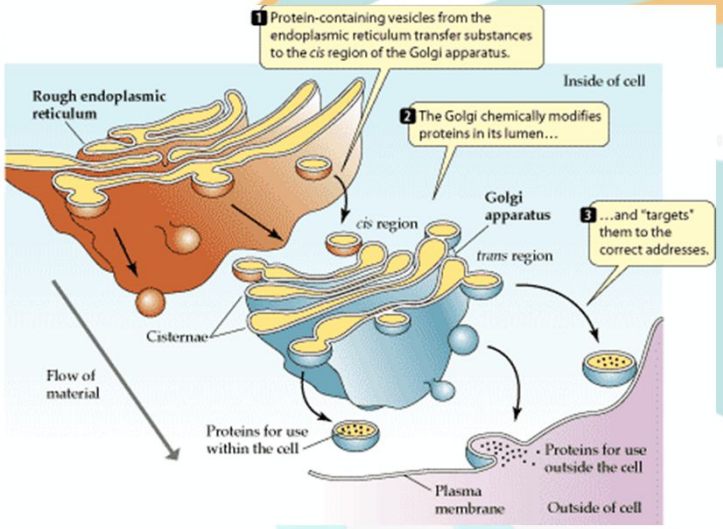
END

The Endoplasmic Reticulum

The Endoplasmic Reticulum

RER + SER

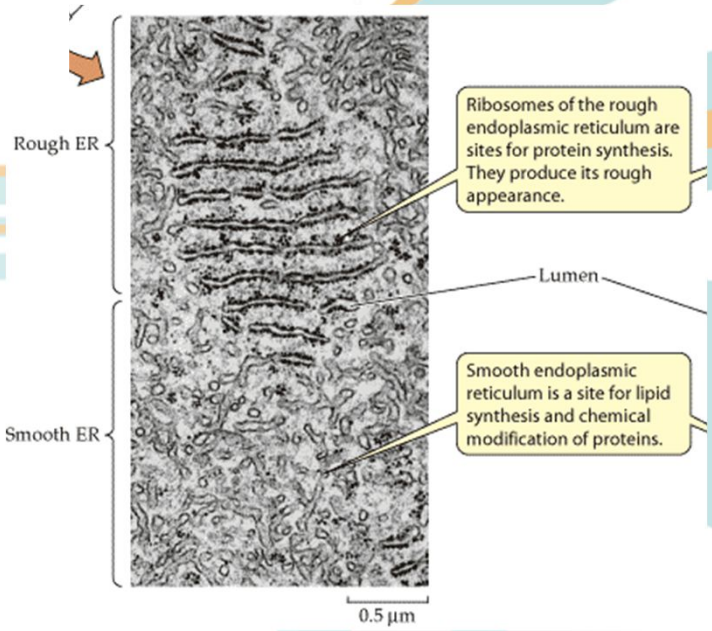




The Endoplasmic Reticulum

RER

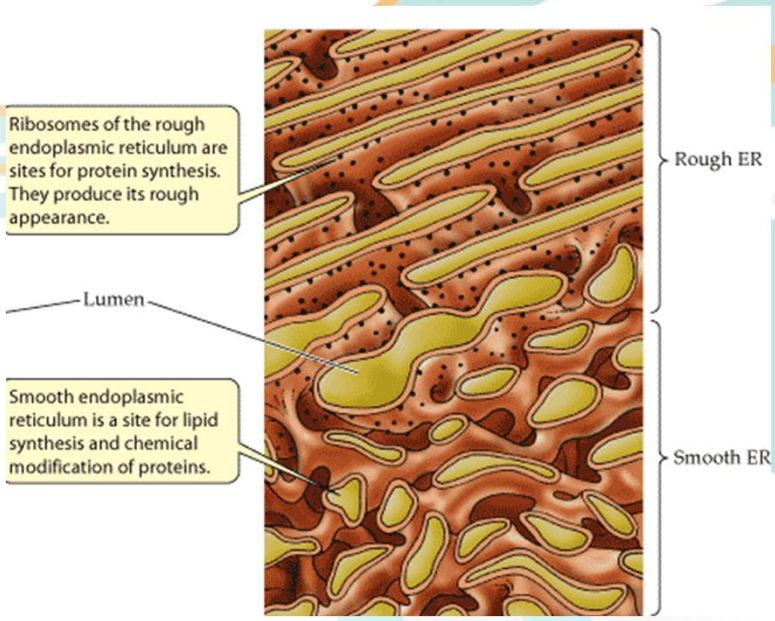
Segregates newly synthesized Proteins



The Endoplasmic Reticulum

RER

Chemically modifies proteins (Glycosylation address)

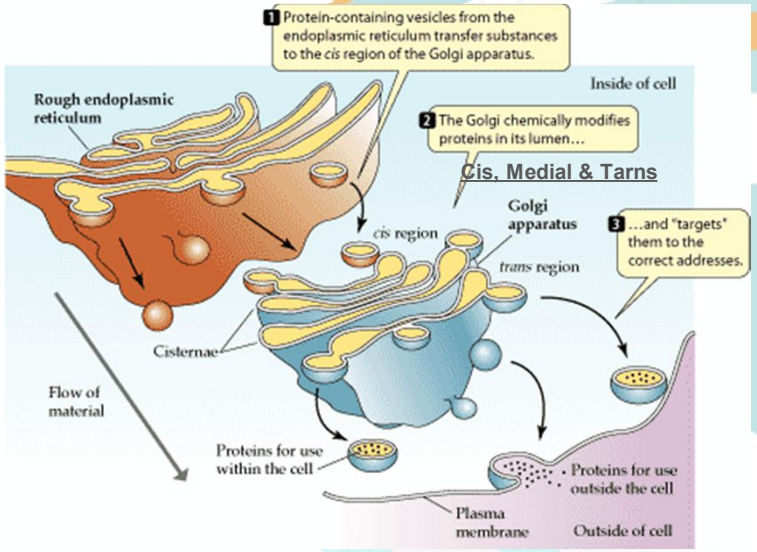


The Endoplasmic Reticulum

SER

Glycogen hydrolysis

Secreting cells have abundant SER



END

CELL BIOLOGY

Golgi & Lysosomes

Golgi

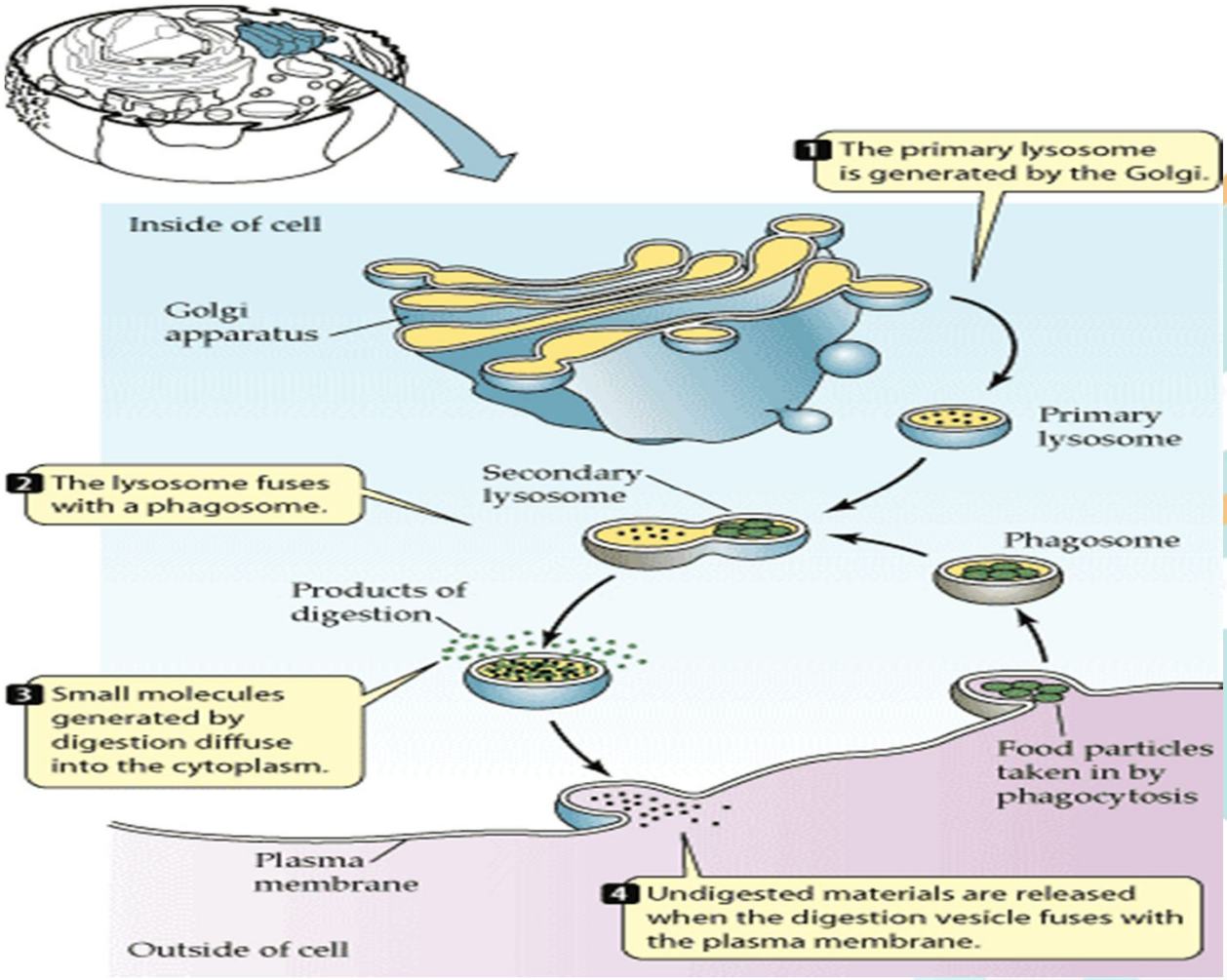
Golgi Apparatus

Receives materials from the rough ER and modifies them.

Concentrates sorts and packages proteins and sends them to other destinations.

Manufactures poly-saccharides for the plant wall

Golgi


1 The primary lysosome is generated by the Golgi.
Inside of cell
Golgi apparatus
Primary lysosome
2 The lysosome fuses with a phagosome.
Secondary lysosome
Phagosome
Products of digestion
3 Small molecules generated by digestion diffuse into the cytoplasm.
Food particles taken in by phagocytosis
Plasma membrane
4 Undigested materials are released when the digestion vesicle fuses with the plasma membrane.
Outside of cell

Lysosomes

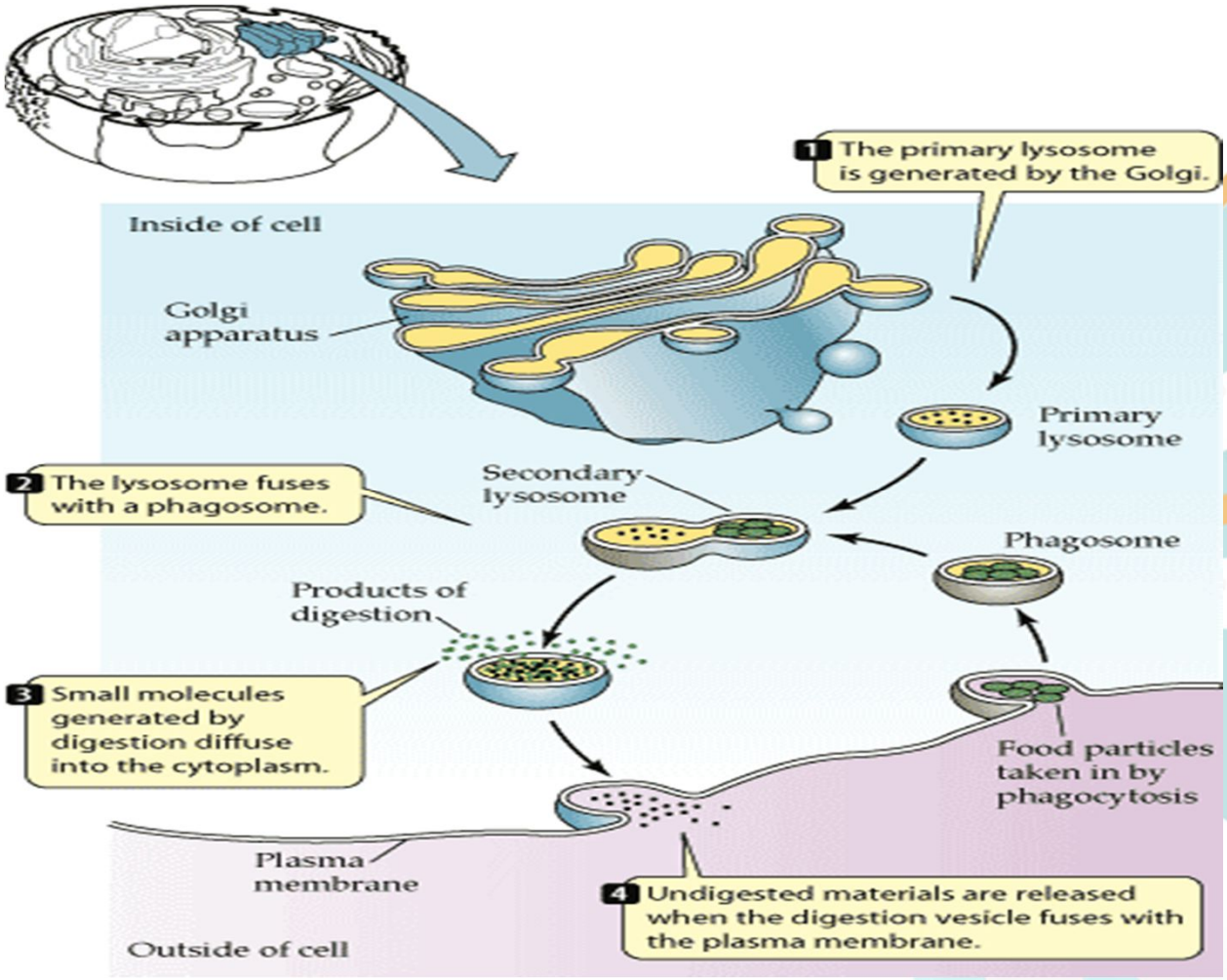
Garbage Disposal

Contain many digestive enzymes.

Fuse with phagosomes (autophagy).

Secondary lysosomes, where engulfed materials are digested phagocytosis.

Lysosomes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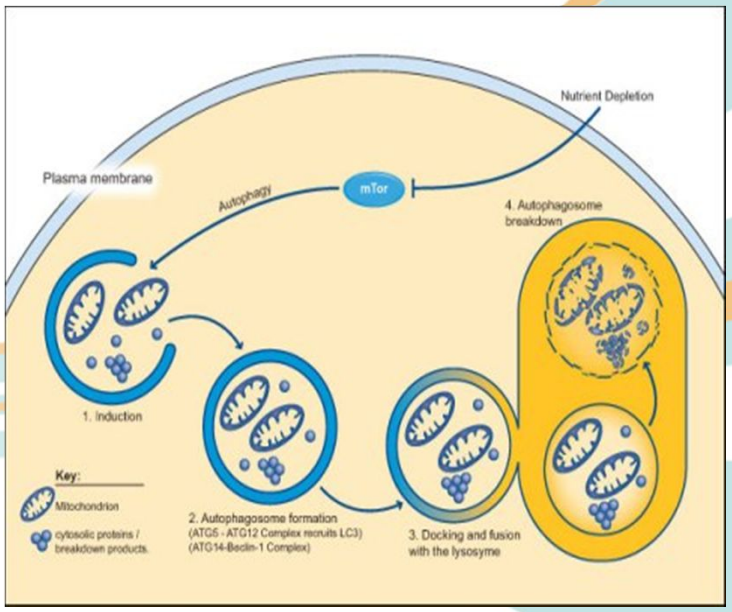
Outside of cell

Lysosomes



END

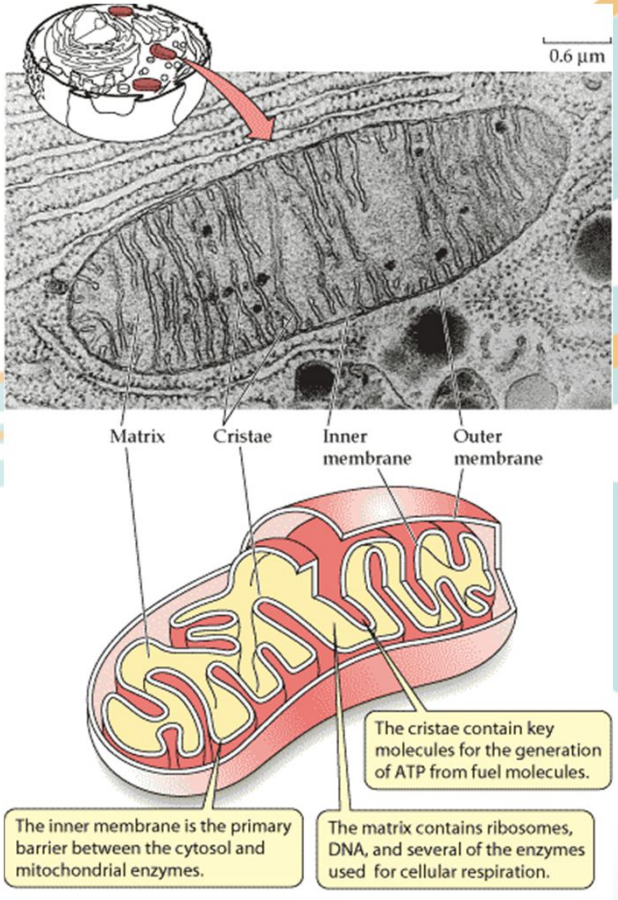
CELL BIOLOGY

Mitochondria

Mitochondria

Make ATP

Breakdown fuel



Mitochondria

Make ATP

Breakdown fuel

Some protists have one, liver more than a thousan and egg a few hundred thousand MT.

Mitochondria and chloroplasts contain their own DNA and ribosomes for making some of their own proteins

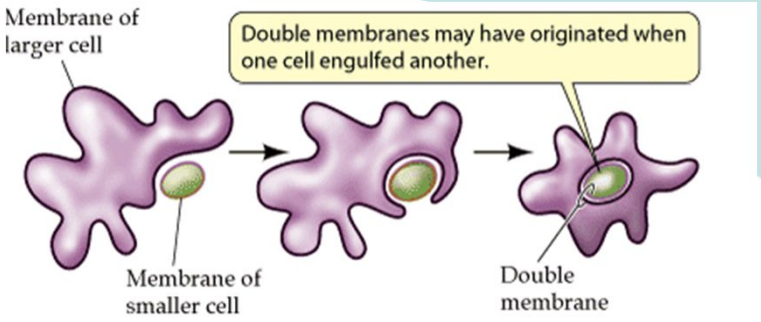
Mitochondria

Endosymbiosis

Mutual benefits permitted this symbiotic relationship

Photosynthesis

Detoxification of O_2



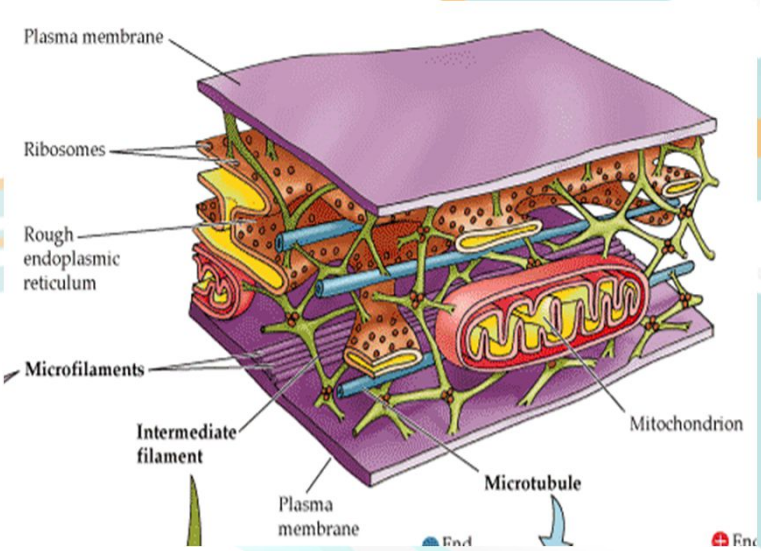
END

CELL BIOLOGY

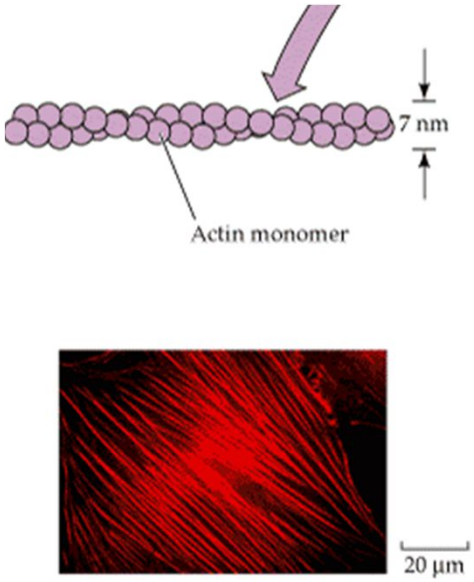
Cytoskeleton

Cytoskeleton

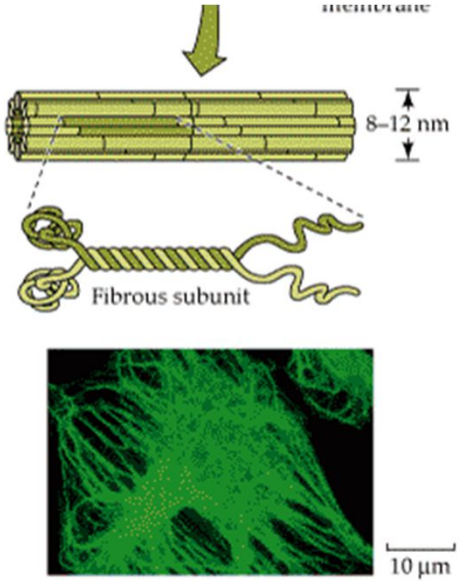
Shape & Function



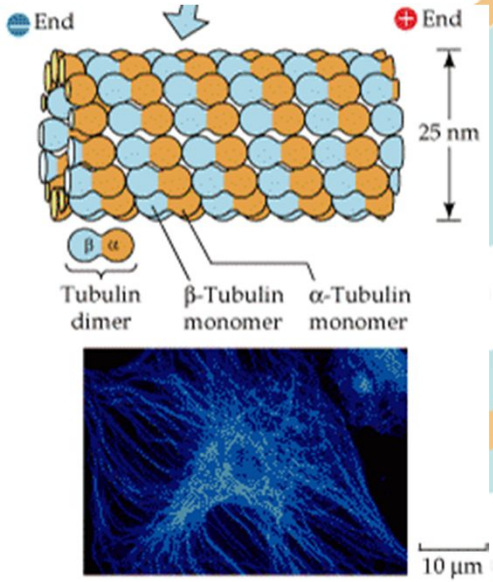
Cytoskeleton



Microfilaments are made up of strands of the protein actin and often interact with strands of other proteins. They change cell shape and drive cellular motion, including contraction, cytoplasmic streaming, and the "pinched" shape changes that occur during cell division. Microfilaments and myosin strands together drive muscle action.



Intermediate filaments are made up of fibrous proteins organized into tough, ropelike assemblages that stabilize a cell's structure and help maintain its shape. Some intermediate filaments help to hold neighboring cells together. Others make up the nuclear lamina.



Microtubules are long, hollow cylinders made up of many molecules of the protein tubulin. Tubulin consists of two subunits, α-tubulin and β-tubulin. Microtubules lengthen or shorten by adding or subtracting tubulin dimers. Microtubule shortening moves chromosomes. Interactions between microtubules drive the movement of cells. Microtubules serve as "tracks" for the movement of vesicles.

Cytoskeleton

Actin

Actin (cortical)
Stabilizes cell shape.

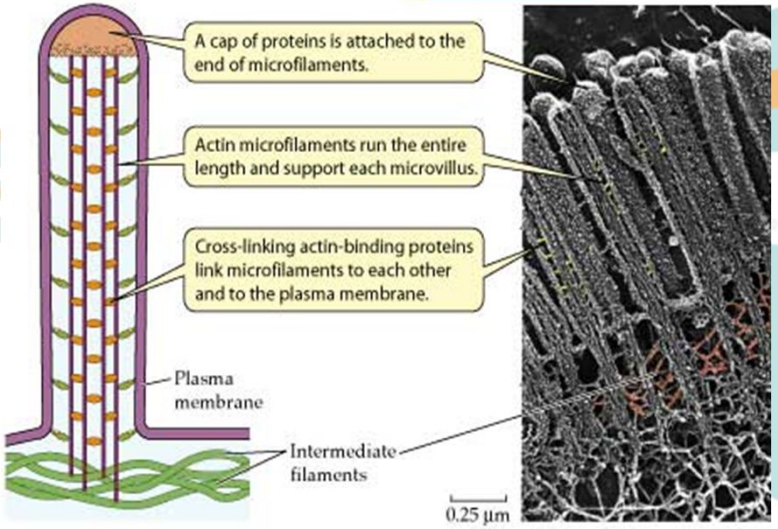
Generates movement local.

Muscle contraction.

Constriction ring in cell divission (Cytokinesis).
Cytoplasmic strreaming.

Cytoskeleton

Actin for Support



CELL BIOLOGY

Cytoskeleton 2

Cytoskeleton

Effort By: AARISH ALI

IF

IF stabilize cell structure (hold organelles).

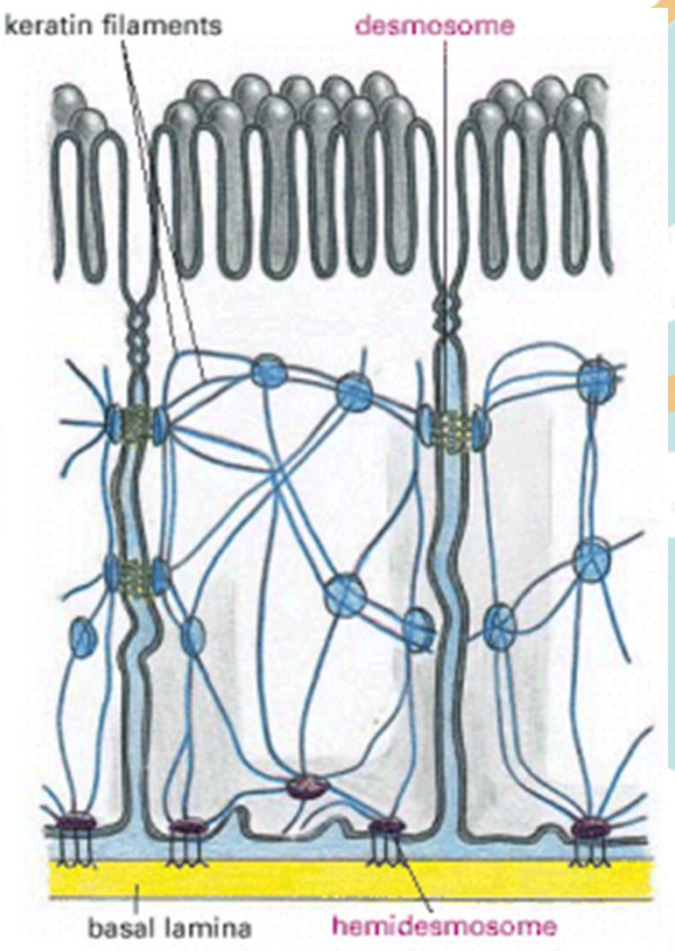
Stabilize and maintain tissue rigidity (desmosomes).

Lamins in nucleus.

Resist tention.

Cytoskeleton

IF



Cytoskeleton

Microtubules

Rigid internal skeleton resist compression.

Tracks along for motor proteins.

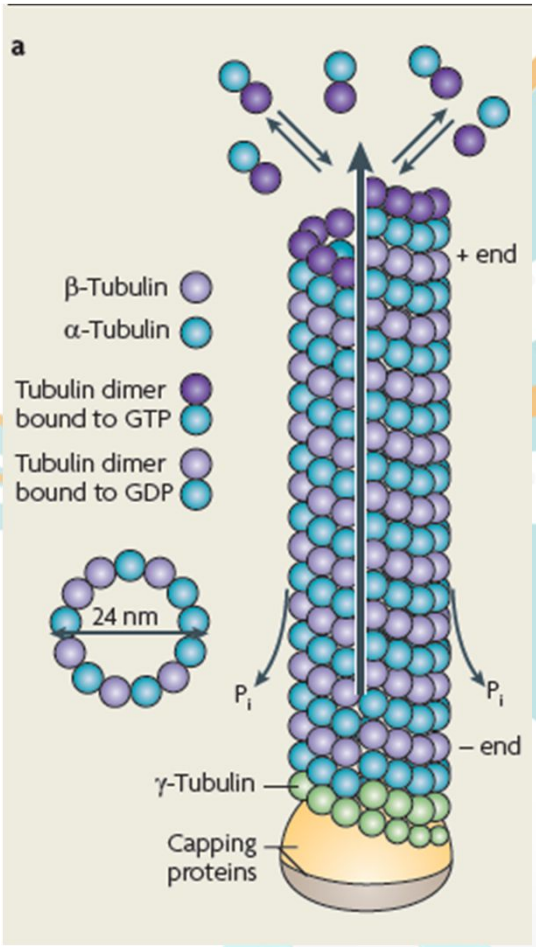
Tubulin is a dimer of

α-tubulin and β-tubulin.

The two ends different plus (+) end & minus (–) end (cat Tail & dynamic).

Cytoskeleton

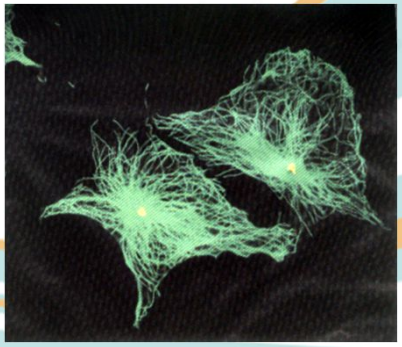
Microtubules 13

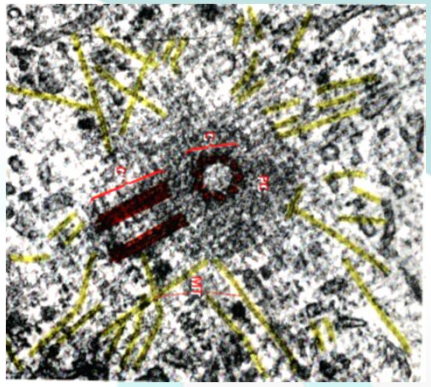


Cytoskeleton

Centrosome

Microtubule organizing center





END

CELL BIOLOGY

Motor Proteins

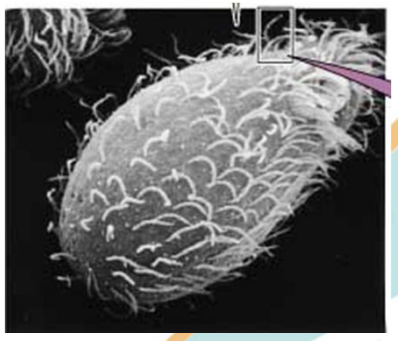
Motor Proteins

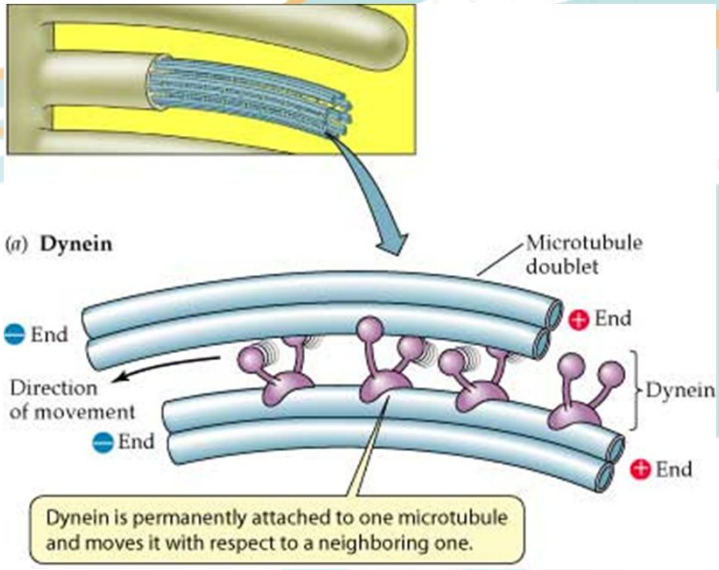
Dynein & Kinesin

Cargo & Motor Domain

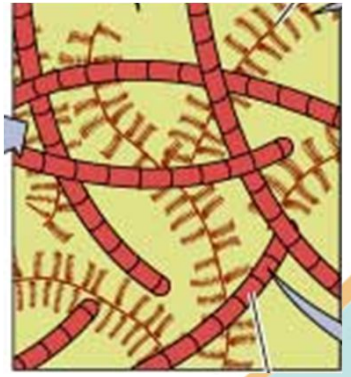
Motor Proteins

Cilia and Flagella





ECM



Holds cells in tissues.

Physical properties of tissues cartilage, skin.

Orients cell movements during development/injury

Chemical signaling from one cell to another.

END

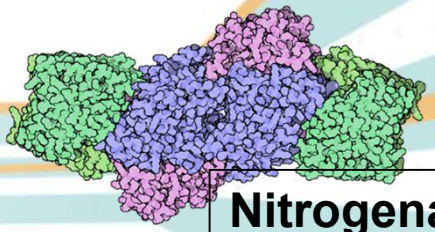
CELL BIOLOGY

The Miracle Workers

The Miracle Workers

The reaction is carried out under conditions of 150-250 atmospheres (atm), 450-500 °C; resulting in a yield of 10-20%

$$N_2 + 3H_2 \longrightarrow 2NH_3$$



Nitrogenase

$$N_2 + 8H^+ + 8e^- + 16ATP \longrightarrow 2NH_3 + H_2 + 16ADP + 16\ Pi$$

Energy Conversions & Enzymes

The Miracle Workers

One of biggest miracles of life is that it can perform slow reactions and the ones requiring extreme conditions, quickly and at physiological conditions

CELL BIOLOGY

Energy

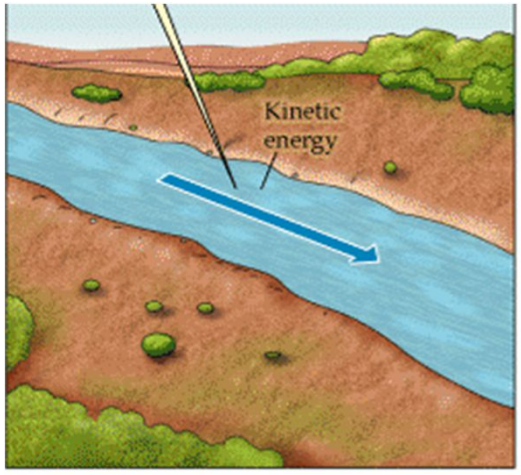
Energy

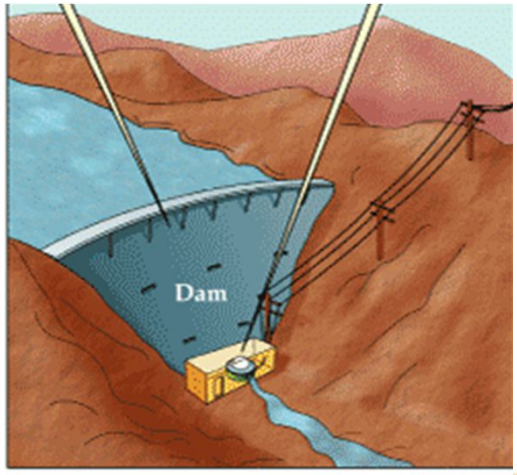
Capacity to do work

Potential energy is the energy of state or position; it includes the energy stored in chemical bonds.

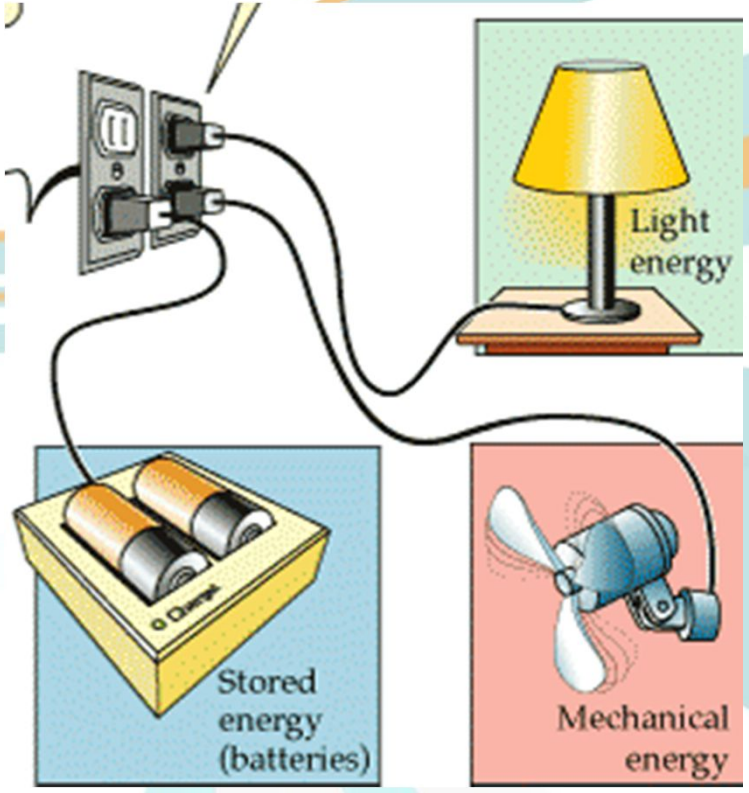
Kinetic energy is the energy of motion (and related forms such as electric energy, light, and heat).

Energy Conversions





Efficiency Not 100%

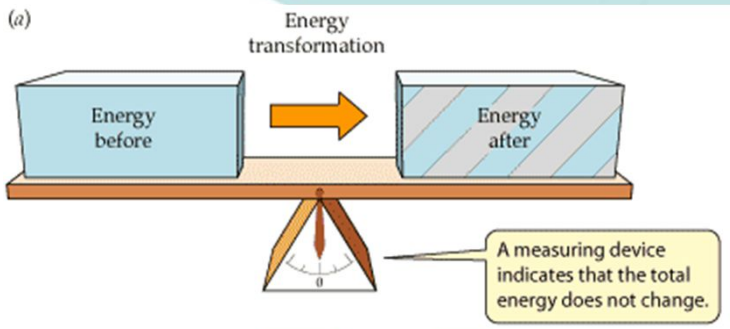


The Laws of Thermodynamics

The First Law

Energy cannot be created or destroyed.

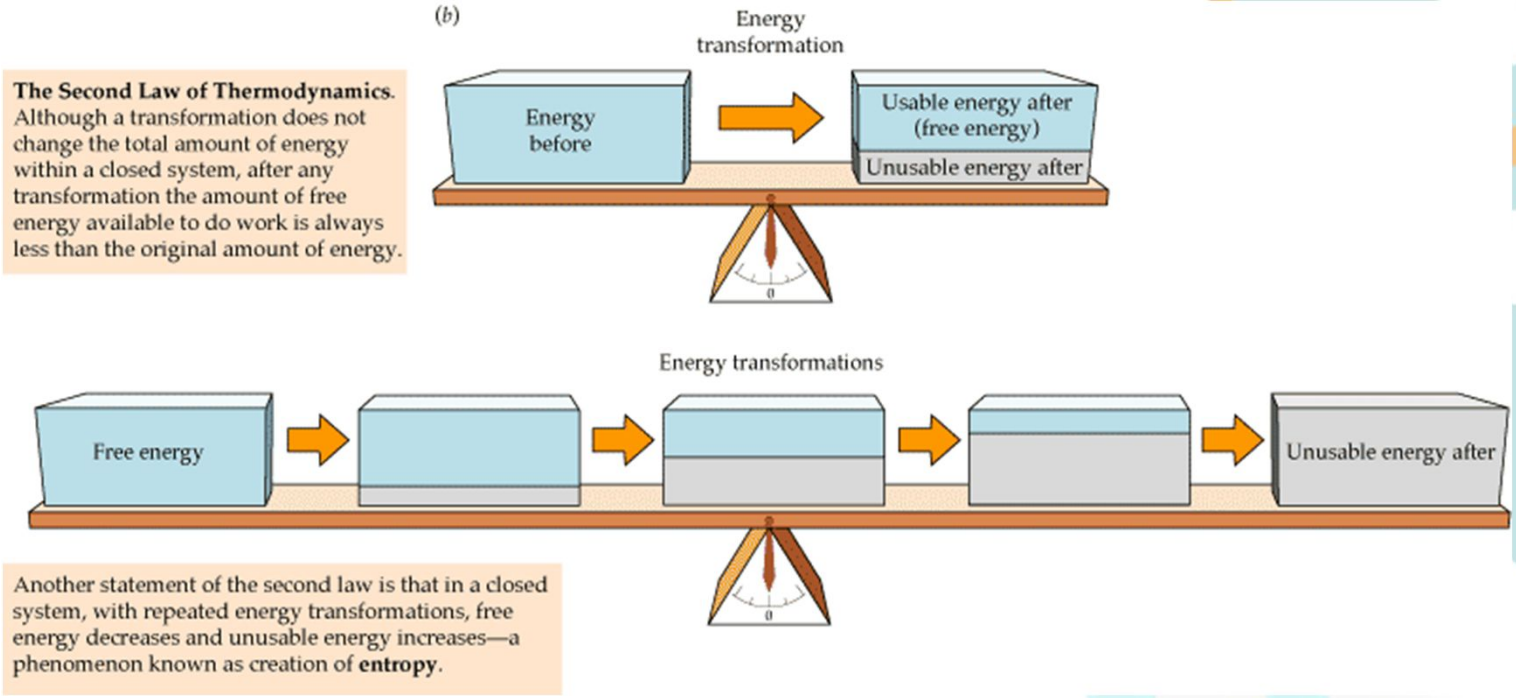
The First Law of Thermodynamics. The total amount of energy before a transformation equals the total amount after a transformation. No new energy is created, and no energy is lost.



The Laws of Thermodynamics

The Second Law

The quantity of energy available to do work (free energy) decreases and unusable energy (associated with entropy) increases.



The Second Law of Thermodynamics. Although a transformation does not change the total amount of energy within a closed system, after any transformation the amount of free energy available to do work is always less than the original amount of energy.

Another statement of the second law is that in a closed system, with repeated energy transformations, free energy decreases and unusable energy increases—a phenomenon known as creation of **entropy**.

Free Energy (ΔG)

The Second Law

total energy =

usable energy +
unusable energy

$H = G + TS$

(H = Enthalpy, G = free energy, T= absolute temperature, S = entropy)

$G = H - TS$

END

Free Energy (ΔG)

Free Energy (ΔG)

The Second Law

total energy =

usable energy +
unusable energy

$H = G + TS$

(H = Enthalpy, G = free energy, T= absolute temperature, S = entropy)

$G = H - TS$

Free Energy (ΔG)

Δ

Absolute G, H or S can not be measured

Changes in free energy, total energy, and entropy are related by the equation.

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G \text{ (reaction)} = G \text{ (products)} - G \text{ (reactants)}$$

$$\Delta H \text{ (reaction)} = H \text{ (products)} - H \text{ (reactants)}$$

$$\Delta S \text{ (reaction)} = S \text{ (products)} - S \text{ (reactants)}$$

ΔG of Combustion

Reactants		Products
$CH_4 + 2O_2$	⟶	$CO_2 + 2H_2O$
H = G + TS		H = G + TS
500 = 400 + 100		200 = 100 + 100

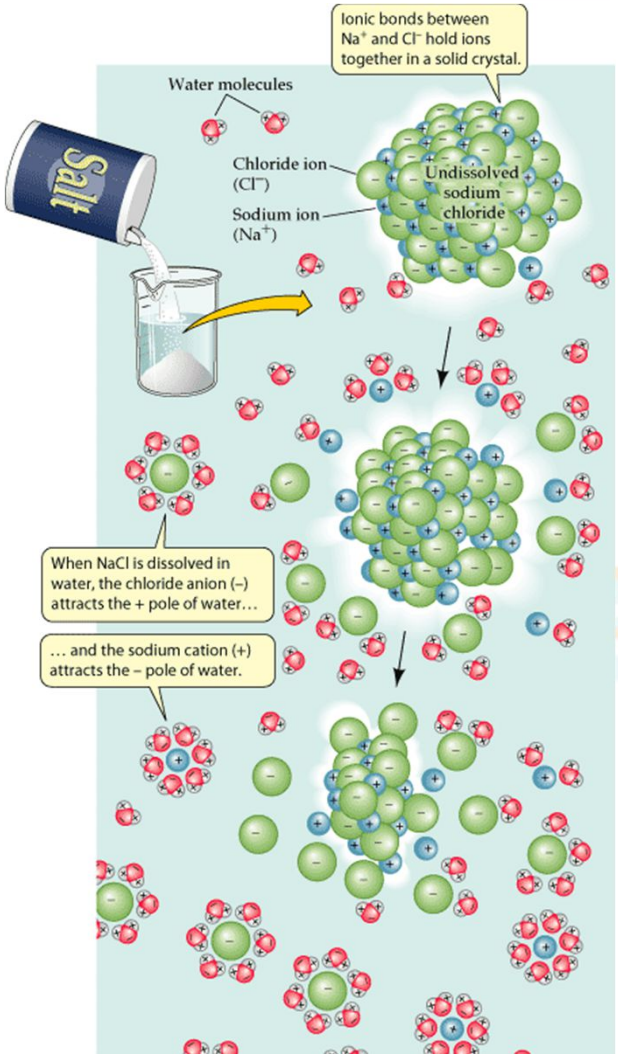
Reactants		Products
G = H – TS	⟶	G = H – TS
400 = 500 - 100		100 = 200 - 100

$$\Delta G_{(reaction)} = G_{(products)} - G_{(reactants)}$$

$$-300 = 100 - 400$$

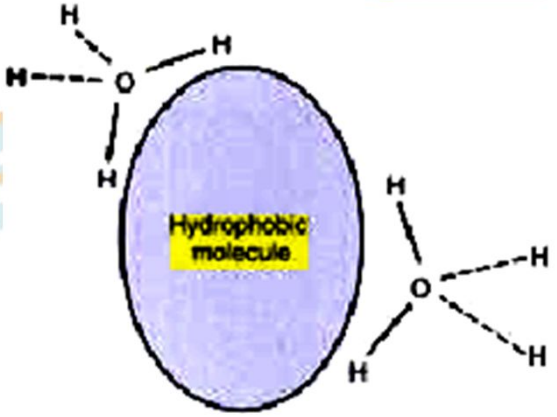
$$\Delta G / \Delta H = -300$$

Hydrophobicity Counterintuitive



????

H_2O adjacent to hydrophobic molecule are restricted to form hydrogen bonds with other water molecules.



Bonding (hydrogen) also reduces energy state.

Free Energy (ΔG)

$^{+}$ve or $^{-}$ve

If ΔG is $^{-}$ve energy is released

If ΔG is $^{+}$ve free energy is required

In living systems magnitude and sign of ΔG can depend significantly on changes in entropy ΔS

ΔS = +ve in hydrolysis

END

CELL BIOLOGY

Exergonic & Endergonic Reactions

Exergonic & Endergonic Reactions

ΔG $^{+}$ve or $^{-}$ve

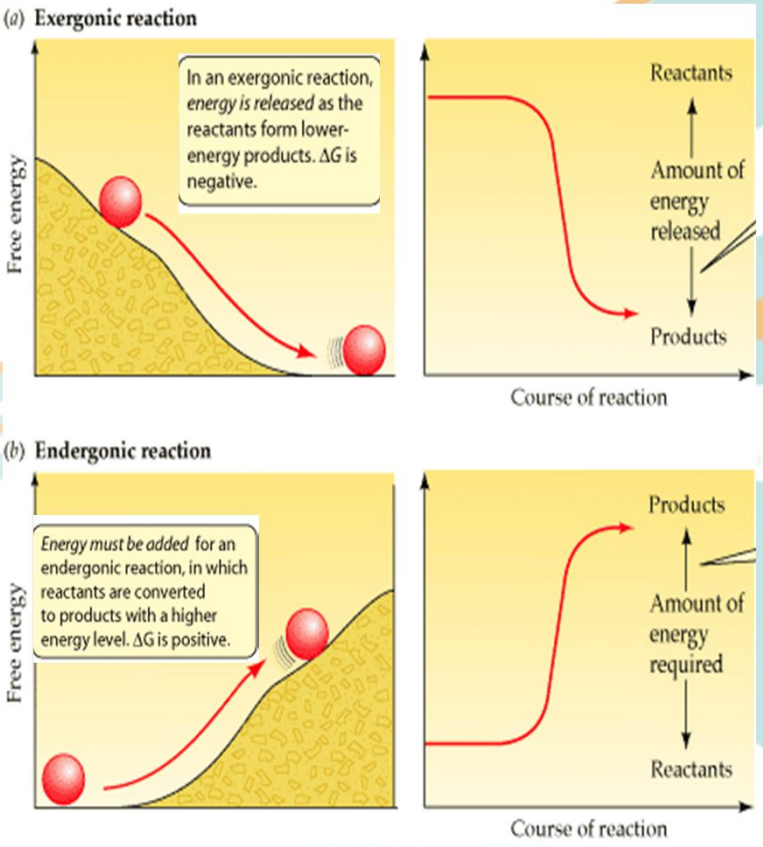
Endergonic reactions take up free energy and have a positive ΔG.

Endergonic reactions proceed only if free energy is provided

Exergonic reactions release free energy and have a negative ΔG.

Exergonic & Endergonic Reactions

ΔG $^{+}$ve or $^{-}$ve



Reversible Reactions

Equilibrium

Point at which the forward and reverse reactions proceed at the same rate.

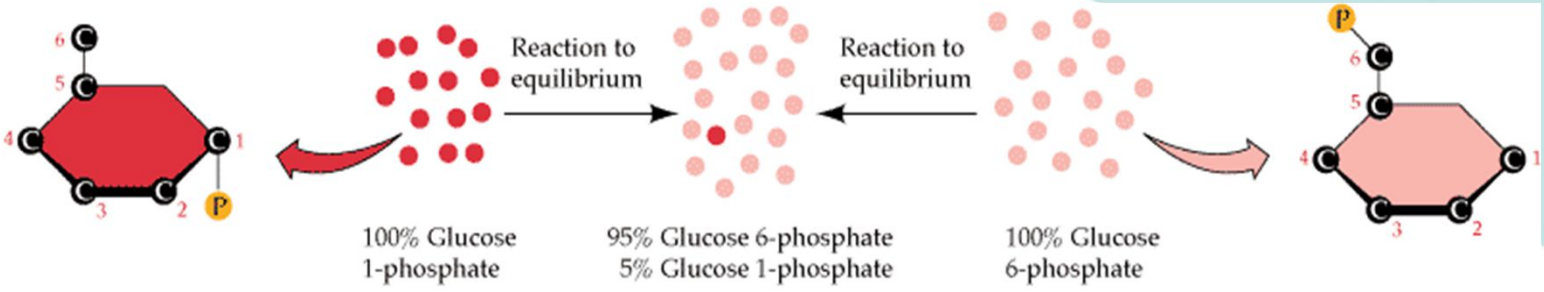
30%A
70%B
ΔG

Reversible Reactions

Equilibrium

The change in free energy (ΔG) of a reaction determines its point of chemical equilibrium,

For exergonic reactions, the equilibrium point lies toward completion (the conversion of all reactants into products).



ΔG = -1.7 kcal/mol

@ 0.02M, 25C & pH 7

Reversible Reactions

Equilibrium

At equilibrium
Amount reactant and product molecules stays constant
Although individual moles molecule keep changing their states
$\Delta G = 0$

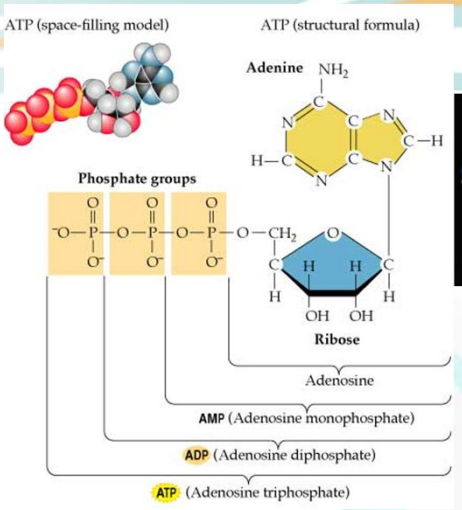
END

ATP Hydrolysis

ATP Hydrolysis

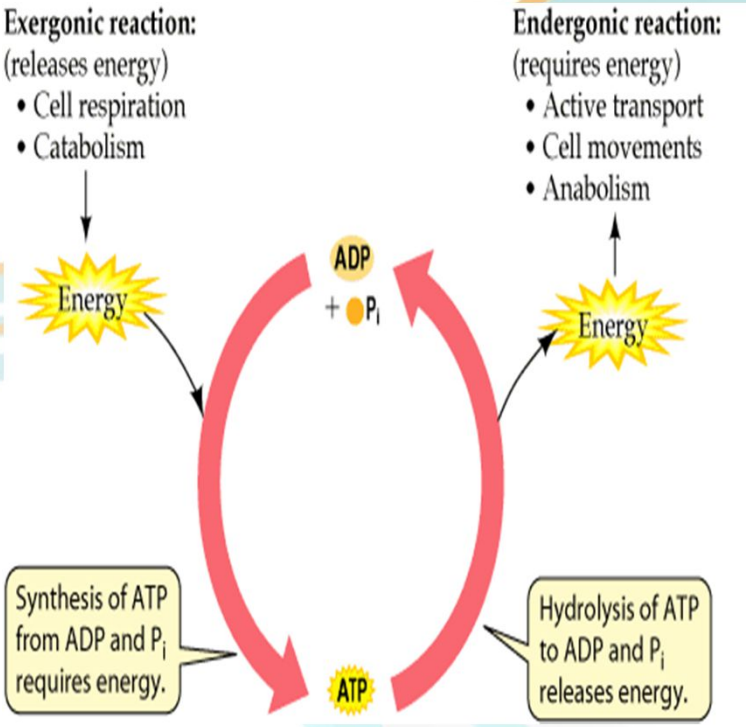
Energy Currency

Hydrolysis of ATP releases a relatively large amount of free energy (-12 kcal/mol).



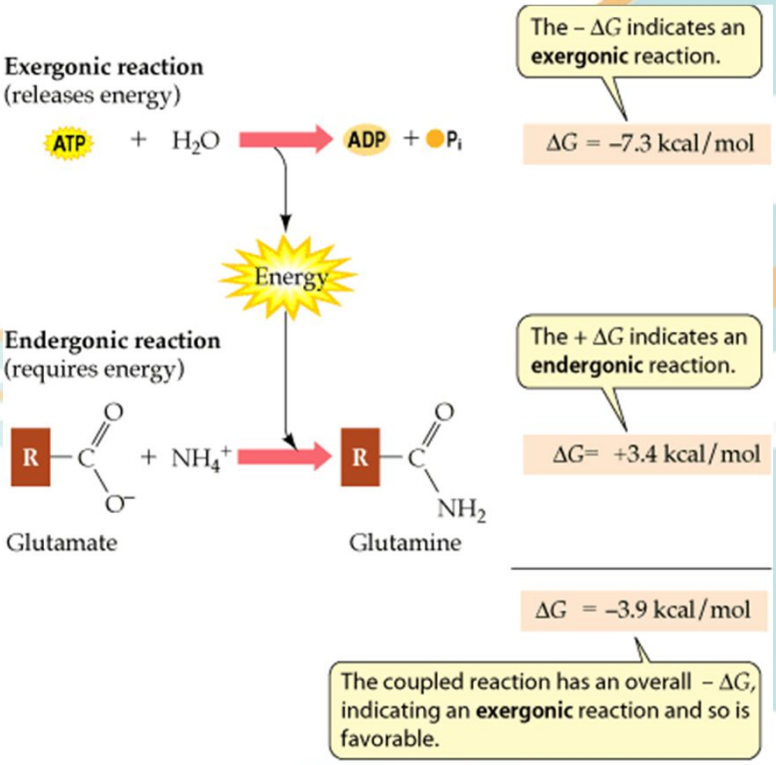
ATP Hydrolysis

Coupling



ATP Hydrolysis

Coupling



Speed of a Reaction

ΔG Independent

50% 600 poly arginine degradation in 7years Carboxypeptidase takes half a second

Reactants Need to cross an energy barrier small amount of added energy (Activation Energy E_a)

e.g. CH4 + O2

END

CELL BIOLOGY

Enzymes

Biological Catalysts

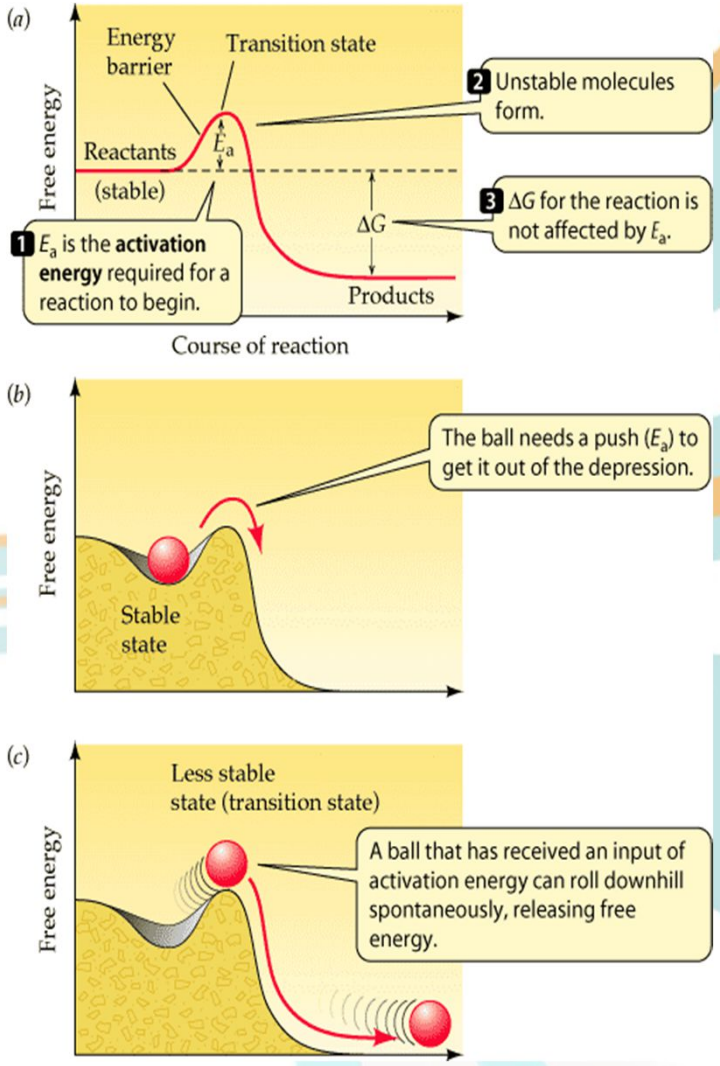
Enzymes

The more -ve ΔG is, the further the reaction proceeds to completion.

ΔG does not tell us the rate of a reaction (the speed at which it moves towards equilibrium).

Most Enzymes are proteins (ribozymes are RNA)

Activation Energy

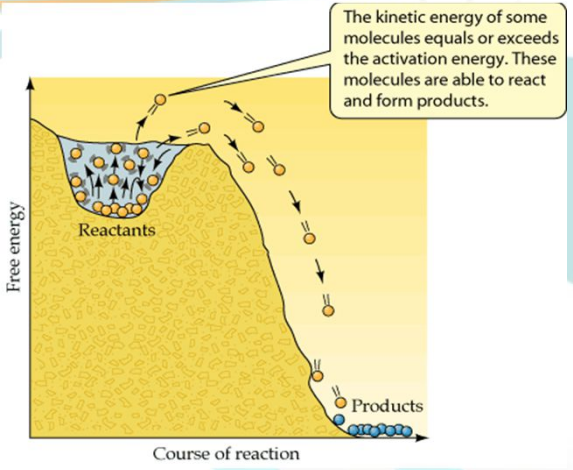


Activation Energy

Enzymes

At normal temperatures few molecules have enough kinetic energy for this transformation.

Enzymes lower the energy barrier.



END

CELL BIOLOGY

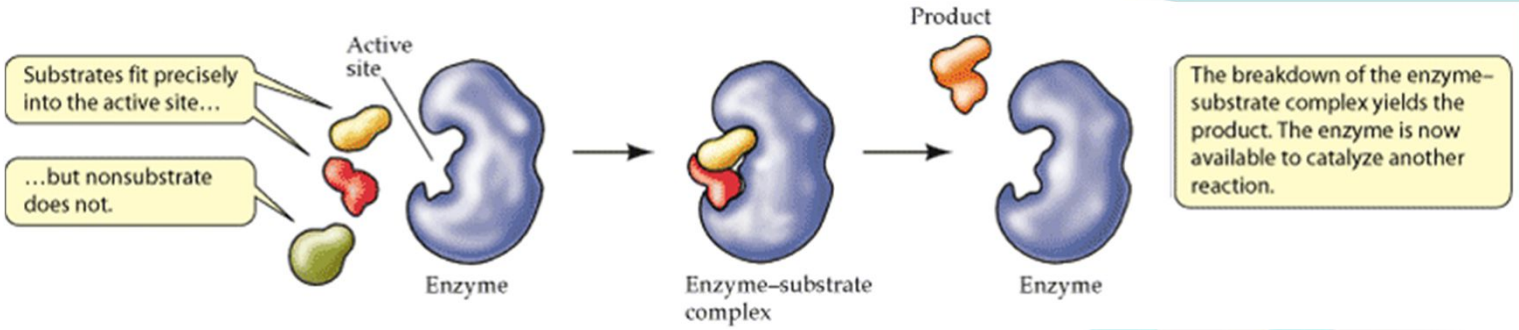
Enzymes
Mechanisms

Enzymes Specificity

Reactants (substrates) bind the a particular (active) site of the respective enzyme.

The active site = the catalytic + binding sites

The specificity of this binding is a function of three-dimensional shape, structure and environment of its active site.

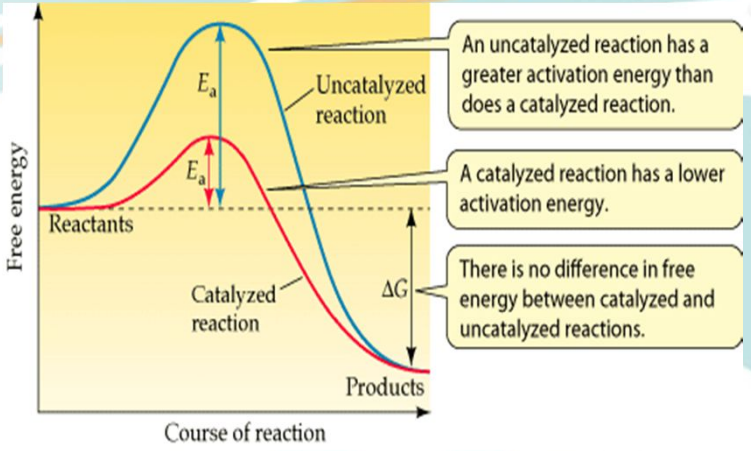


Enzymes Lower the Energy Barrier

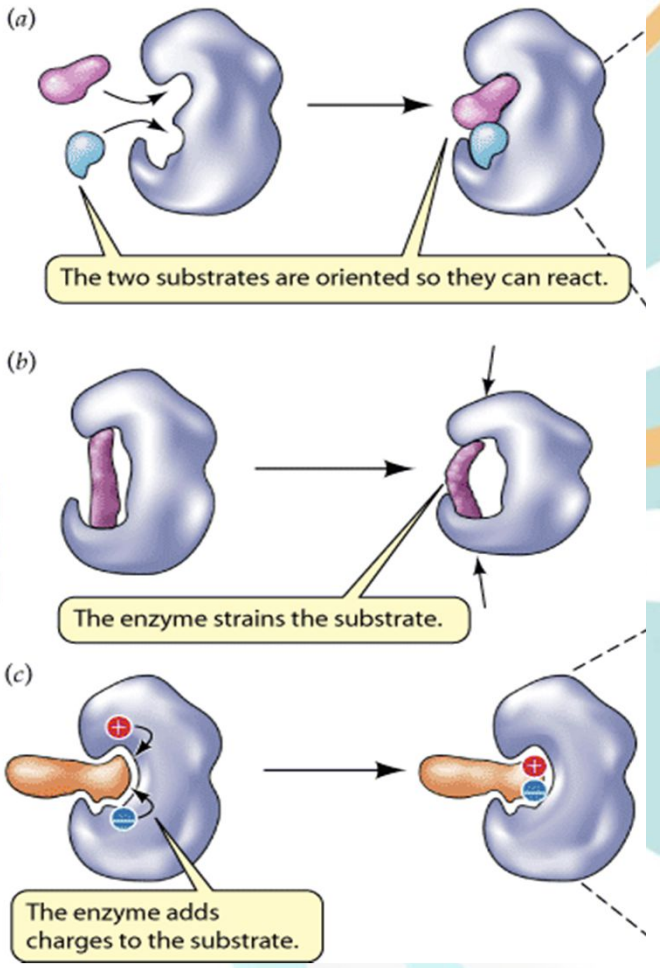
E_a

Enzymes Lower the Energy Barrier (forward and reverse reaction).

Equilibrium and ΔG are not changed



How do Enzymes Work



Induced Fit

Conformational Change

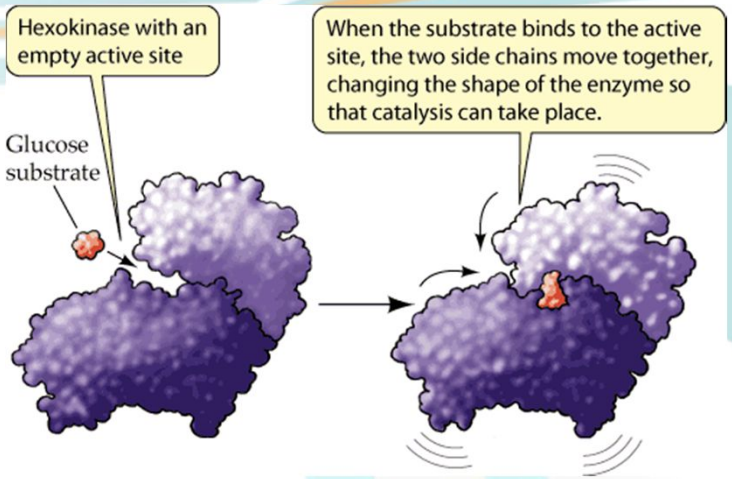
Upon binding to substrate, some enzymes change shape, facilitating catalysis.

The shape changes (Induced fitting) modifies active site and exposes/ aligns those regions that perform catalysis

Induced Fit

Conformational Change

Water exclusion at active site results in transfer of Pi from ATP to glucose preventing APT hydrolysis to ADP



CELL BIOLOGY

Enzyme Helpers

Enzyme Helpers

6.1 ***A Few Examples of Nonprotein "Partners" of Enzymes***

TYPE OF MOLECULE	ROLE IN CATALYZED REACTIONS
Cofactors	
Iron (Fe^{2+} or Fe^{3+})	Oxidation/reduction
Copper (Cu^{+} or Cu^{2+})	Oxidation/reduction
Zinc (Zn^{2+})	Helps bind NAD
Coenzymes	
Biotin	Carries $—COO^{-}$
Coenzyme A	Carries $—CH_2—CH_3$
NAD	Carries electrons
FAD	Carries electrons
ATP	Provides/extracts energy
Prosthetic groups	
Heme	Binds ions, O_2, and electrons; contains iron cofactor
Flavin	Binds electrons
Retinal	Converts light energy

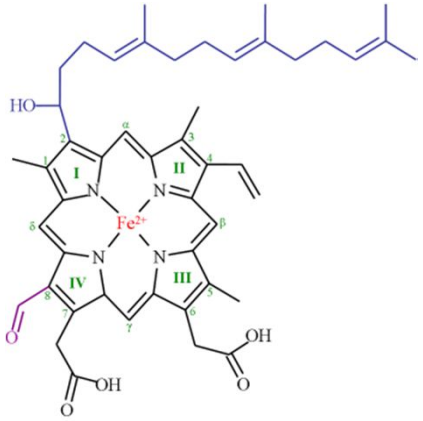
Enzyme Helpers

Cofactors: Inorganic ions (Cu, Zn & Fe) bound to some enzymes and essential to their function.

Coenzymes: Carbon containing molecules bind enzymes temporarily chemically changed during reaction (substrate).

Enzyme Helpers

Prosthetic groups: non-protein component permanently bound to enzymes (heme group), without its prosthetic group apoprotein vs. holoprotein.

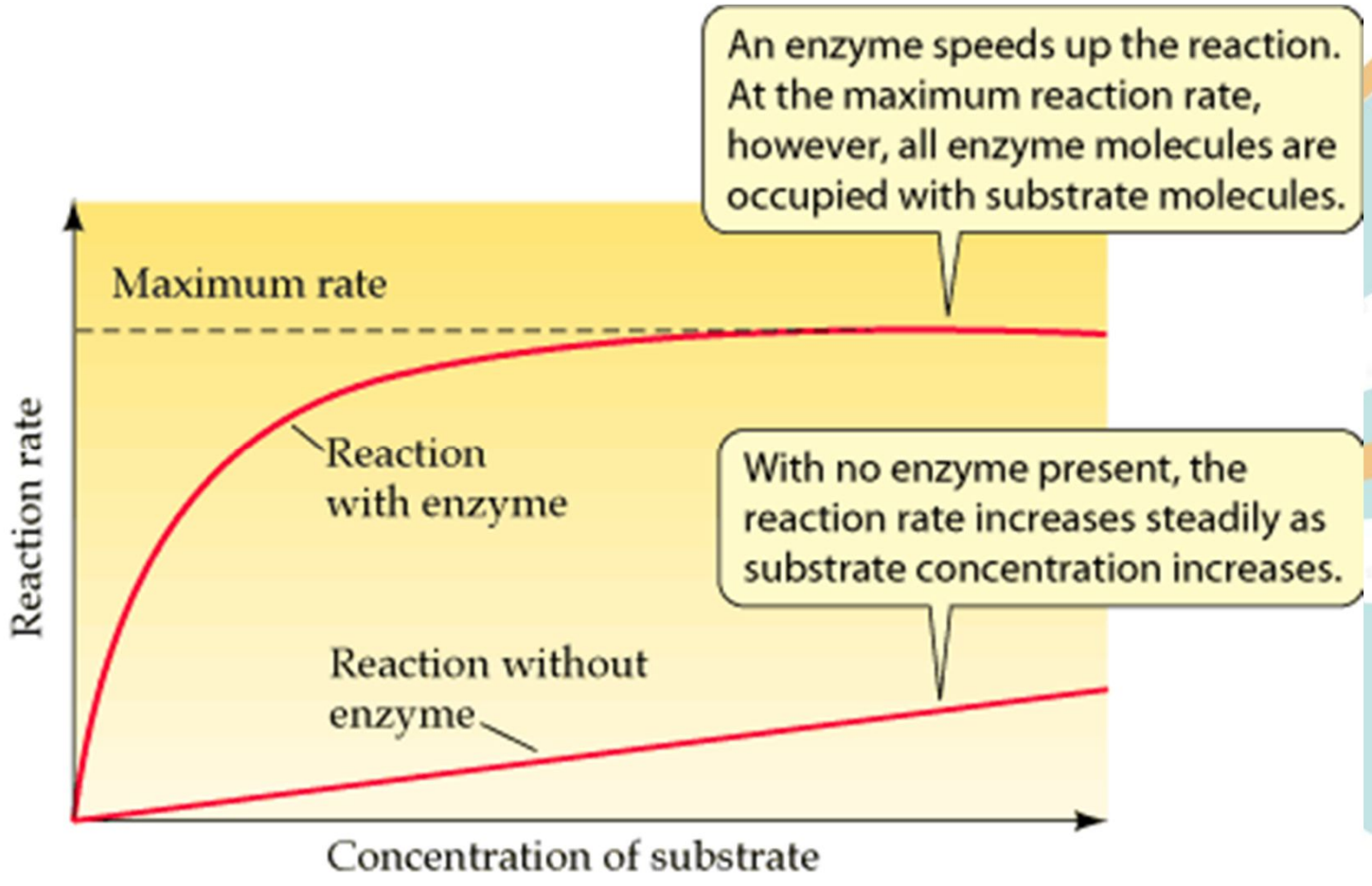


END

CELL BIOLOGY

Enzyme Regulation

Substrate Concentration



Enzyme Regulation

The Inhibitors

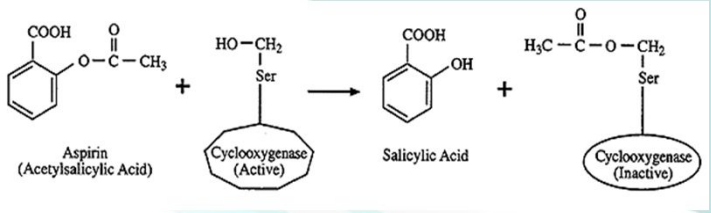
Some factors can activate enzymes and other can deactivate them (inhibit their function).

1. Irreversible inhibitors
2. Competitive inhibiters
3. Noncompetitive inhibiters

Enzyme Regulation

Irreversible inhibitors

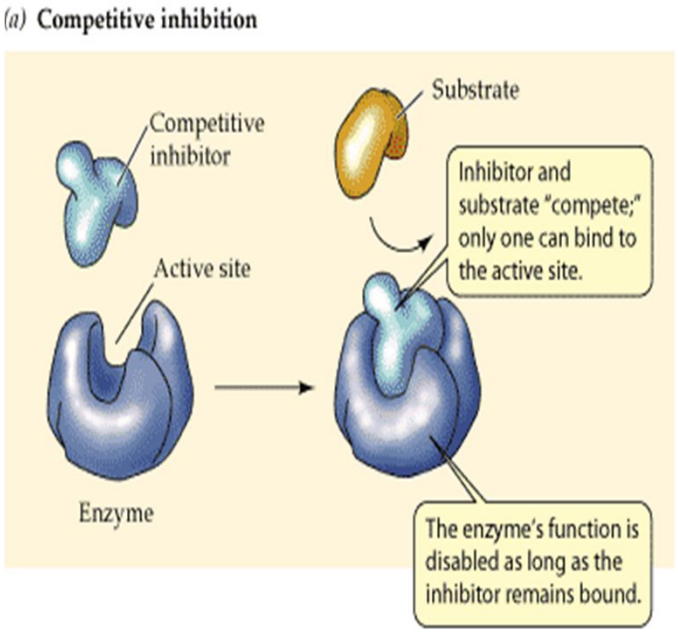
Permanently inhibit enzymes
function generally by modifying the active site

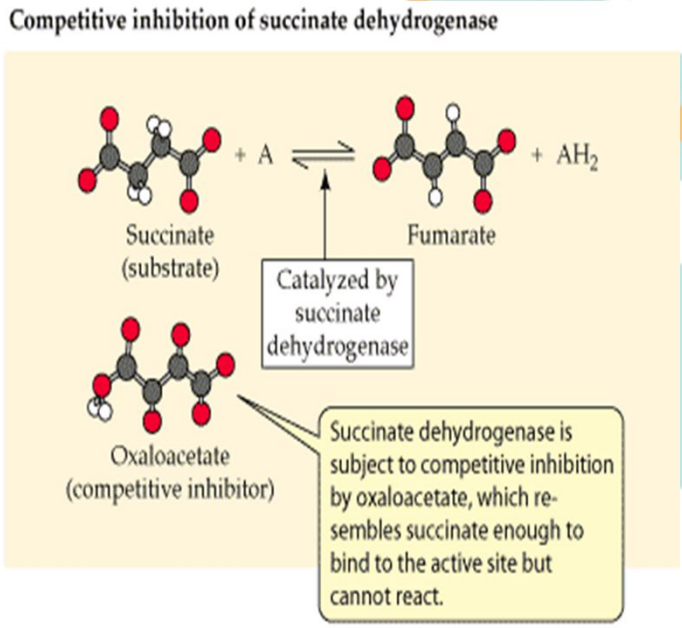


Enzyme Regulation

Reversible inhibitors

Competitive inhibiters are similar to the natural substrate (bind the active site) however they are different enough that no reaction is catalyzed.

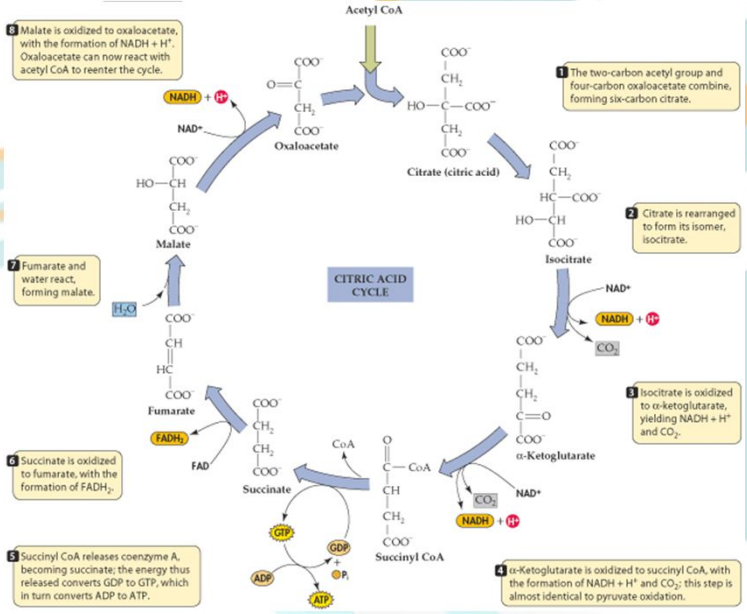




Enzyme Regulation

The Citric Acid Cycle

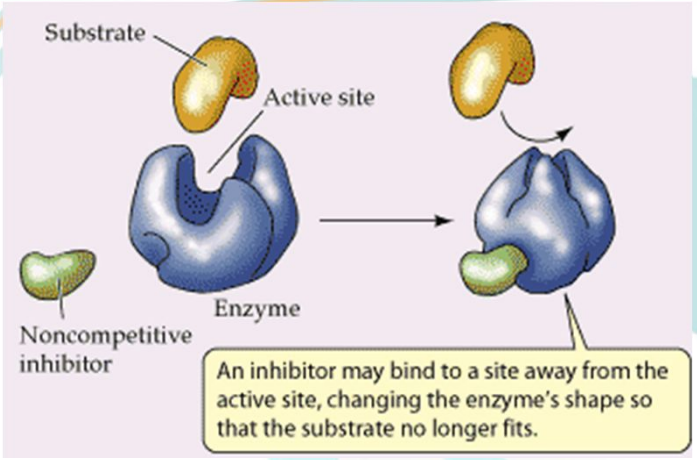
Feed Back/End Product Inhibition



Enzyme Regulation

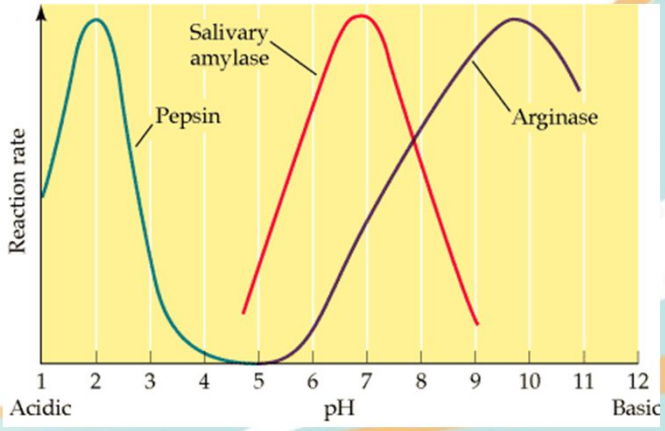
Noncompetitive inhibiters

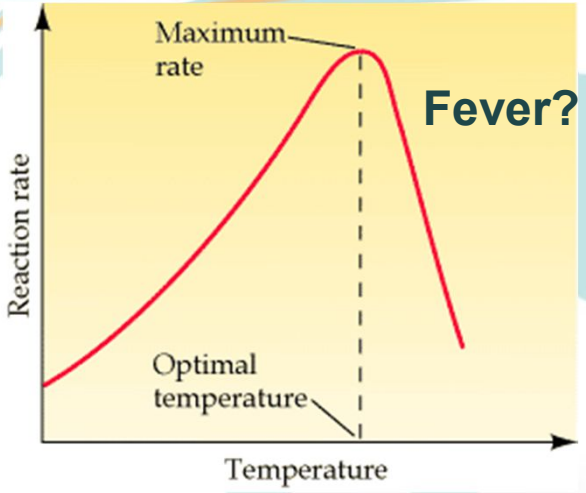
Bind the enzyme at nonactive site cause a conformational change prevents enzyme from binding its substrate.



Physical Factors

Influence Enzymes

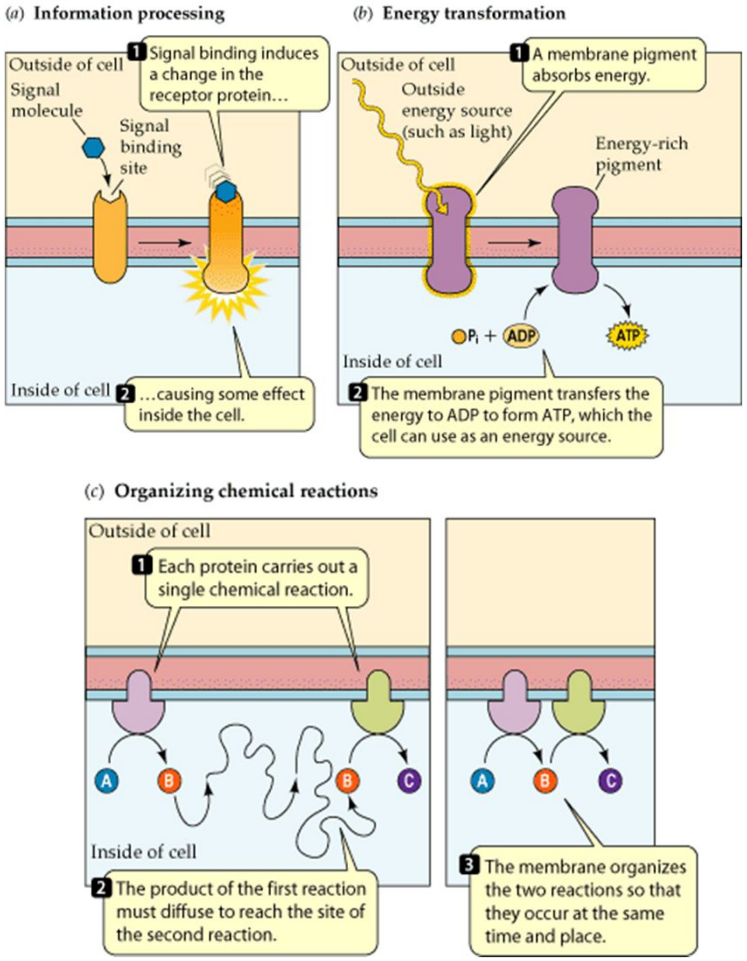


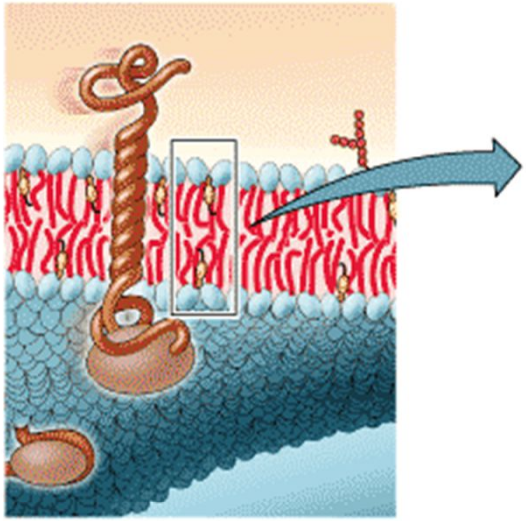


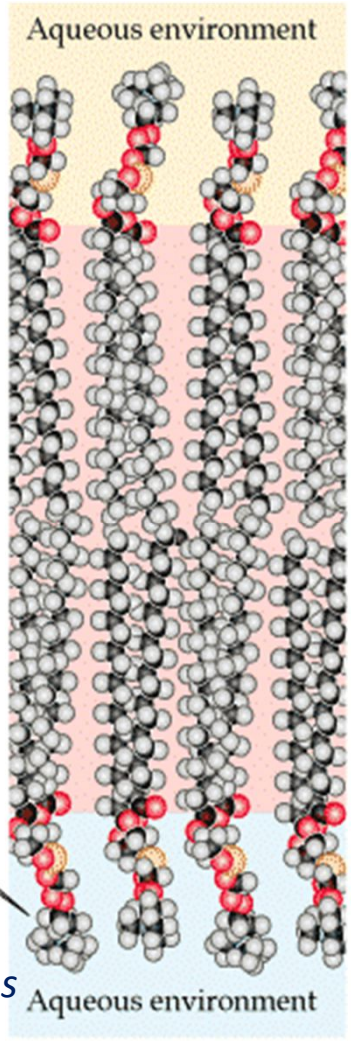
END

Membranes

More Than Barriers



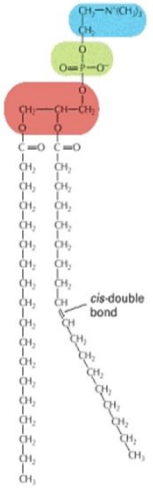


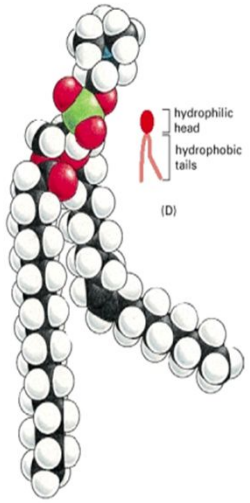


The nonpolar, hydrophobic fatty acid "tails" interact with one another in the interior of the bilayer.

The charged, or polar, hydrophilic "head" portions interact with polar water.

Amphipathic: *Hydrophilic regions + Hydrophobic regions*



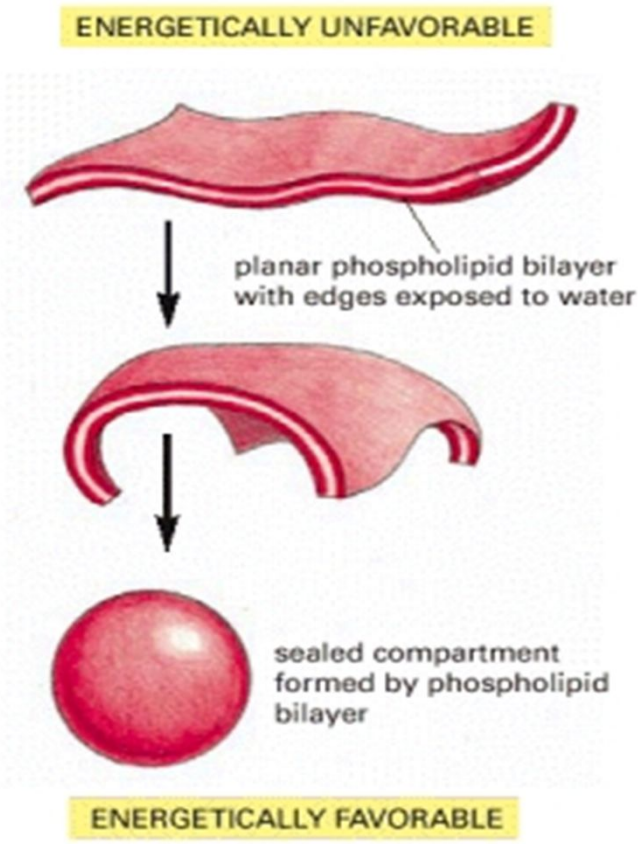


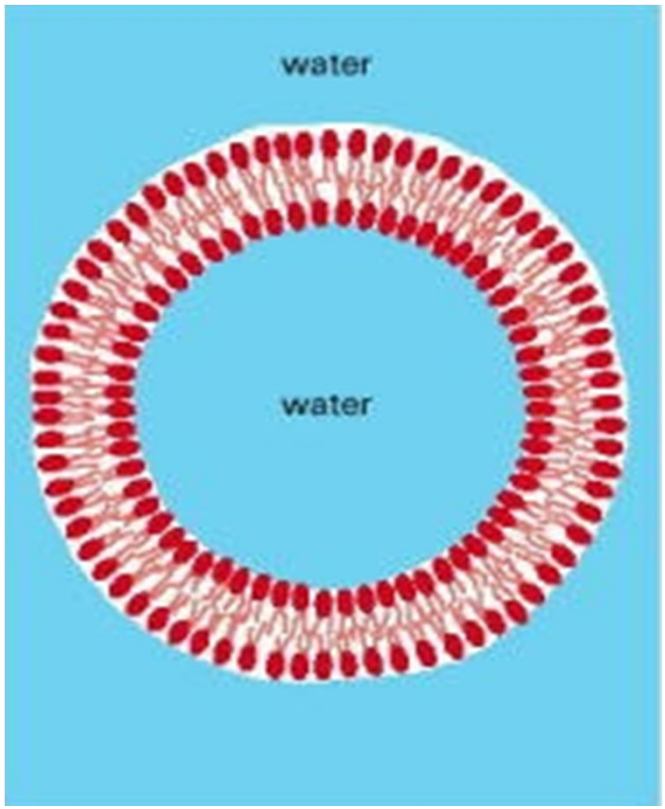
Fatty Acid Length: 14-24 Cs

One Tail Unsaturated Cis

Spontaneous Assembly

Free Energy Dictates





Features

Phospholipids coexist with water form a bilayer

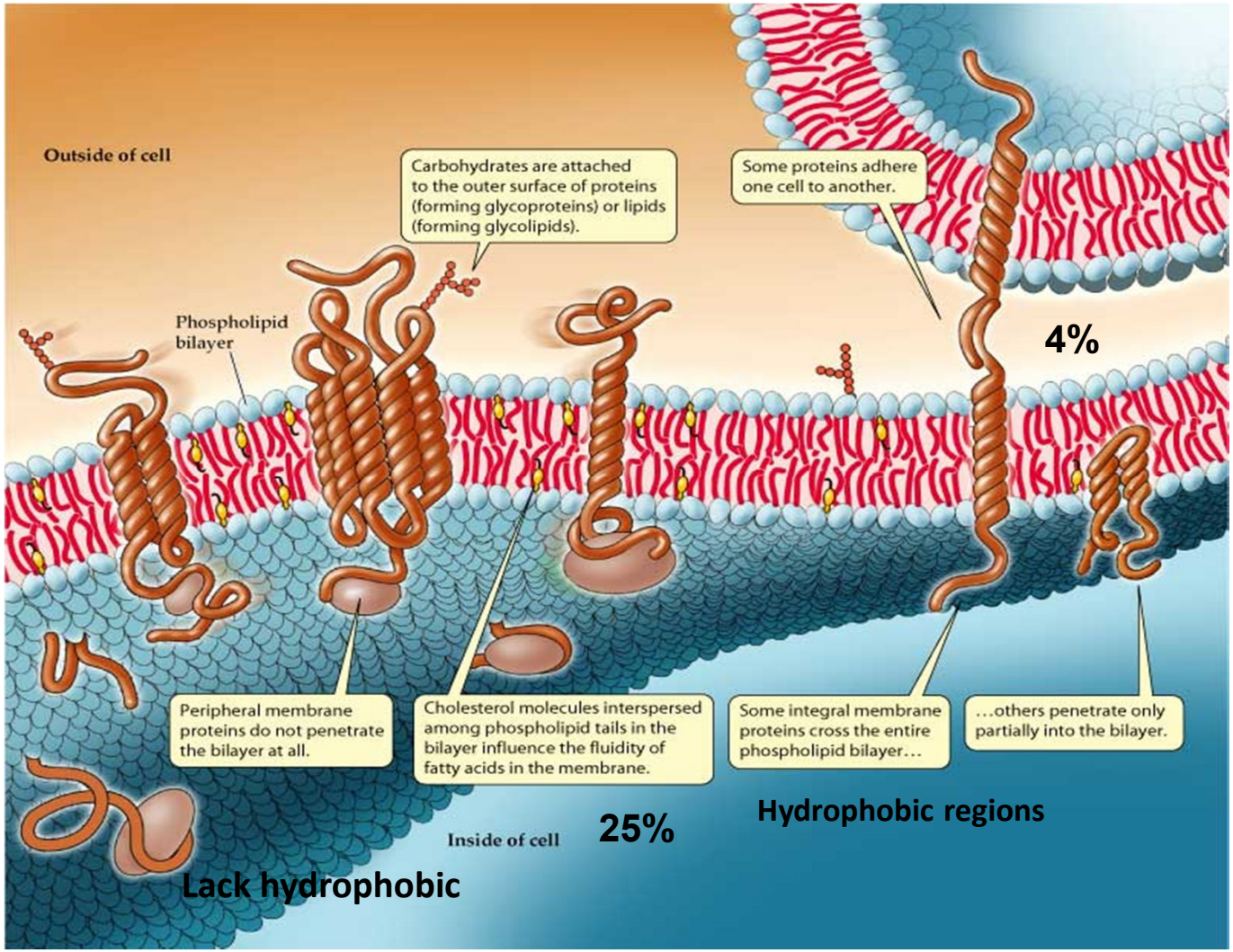
Dynamic structures: parts move & change.

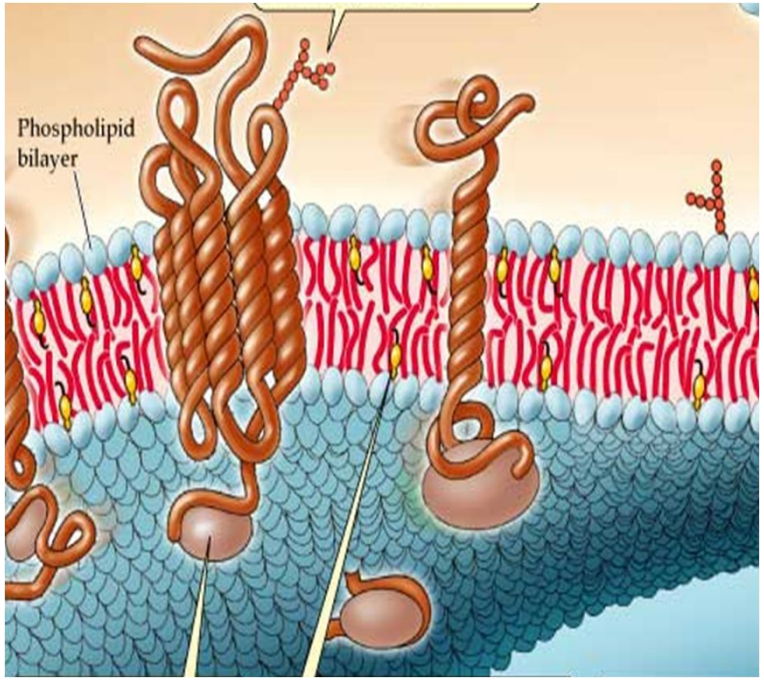
Vital physiological roles: cell cell & cell interaction with environmental

Physical integrity: lipids, create a barrier to hydrophilic materials such as water and ions

END

Membranes Components


Outside of cell
Carbohydrates are attached to the outer surface of proteins (forming glycoproteins) or lipids (forming glycolipids).
Some proteins adhere one cell to another.
Phospholipid bilayer
4%
Peripheral membrane proteins do not penetrate the bilayer at all.
Cholesterol molecules interspersed among phospholipid tails in the bilayer influence the fluidity of fatty acids in the membrane.
Some integral membrane proteins cross the entire phospholipid bilayer…
…others penetrate only partially into the bilayer.
Hydrophobic regions
Inside of cell
25%
Lack hydrophobic

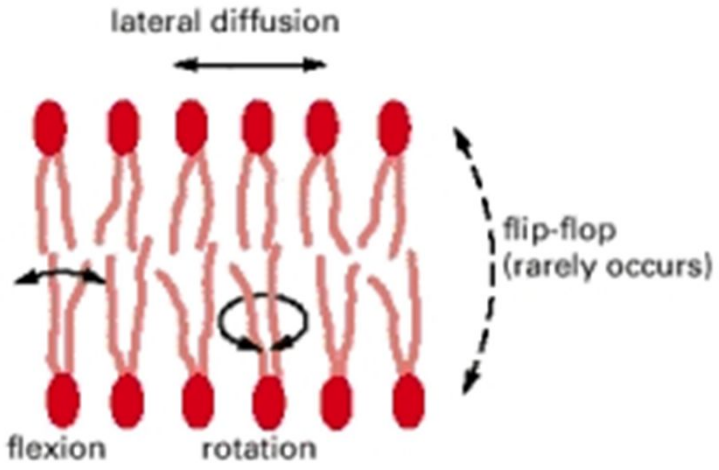


Fatty acids interact with each other & the polar regions face aqueous environment outside

Lipid “lake” in which a variety of proteins “float”

Each membrane has a set of proteins suitable to the specialized function of the cell or organelle it surrounds.

Carbohydrates attached either to lipids/protein molecules



Movement

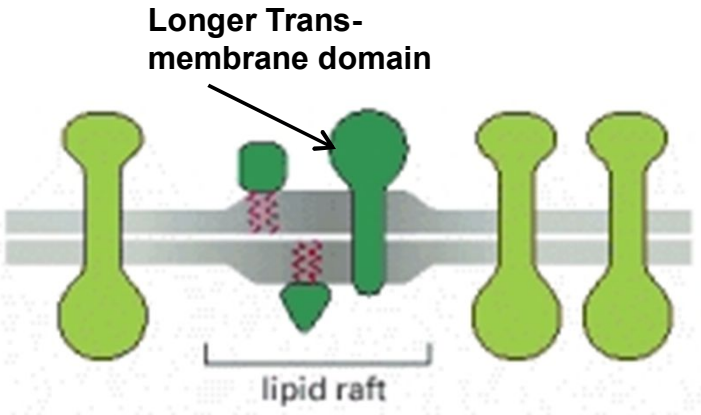
A phospholipid molecule travels from one end the other in about 1 second

Phospholipid molecule flip overs seldom (polar part of molecule hardly ever moves through the hydrophobic interior

Inner and outer halves of the bilayer have different phospholipids

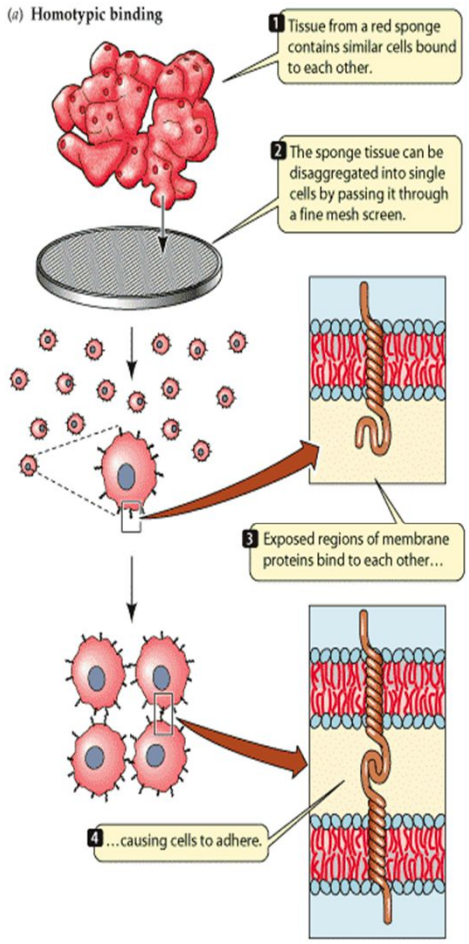
Lipid Rafts

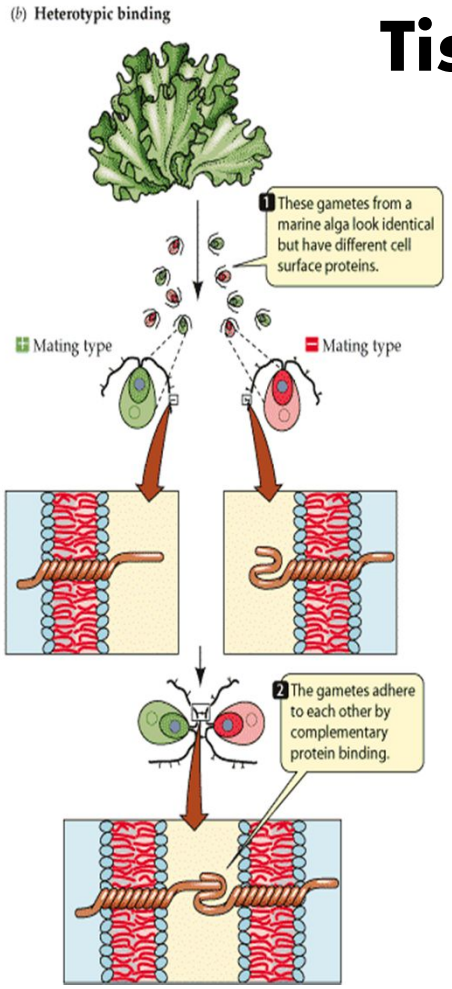
Some proteins migrate freely others “anchored” to specific semisolid region different lipid composition (very long fatty acid chains)



END

Recognition & Adhesion

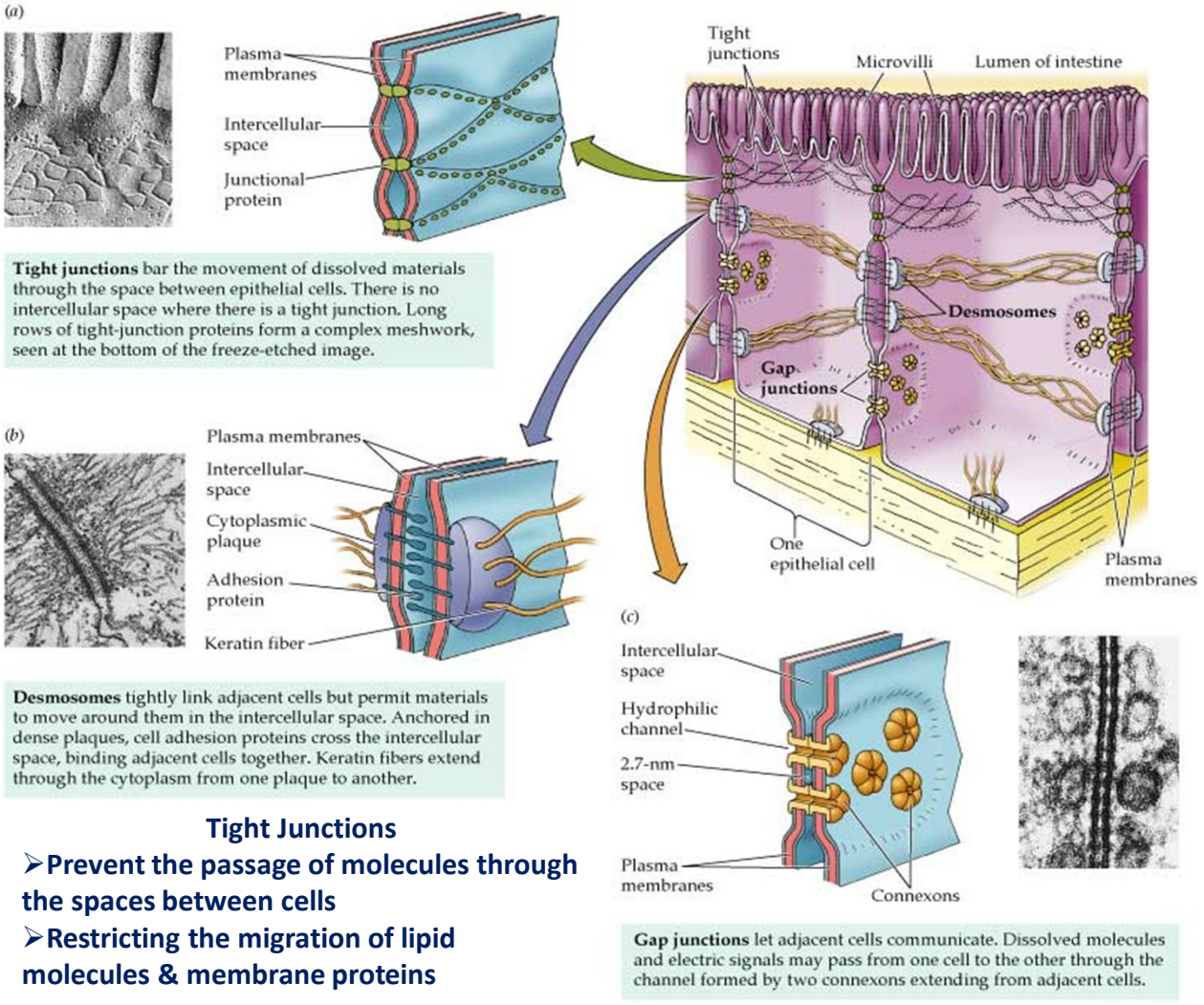




Tissue Formation

Cell recognition: One cell specifically binds to another cell

Cell adhesion: relationship between the two cells is “cemented”



Tight junctions bar the movement of dissolved materials through the space between epithelial cells. There is no intercellular space where there is a tight junction. Long rows of tight-junction proteins form a complex meshwork, seen at the bottom of the freeze-etched image.

Desmosomes tightly link adjacent cells but permit materials to move around them in the intercellular space. Anchored in dense plaques, cell adhesion proteins cross the intercellular space, binding adjacent cells together. Keratin fibers extend through the cytoplasm from one plaque to another.

Gap junctions let adjacent cells communicate. Dissolved molecules and electric signals may pass from one cell to the other through the channel formed by two connexons extending from adjacent cells.

Tight Junctions

- **Prevent the passage of molecules through the spaces between cells**
- **Restricting the migration of lipid molecules & membrane proteins**

Effort By:
AARISH ALI

Passive Transport

Does not require any input of energy to drive the process

Active Transport

Requires input of energy to drive the process

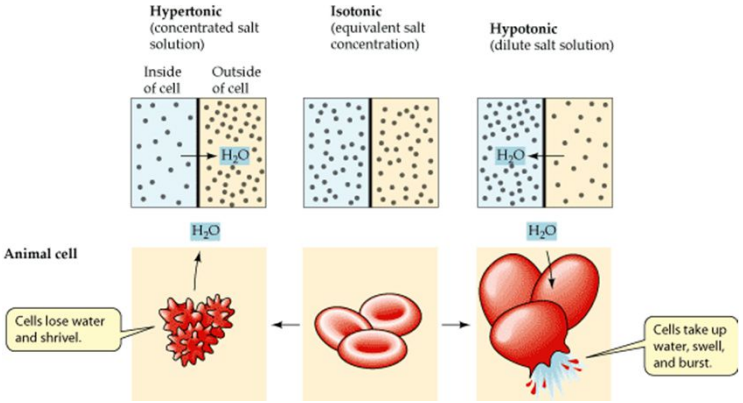
END

Transport Across Membrane

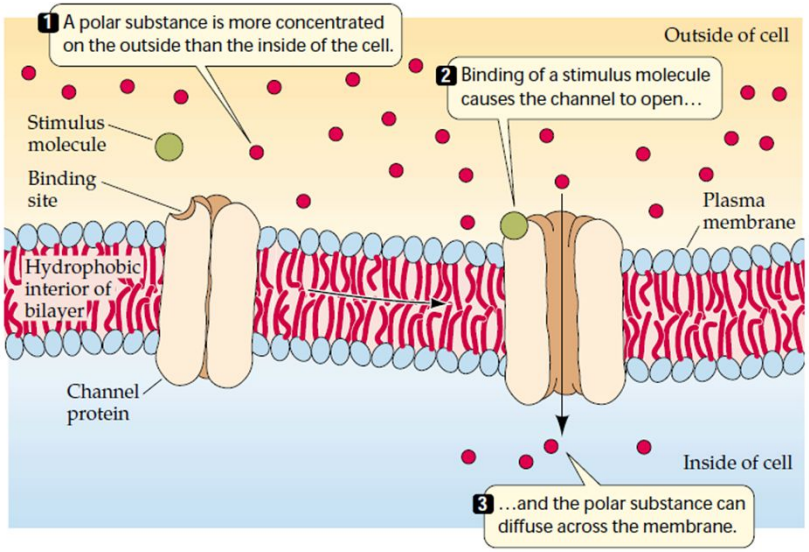
Aqua-porins:

Water molecules move through membranes by Osmosis (diffusion).

Number of solute particles not on the kinds of particles.



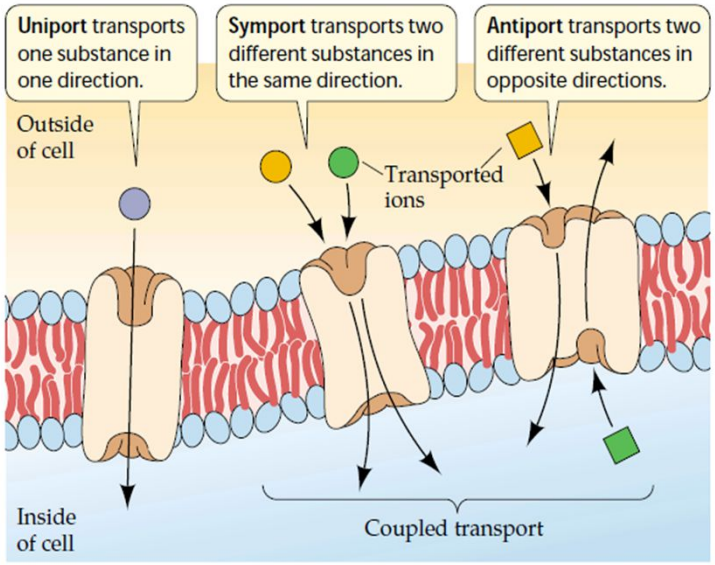
The Two Domains



Polar amino acids and water on the inside of the channel pore (to bind to the polar or charged substance and allow it to pass through)

Nonpolar amino acids on the outside of the macromolecule (to allow the channel protein to insert itself into the lipid bilayer)

Primary active transport: ATP required directly. Secondary active transport: ATP not used directly; ion conc gradient (PAT).



END

Cell Division

Life Span & Continuity

All living organisms have a limited life span.

The essence of life is to propagate.

Cells are the basic unit of life.

Cell division is the basic process that ensures the continuity of life.

The Necessity

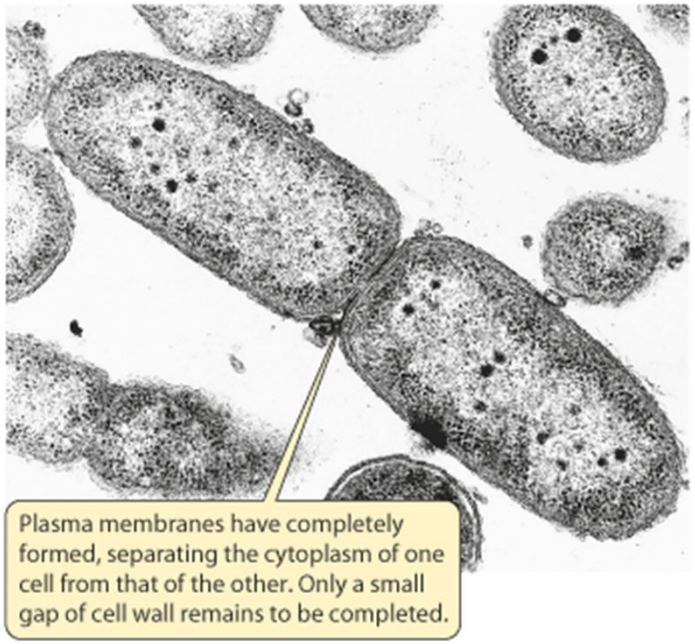
Unicellular organisms propagate by splitting into two cells.

Multicellular organisms: replace damaged or senescent cells.

Billions of cells die each day in humans, (blood epithelia)

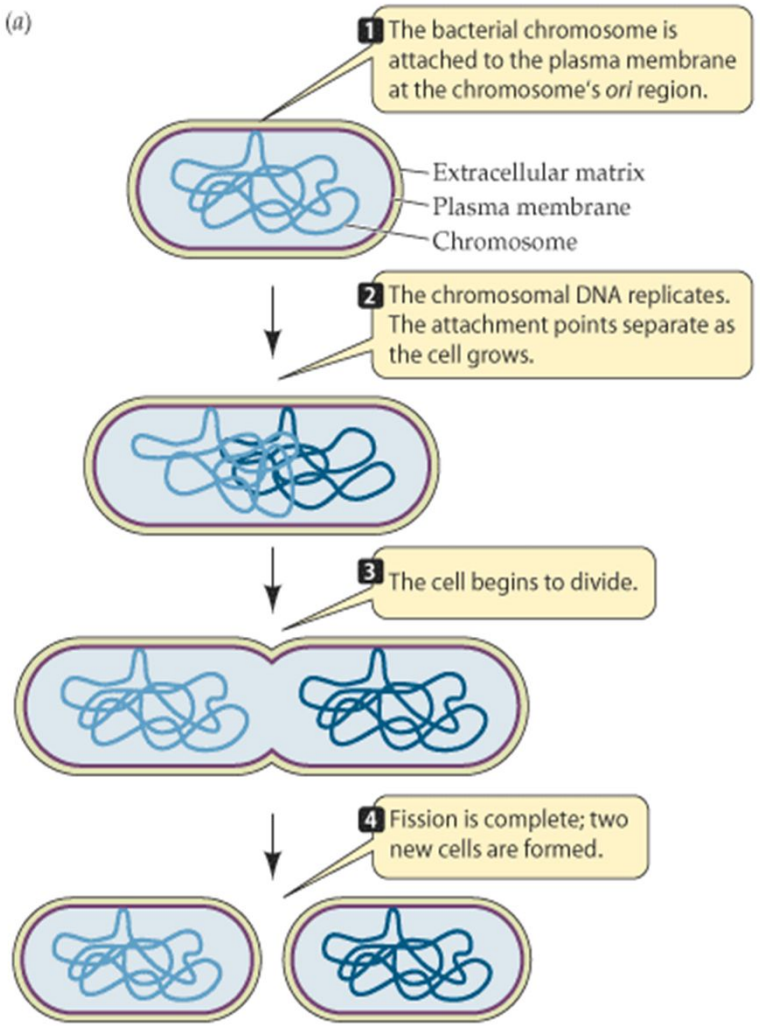
Meiosis: Organisms shuffle genes.

Even the Prokaryotes Do It



Favorable Conditions

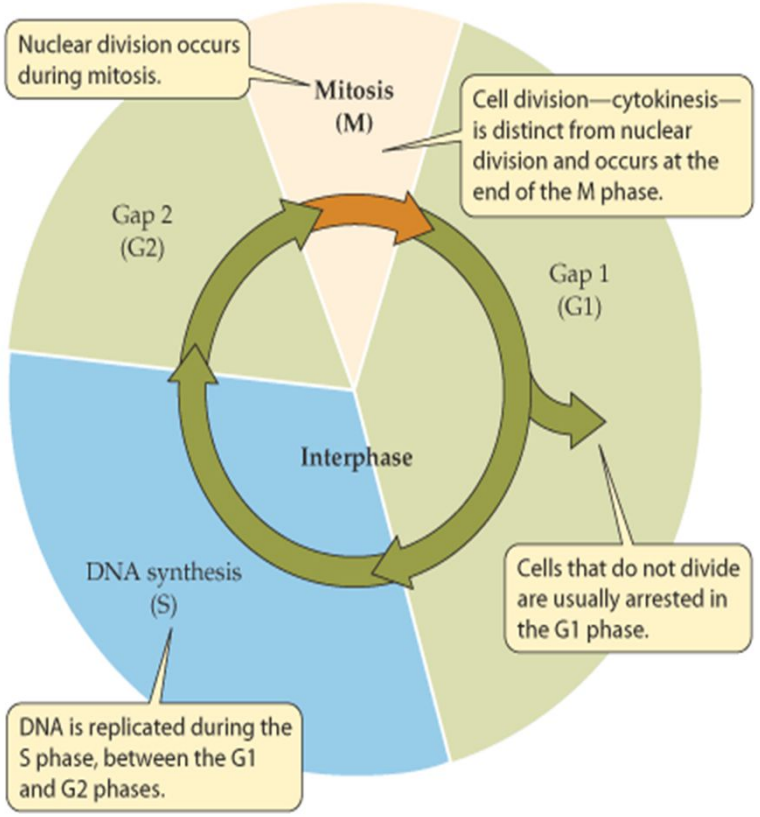
20 - 40 min

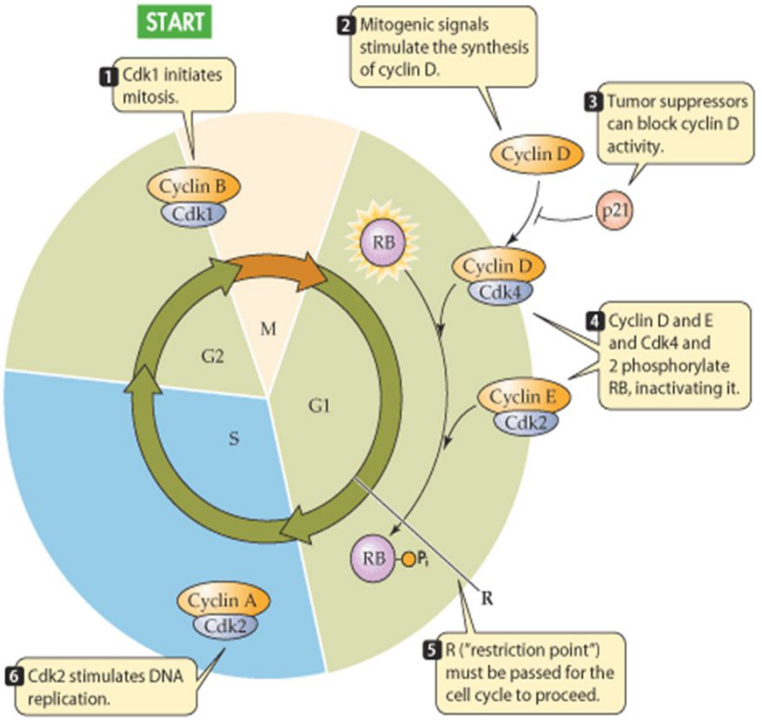

(a)
1 The bacterial chromosome is attached to the plasma membrane at the chromosome's *ori* region.
Extracellular matrix
Plasma membrane
Chromosome
2 The chromosomal DNA replicates. The attachment points separate as the cell grows.
3 The cell begins to divide.
4 Fission is complete; two new cells are formed.

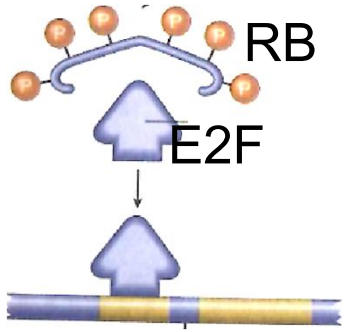
END

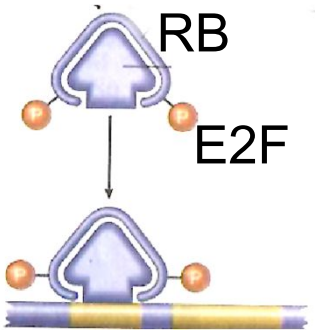
Cell Cycle

Cell Cycle


Nuclear division occurs during mitosis.
Mitosis (M)
Cell division—cytokinesis—is distinct from nuclear division and occurs at the end of the M phase.
Gap 2 (G2)
Gap 1 (G1)
Interphase
DNA synthesis (S)
Cells that do not divide are usually arrested in the G1 phase.
DNA is replicated during the S phase, between the G1 and G2 phases.


START
1 Cdk1 initiates mitosis.
2 Mitogenic signals stimulate the synthesis of cyclin D.
3 Tumor suppressors can block cyclin D activity.
Cyclin D
p21
Cyclin B
Cdk1
RB
Cyclin D
Cdk4
M
G2
G1
S
4 Cyclin D and E and Cdk4 and 2 phosphorylate RB, inactivating it.
Cyclin E
Cdk2
RB
Pi
R
Cyclin A
Cdk2
6 Cdk2 stimulates DNA replication.
5 R ("restriction point") must be passed for the cell cycle to proceed.

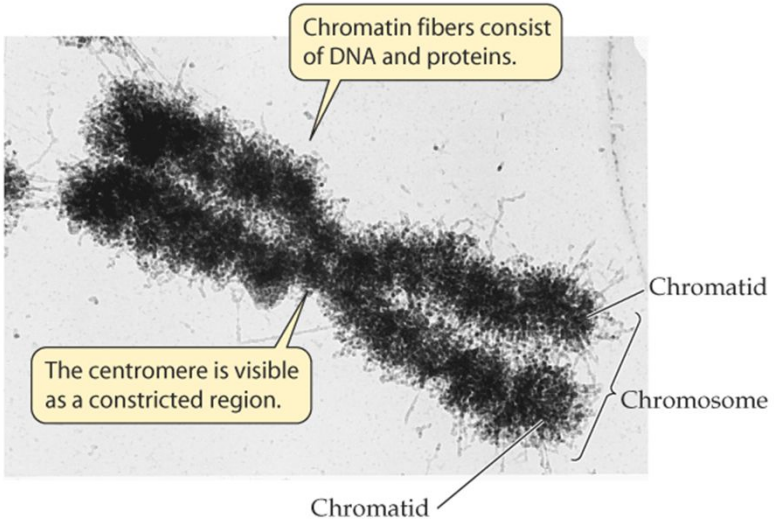

RB
E2F


RB
E2F

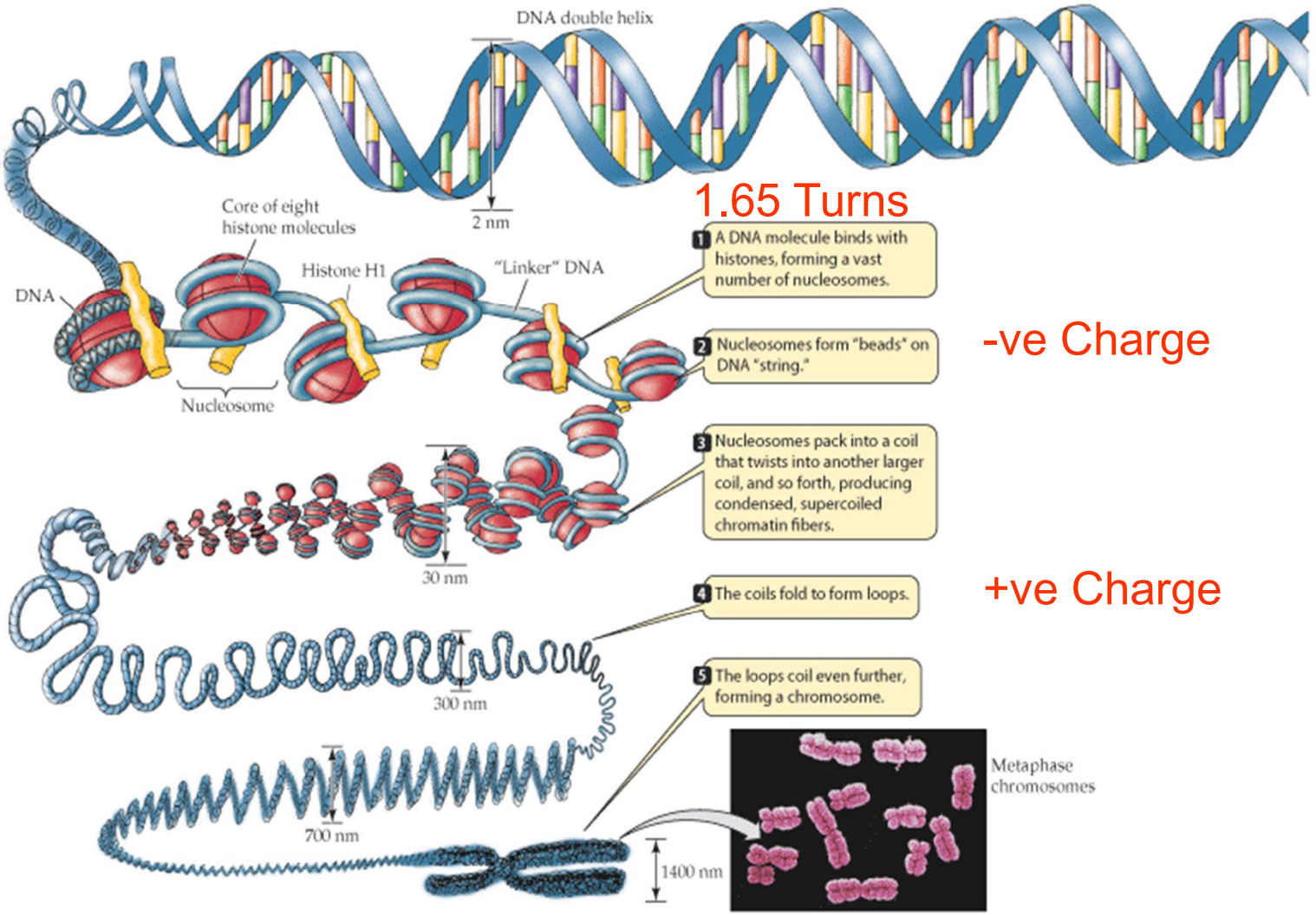
END

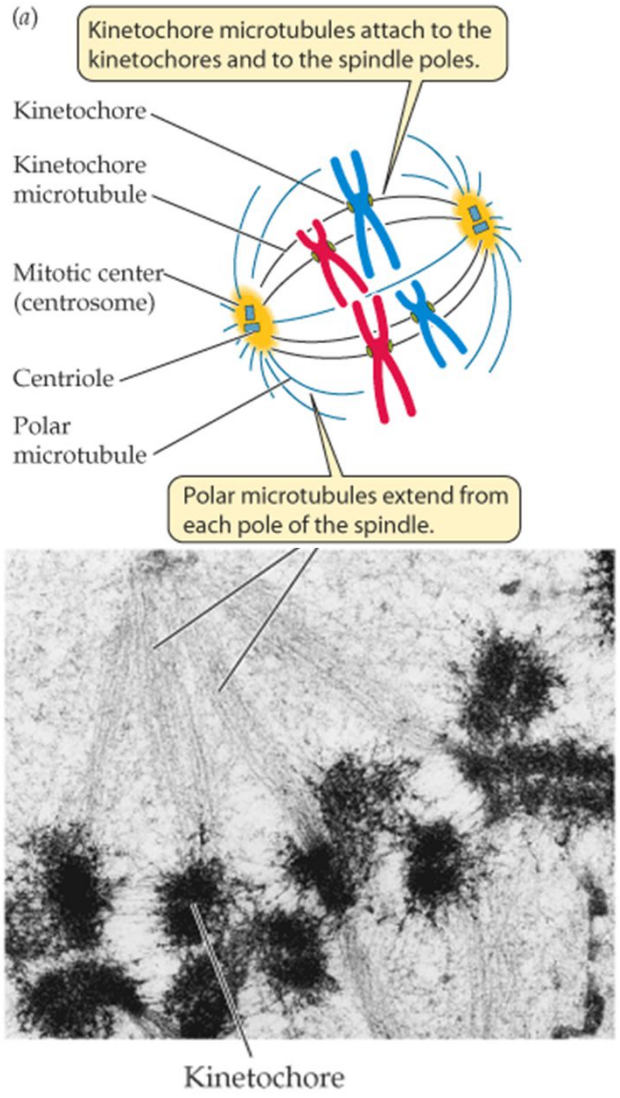
Chromosome

Anatomy



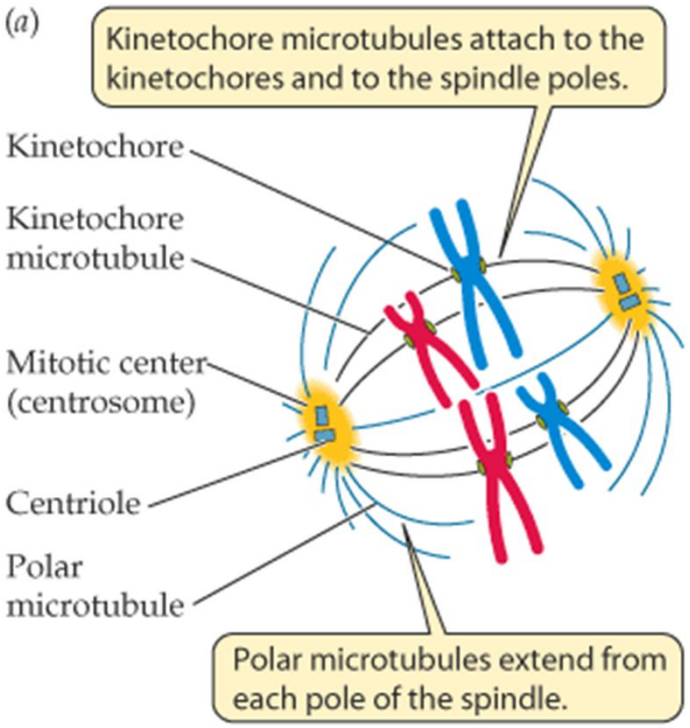
A chromosome is an organized structure of DNA and protein

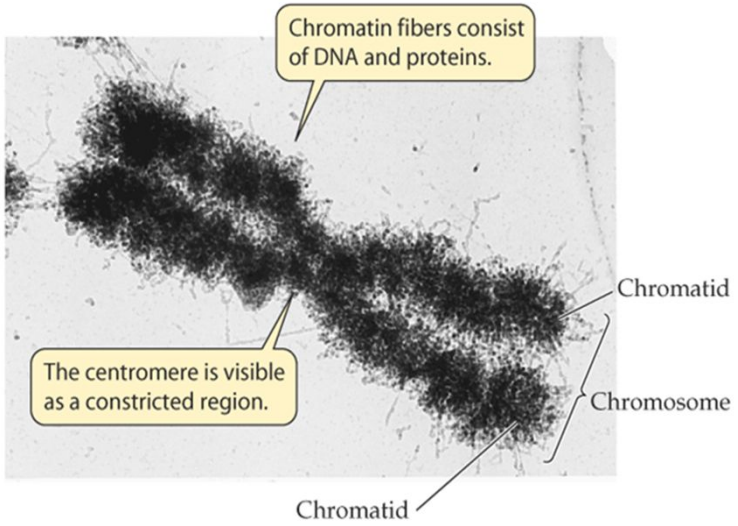

DNA double helix
2 nm
1.65 Turns
Core of eight histone molecules
Histone H1
"Linker" DNA
DNA
Nucleosome
1 A DNA molecule binds with histones, forming a vast number of nucleosomes.
2 Nucleosomes form "beads" on DNA "string."
-ve Charge
3 Nucleosomes pack into a coil that twists into another larger coil, and so forth, producing condensed, supercoiled chromatin fibers.
30 nm
+ve Charge
4 The coils fold to form loops.
300 nm
5 The loops coil even further, forming a chromosome.
700 nm
1400 nm
Metaphase chromosomes


(a)
Kinetochore microtubules attach to the kinetochores and to the spindle poles.
Kinetochore
Kinetochore microtubule
Mitotic center (centrosome)
Centriole
Polar microtubule
Polar microtubules extend from each pole of the spindle.
Kinetochore

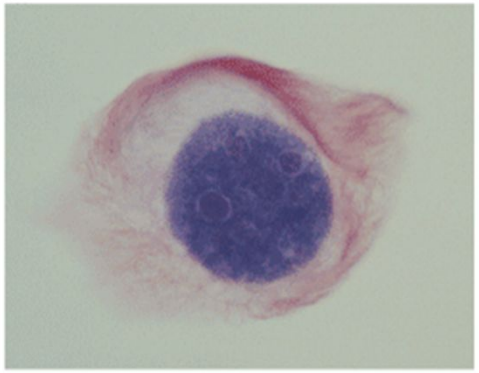
END

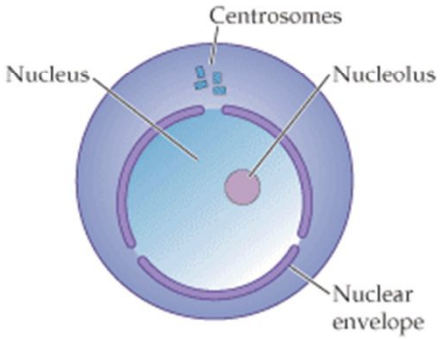
Mitosis


(a)
Kinetochore microtubules attach to the kinetochores and to the spindle poles.
Kinetochore
Kinetochore microtubule
Mitotic center (centrosome)
Centriole
Polar microtubule
Polar microtubules extend from each pole of the spindle.


Chromatin fibers consist of DNA and proteins.
The centromere is visible as a constricted region.
Chromatid
Chromosome
Chromatid

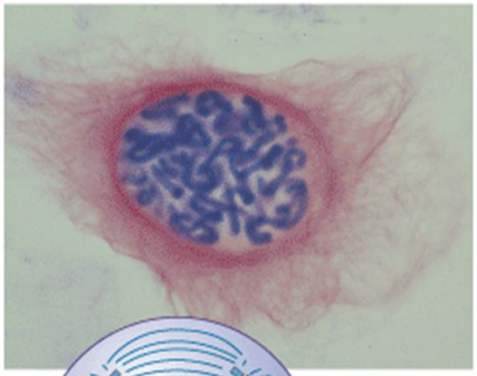
Interphase

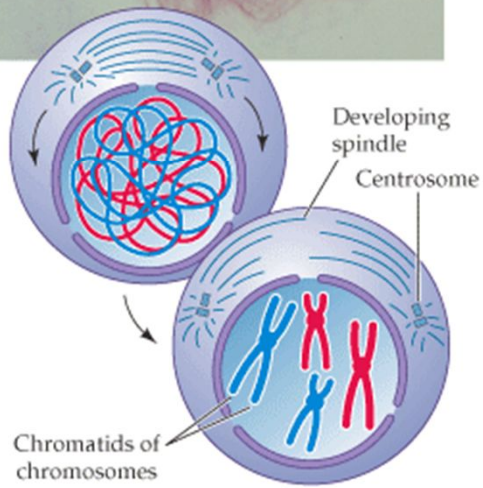




1 During the S phase of interphase, the nucleus replicates its DNA and centrosomes.

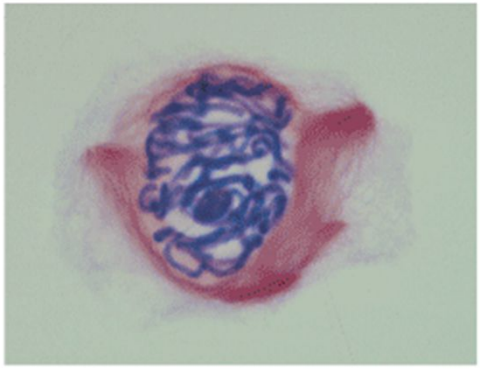
Prophase

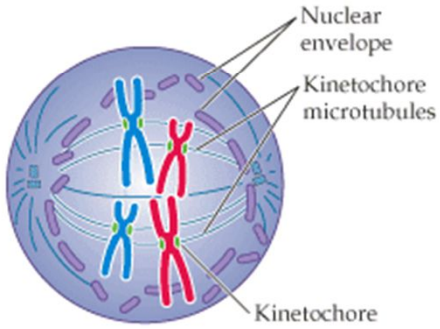




2 The chromatin coils and supercoils, becoming more and more compact and eventually condensing into visible chromosomes. The chromosomes consist of identical, paired sister chromatids.

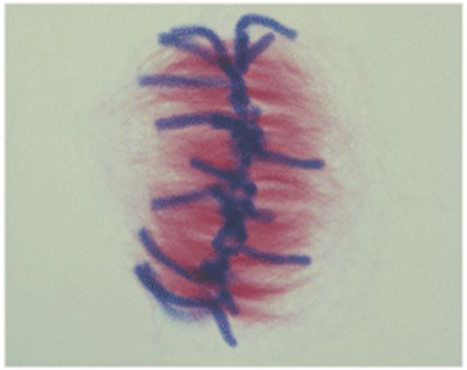
Prometaphase

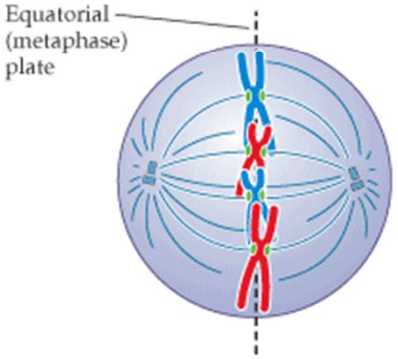




3 The nuclear envelope breaks down. Kinetochore microtubules appear and connect the kinetochores to the microtubule organizing centers.

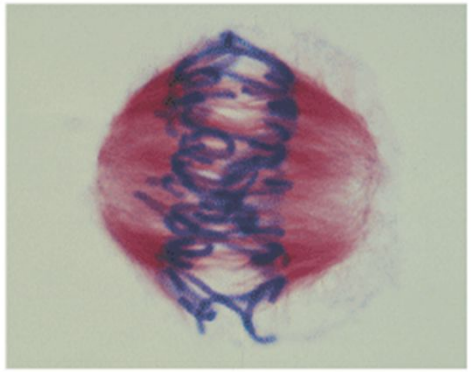
Metaphase

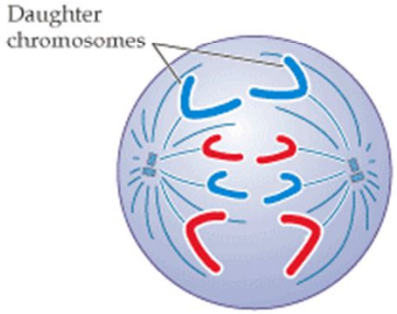




4 The centromeres (regions connecting paired chromatids) become aligned in a plane at the cell's equator.

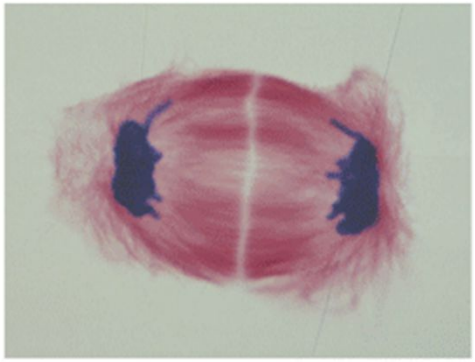
Anaphase

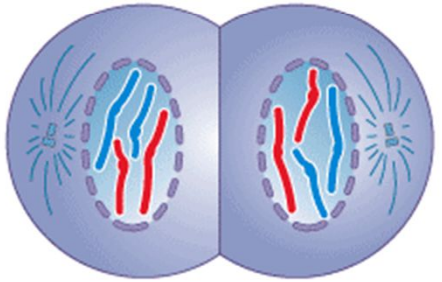




5 The paired sister chromatids separate, and the new daughter chromosomes begin to move toward the poles.

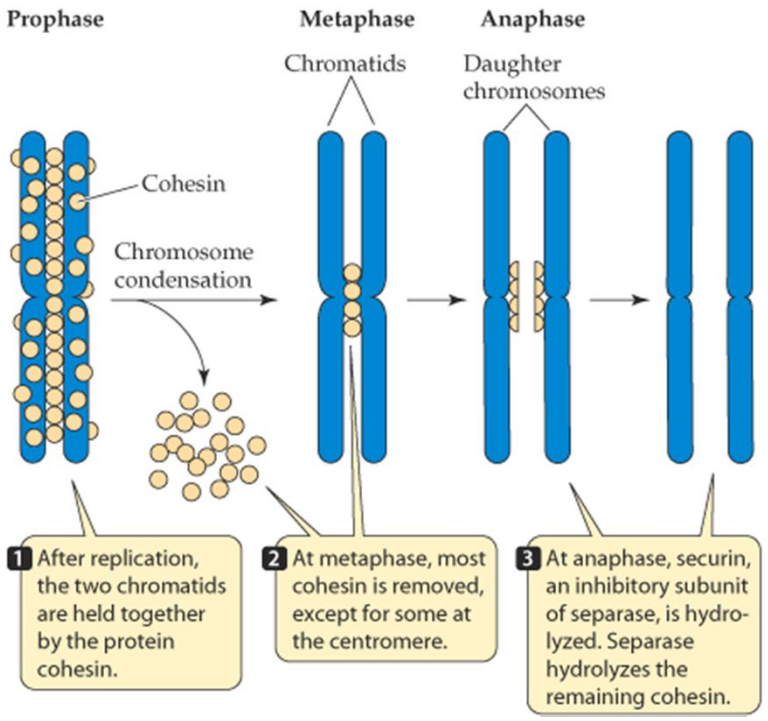
Telophase



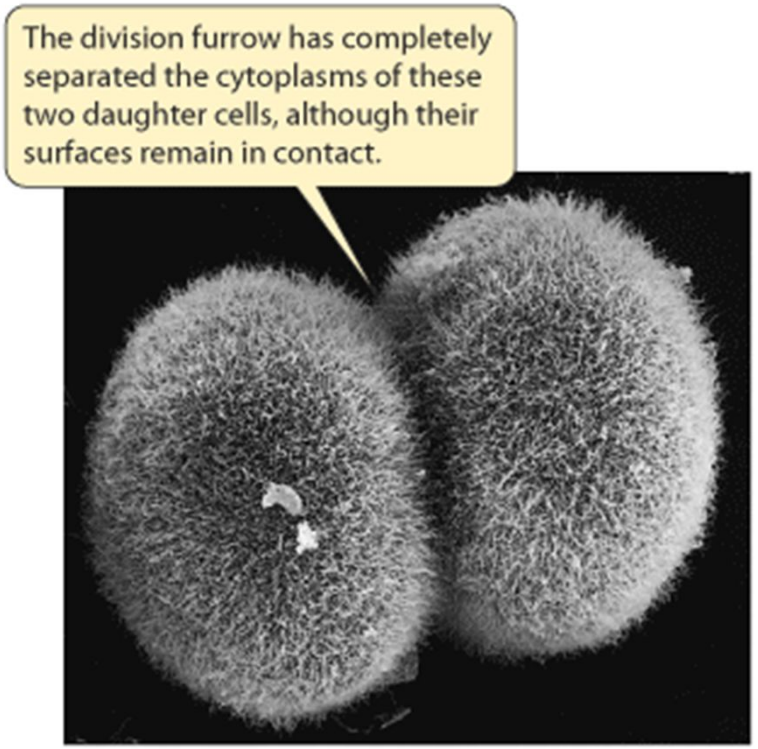


6 The daughter chromosomes reach the poles. Telophase passes into the next interphase as the nuclear envelopes and nucleoli re-form and the chromatin becomes diffuse.

The Glue


Prophase
Metaphase
Anaphase
Chromatids
Daughter chromosomes
Cohesin
Chromosome condensation
1 After replication, the two chromatids are held together by the protein cohesin.
2 At metaphase, most cohesin is removed, except for some at the centromere.
3 At anaphase, securin, an inhibitory subunit of separase, is hydrolyzed. Separase hydrolyzes the remaining cohesin.

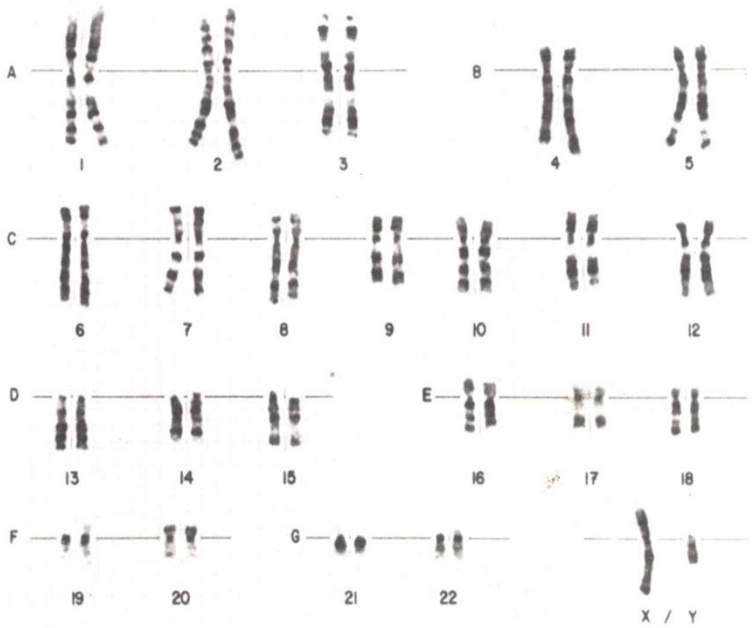
Cytokinesis



END

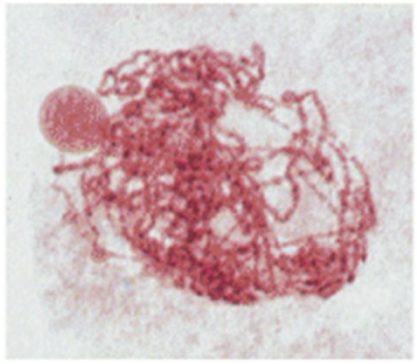
Meiosis I

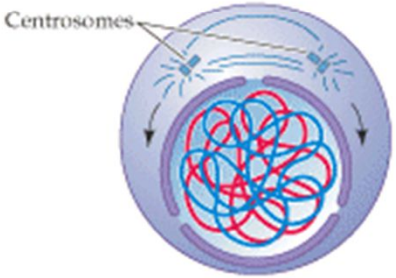
Karyotype



MEIOSIS I

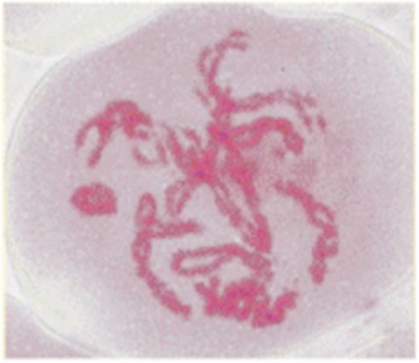
Early Prophase I

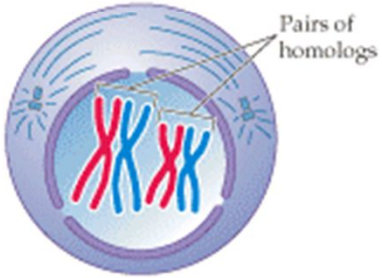




1 The chromatin begins to condense following interphase.

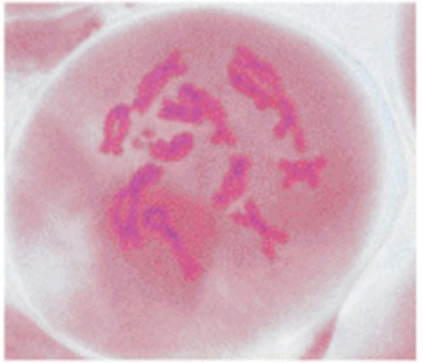
Mid-Prophase I





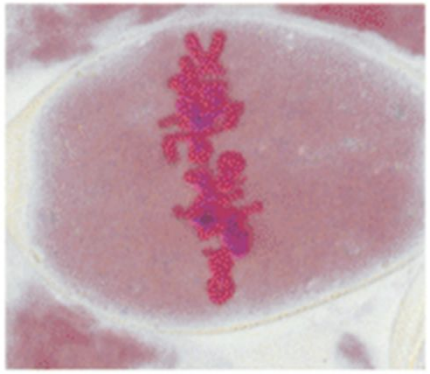
2 Synapsis aligns homologs, and chromosomes condense. Homologs are shown in different colors indicating those coming from each parent. In reality, their differences are very small, usually comprising different alleles of some genes.

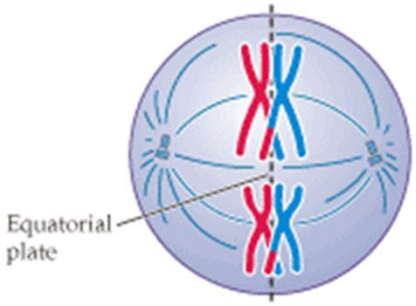
Late Prophase I–Prometaphase



3 The chromosomes continue to coil and shorten. Crossing over results in an exchange of genetic material. In prometaphase the nuclear envelope breaks down.

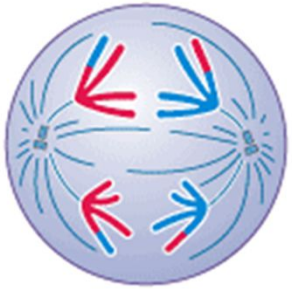
Metaphase I





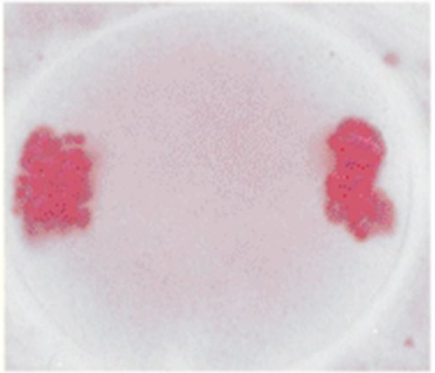
4 The homologous chromosomes line up on the equatorial (metaphase) plate.

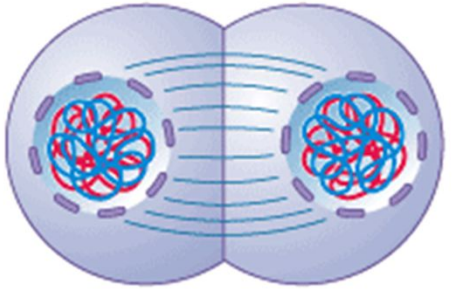
Anaphase I



5 The homologous chromosomes (each with two chromatids) move to opposite poles of the cell.

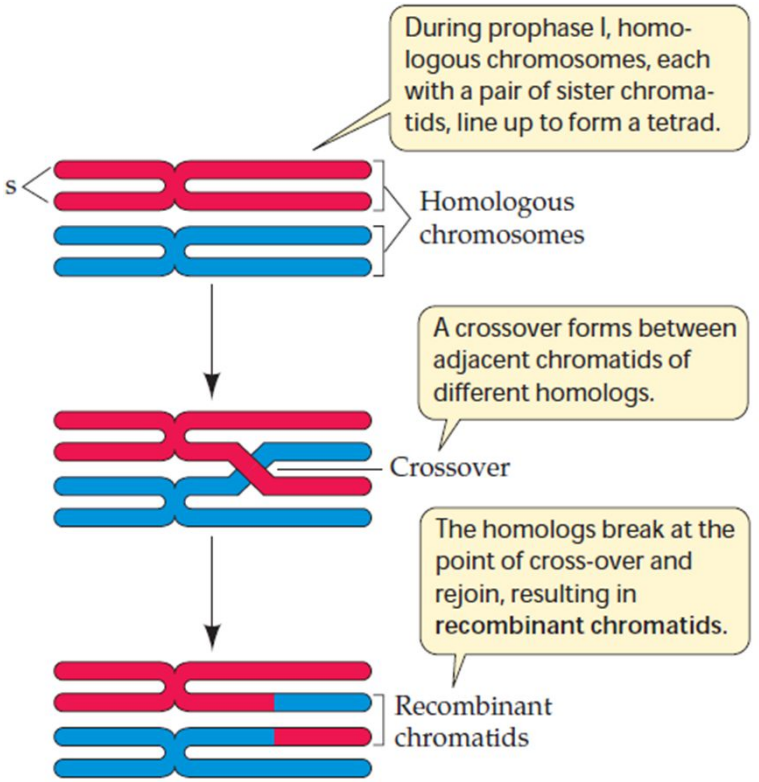
Telophase I





6 The chromosomes gather into nuclei, and the original cell divides.

Genetically Diverse Chromosomes

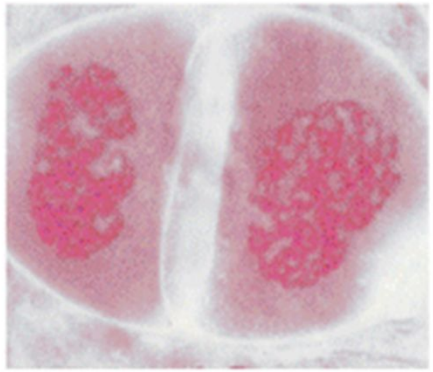


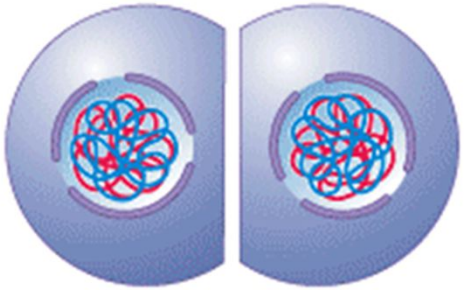
END

Meiosis II

MEIOSIS II

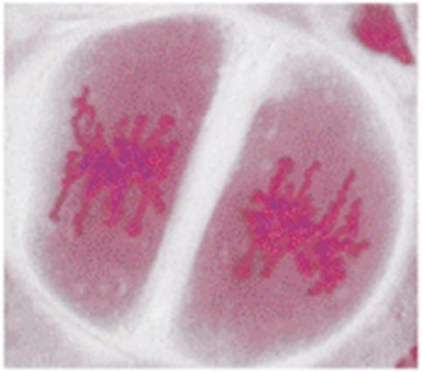
Prophase II

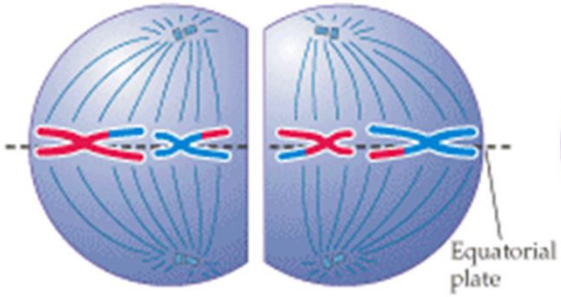




7 The chromosomes condense again, following a brief interphase (interkinesis) in which DNA does not replicate.

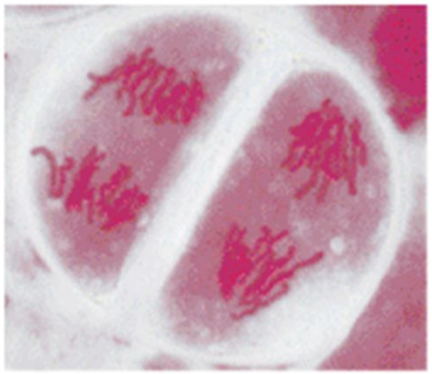
Metaphase II

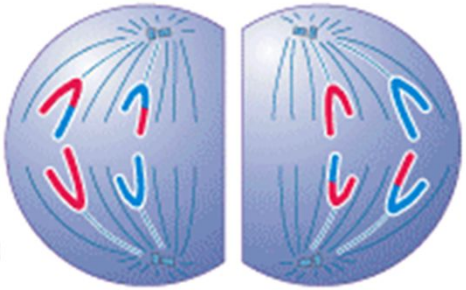




8 Kinetochores of the paired chromatids line up across the equatorial plates of each cell.

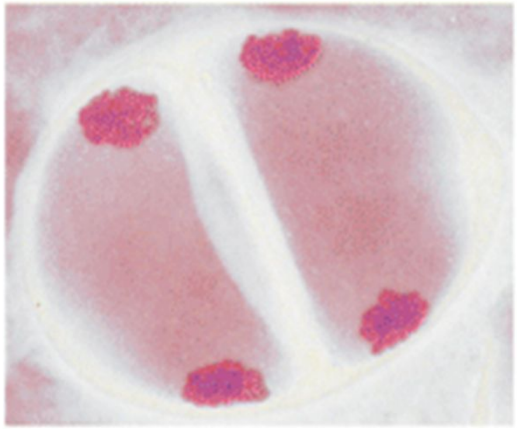
Anaphase II

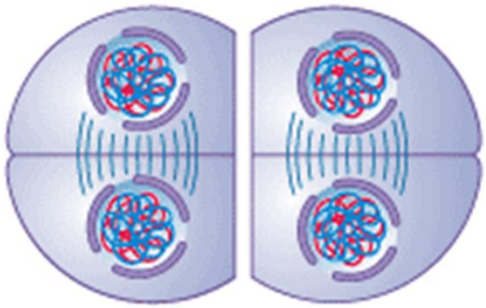




9 The chromatids finally separate, becoming chromosomes in their own right, and are pulled to opposite poles. Because of crossing over in prophase I, each new cell will have a different genetic makeup.

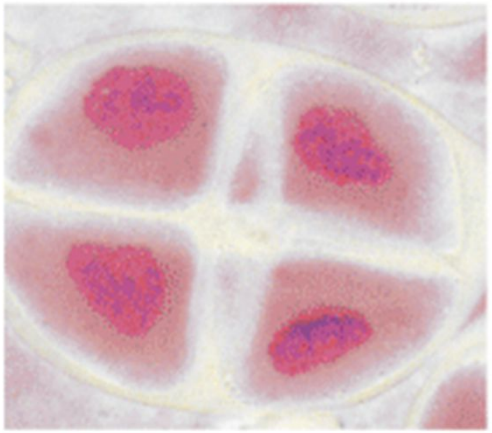
Telophase II

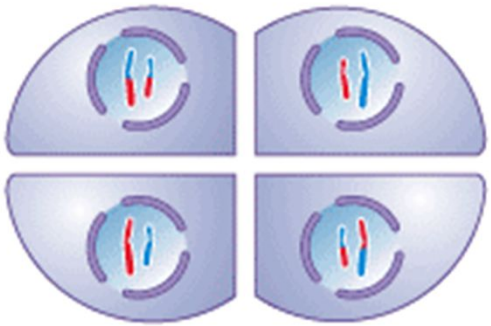




10 The chromosomes gather into nuclei, and the cells divide.

Products





11 Each of the four cells has a nucleus with a haploid number of chromosomes.

Features

Reduces chromosome number from diploid to haploid

Ensures each of the haploid products has a complete set of chromosomes

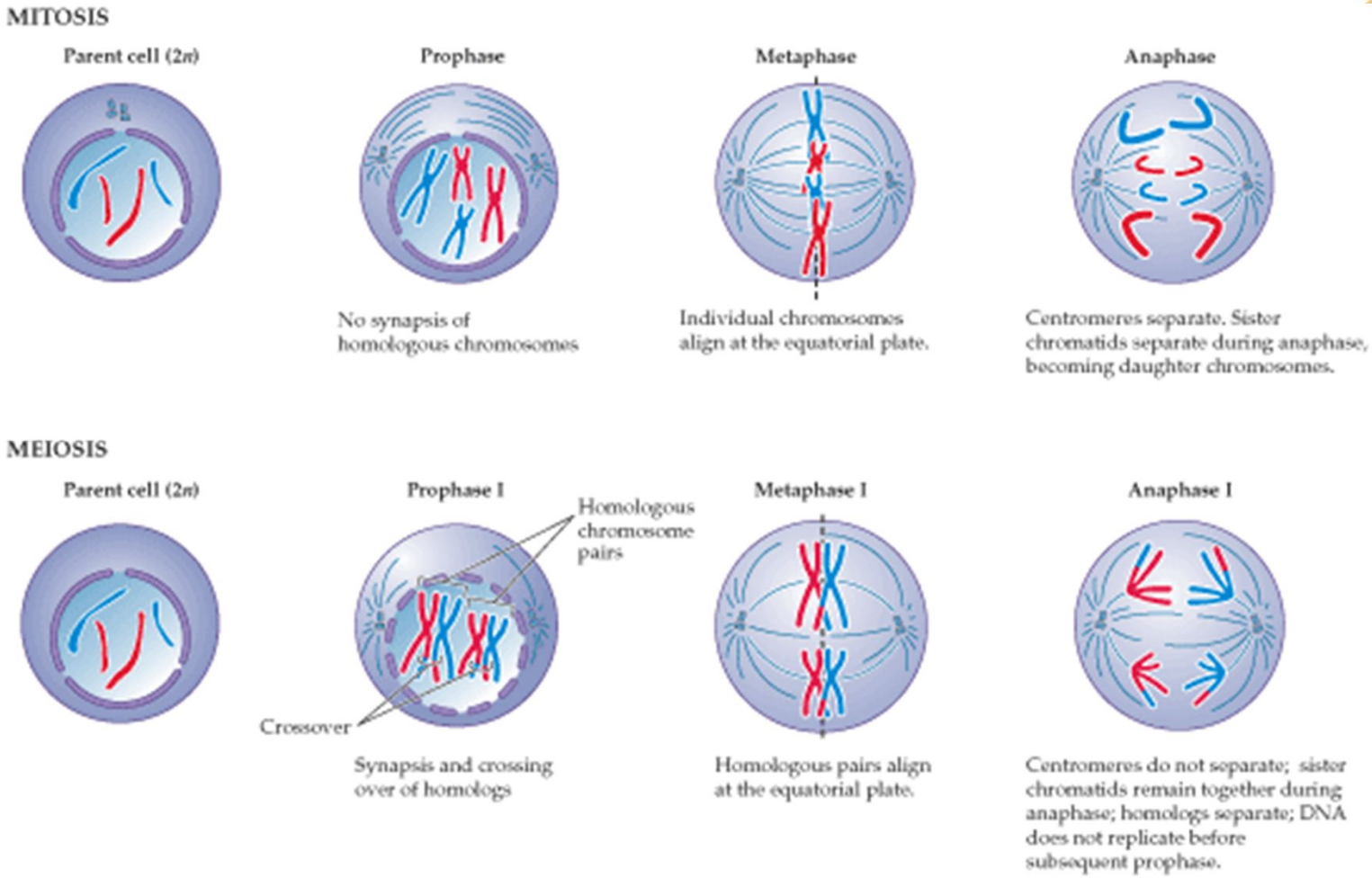
Promotes genetic diversity among the products.

END

Cell Biology

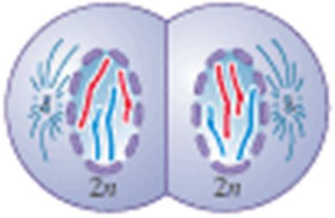
The Comparison

The Comparison



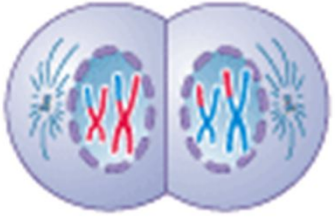
The Comparison

Two daughter cells (each 2n)

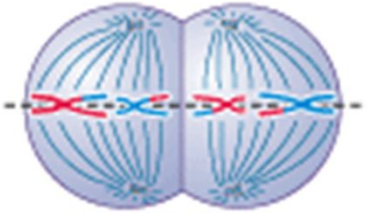


Mitosis is a mechanism for constancy: The parent nucleus produces two identical daughter nuclei.

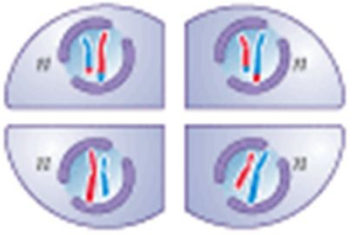
Telophase I



Metaphase II



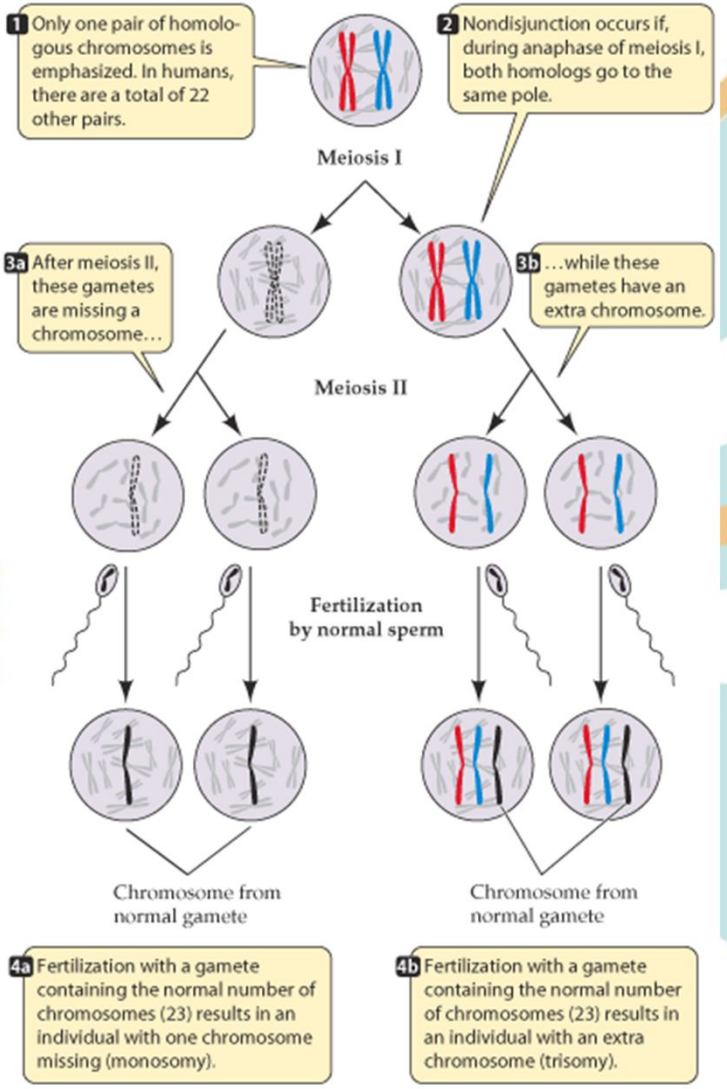
Four daughter cells (each n)



Chromatids separate.

Meiosis is a mechanism for diversity: The parent nucleus produces four different haploid daughter nuclei.

The Common Mistake



END

Genetics Introduction

Historic Background

Blending VS Particulate

Definitions

A character is an observable feature, such as flower color.

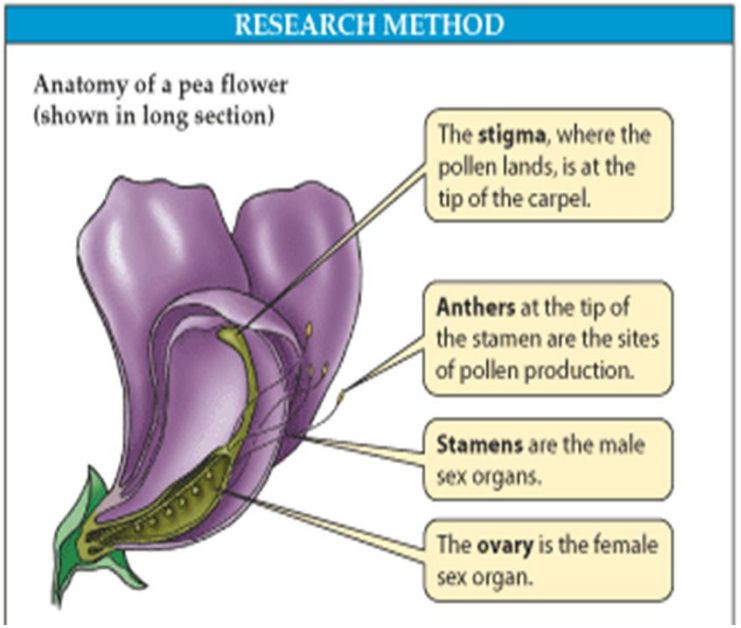
A trait is a particular form of a character, such as white flowers.

A heritable character trait is one that is passed from parent to offspring.

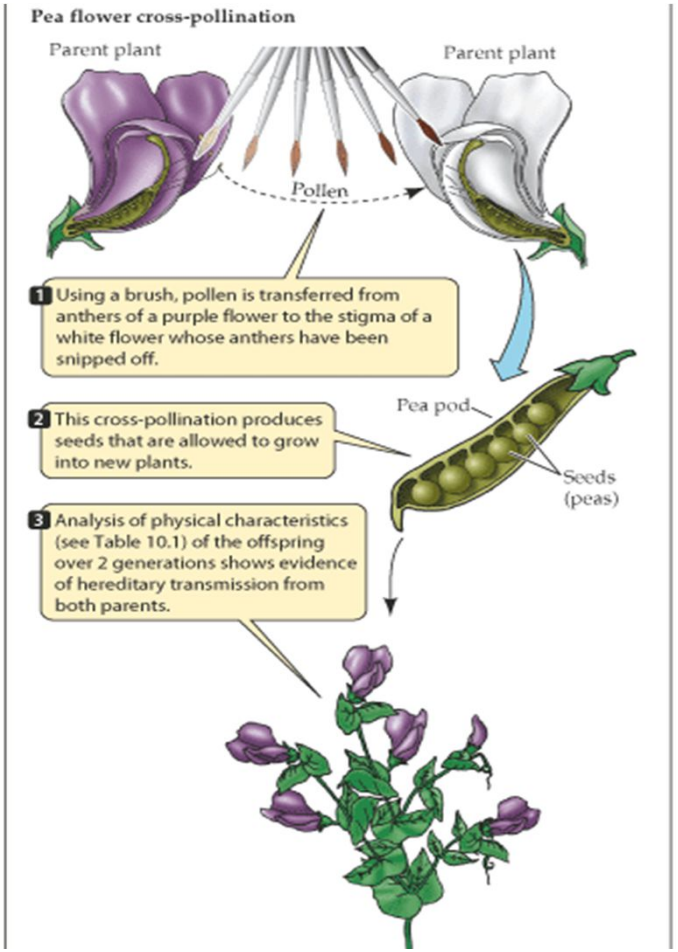
END

The Question

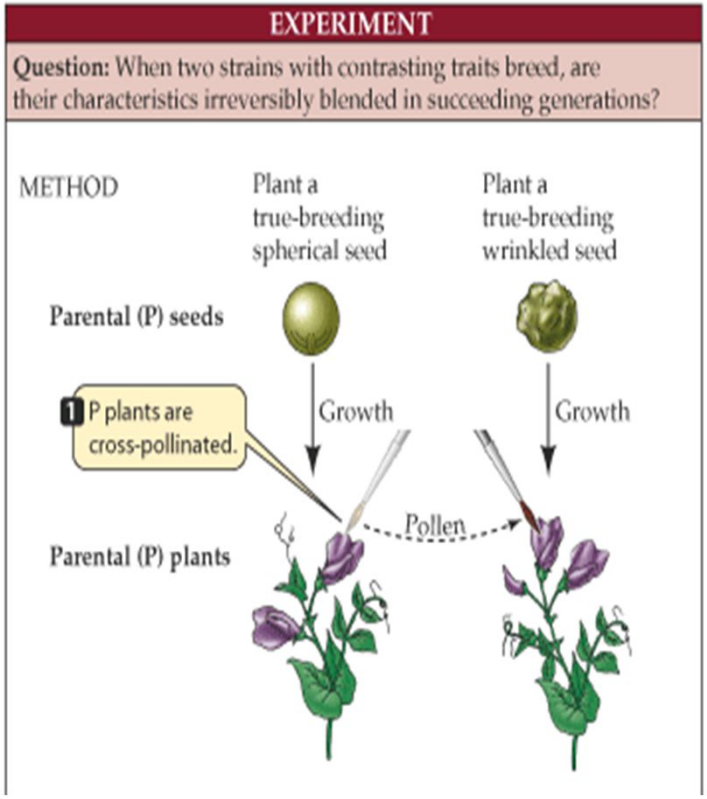
The Anatomy of the Subject



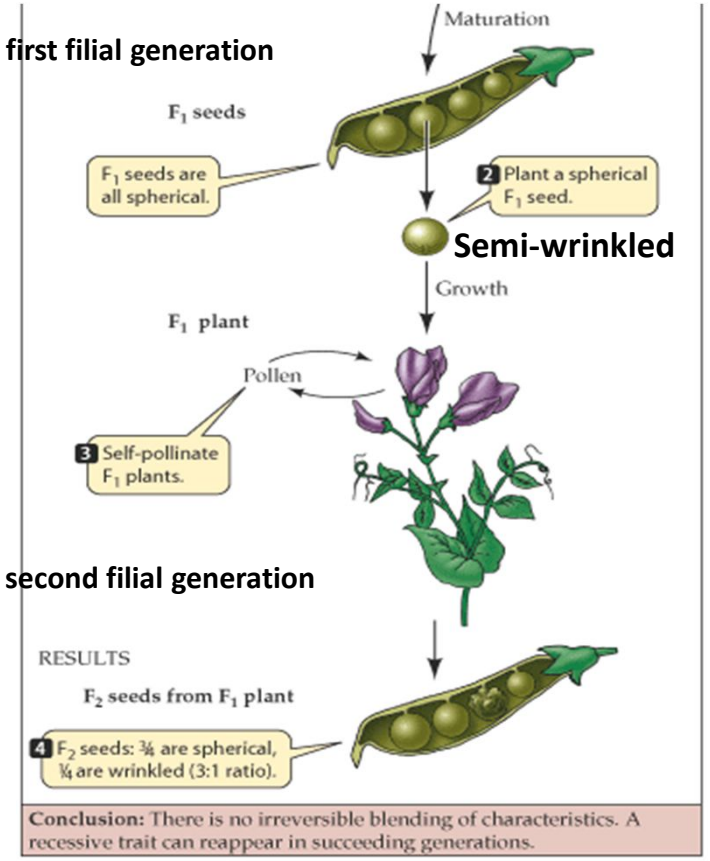
The Technique



The Monohybrid Cross

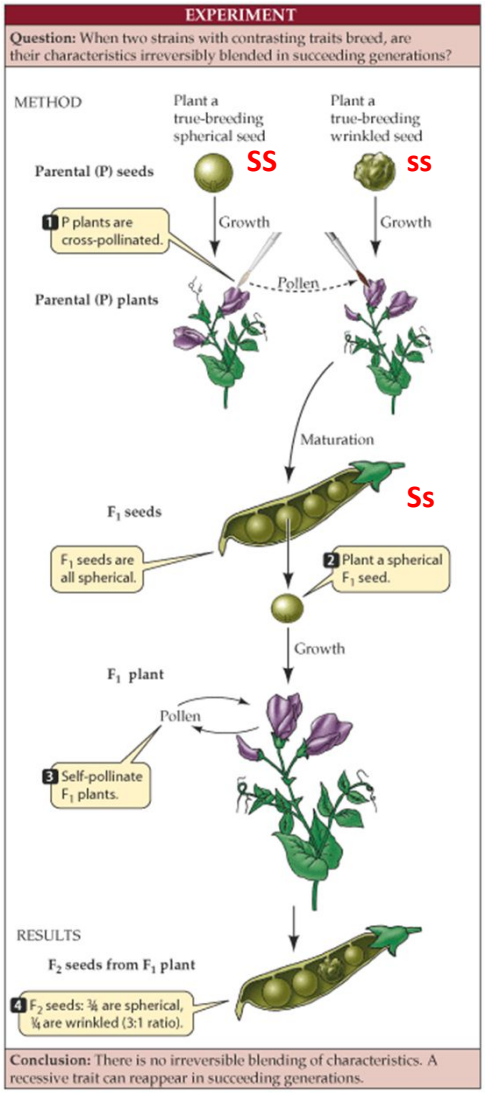


The Answer



END

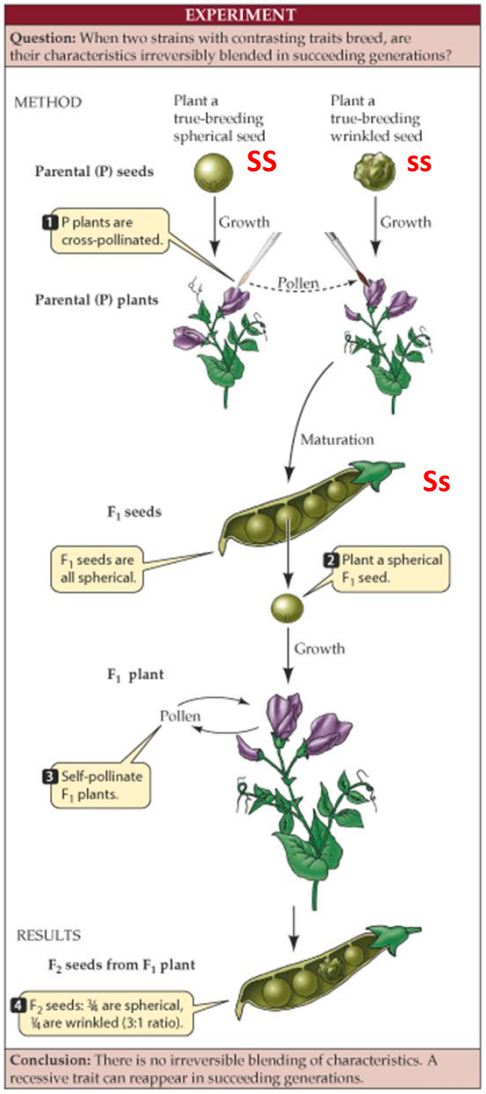
The Lessons Learned



ss trait not expressed in the F1 generation, (reappeared in F2) Mendal concluded SS trait domin-ant to the ss trait (recessive).

Mendel worked with his data mathematically & concluded each pea plant has two units of inheritance for each character, 1 from each parent.

During gamete production only 1 of these paired units is given to a gamete. Resulting zygote contains 2, (produced fusion of 2 gametes).

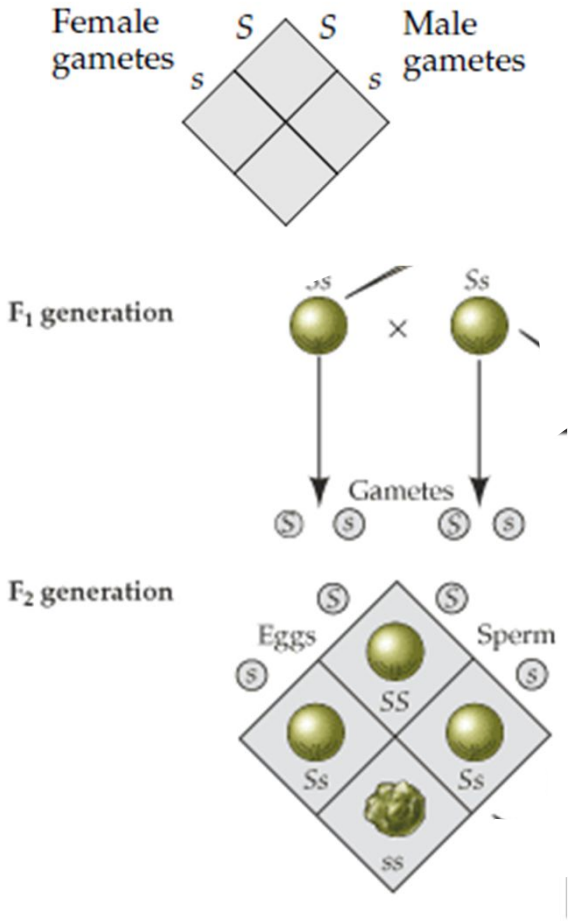


Mendel's unit of inheritance is now called a gene. Different forms of a gene (S & s) are called alleles.

Homozygous have two copies of the same allele (ss). Heterozygous have two different alleles of the gene in question (in this case, Ss).

The physical appearance of an organism is its phenotype. Phenotype to be the result of the genotype, or genetic constitution.

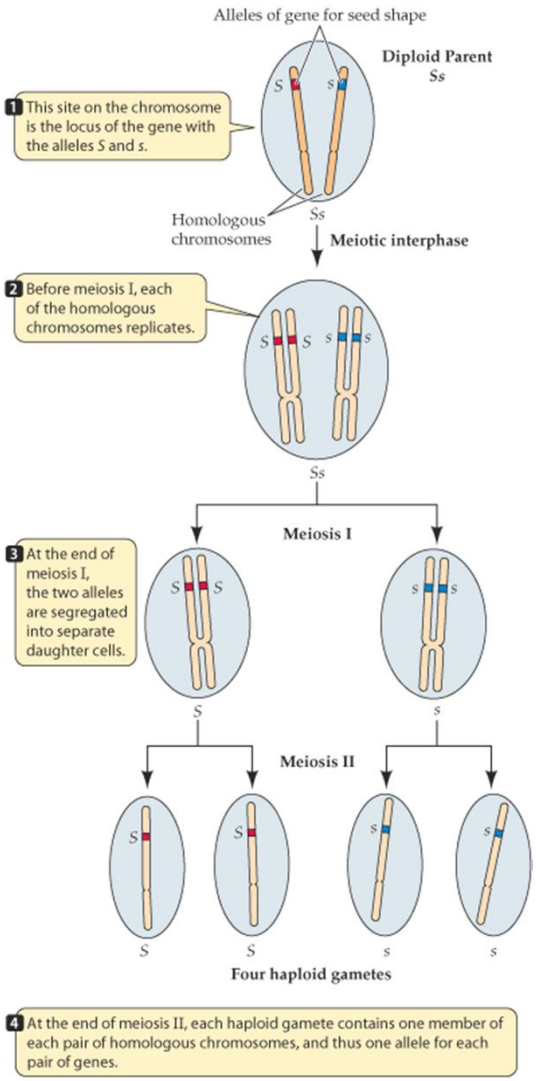
The Punnett Square

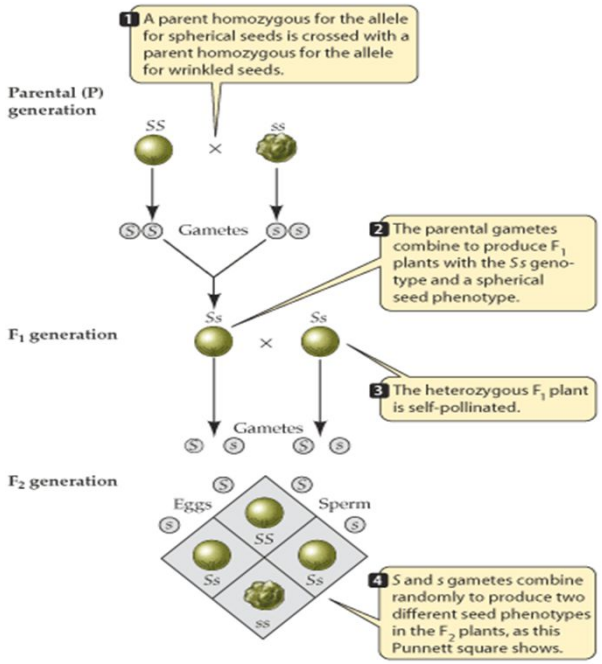


END

The First Law

When any individual produces gametes, the two alleles separate, so that each gamete receives only one member of the pair of alleles.



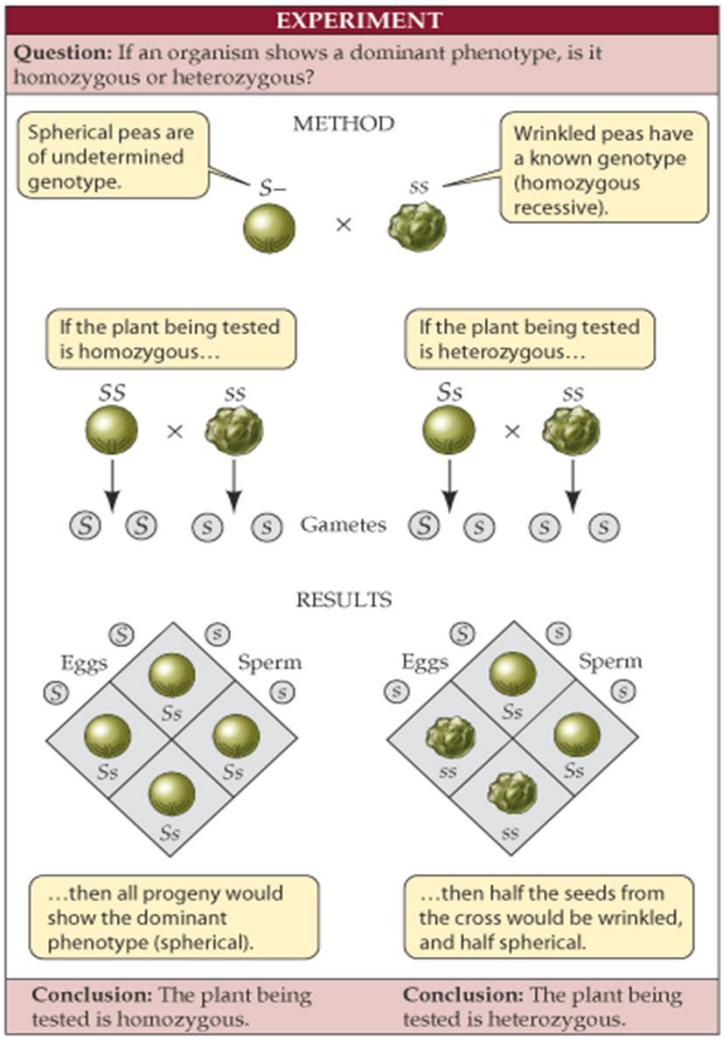


10.1 Mendel's Results from Monohybrid Crosses

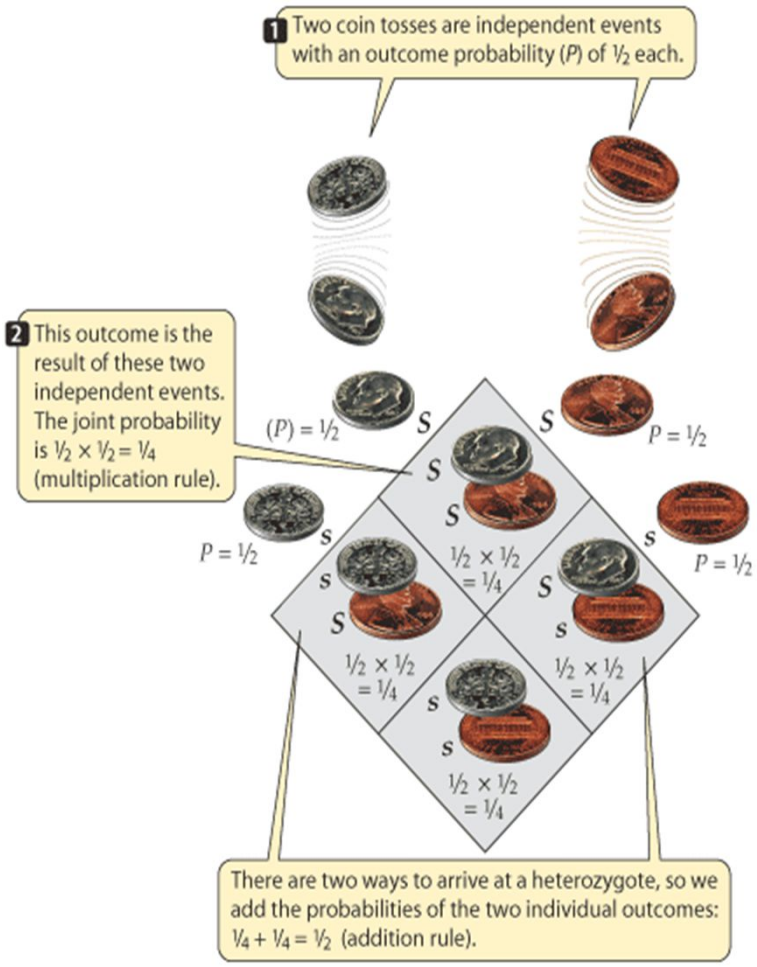
each member of the P generation is true-breeding

	PARENTAL GENERATION PHENOTYPES			F_2 GENERATION PHENOTYPES			
	DOMINANT		RECESSIVE	DOMINANT	RECESSIVE	TOTAL	RATIO
	Spherical seeds	×	Wrinkled seeds	5,474	1,850	7,324	2.96:1
	Yellow seeds	×	Green seeds	6,022	2,001	8,023	3.01:1
	Purple flowers	×	White flowers	705	224	929	3.15:1
	Inflated pods	×	Constricted pods	882	299	1,181	2.95:1
	Green pods	×	Yellow pods	428	152	580	2.82:1
	Axial flowers	×	Terminal flowers	651	207	858	3.14:1
	Tall stems (1 m)	×	Dwarf stems (0.3 m)	787	277	1,064	2.84:1

END


EXPERIMENT
Question: If an organism shows a dominant phenotype, is it homozygous or heterozygous?
METHOD
Spherical peas are of undetermined genotype.
S–
×
ss
Wrinkled peas have a known genotype (homozygous recessive).
If the plant being tested is homozygous…
SS
×
ss
S S s s
Gametes
If the plant being tested is heterozygous…
Ss
×
ss
S s s s
RESULTS
Eggs
Sperm
Ss
Ss
Ss
Ss
…then all progeny would show the dominant phenotype (spherical).
Eggs
Sperm
Ss
ss
Ss
ss
…then half the seeds from the cross would be wrinkled, and half spherical.
Conclusion: The plant being tested is homozygous.
Conclusion: The plant being tested is heterozygous.

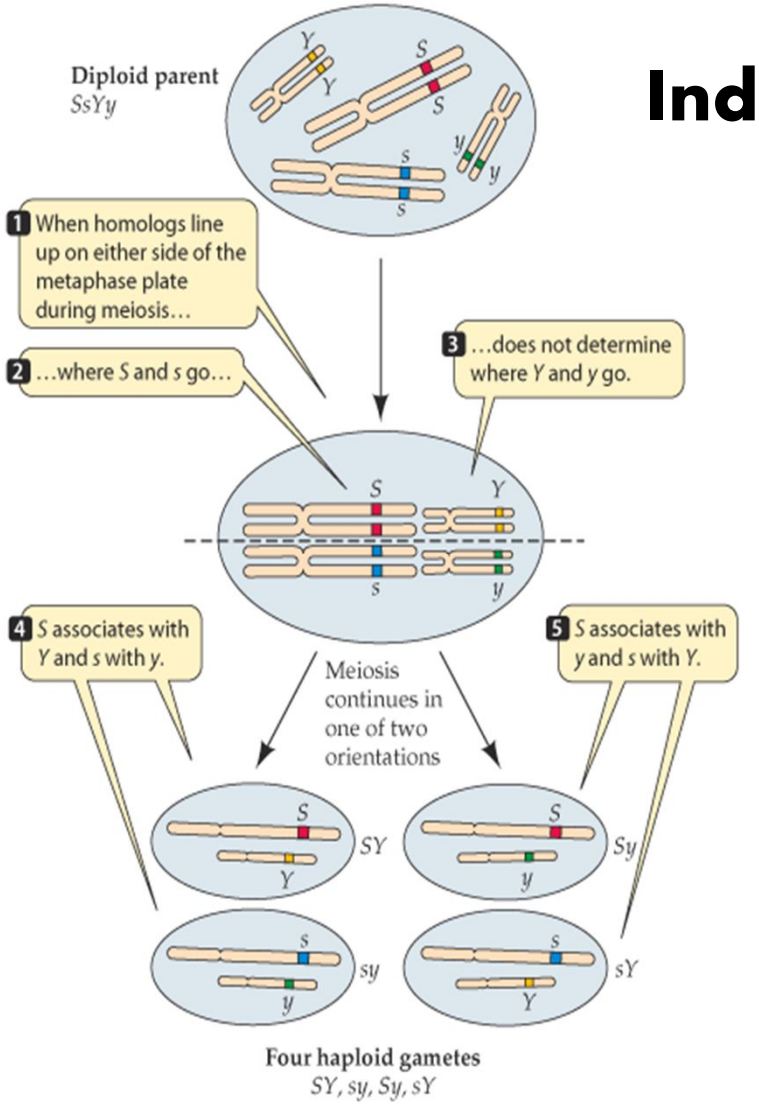
The Second Law



Probabilities

If an event is absolutely certain to happen, its probability is 1.

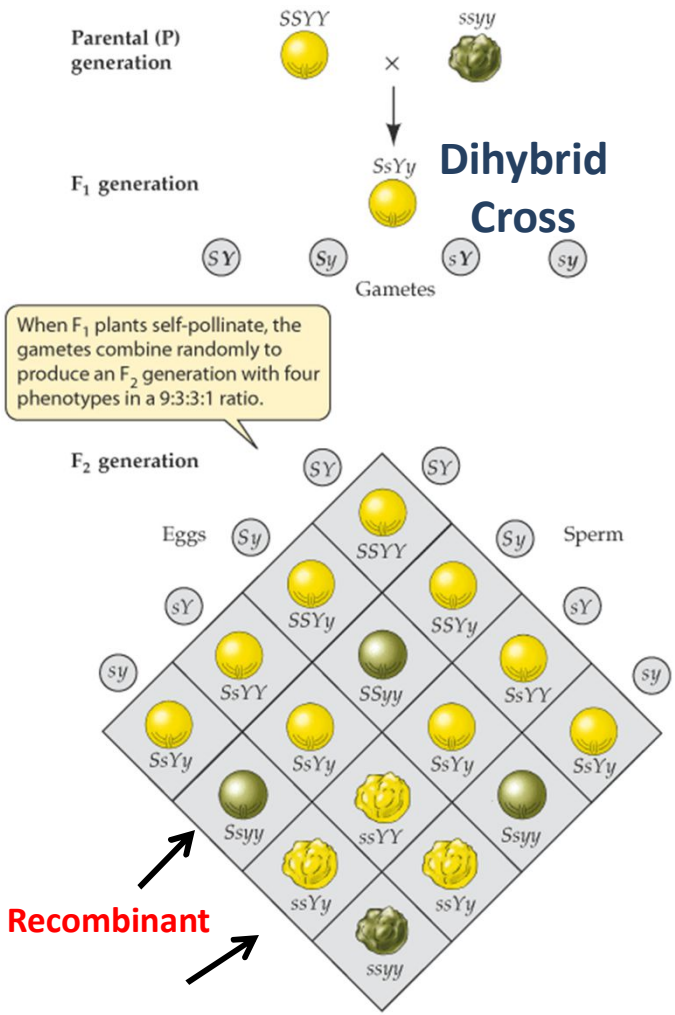
If it cannot possibly happen, its probability is 0. Otherwise, its probability lies between 0 and 1.



Independent Assortment

Alleles of different genes assort independently of one another during gamete formation.

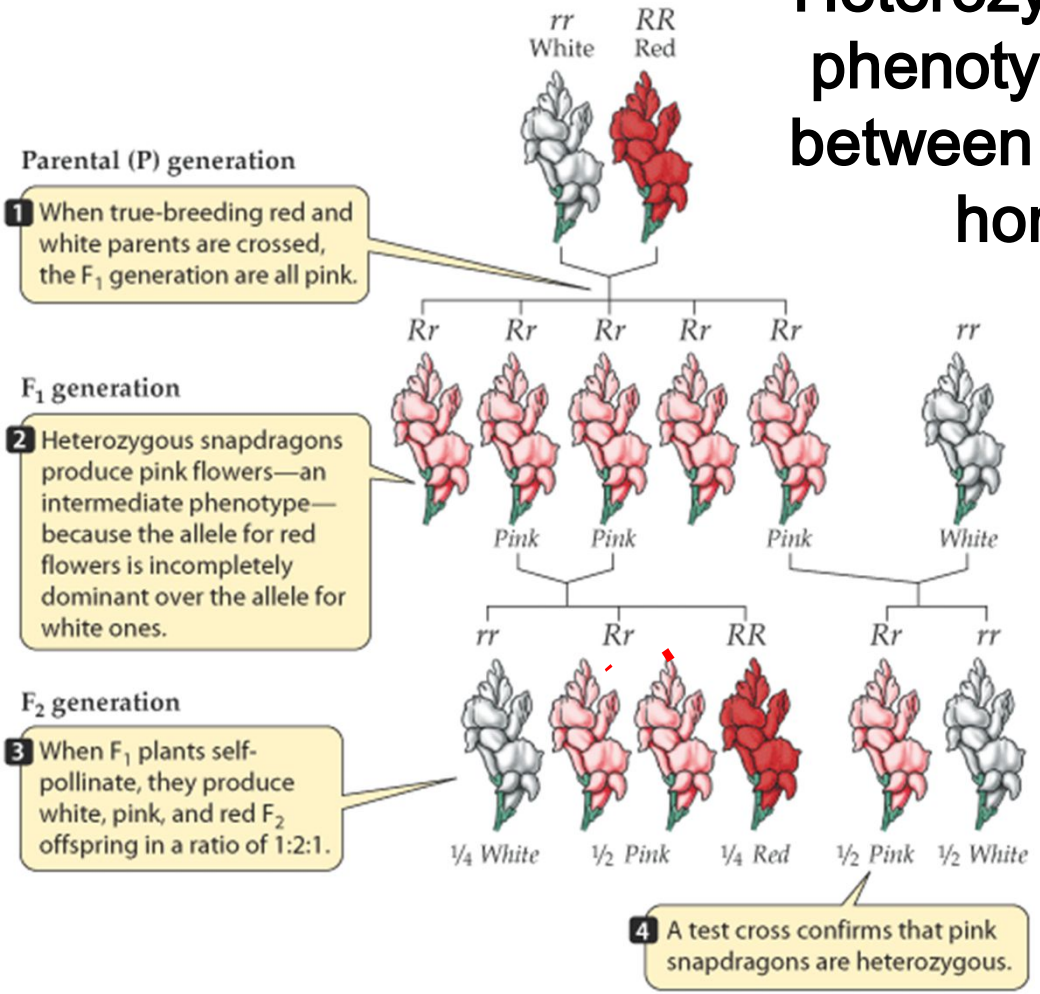
That is, the segregation of the alleles of gene A is independent of the segregation of alleles of gene B.


Parental (P) generation
SSYY
ssyy
×
F1 generation
SsYy
Dihybrid Cross
SY
Sy
sY
sy
Gametes
When F1 plants self-pollinate, the gametes combine randomly to produce an F2 generation with four phenotypes in a 9:3:3:1 ratio.
F2 generation
Eggs
Sperm
SSYY
SSYy
SSYy
SsYY
SSyy
SsYY
SsYy
SsYy
SsYy
SsYy
Ssyy
ssYY
Ssyy
ssYy
ssYy
ssyy
Recombinant

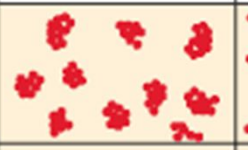
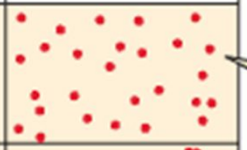
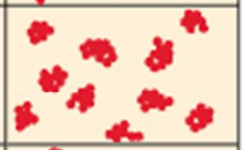
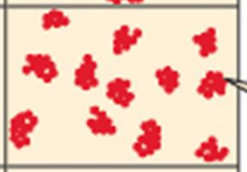
END

Aberrant Phenomenon

Heterozygotes show a phenotype intermediate between those of the two homozygotes



Both alleles appear in heterozygotes

Blood type of cells	Genotype	Antibodies made by body	Reaction to added antibodies	
			Anti-A	Anti-B
A	I^AI^A or I^Ai^O	Anti-B		
B	I^BI^B or I^Bi^O	Anti-A		
AB	I^AI^B	Neither anti-A nor anti-B		
O	i^Oi^O	Both anti-A and anti-B		

Red blood cells that do not react with antibody remain evenly dispersed.

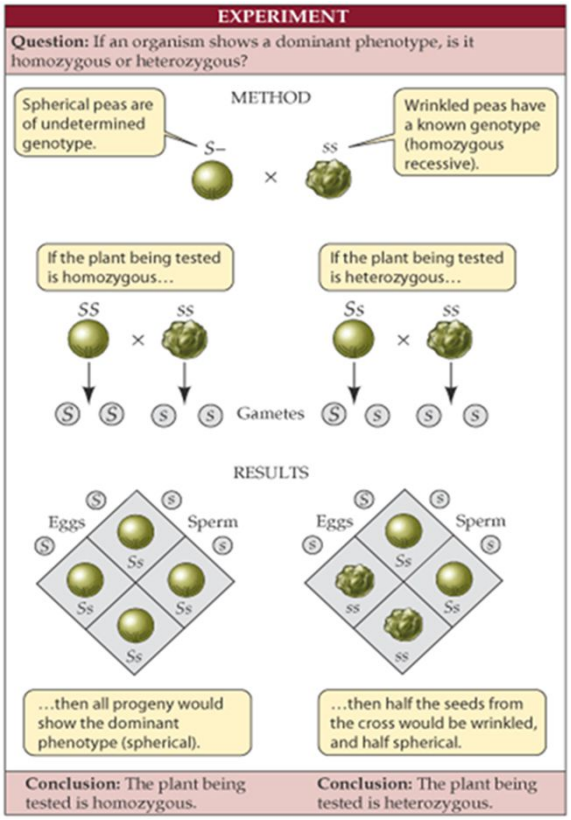
Red blood cells that react with antibody clump together (speckled appearance).

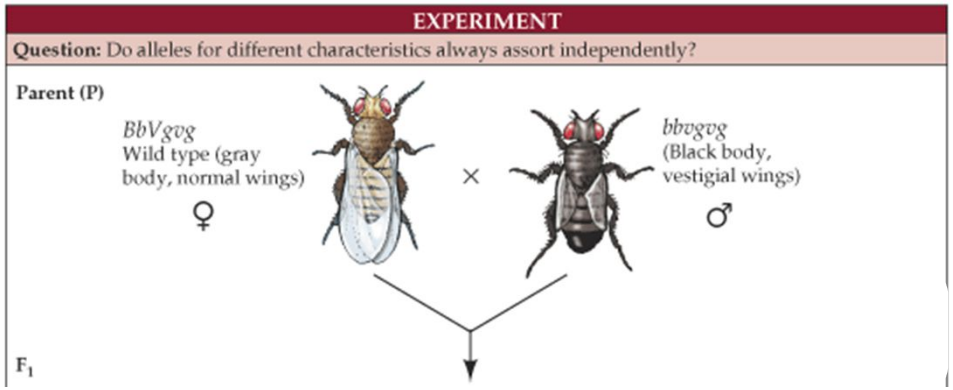
Effect of one gene dependent on the presence of one or more 'modifier genes'

END

Linkage & Recombination

Independent Assortment

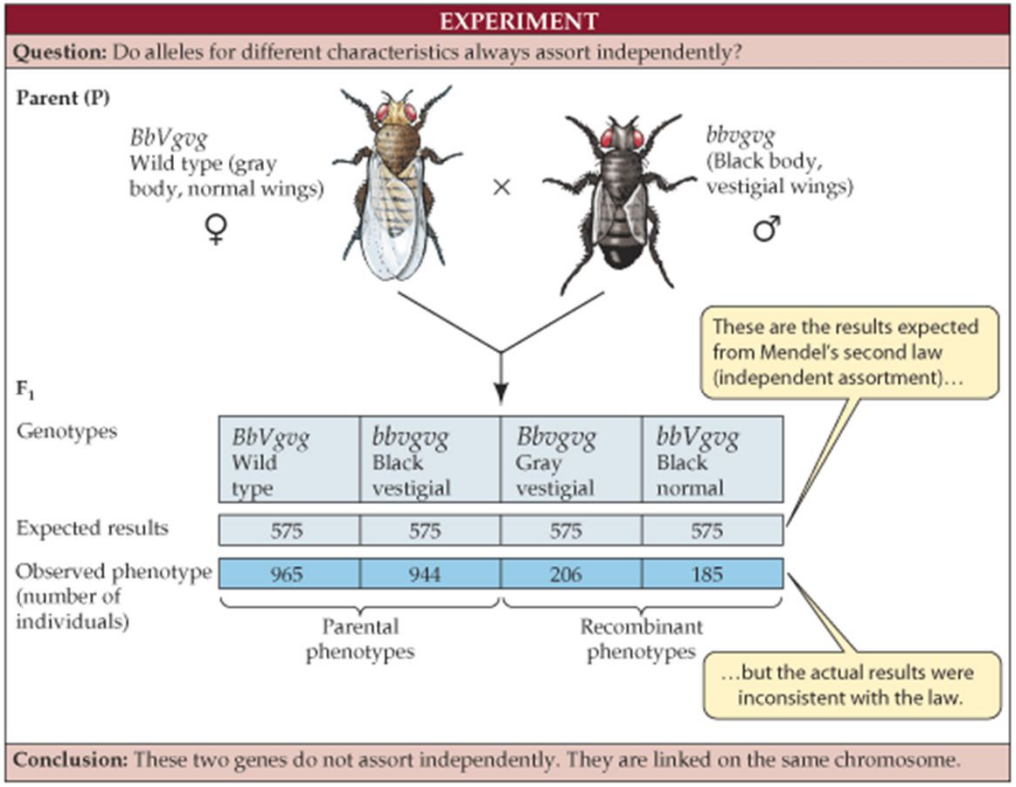


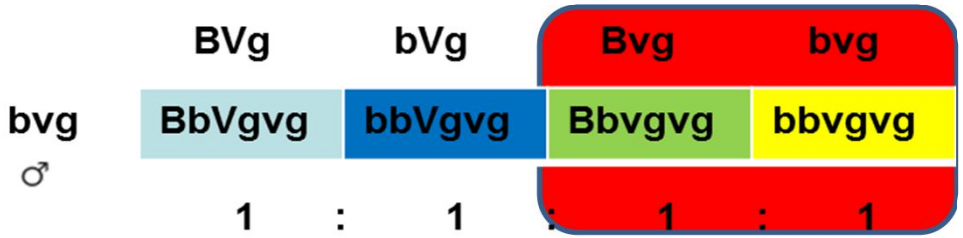


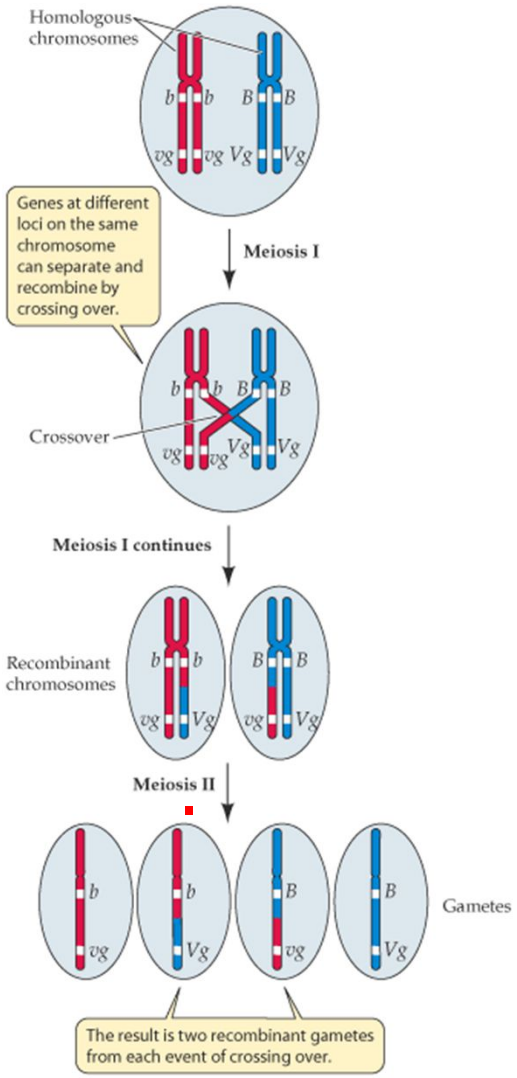
♂ \ ♀	BVg	bVg	Bvg	bvg
bvg	BbVgvg	bbVgvg	Bbvgvg	bbvgvg

1 : 1 : 1 : 1

No Independent Assortment

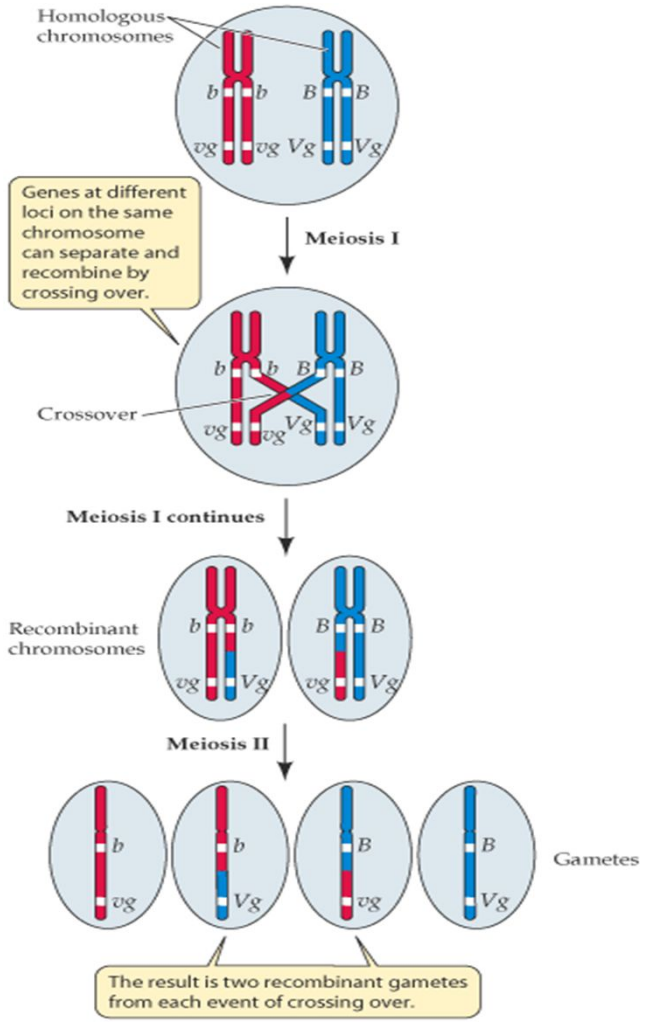






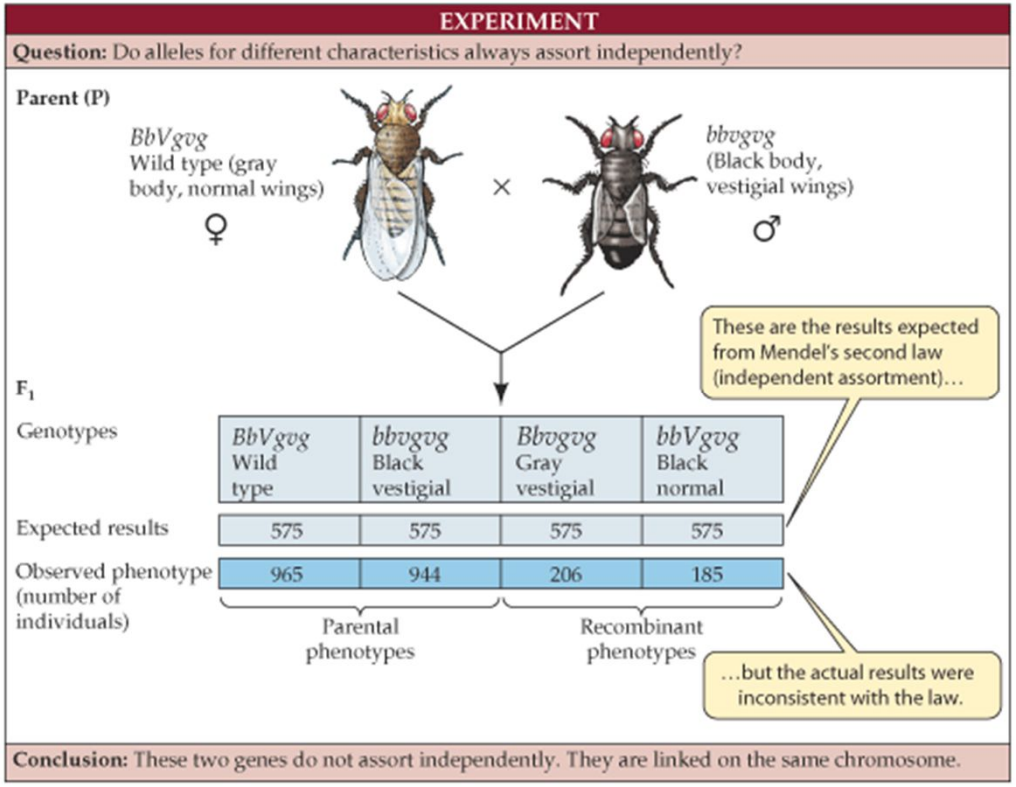
The number of linkage groups in a species equals the number of homologous chromosome pairs.

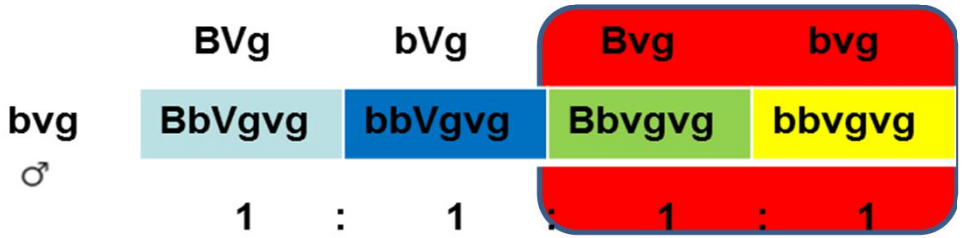
Number of genes in a cell far exceeds Number of chromosomes, (1 chromo must contain many genes)

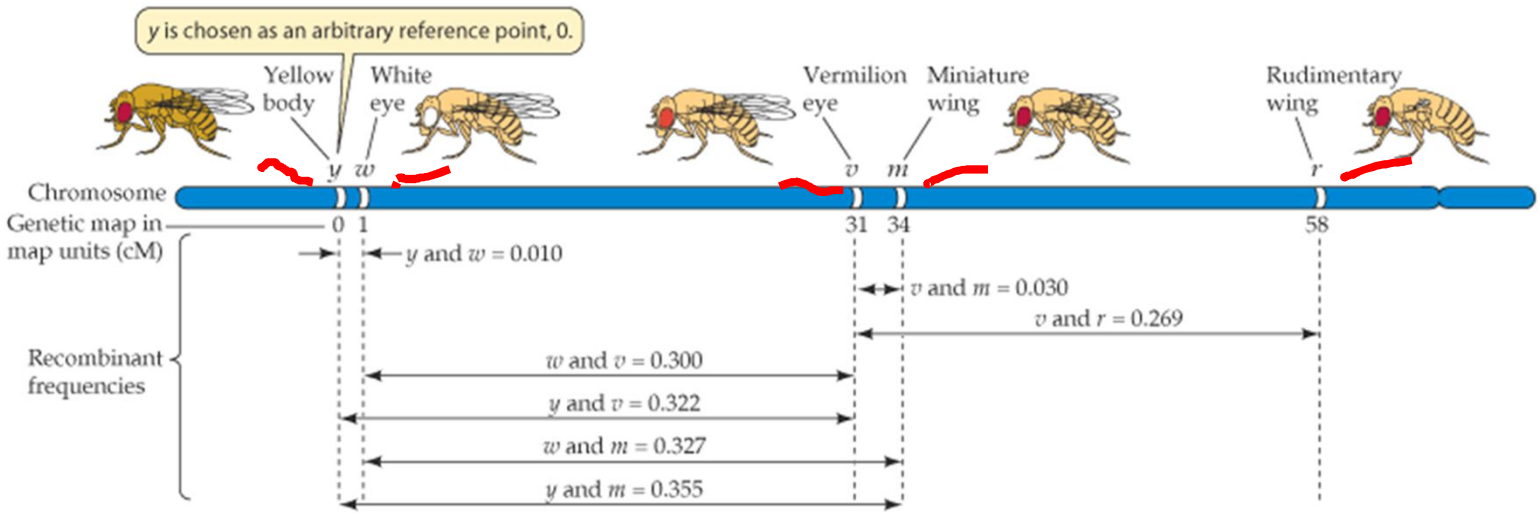

Homologous chromosomes
b b
vg vg
B B
Vg Vg
Genes at different loci on the same chromosome can separate and recombine by crossing over.
Meiosis I
Crossover
Meiosis I continues
Recombinant chromosomes
Meiosis II
Gametes
The result is two recombinant gametes from each event of crossing over.

Genetics Map

No Independent Assortment



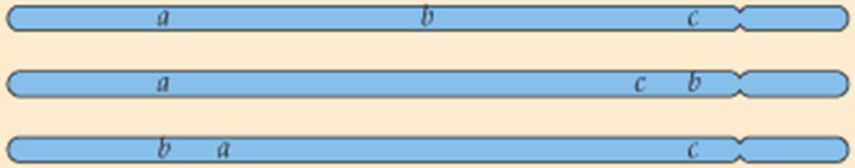




A map unit corresponds to a recombinant frequency of 0.01; it is also referred to as a *centimorgan* (cM)

If two loci are very close together on a chromosome, the odds of crossing over between them are small. In contrast, if two loci are far apart, crossing over could occur between them at many points.

1 At the outset, we have no idea of the individual distances, and there are several possible sequences (*a*-*b*-*c*, *a*-*c*-*b*, *b*-*a*-*c*).

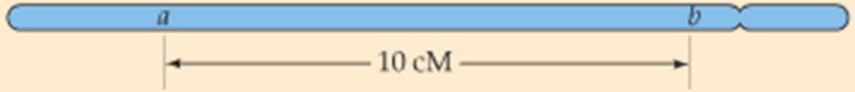


We make a cross *AABB* × *aabb*, and obtain an F_1 generation with a genotype *AaBb*. We test cross these *AaBb* individuals with *aabb*. Here are the genotypes of the first 1,000 progeny:

450 *AaBb*, 450 *aabb*, 50 *Aabb*, and 50 *aaBb*.

2 **How far apart are the *a* and *b* genes?** Well, what is the recombinant frequency? Which are the recombinant types, and which are the parental types?

Recombinant frequency (*a* to *b*) = (50 + 50)/1,000 = 0.1
So the map distance is
Map distance = 100 × recombinant frequency =
100 × 0.1 = 10 cM

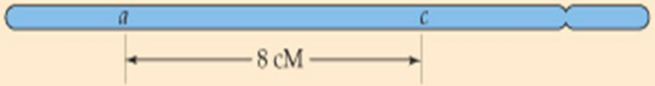


3 Now we make a cross *AACC* × *aacc*, obtain an F_1 generation, and test cross it, obtaining:

460 *AaCc*, 460 *aacc*, 40 *Aacc*, and 40 *aaCc*.

How far apart are the *a* and *c* genes?

Recombinant frequency (*a* to *c*) = (40 + 40)/1,000 = 0.08
Map distance = 100 × recombinant frequency =
100 × 0.08 = 8 cM



4 **How far apart are the *b* and *c* genes?**

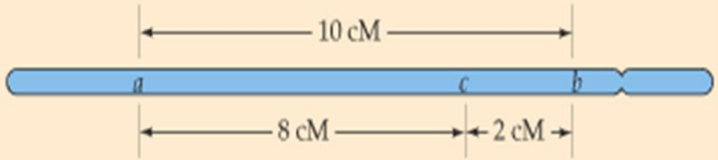
We make a cross *BBCC* × *bbcc*, obtain an F_1 generation, and test cross it, obtaining:

490 *BbCc*, 490 *bbcc*, 10 *Bbcc*, and 10 *bbCc*.

Determine the map distance between *b* and *c*.

Recombinant frequency (*b* to *c*) = (10 + 10)/1,000 = 0.02
Map distance = 100 × recombinant frequency =
100 × 0.02 = 2 cM

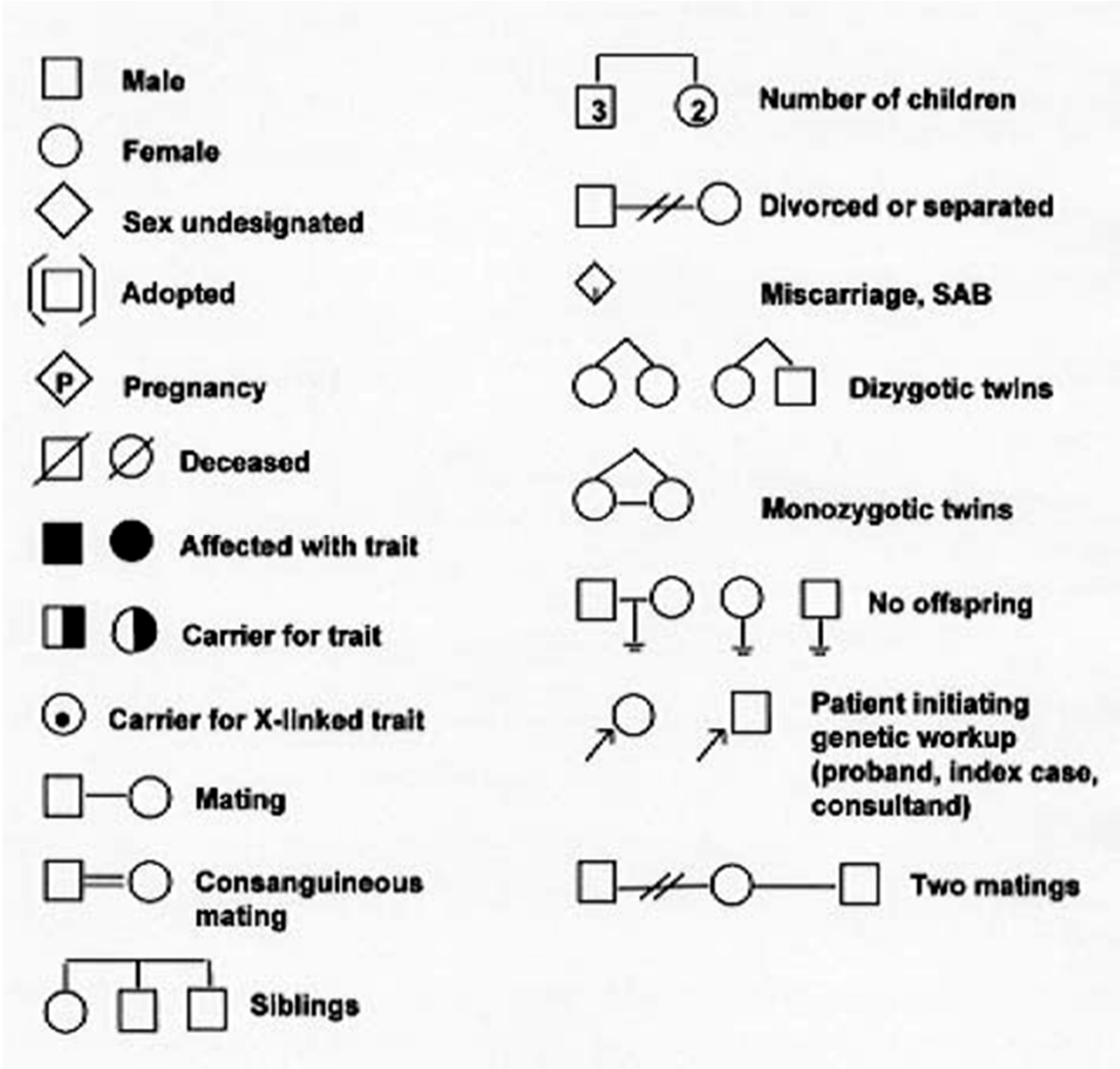
5 **Which of the three genes is between the other two?**
Because *a* and *b* are the farthest apart, *c* must be between them.



These numbers add up perfectly, but in most real cases they don't add up perfectly because of multiple crossovers.

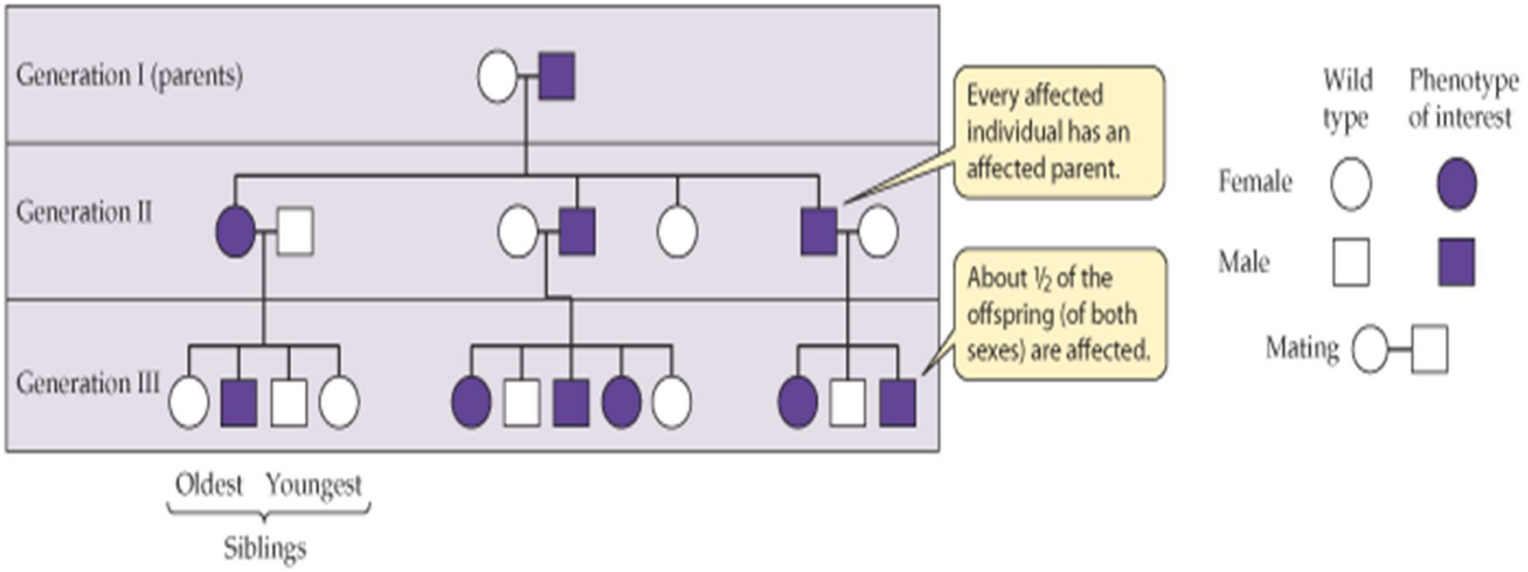
END

Clinical Genetics I



Pedigree Symbols

What Is My Mode of Inheritance



Features

Manifested in the heterozygous state

At least one parent of an index case is affected

Both males and females transmit the condition

When an effected person marries an unaffected child has one chance in two of having the disease

Reduce Penetrance: Some individuals inherit the mutant allele but are phenoltypically normal

Variable Expressivity: gene expression is variable e.g. neurofibromatosis type 1 range from brown spots to multiple tumors and skeletal deformities

Age of on set can delayed e.g. Huntington

Neurofibromatosis

Neurofibromatosis

Clinical Genetics II

Features

Manifested in the homozygous state

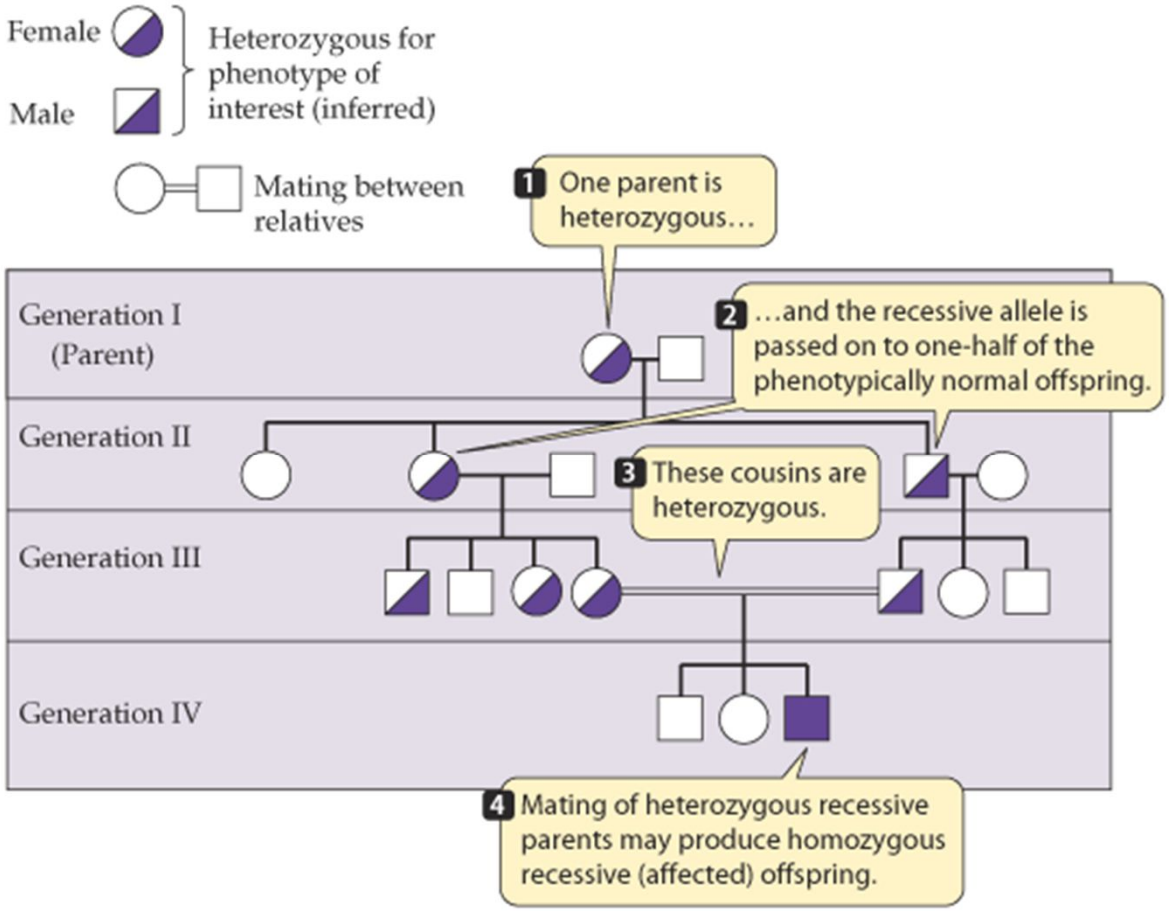
Parents usually not affected but the siblings may show the trait 1/4 (25%)

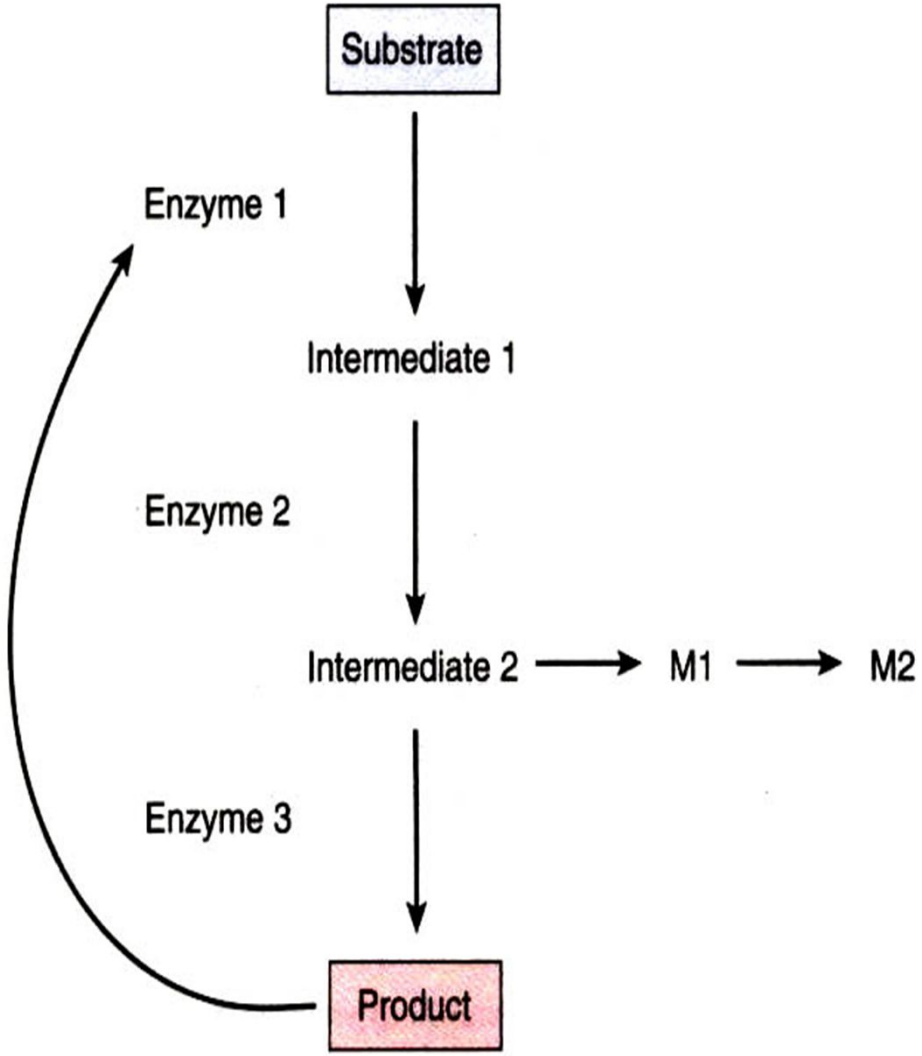
Affected individual may be product of consanguineous marriage (low frequency alleles)

The expression of the defect more uniform

Mostly enzyme proteins affected (half of normal enzymes ensures margin of safety)

Pedigree


Female
Male
Heterozygous for phenotype of interest (inferred)
Mating between relatives
1 One parent is heterozygous…
2 …and the recessive allele is passed on to one-half of the phenotypically normal offspring.
3 These cousins are heterozygous.
4 Mating of heterozygous recessive parents may produce homozygous recessive (affected) offspring.
Generation I (Parent)
Generation II
Generation III
Generation IV

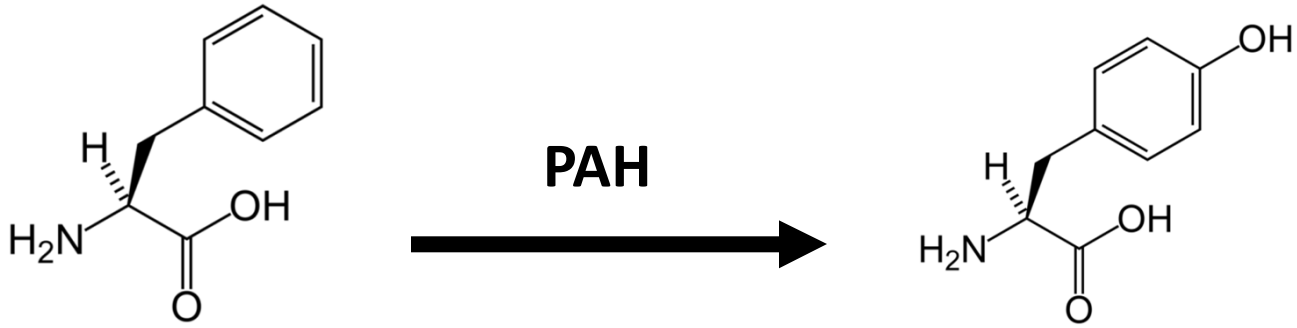

Substrate
Enzyme 1
Intermediate 1
Enzyme 2
Intermediate 2
M1
M2
Enzyme 3
Product

Mechanism

Phenylketonuria (PKU) is an autosomal recessive genetic disorder characterized by a deficiency in the enzyme phenylalanine hydroxylase (PAH)

This enzyme is necessary to metabolize the amino acid phenylalanine to the amino acid tyrosine

When PAH is deficient, phenylalanine accumulates and is converted into phenylketones, which are detected in the urine



Untreated PKU can cause problems with brain development, leading to progressive mental retardation and seizures

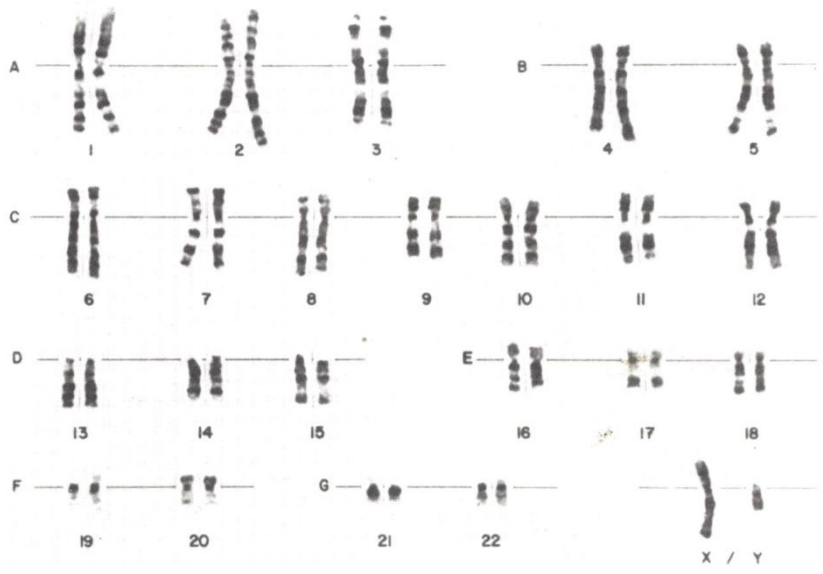
PKU is one of the few genetic diseases that can be controlled by diet

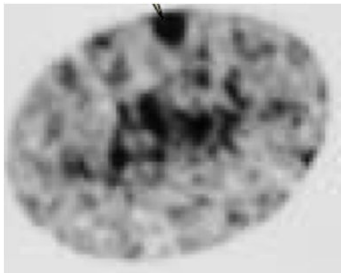
A diet low in phenylalanine and high in tyrosine can bring about a nearly total cure

the disease may present clinically with seizures, albinism (excessively fair hair and skin), and a "musty odor" to the baby's sweat and urine (due to phenylacetate, one of the ketones produced

H
H2N
OH
O
PAH
OH
H
H2N
OH
O

X-Linked Disorders





All sex-linked disorders are X-linked

No Y-linked diseases known

Transmitted by heterozygous female carriers to sons (hemizygous)

Hetrozygous female rarely express full phenotype

Partial expression due to X chromosome inactivation

Features

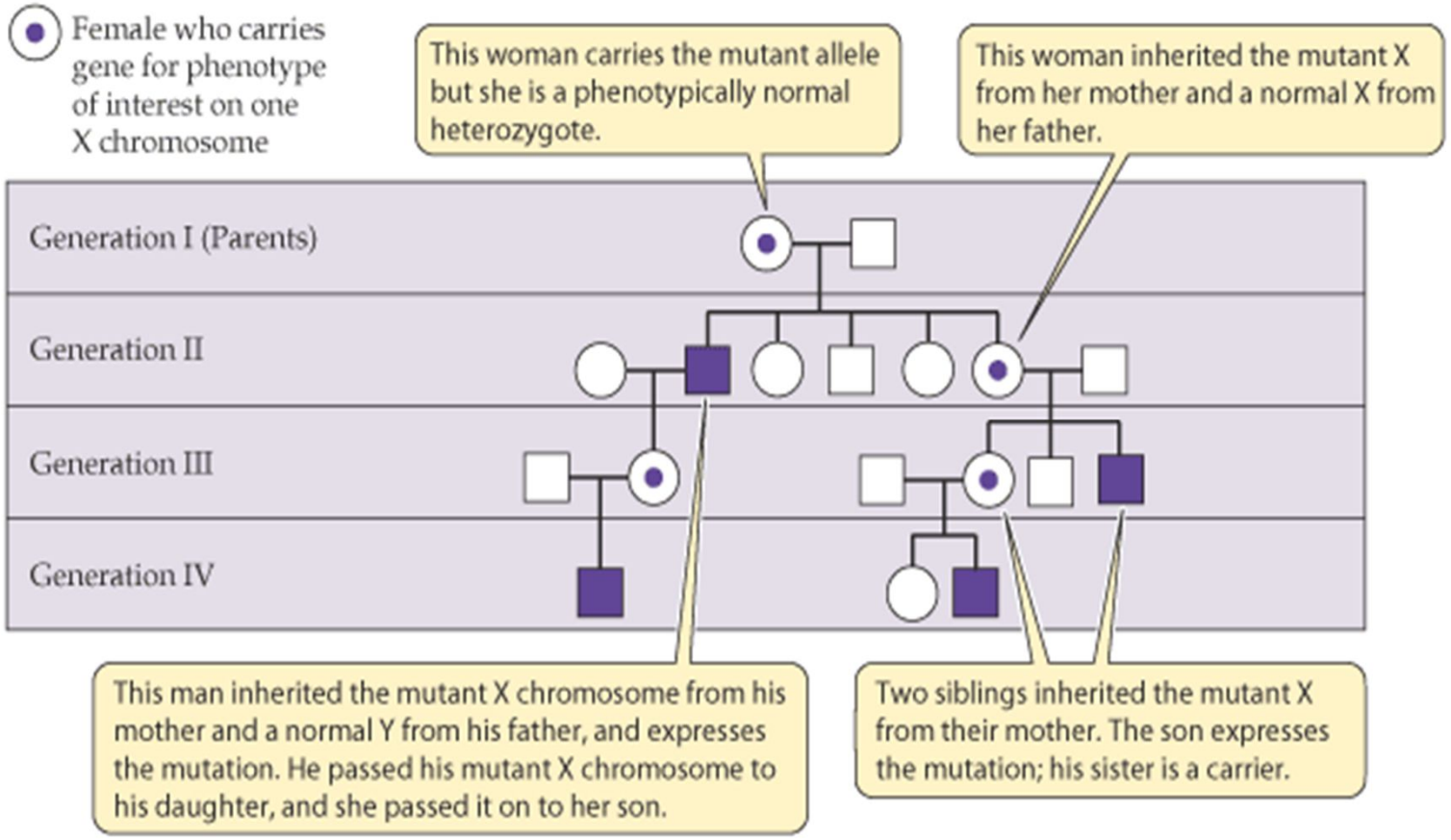
An affected male does not transmit the disorder to sons but all his daughters are carriers

Sons of heterozygous females have one chance in two of receiving mutant allele

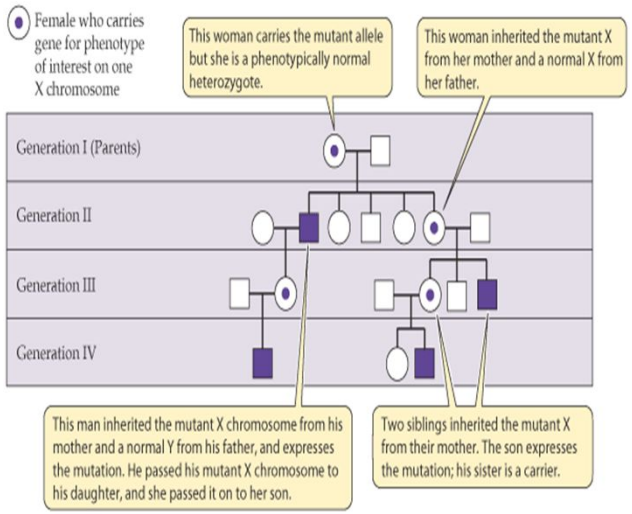
There are very few X-linked dominant diseases, inheritance pattern is characterized by transmission of the disease to 50% of the sons and daughters of an affected female

An affected male cannot transmit the disease to his sons but all daughters are affected VDRR

Pedigree

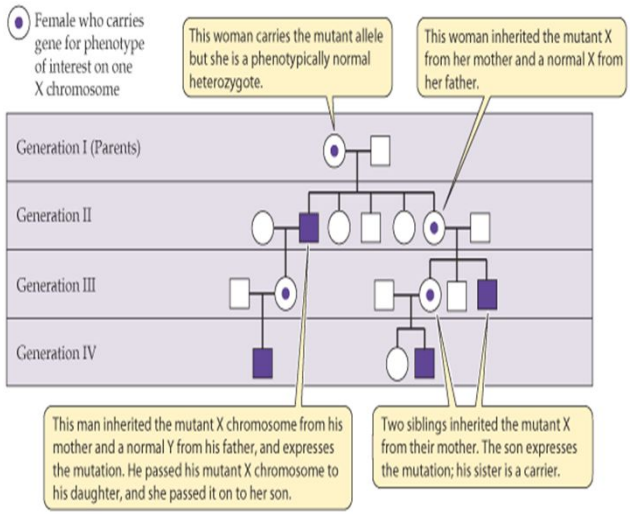

Female who carries gene for phenotype of interest on one X chromosome
This woman carries the mutant allele but she is a phenotypically normal heterozygote.
This woman inherited the mutant X from her mother and a normal X from her father.
Generation I (Parents)
Generation II
Generation III
Generation IV
This man inherited the mutant X chromosome from his mother and a normal Y from his father, and expresses the mutation. He passed his mutant X chromosome to his daughter, and she passed it on to her son.
Two siblings inherited the mutant X from their mother. The son expresses the mutation; his sister is a carrier.

Pedigree



END

Pedigree



END

Multifactorial Disorders

Features

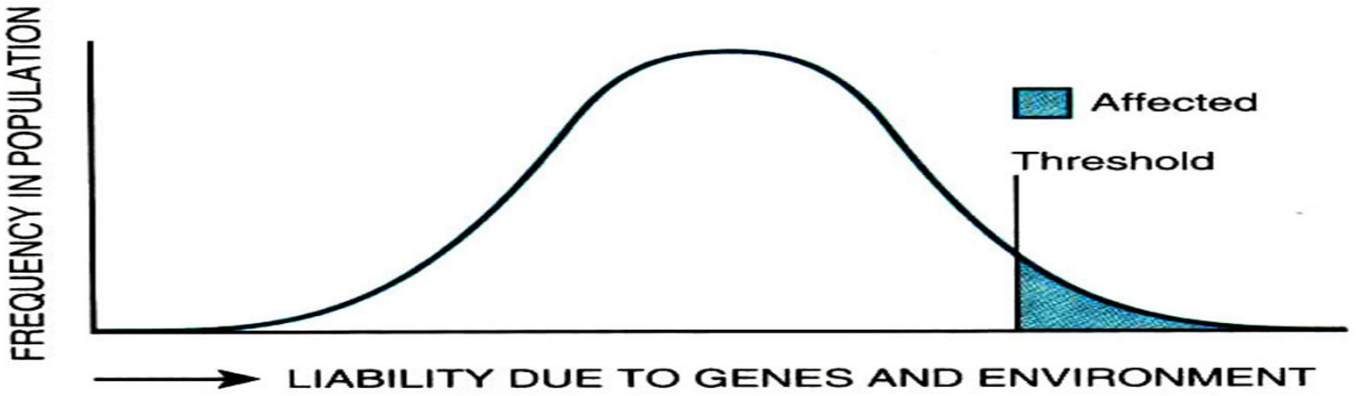
A multifactorial trait is governed by the additive effect of two or more genes of small effect but conditioned by environmental, nongenetic influences

Many physiologic characteristics of humans (e.g. diabetes, height, weight, blood pressure, hair color, gout, etc.)

Monozygous twins reared separately may achieve different heights because of nutritional/other environmental influences

Features

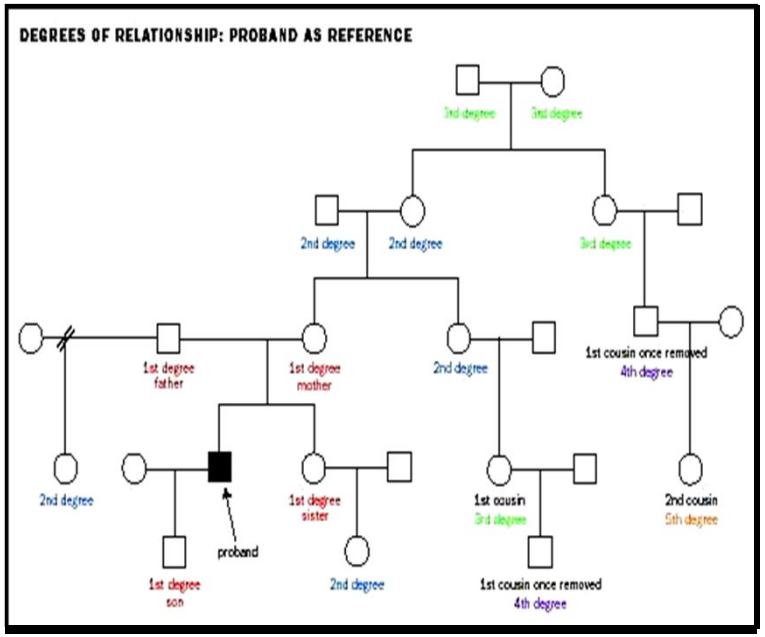
In a large population phenotypic attributes governed by multifactorial inheritance fall on a continuous Gaussian distribution



A disorder manifests itself only when a certain number of effector genes, as well as conditioning environmental influences are involved

this phenomenon is termed as threshold effect

Recurrence rate 2%-7% in for first degree relatives and 20%-40% for identical twins


DEGREES OF RELATIONSHIP: PROBAND AS REFERENCE
1st degree father
1st degree mother
2nd degree
1st cousin once removed
4th degree
2nd degree
proband
1st degree sister
1st cousin
3rd degree
2nd cousin
5th degree
1st degree son
2nd degree
1st cousin once removed
4th degree

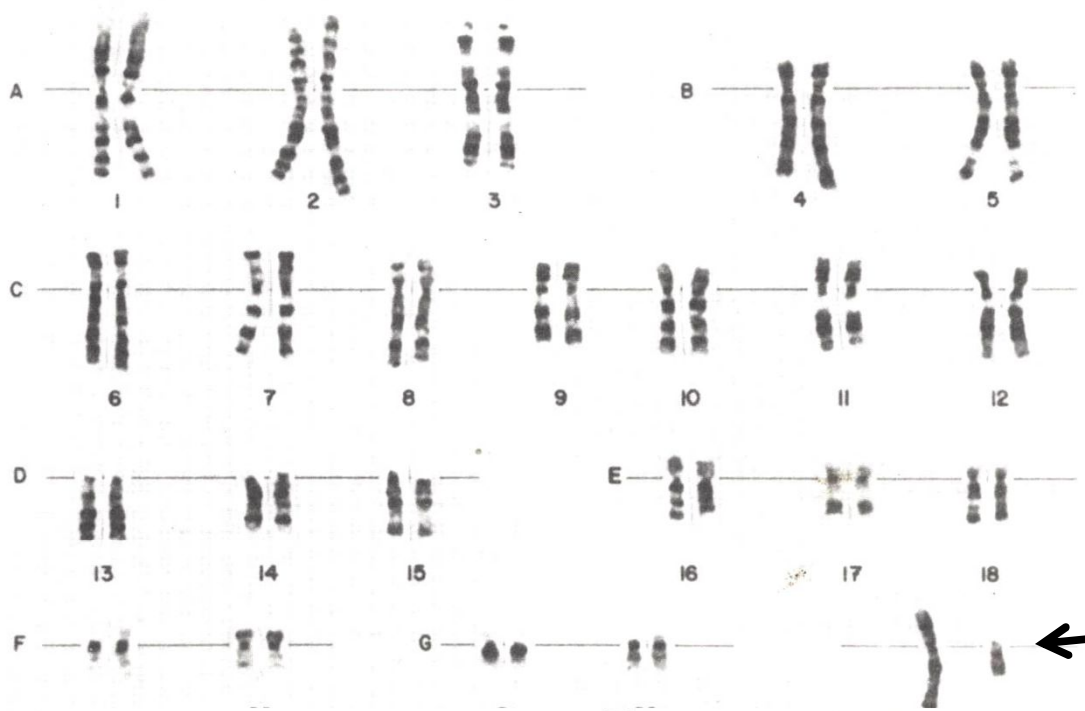
END

Cytogenetic Disorders

Karyotyping is used to diagnose Cytogenetic disorders

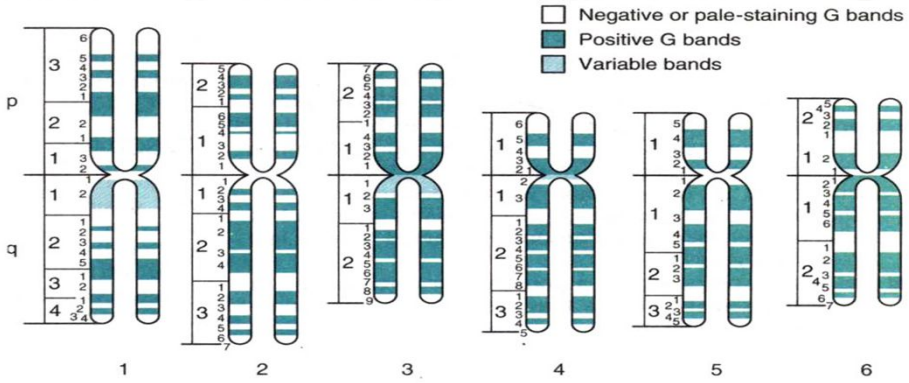
Karyotype is a photographic representation of a (G banded) stained metaphase chromosome spread arranged in order of decreasing length

1in 2000 newborns and 50% of first trimester abortions have chromosomal aberrations



Karyotype

Sex Chromosomes



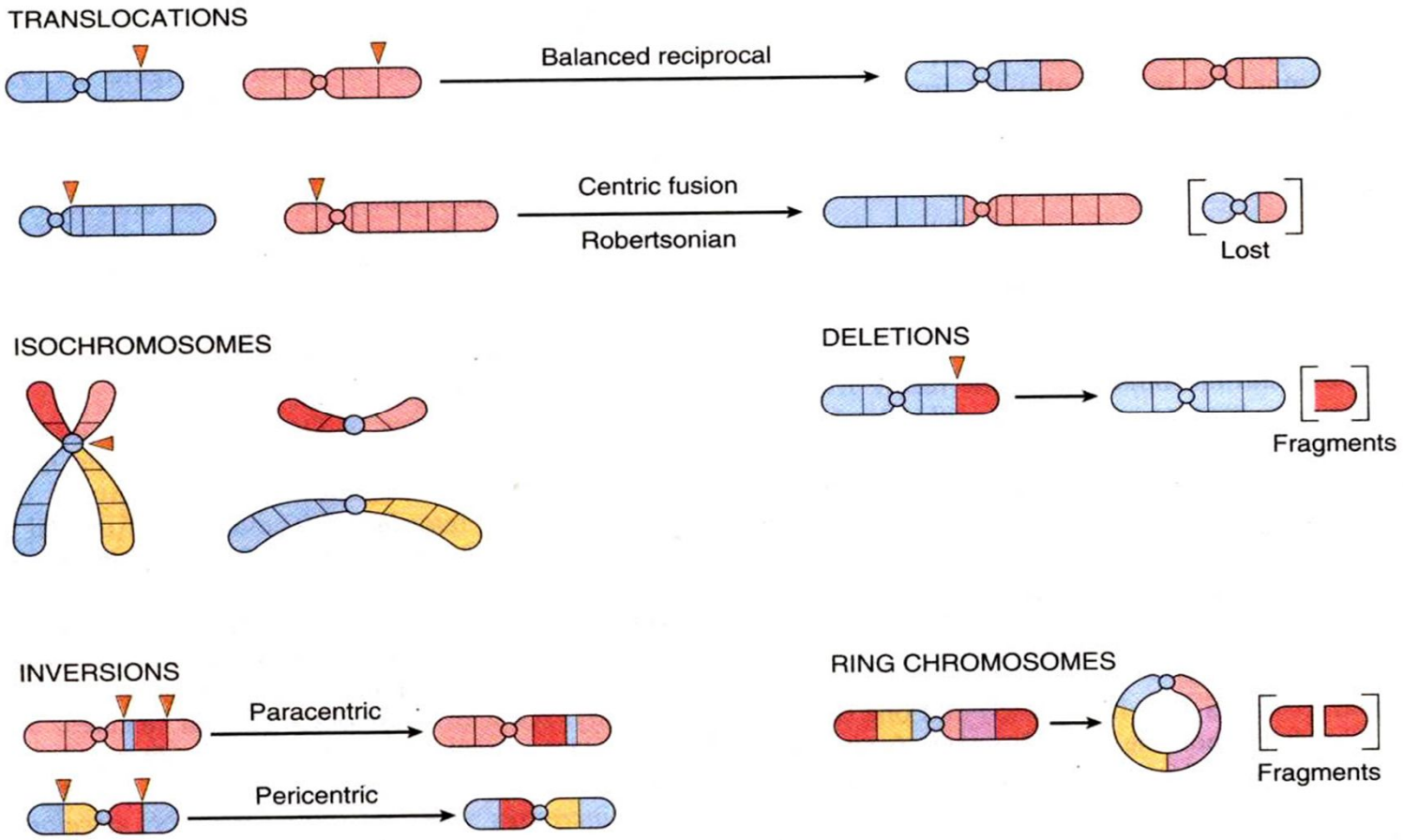
Chromosomal Nomenclature

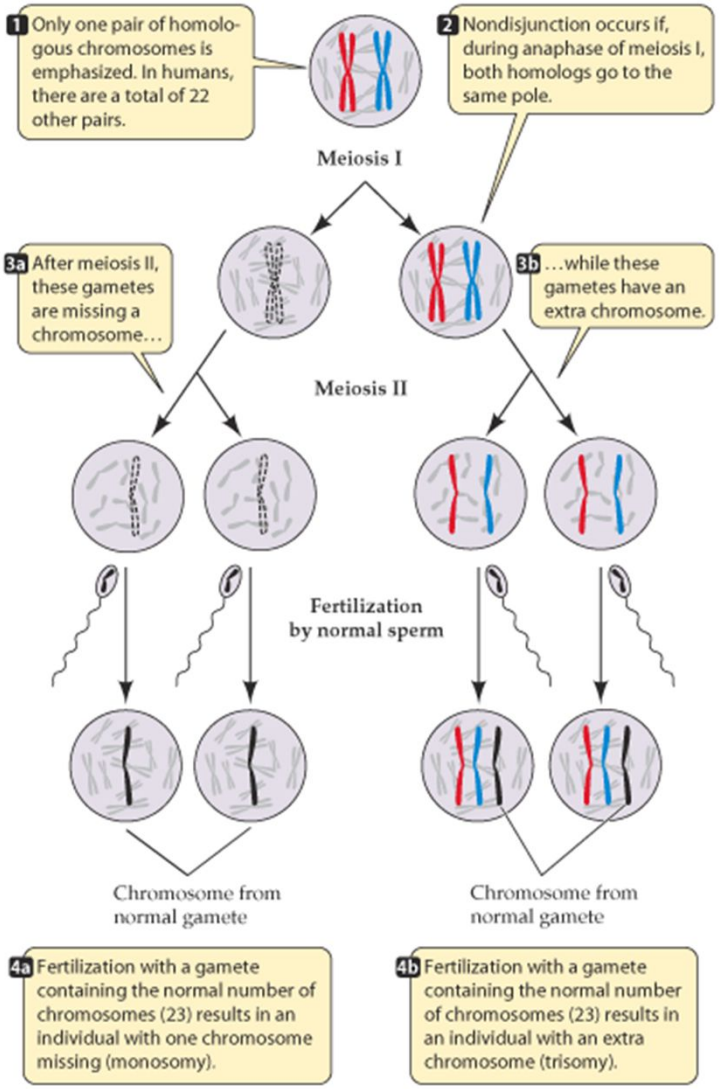
Any exact multiple of haploid (n) number of chromosomes (other than diploid 2n) are called Polyploid e.g. 3n, 4n etc

A chromosomal number that is not exact multiple of n is called aneuploid

Aneuploidy can result from non disjunction of homologous pair of chromosomes (first meiotic division) or failure of sister chromatids to separate (second meiotic division)

The result is monosomy (2n-1) or trisomy (2n+1)


TRANSLOCATIONS
Balanced reciprocal
Centric fusion
Robertsonian
Lost
ISOCHROMOSOMES
DELETIONS
Fragments
INVERSIONS
Paracentric
Pericentric
RING CHROMOSOMES
Fragments


1 Only one pair of homologous chromosomes is emphasized. In humans, there are a total of 22 other pairs.
2 Nondisjunction occurs if, during anaphase of meiosis I, both homologs go to the same pole.
Meiosis I
3a After meiosis II, these gametes are missing a chromosome…
3b …while these gametes have an extra chromosome.
Meiosis II
Fertilization by normal sperm
Chromosome from normal gamete
Chromosome from normal gamete
4a Fertilization with a gamete containing the normal number of chromosomes (23) results in an individual with one chromosome missing (monosomy).
4b Fertilization with a gamete containing the normal number of chromosomes (23) results in an individual with an extra chromosome (trisomy).

END

Autosomal Cytogenetic Disorders

Trisomy 21 (Down Syndrome)

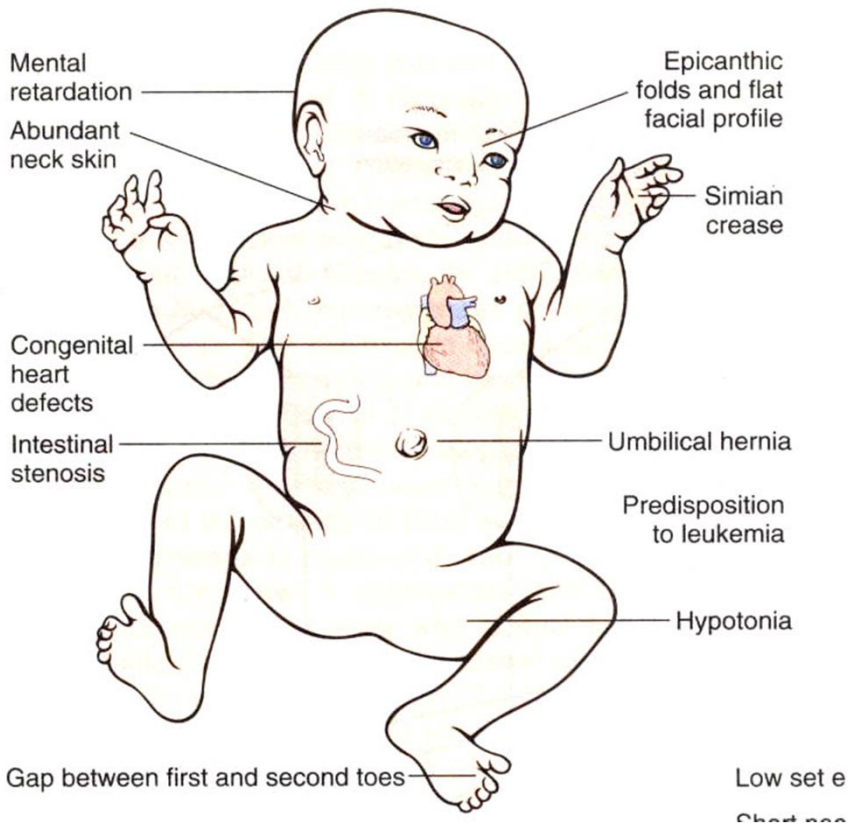
Arises from meiotic nondisjunction

Normal Parents 95% of time the extra chromosome comes from mother

Maternal age plays a significant role (1 in 1550 for women under 20 and 1 in 25 for women over 45)

4% of patients have 14:21 translocation

1% of patients are mosaic



Single palmar fold (also known as a simian crease), Epicanthic Folds, Flat Facial Profile, IQ 25-50, Cardiac Malformations

END

Cytogenetic Disorders of the Sex Chromosomes

Sex Chromosomes Disorders

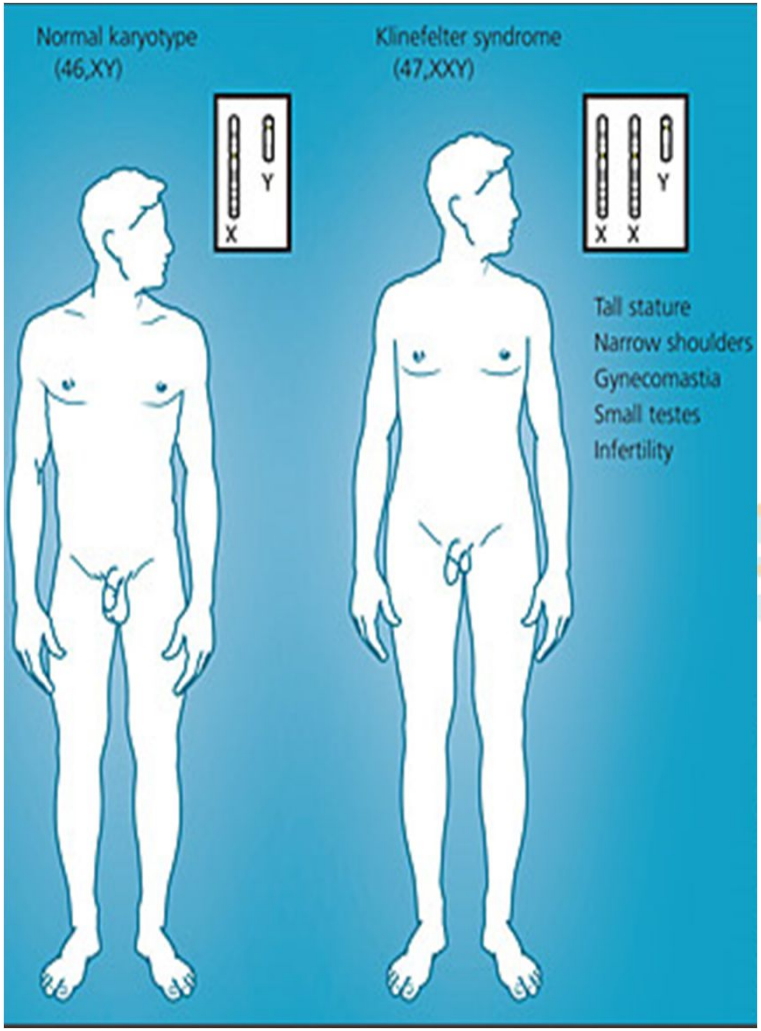
Trisomy 21 (Down Syndrome)

Numerical aberrations of sex chromosomal are relatively better tolerated than those caused by autosomes 46X-46XXXY

Only one X chromosome is active (females are mosaic) extra chromosome forms Barr body

People with extra Y chromosome are phenotypically normal

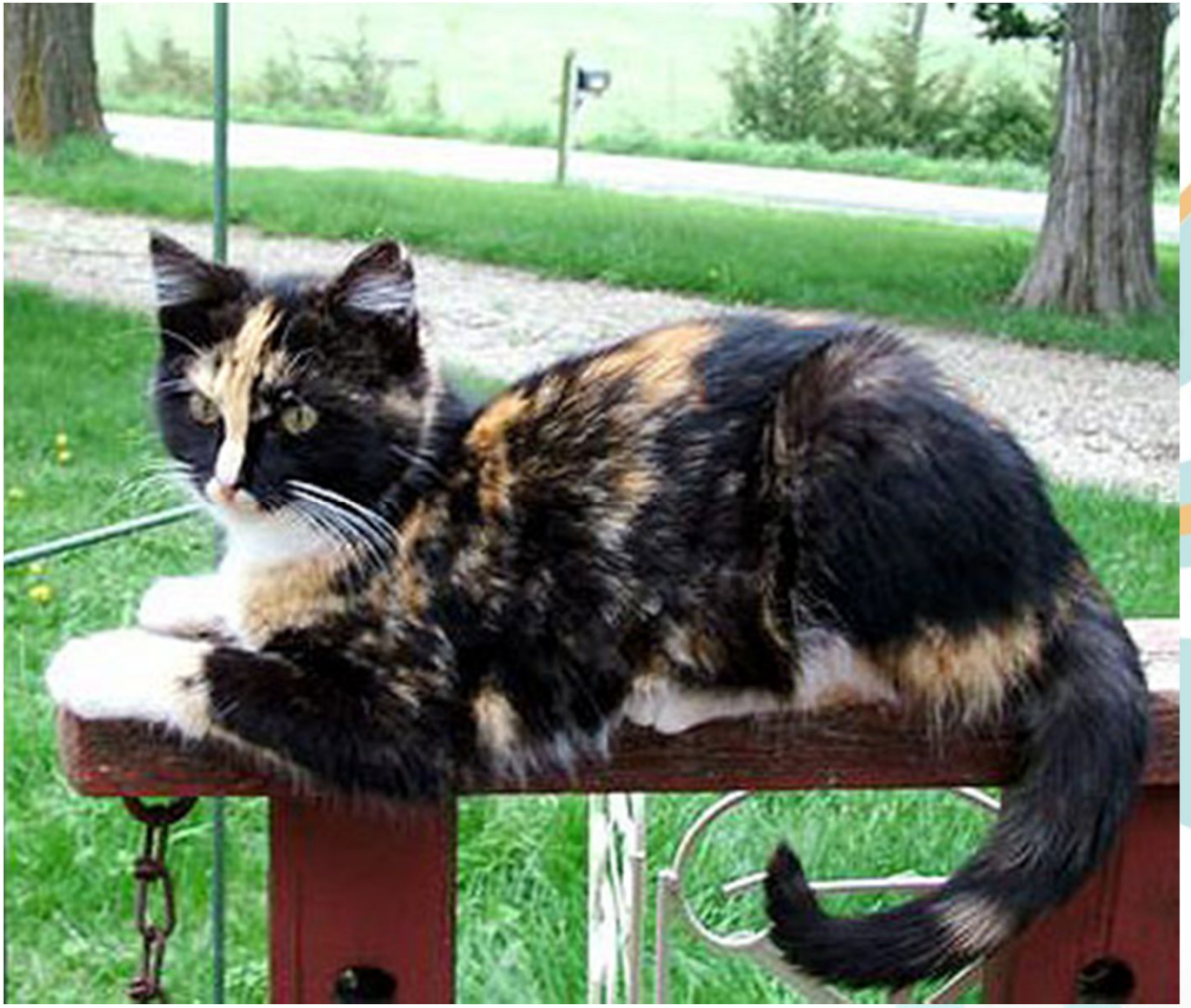
Klinefelter Syndrome (47 XXY)



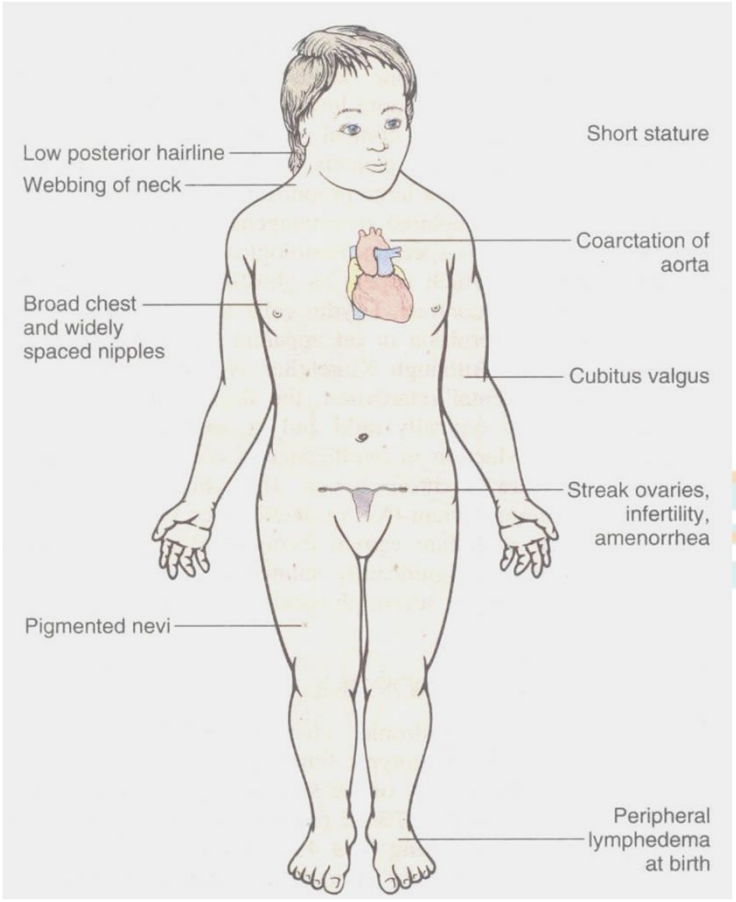
Extra chromosome can be maternal or paternal (advance maternal age or irradiation), 15% patients show mosaicism

Microorchidism, hypogonadism, longer legs, gynecomastia, infertility, may be mildly mentally retarded, reduced facial & pubic hair

Klinefelter Syndrome (47 XXY)



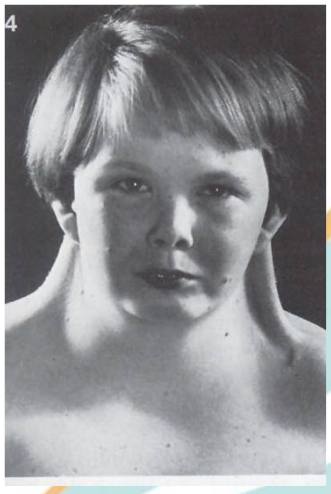
Turner Syndrome

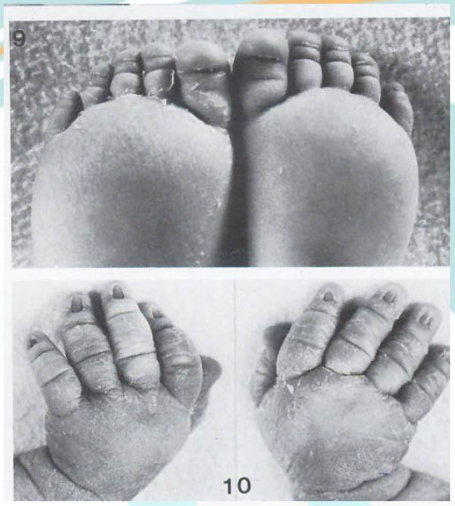


Turner syndrome encompasses several chromosomal abnormalities, of which monosomy X is the most common 57% rest may have i (X)
43% of patients mosaic

Short stature
Lymphoedema
Broad chest
Reproductive sterility
Obesity

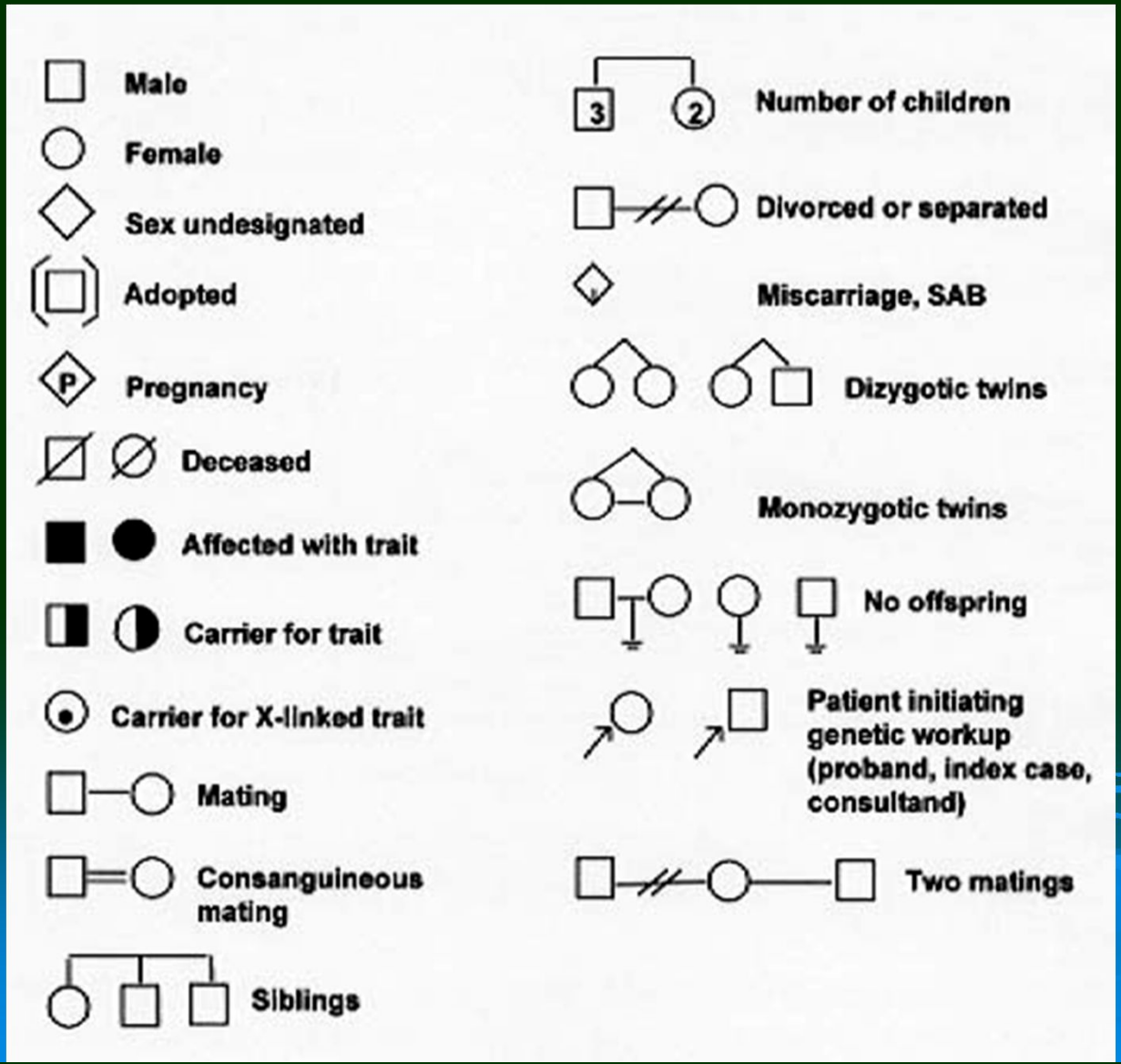
Turner Syndrome

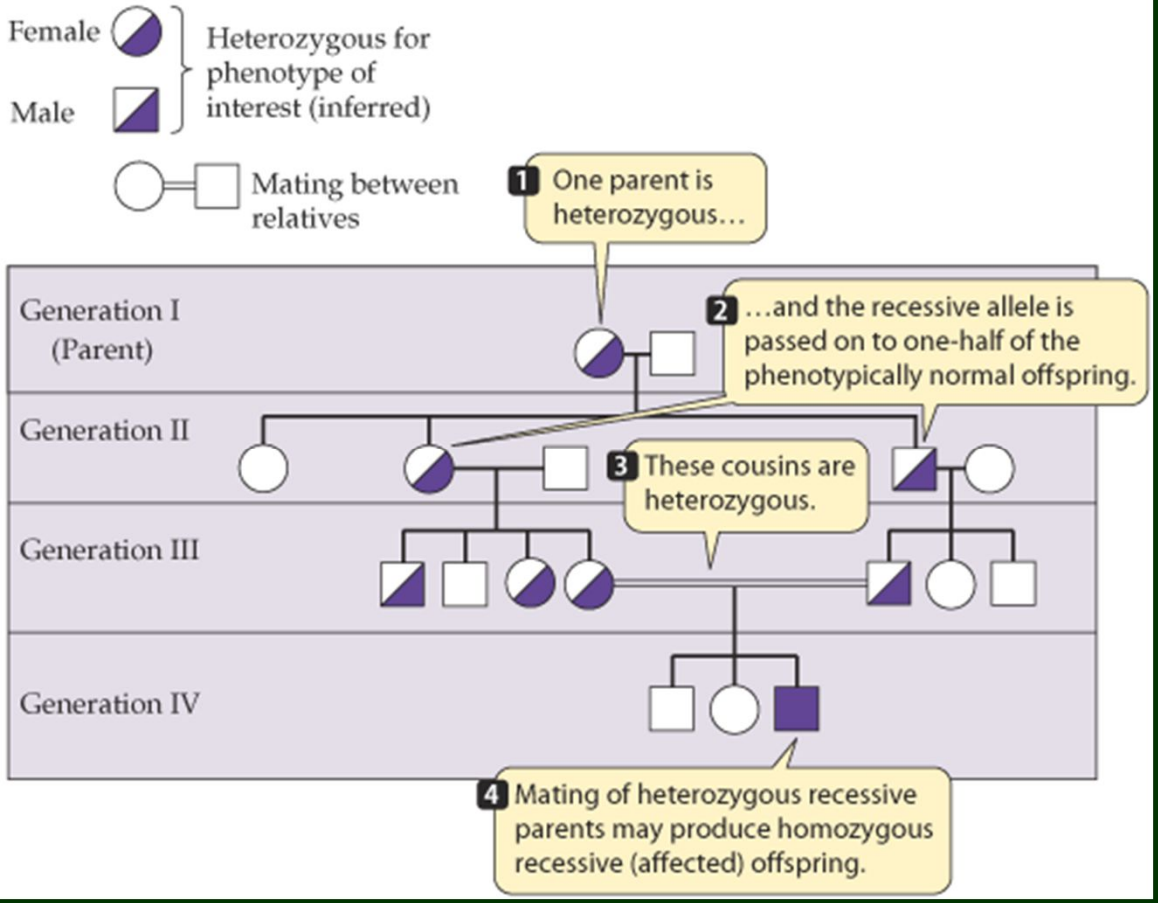




END

Pedigree Symbols




Female
Male
Heterozygous for phenotype of interest (inferred)
Mating between relatives
1 One parent is heterozygous…
2 …and the recessive allele is passed on to one-half of the phenotypically normal offspring.
3 These cousins are heterozygous.
4 Mating of heterozygous recessive parents may produce homozygous recessive (affected) offspring.
Generation I (Parent)
Generation II
Generation III
Generation IV

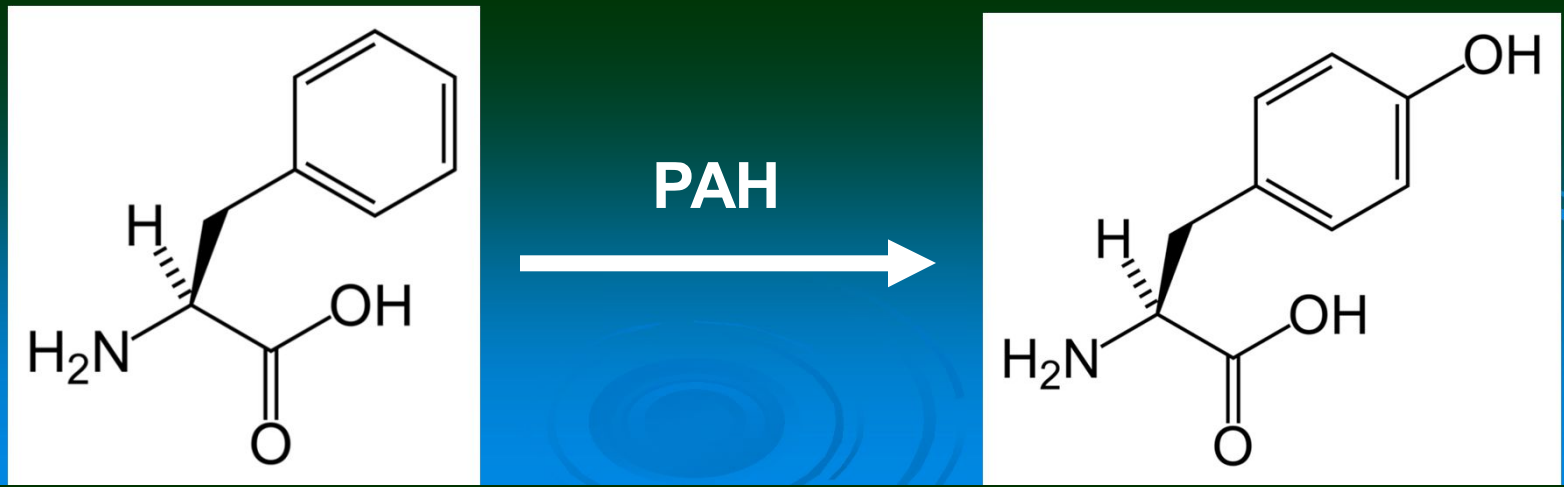
Pedigrees

Phenylketonuria (PKU)

Phenylketonuria (PKU) is an autosomal recessive genetic disorder characterized by a deficiency in the enzyme phenylalanine hydroxylase (PAH)

This enzyme is necessary to metabolize the amino acid phenylalanine to the amino acid tyrosine

When PAH is deficient, phenylalanine accumulates and is converted into phenylketones, which are detected in the urine



PKU

Untreated PKU can cause problems with brain development, leading to progressive mental retardation and seizures

PKU is one of the few genetic diseases that can be controlled by diet

A diet low in phenylalanine and high in tyrosine can bring about a nearly total cure

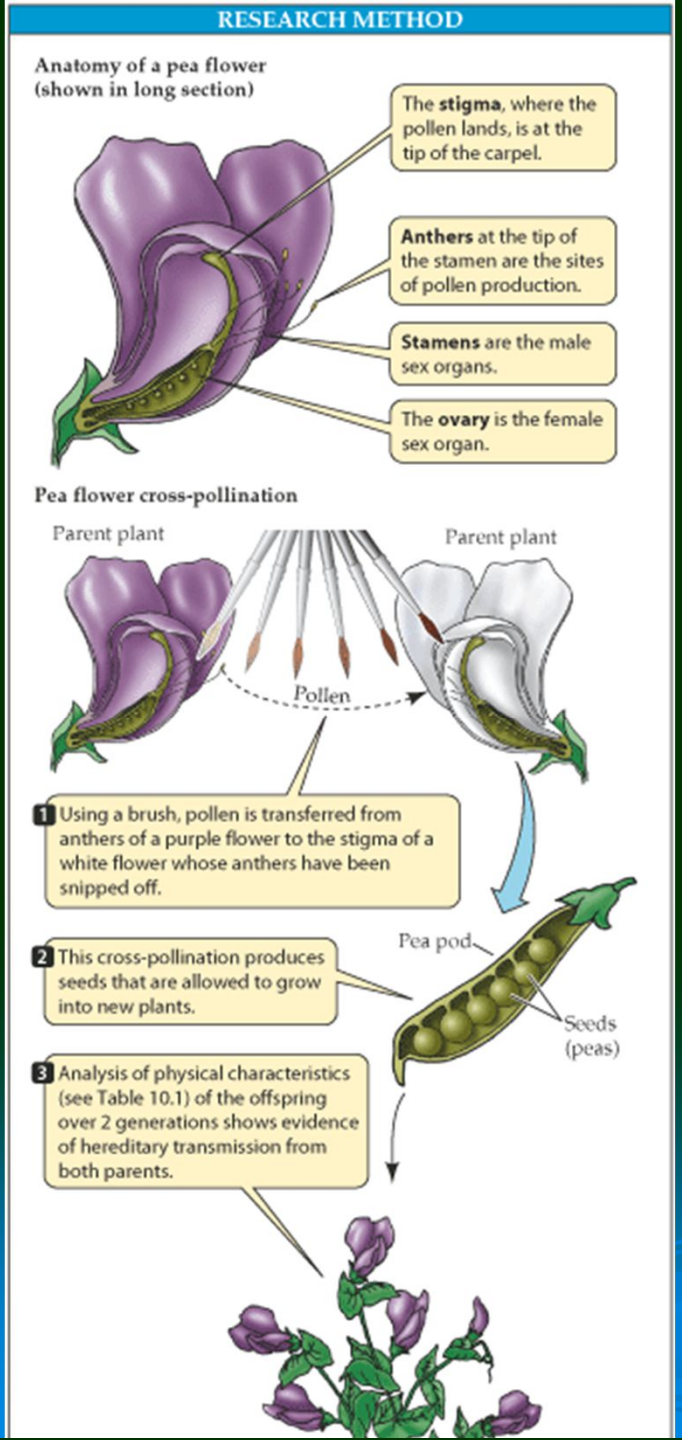
the disease may present clinically with seizures, albinism (excessively fair hair and skin), and a "musty odor" to the baby's sweat and urine (due to phenylacetate, one of the ketones produced

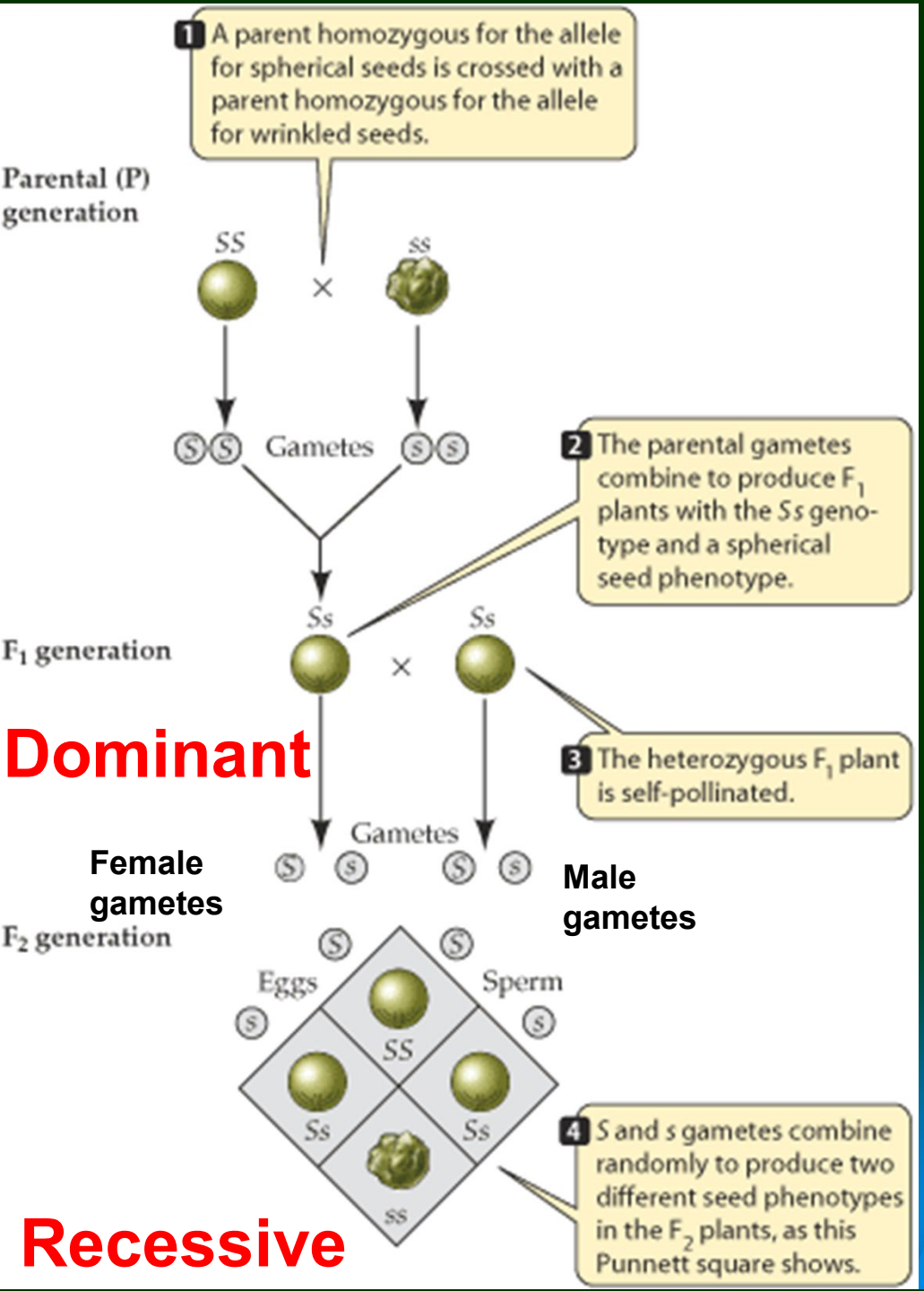
Neurofibromatosis

Neurofibromatosis causes tumors to grow on nerves and is is characterized by spots of increased skin pigmentation(café au lait spots), combined with peripheral nerve tumors and a variety of others dysplastic abnormalities of the skin, nervous system, bones, endocrine organs and blood vessels

Neurofibromin a negative regulator of RAS (OP) is mutated

Effort By: AARISH ALI





Dominant & Recessive

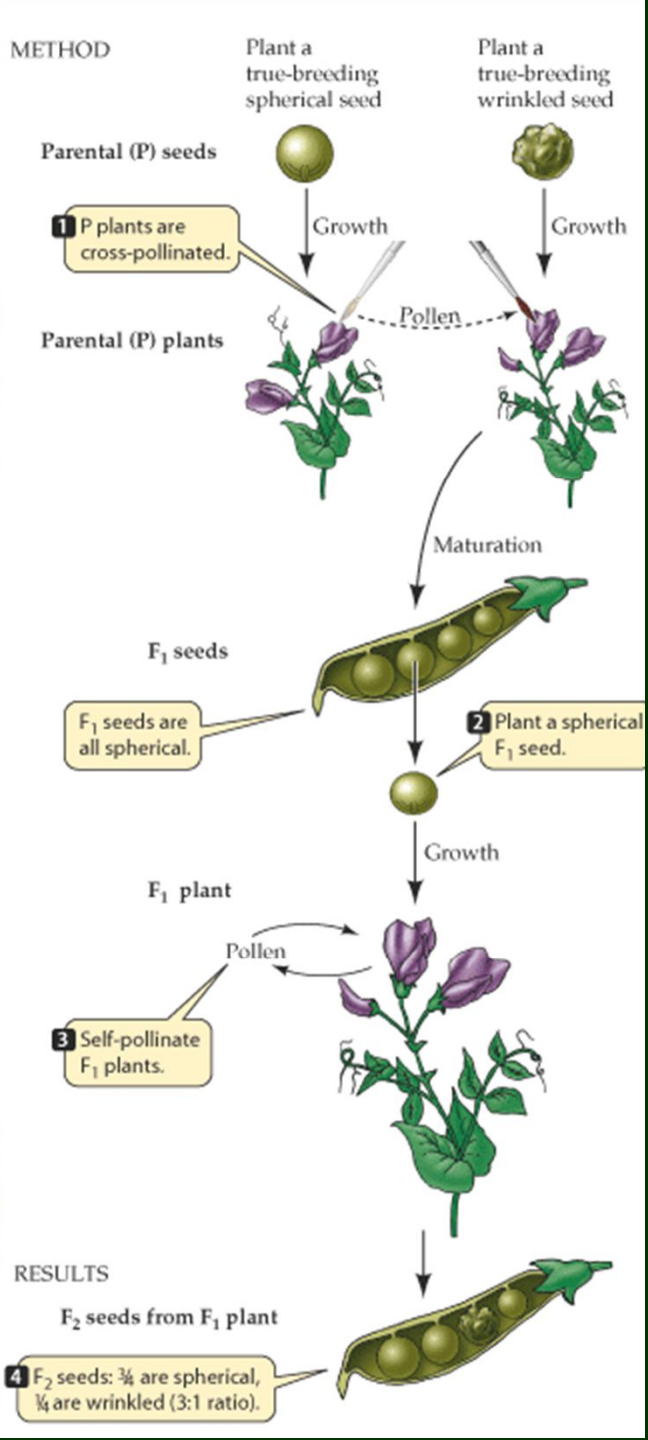
Mendel observed that the wrinkled seed trait was never expressed in the F1 generation, even though it reappeared in the F2 generation

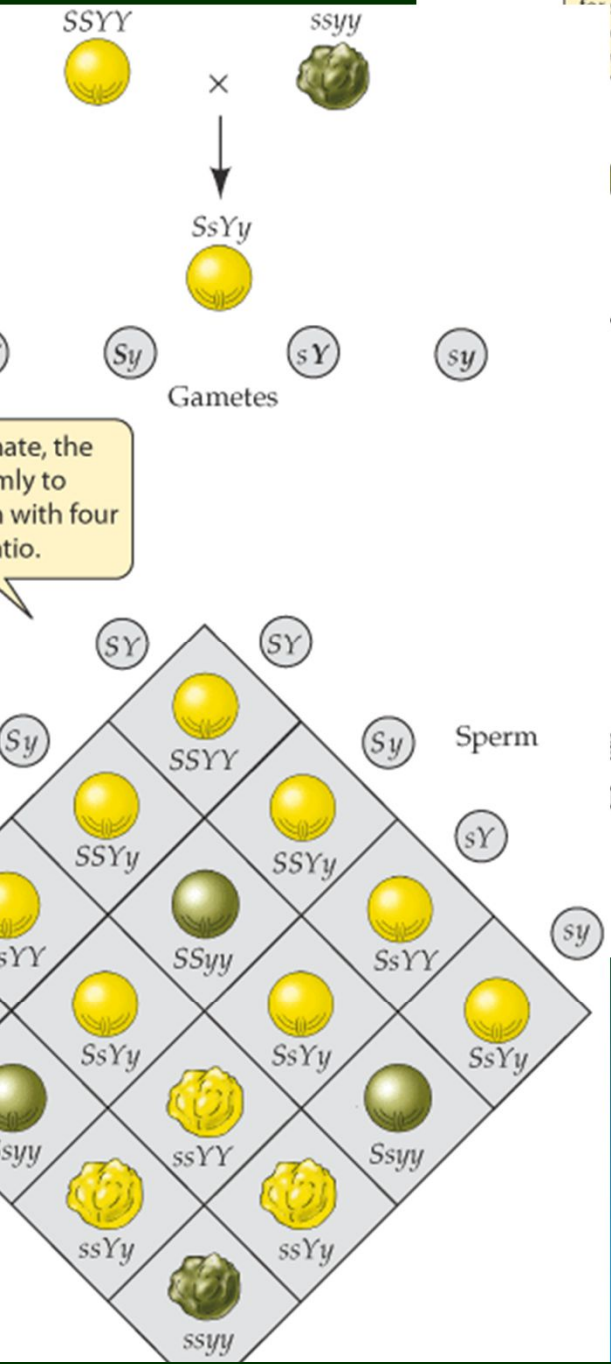
No Blending

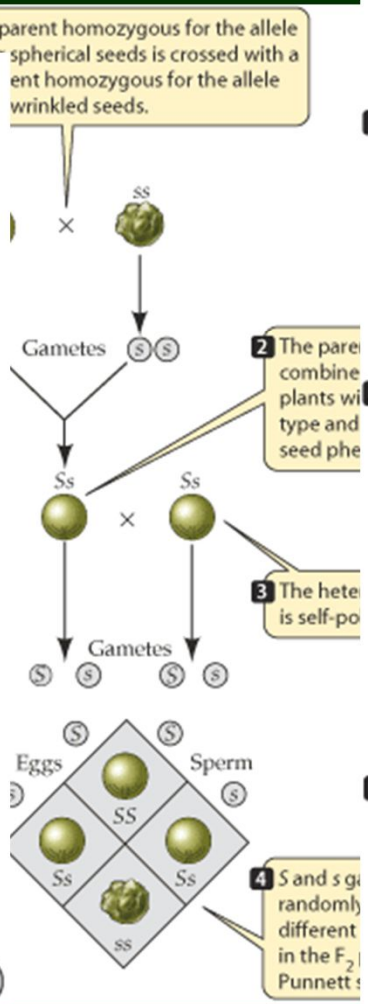
Particulate Theory

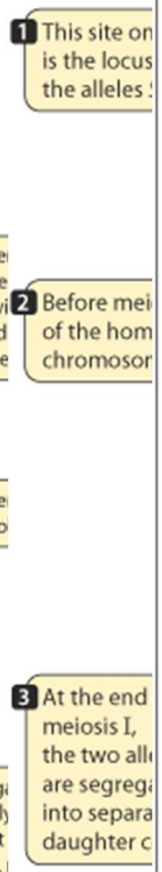
10.1 Mendel's Results from Monohybrid Crosses

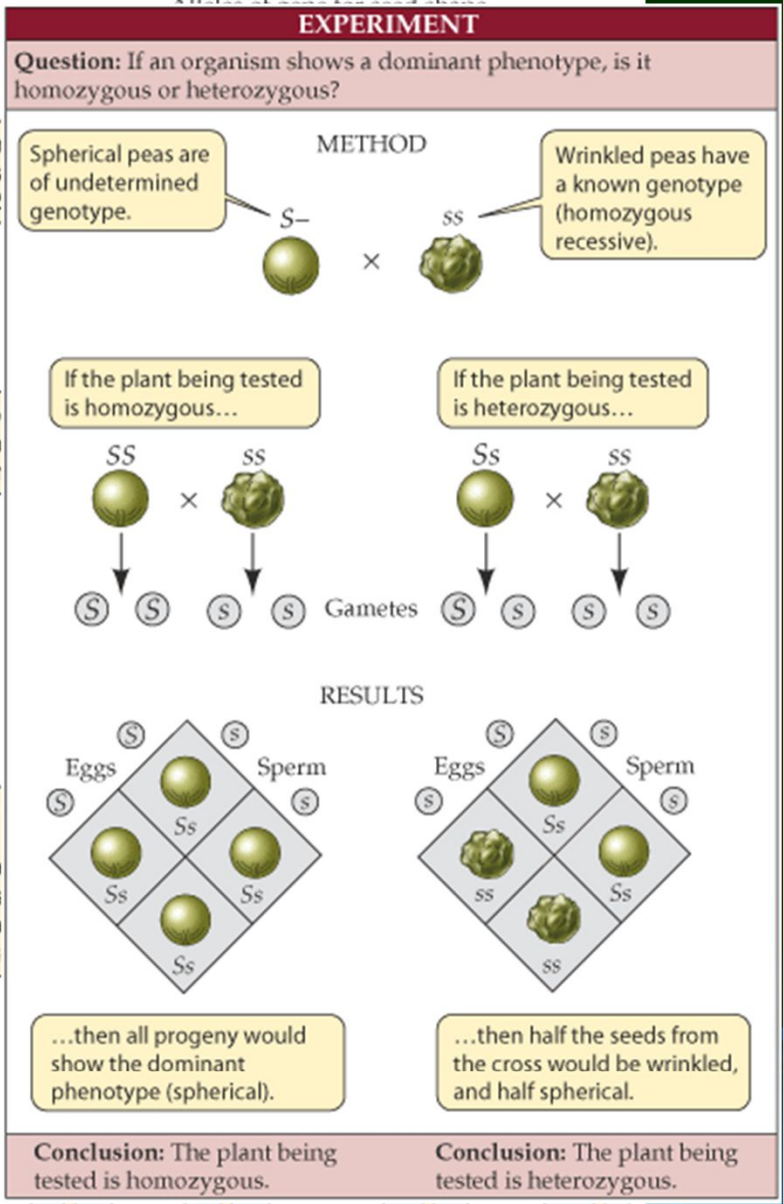
PARENTAL GENERATION PHENOTYPES			F_2 GENERATION PHENOTYP	
	DOMINANT	RECESSIVE	DOMINANT	RECESSI
	Spherical seeds ×	Wrinkled seeds	5,474	1,850
	Yellow seeds ×	Green seeds	6,022	2,001
	Purple flowers ×	White flowers	705	224
	Inflated pods ×	Constricted pods	882	299
	Green pods ×	Yellow pods	428	152
	Axial flowers ×	Terminal flowers	651	207
	Tall stems (1 m) ×	Dwarf stems (0.3 m)	787	277

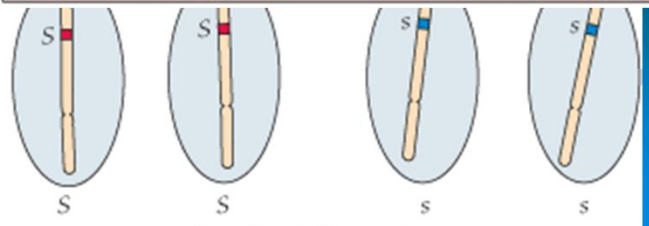


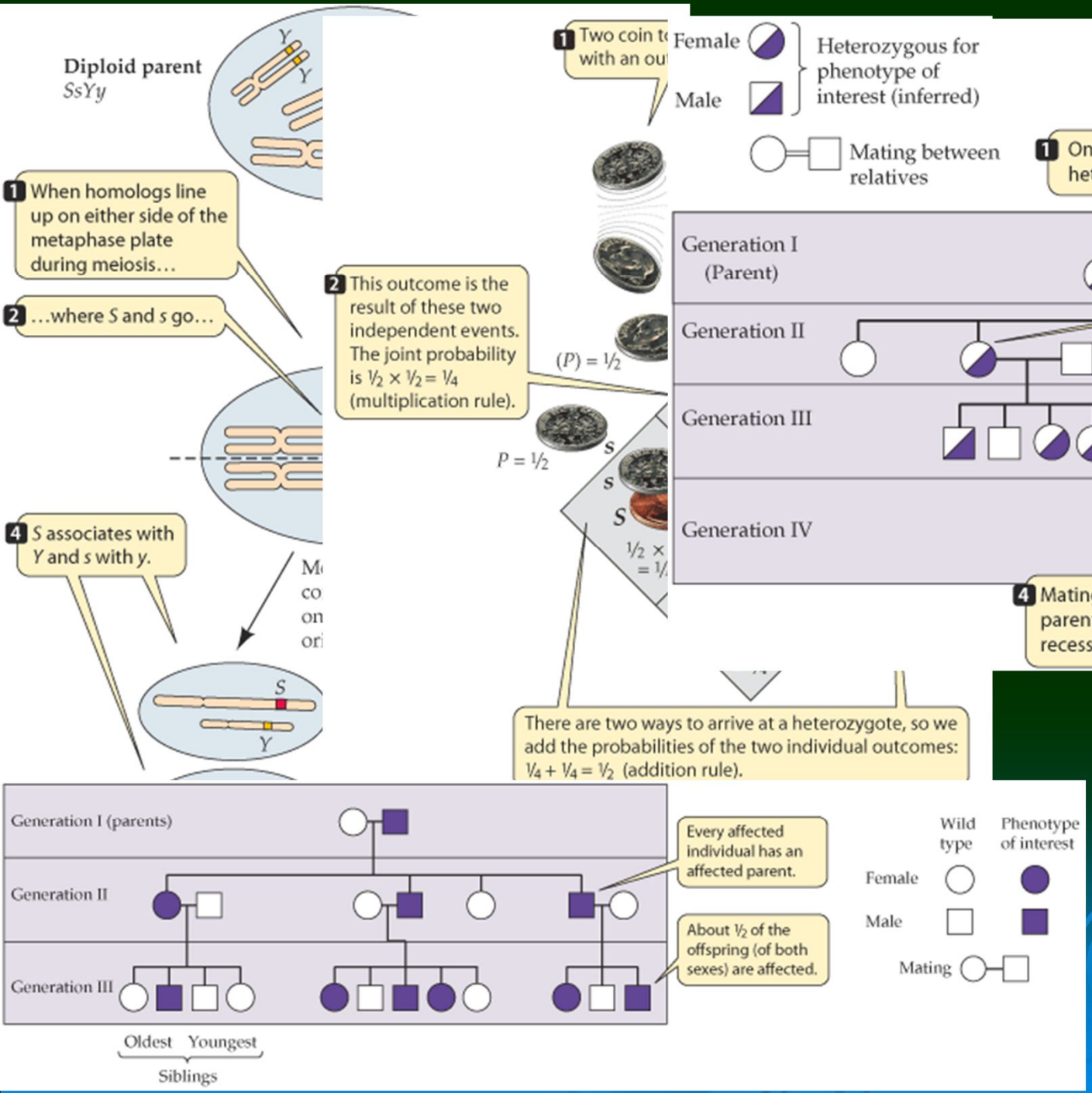

SSYY
ssyy
×
SsYy
Sy
sY
sy
Gametes
ate, the
nly to
with four
tio.
SY
SY
Sy
Sy
Sperm
sY
sy
SSYY
SSYy
SSYy
SSyy
SsYY
SsYy
SsYy
SsYy
ssYY
Ssyy
ssYy
ssYy
ssyy


1 A parent homozygous for the allele
spherical seeds is crossed with a
ent homozygous for the allele
wrinkled seeds.
ss
×
Gametes
S S
2 The pare
combine
plants wi
type and
seed phe
Ss
Ss
×
3 The hete
is self-po
Gametes
S S S S
Eggs
Sperm
S
S
SS
Ss
Ss
ss
4 S and s ga
randomly
different
in the F2
Punnett s

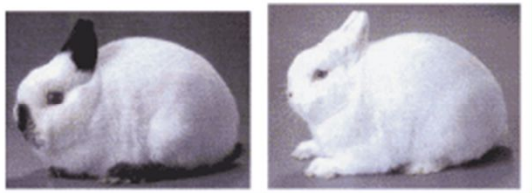

1 This site on
is the locus
the alleles
2 Before mei
of the hom
chromosor
3 At the end
meiosis I,
the two all
are segrega
into separa
daughter c

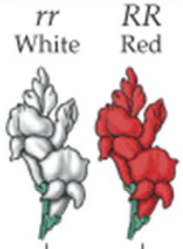

EXPERIMENT
Question: If an organism shows a dominant phenotype, is it homozygous or heterozygous?
METHOD
Spherical peas are of undetermined genotype.
S–
×
ss
Wrinkled peas have a known genotype (homozygous recessive).
If the plant being tested is homozygous…
SS
ss
×
S S
s s
Gametes
If the plant being tested is heterozygous…
Ss
ss
×
S s
s s
Gametes
RESULTS
Eggs
Sperm
S
s
Ss
Ss
Ss
Ss
…then all progeny would show the dominant phenotype (spherical).
Eggs
Sperm
S
s
Ss
Ss
ss
Ss
…then half the seeds from the cross would be wrinkled, and half spherical.
Conclusion: The plant being tested is homozygous.
Conclusion: The plant being tested is heterozygous.


S
S
S
S
s
s
s
s


Diploid parent
SsYy
Y
Y
1 When homologs line up on either side of the metaphase plate during meiosis…
2 …where S and s go…
4 S associates with Y and s with y.
S
Y
1 Two coin t
with an ou
2 This outcome is the result of these two independent events. The joint probability is ½ × ½ = ¼ (multiplication rule).
(P) = ½
P = ½
s
s
S
½ ×
= ½
There are two ways to arrive at a heterozygote, so we add the probabilities of the two individual outcomes: ¼ + ¼ = ½ (addition rule).
Female
Male
Heterozygous for phenotype of interest (inferred)
Mating between relatives
1 One parent is heterozygous…
2 …and the recessive allele is passed on to one-half of the phenotypically normal offspring
Generation I (Parent)
Generation II
Generation III
Generation IV
3 These cousins are heterozygous.
4 Mating of heterozygous recessive parents may produce homozygous recessive (affected) offspring.
Generation I (parents)
Generation II
Generation III
Every affected individual has an affected parent.
About ½ of the offspring (of both sexes) are affected.
Wild type
Phenotype of interest
Female
Male
Mating
Oldest
Youngest
Siblings

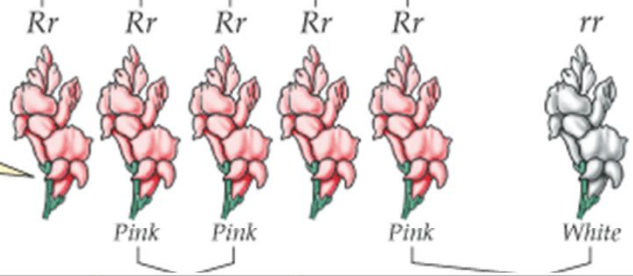
Possible genotypes	CC, Cc^{ch}, Cc^{h}, Cc	$c^{ch}c^{ch}$	$c^{ch}c^{h}$, $c^{ch}c$	$c^{h}c^{h}$, $c^{h}c$	cc
Phenotype	Dark gray	Chinchilla	Light gray	Himalayan	Albino





Parental (P) generation

1 When true-breeding red and white parents are crossed, the F_1 generation are all pink.



F_1 generation

2 Heterozygous snapdragons produce pink flowers—an intermediate phenotype—because the allele for red flowers is incompletely dominant white one

F_2 generat

3 When F_1 pollinate, white, pir offspring

Blood type of cells	Genotype	Antibodies made by body	Reaction to added antibodies	
			Anti-A	Anti-B
A	I^AI^A or I^Ai^O	Anti-B		
B	I^BI^B or I^Bi^O	Anti-A		
AB	I^AI^B	Neither anti-A nor anti-B		
O	i^Oi^O	Both anti-A and anti-B		

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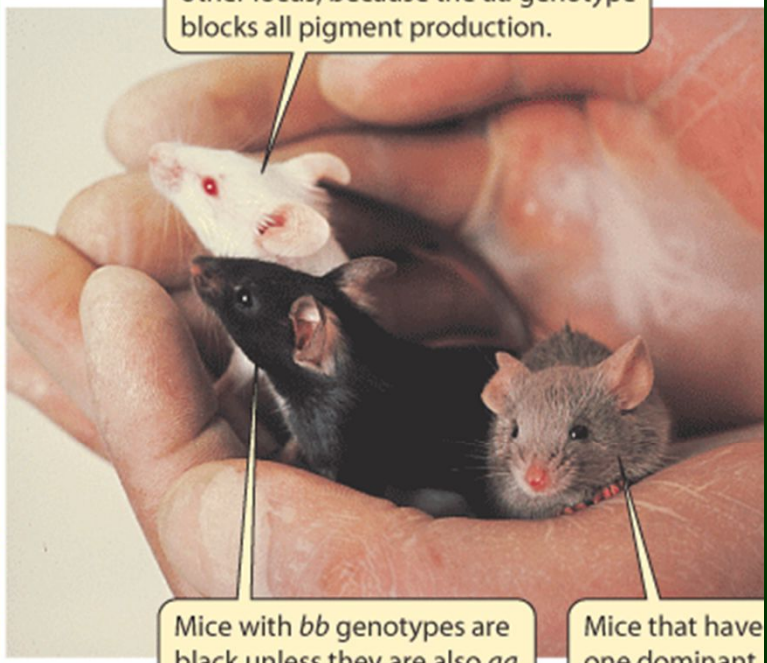


EXPERIMENT

estion: Do alleles for different characteristics always assort independently?

rent (P)

BbVgvg
Wild type (gray body, normal wings)

×

bbvgvg
(Black body, vestigial win

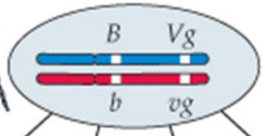
♀ ♂

Gray normal (wild type)

♀

Black vestigial

♂



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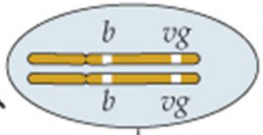
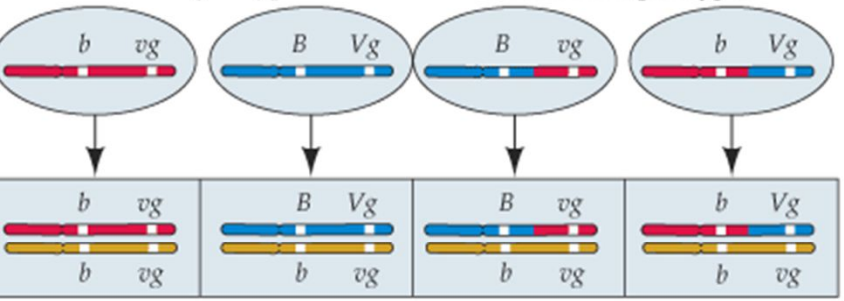


Recombination

Parental genotypes

Recombinant genotypes



Black vestigial	Wild type	Gray vestigial	Black normal
944	965	206	185

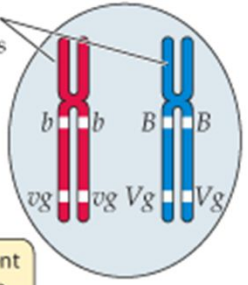
Number of individuals

Parental phenotypes

Recombinant (nonparental) phenotypes

$$\text{Recombinant frequency} = \frac{391 \text{ recombinants}}{2{,}300 \text{ total offspring}} = 0.17$$

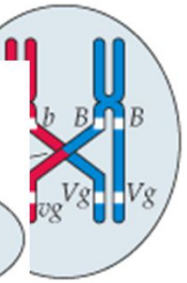
Homologous chromosomes



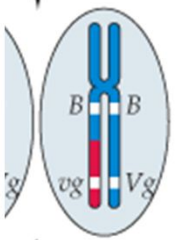
Genes at different loci on the same chromosome can separate and recombine by crossing over.

These are th
from Mendel
(independen

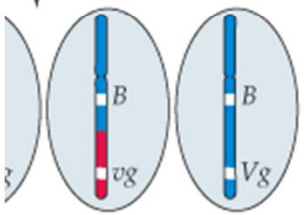
Meiosis I



s

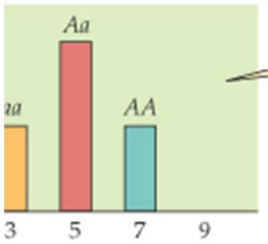


II

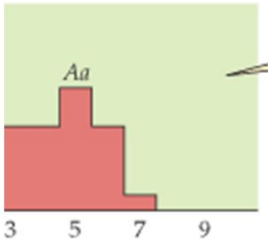


Gametes

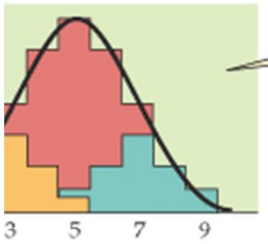
recombinant gametes
t of crossing over.



Quantitativ
3 phenotyp

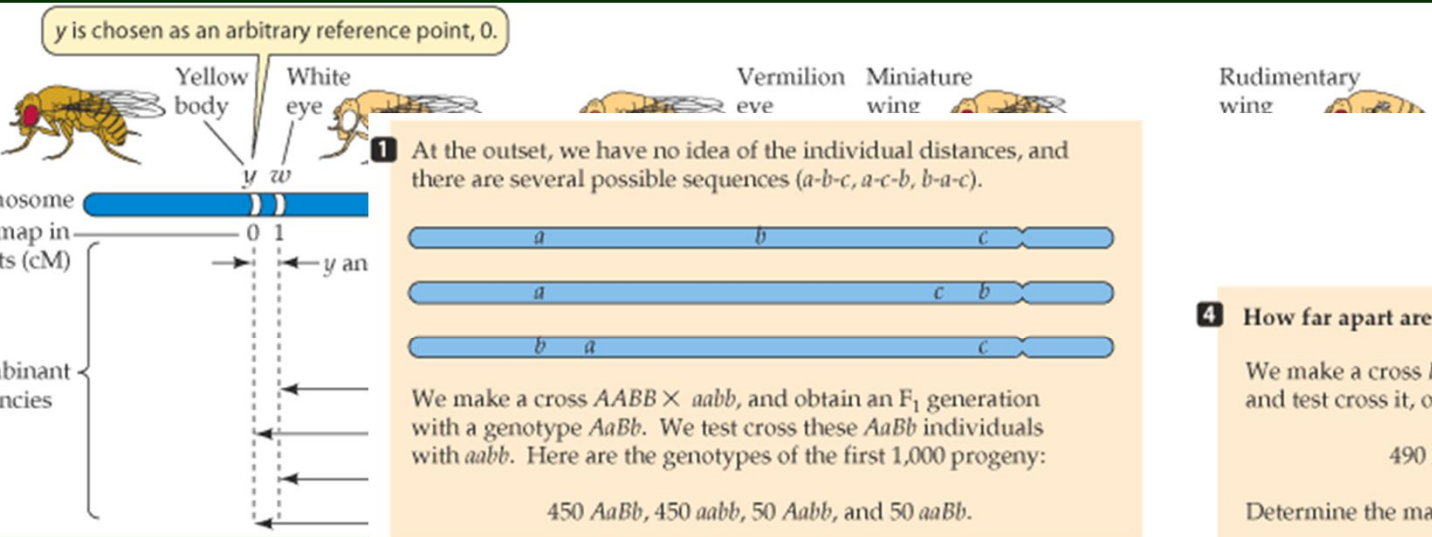


Quantitativ
due to envi



Interaction
environme
continuous

on in phenotype



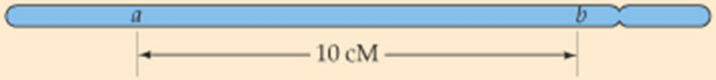
1 At the outset, we have no idea of the individual distances, and there are several possible sequences (*a*-*b*-*c*, *a*-*c*-*b*, *b*-*a*-*c*).

We make a cross *AABB* × *aabb*, and obtain an F_1 generation with a genotype *AaBb*. We test cross these *AaBb* individuals with *aabb*. Here are the genotypes of the first 1,000 progeny:

450 *AaBb*, 450 *aabb*, 50 *Aabb*, and 50 *aaBb*.

2 **How far apart are the *a* and *b* genes?** Well, what is the recombinant frequency? Which are the recombinant types, and which are the parental types?

Recombinant frequency (*a* to *b*) = (50 + 50)/1,000 = 0.1
So the map distance is
Map distance = 100 × recombinant frequency =
100 × 0.1 = 10 cM

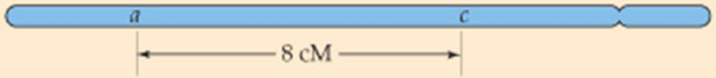


3 Now we make a cross *AACC* × *aacc*, obtain an F_1 generation, and test cross it, obtaining:

460 *AaCc*, 460 *aacc*, 40 *Aacc*, and 40 *aaCc*.

How far apart are the *a* and *c* genes?

Recombinant frequency (*a* to *c*) = (40 + 40)/1,000 = 0.08
Map distance = 100 × recombinant frequency =
100 × 0.08 = 8 cM



4 **How far apart are the *b* and *c* genes?**

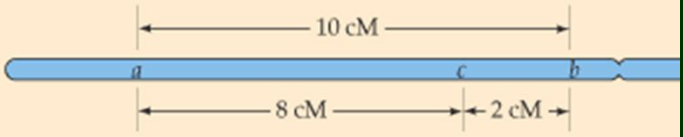
We make a cross *BBCC* × *bbcc*, obtain an F_1 generation, and test cross it, obtaining:

490 *BbCc*, 490 *bbcc*, 10 *Bbcc*, and 10 *bbCc*.

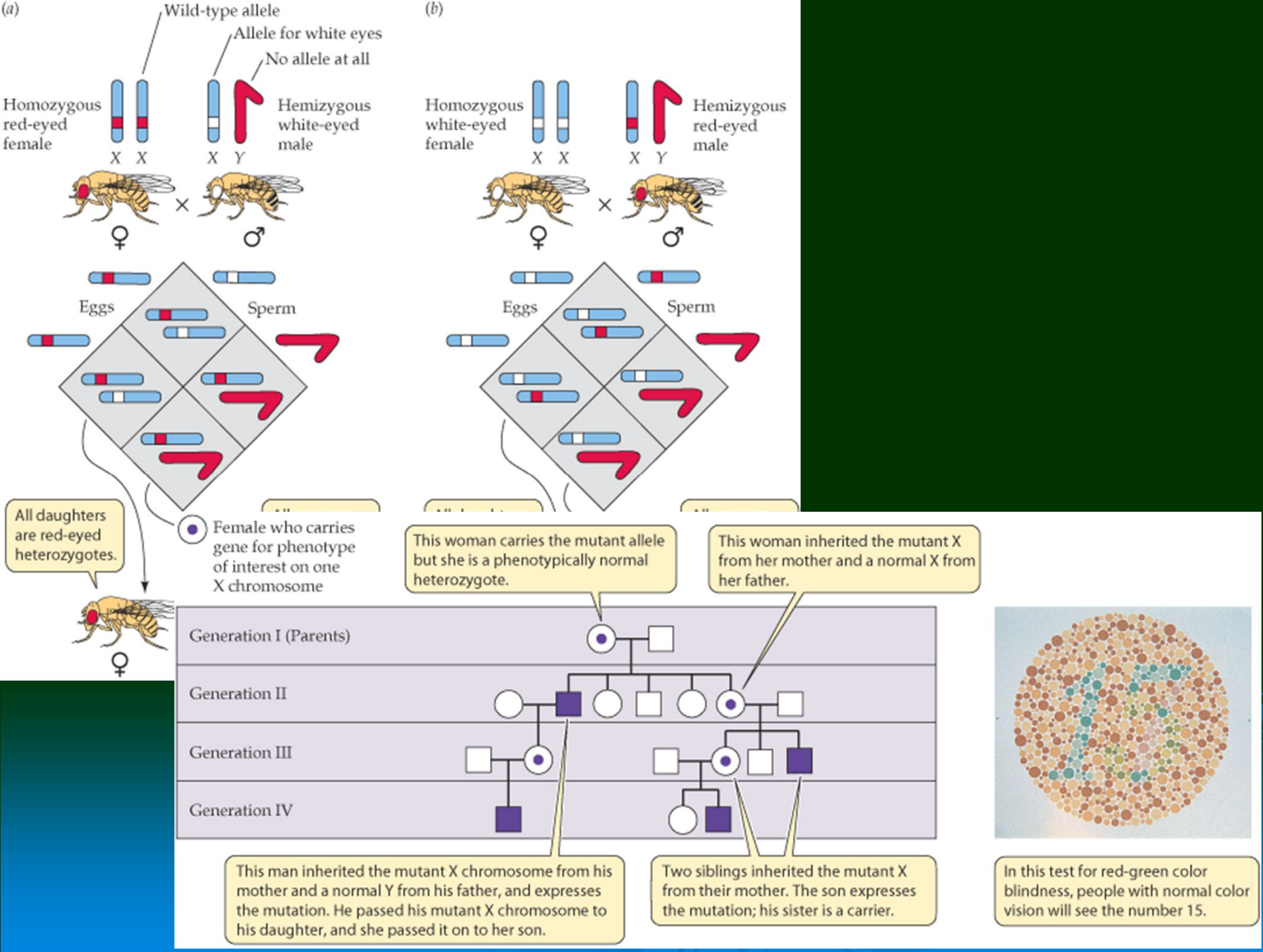
Determine the map distance between *b* and *c*.

Recombinant frequency (*b* to *c*) = (10 + 10)/1,000 = 0.02
Map distance = 100 × recombinant frequency =
100 × 0.02 = 2 cM

5 **Which of the three genes is between the other two?**
Because *a* and *b* are the farthest apart, *c* must be between them.



These numbers add up perfectly, but in most real cases they don't add up perfectly because of multiple crossovers.


(a)
Wild-type allele
Allele for white eyes
No allele at all
Homozygous red-eyed female
Hemizygous white-eyed male
X X
X Y
Eggs
Sperm
All daughters are red-eyed heterozygotes.
Female who carries gene for phenotype of interest on one X chromosome
(b)
Homozygous white-eyed female
Hemizygous red-eyed male
X X
X Y
Eggs
Sperm
This woman carries the mutant allele but she is a phenotypically normal heterozygote.
This woman inherited the mutant X from her mother and a normal X from her father.
Generation I (Parents)
Generation II
Generation III
Generation IV
This man inherited the mutant X chromosome from his mother and a normal Y from his father, and expresses the mutation. He passed his mutant X chromosome to his daughter, and she passed it on to her son.
Two siblings inherited the mutant X from their mother. The son expresses the mutation; his sister is a carrier.
In this test for red-green color blindness, people with normal color vision will see the number 15.

Introduction to Molecular Cell Biology

Hereditary Molecules

Dr. Fridoon Jawad Ahmad

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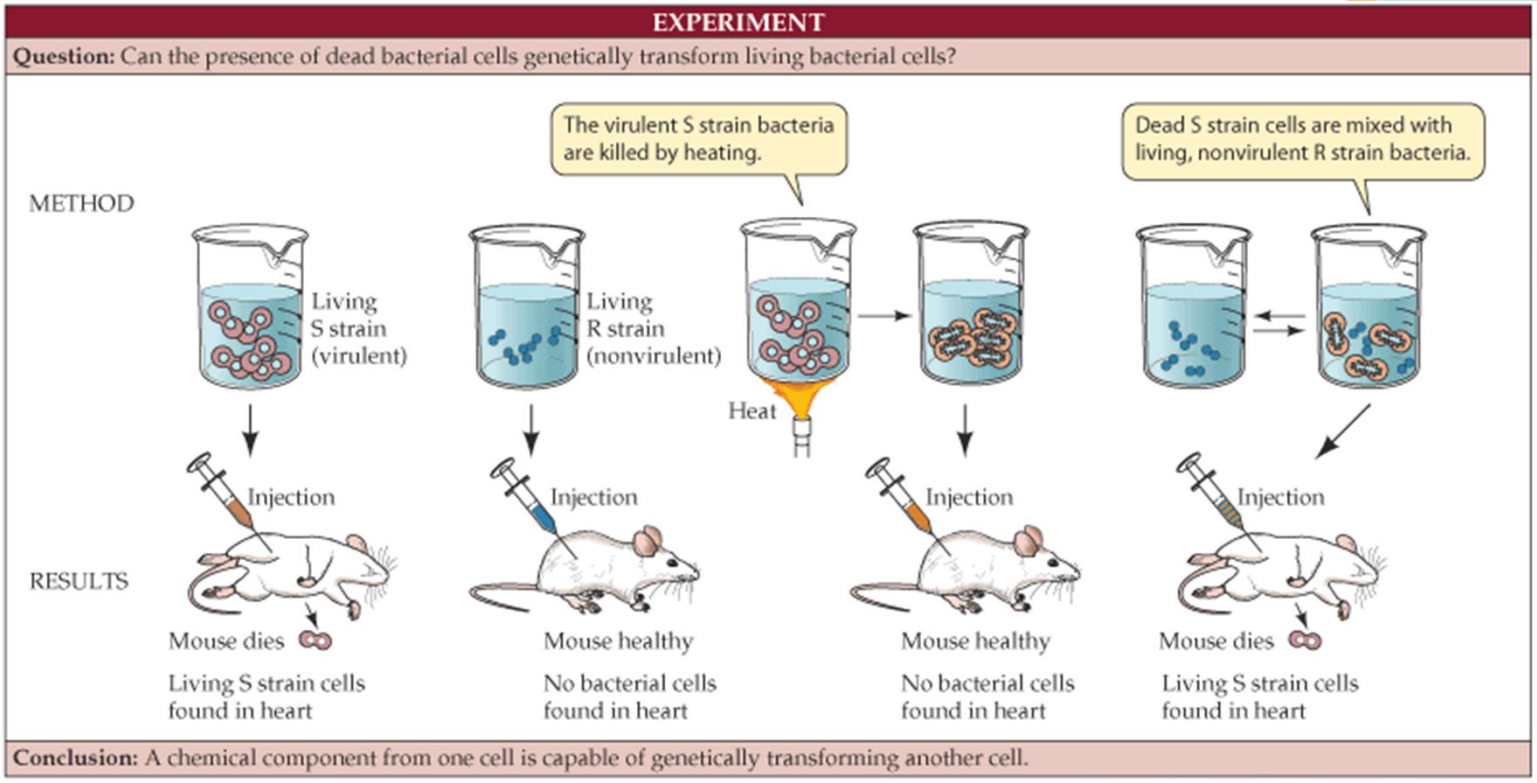

MacLeaod and MacCarty 1944

The nature of "Transforming Principle"

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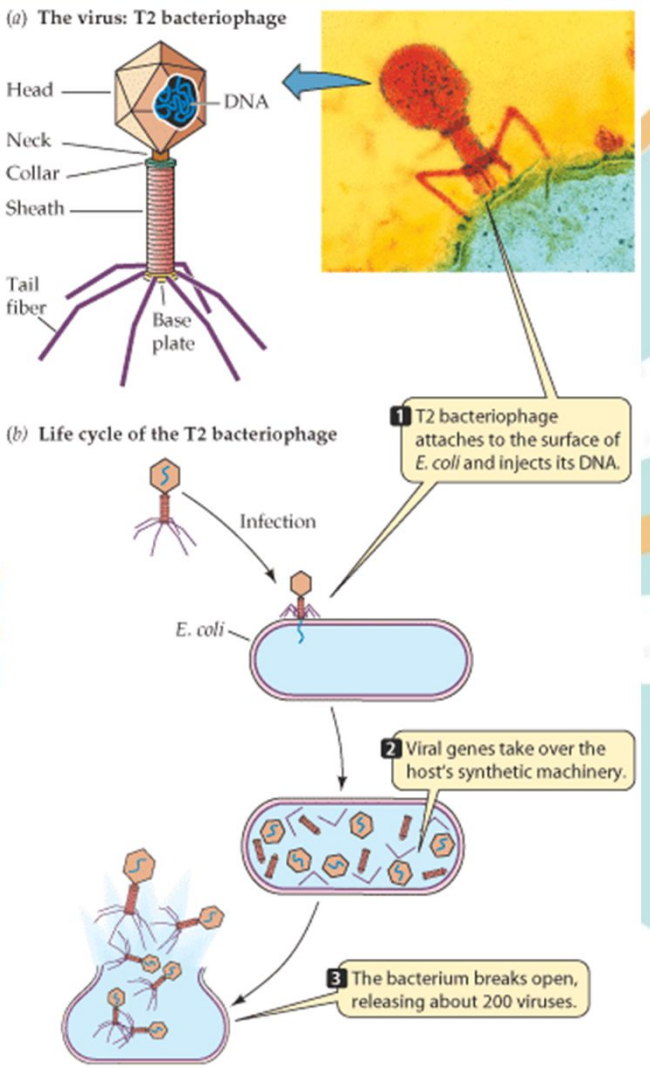
Search for the Heredity Molecule 2

T2 Reproduction Cycle



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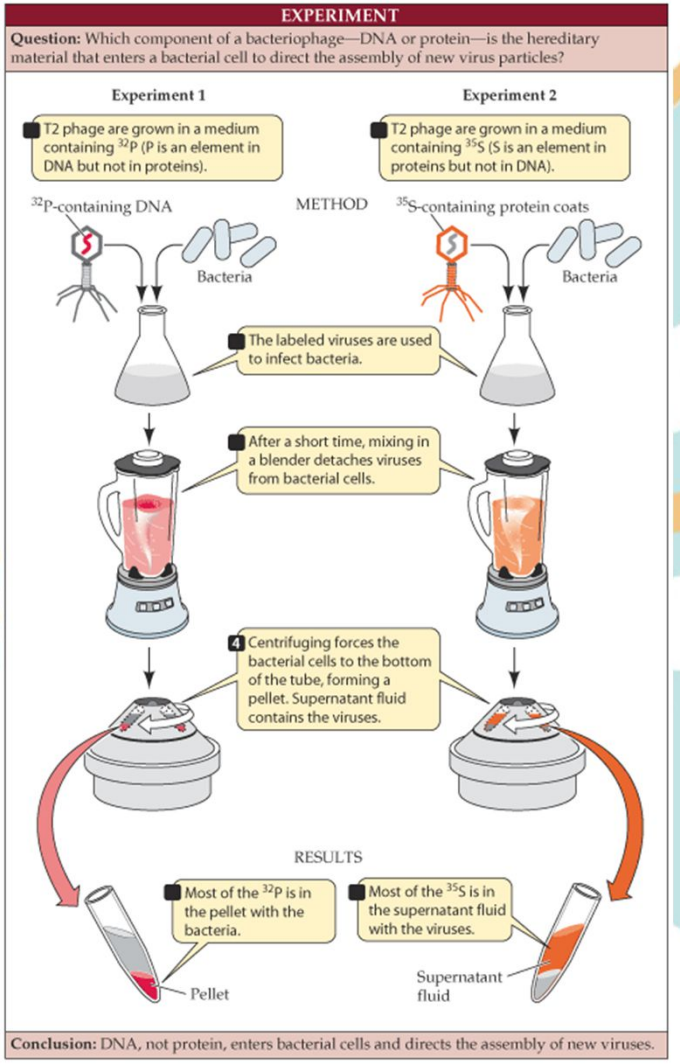
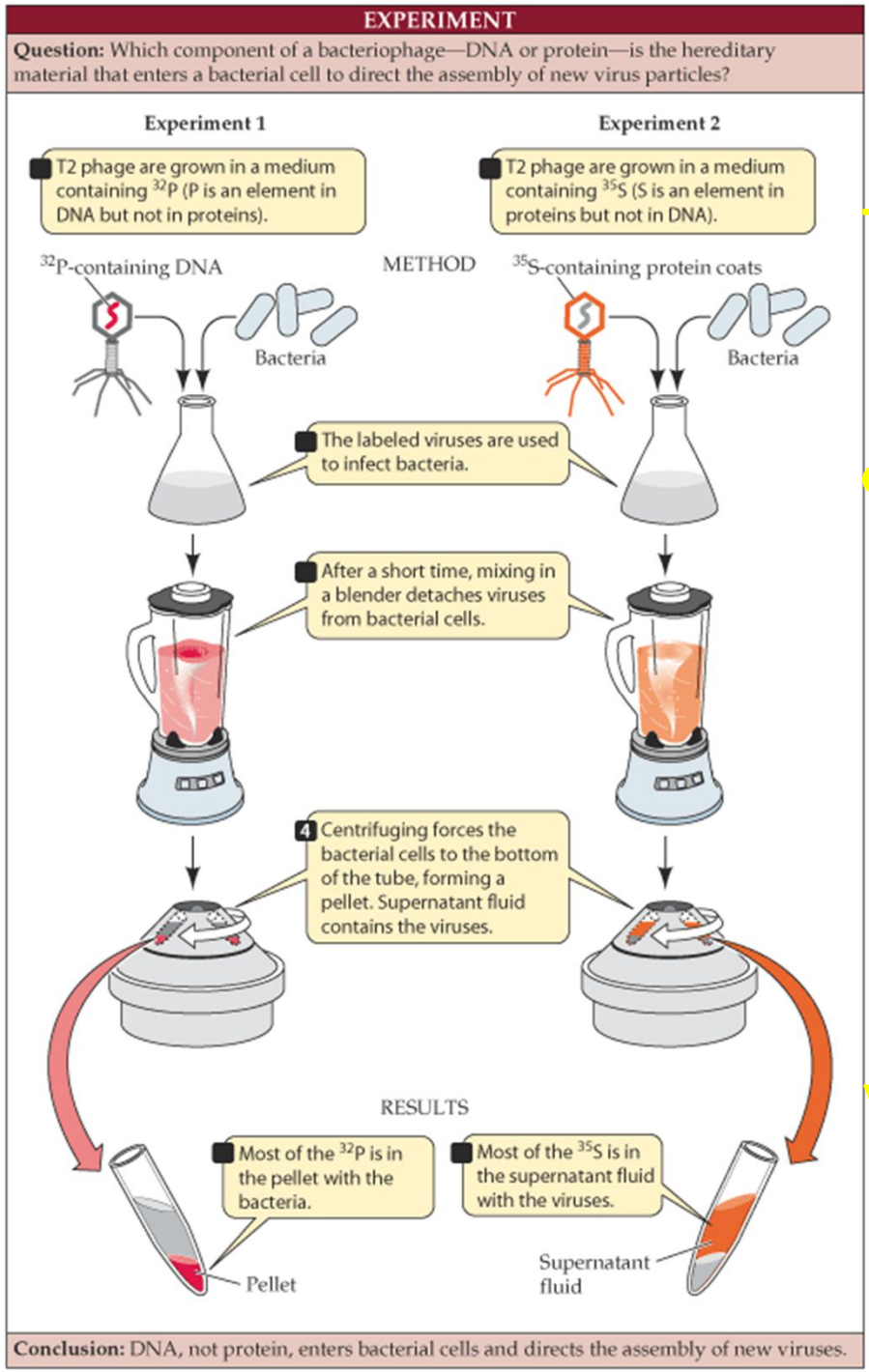


END



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Cell Biology

Investigating:
Heredity Molecule

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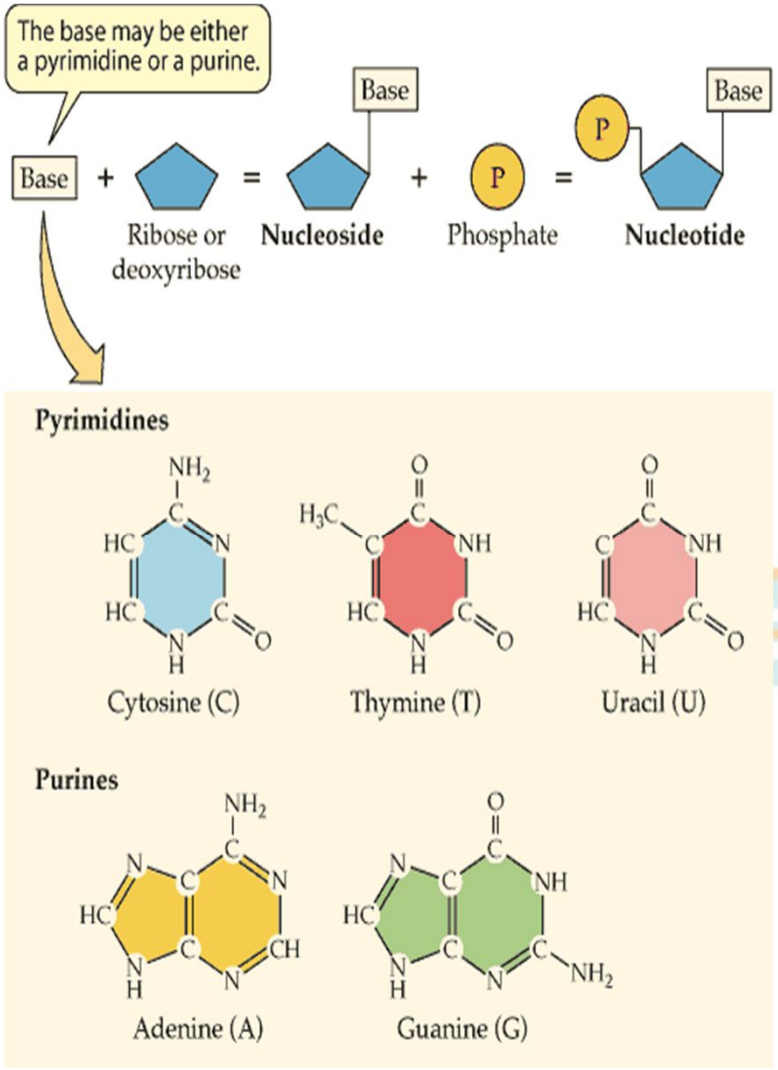

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Investigating: Heredity Molecule

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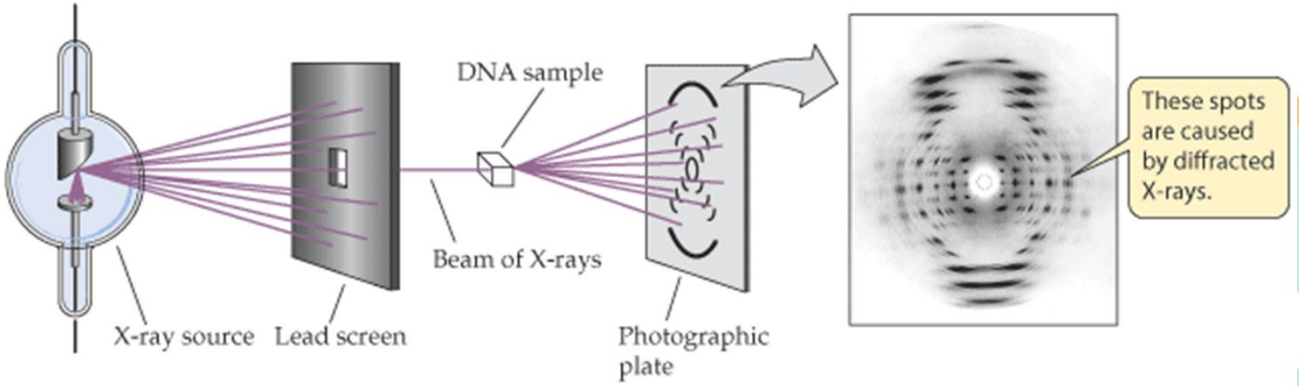

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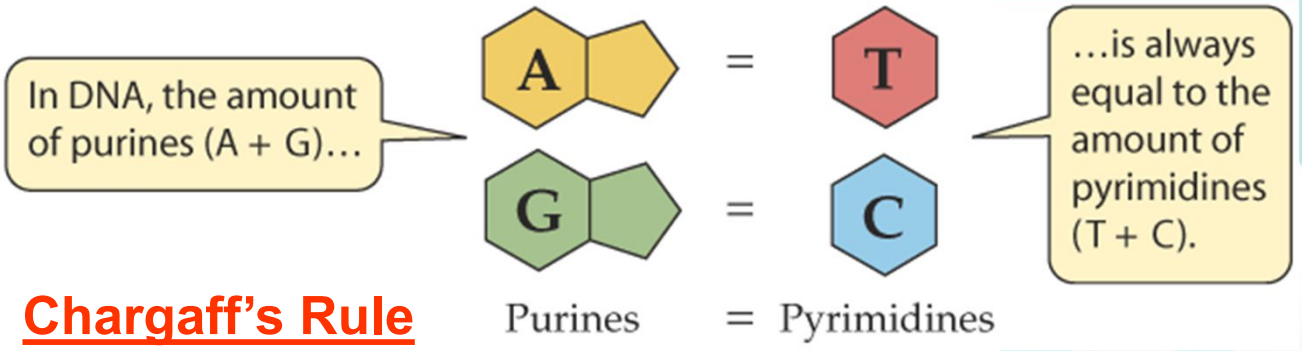
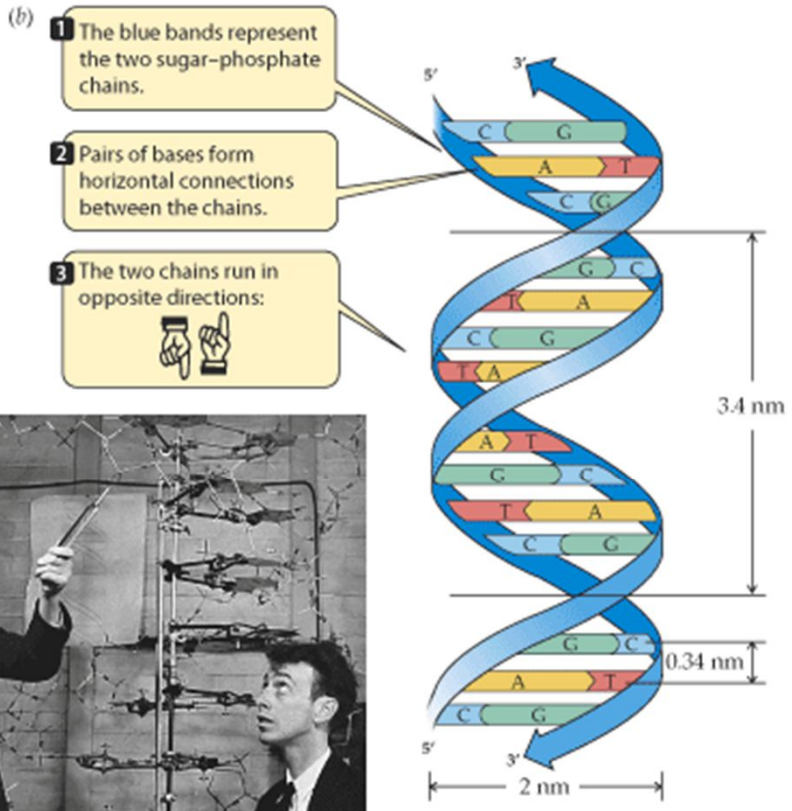
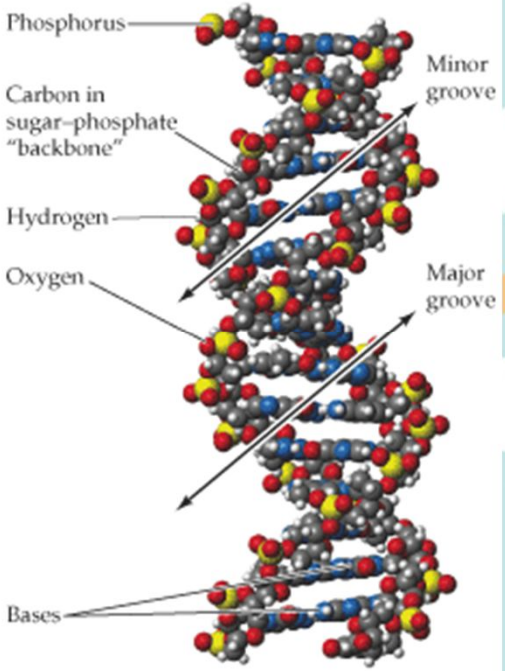


Chargaff's Rule

Investigating: Heredity Molecule





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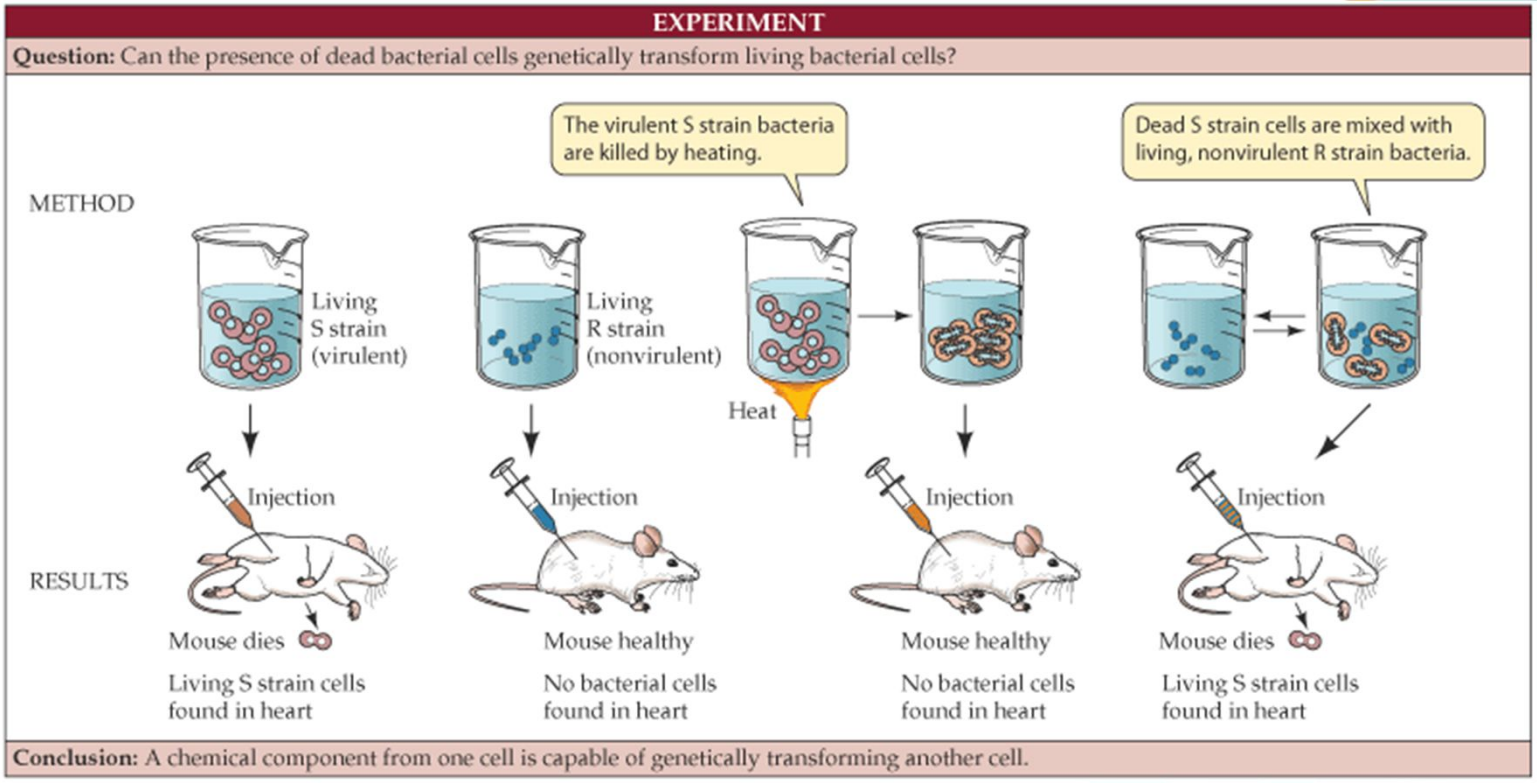

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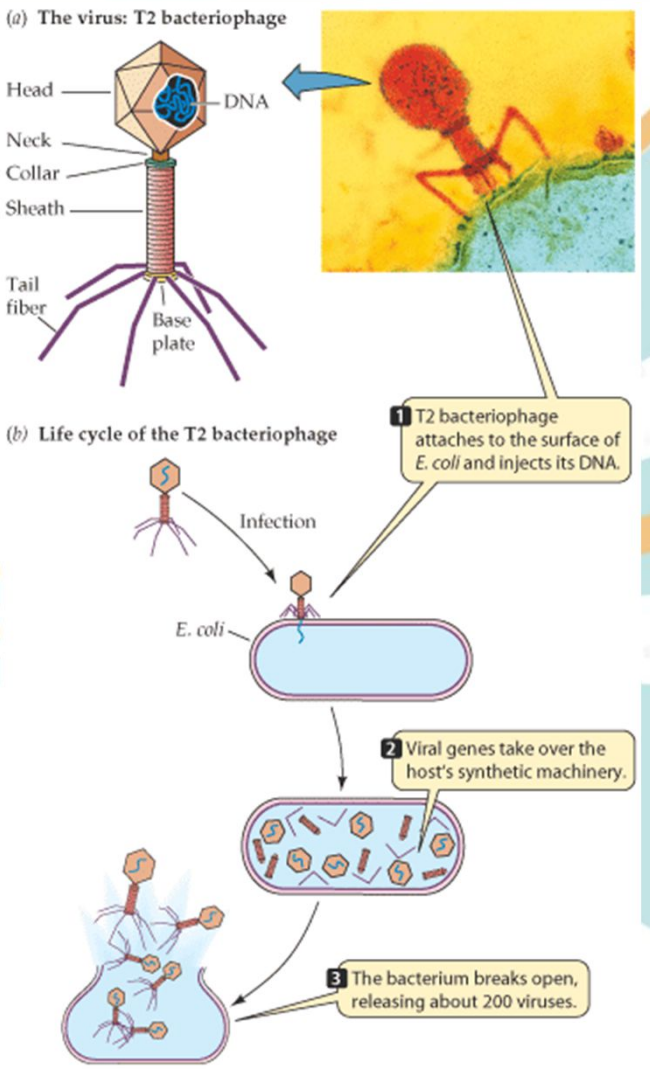

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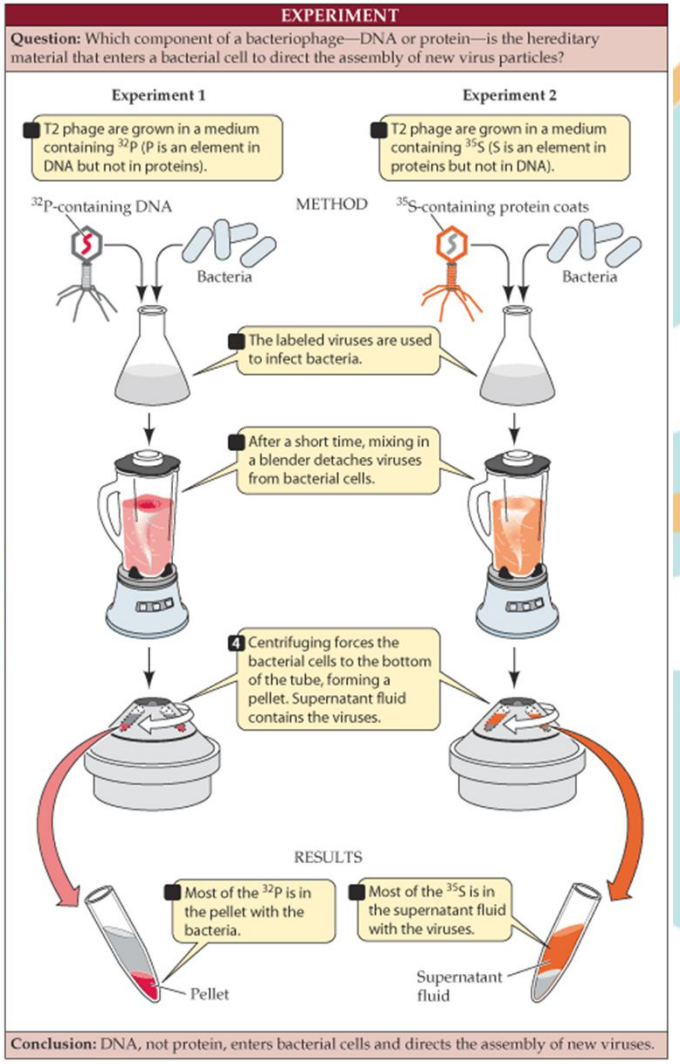

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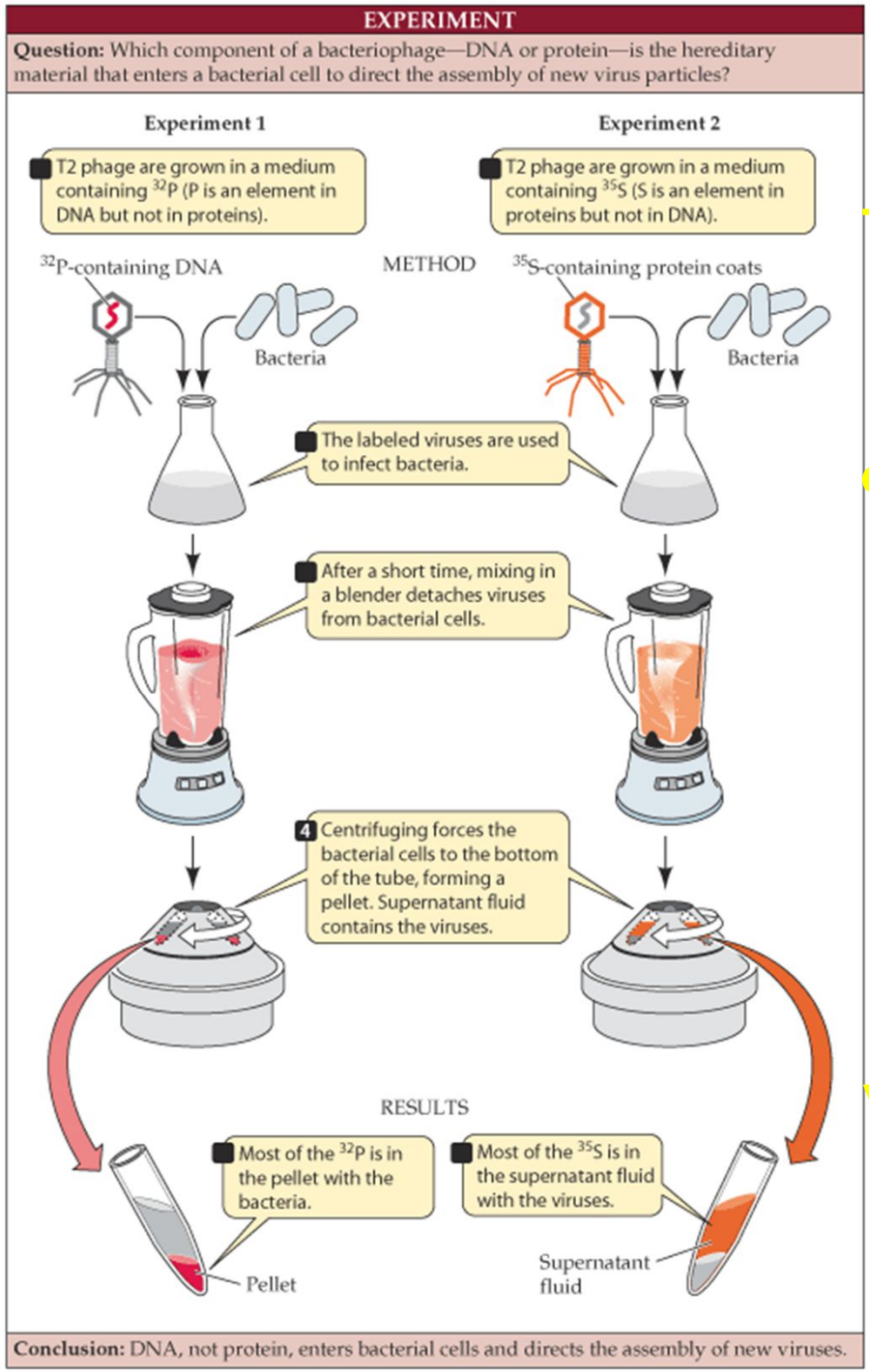

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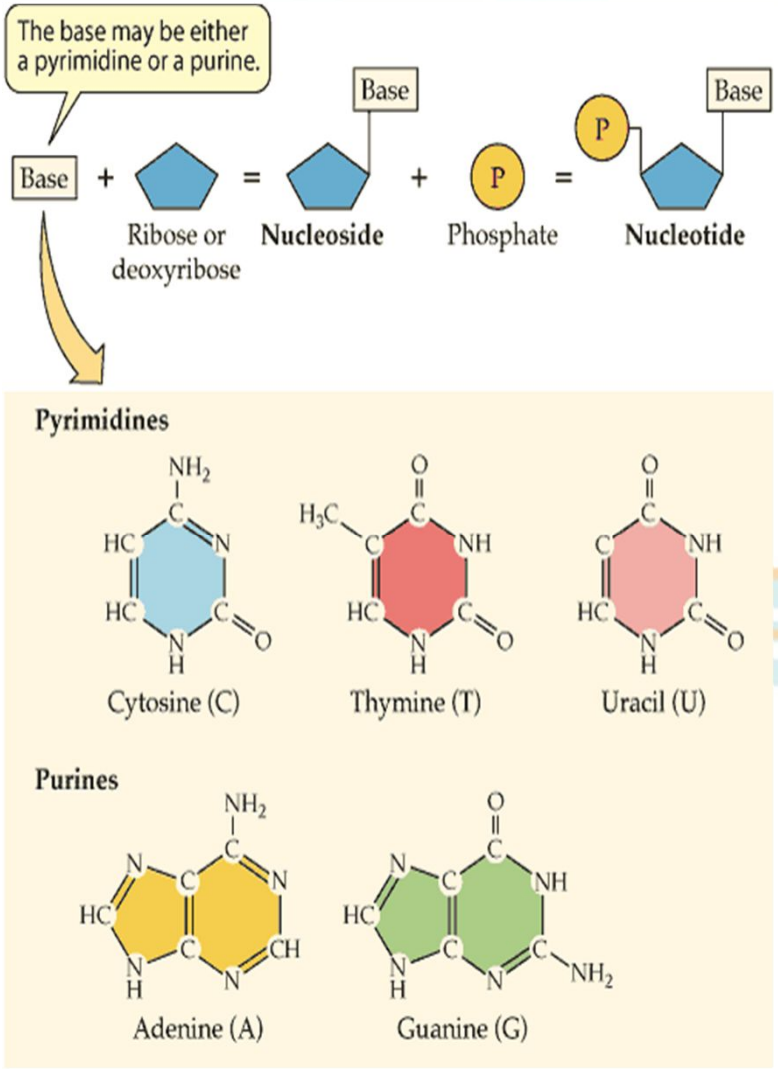

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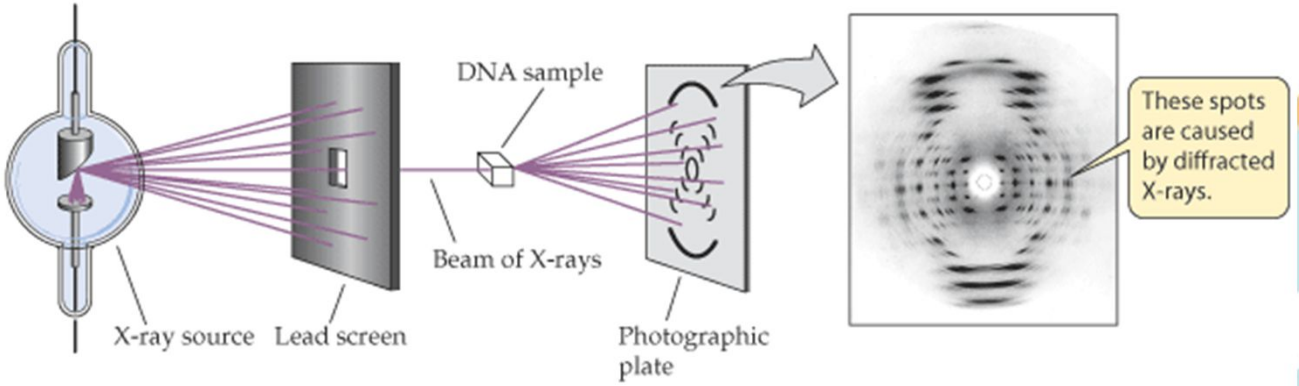

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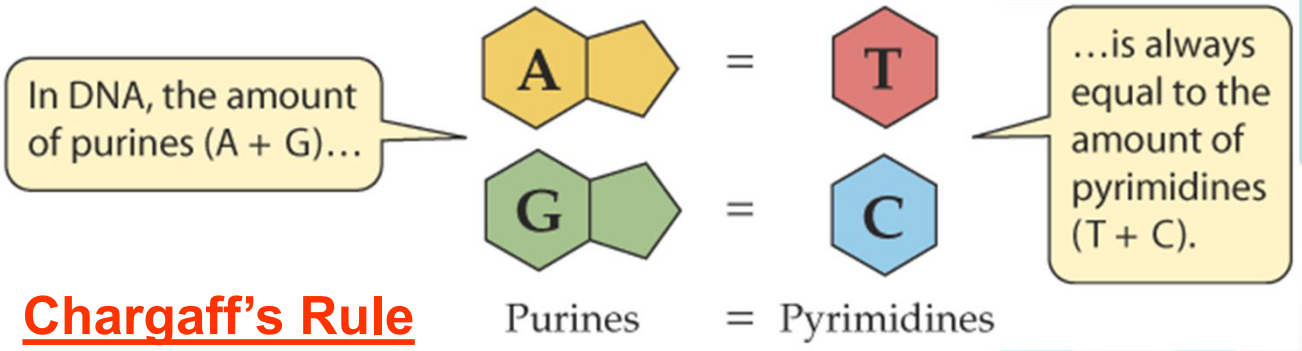

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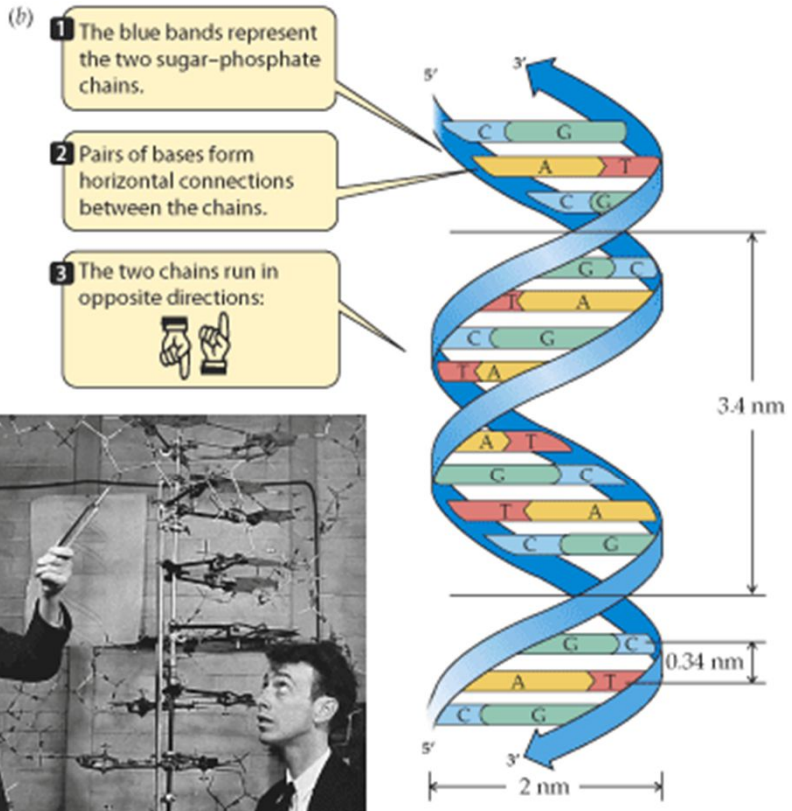

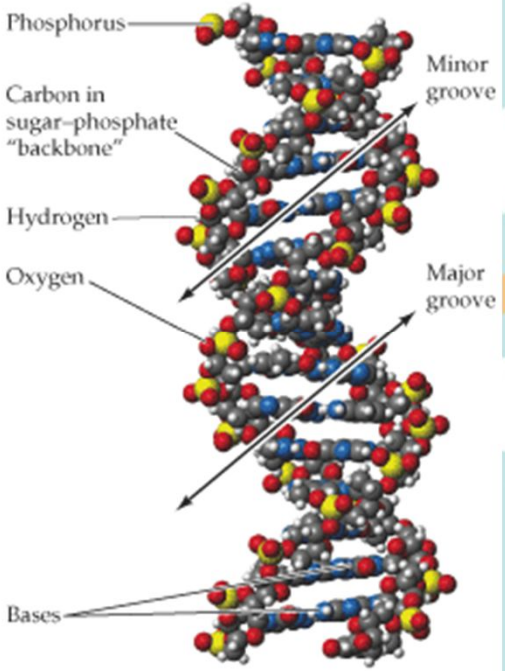

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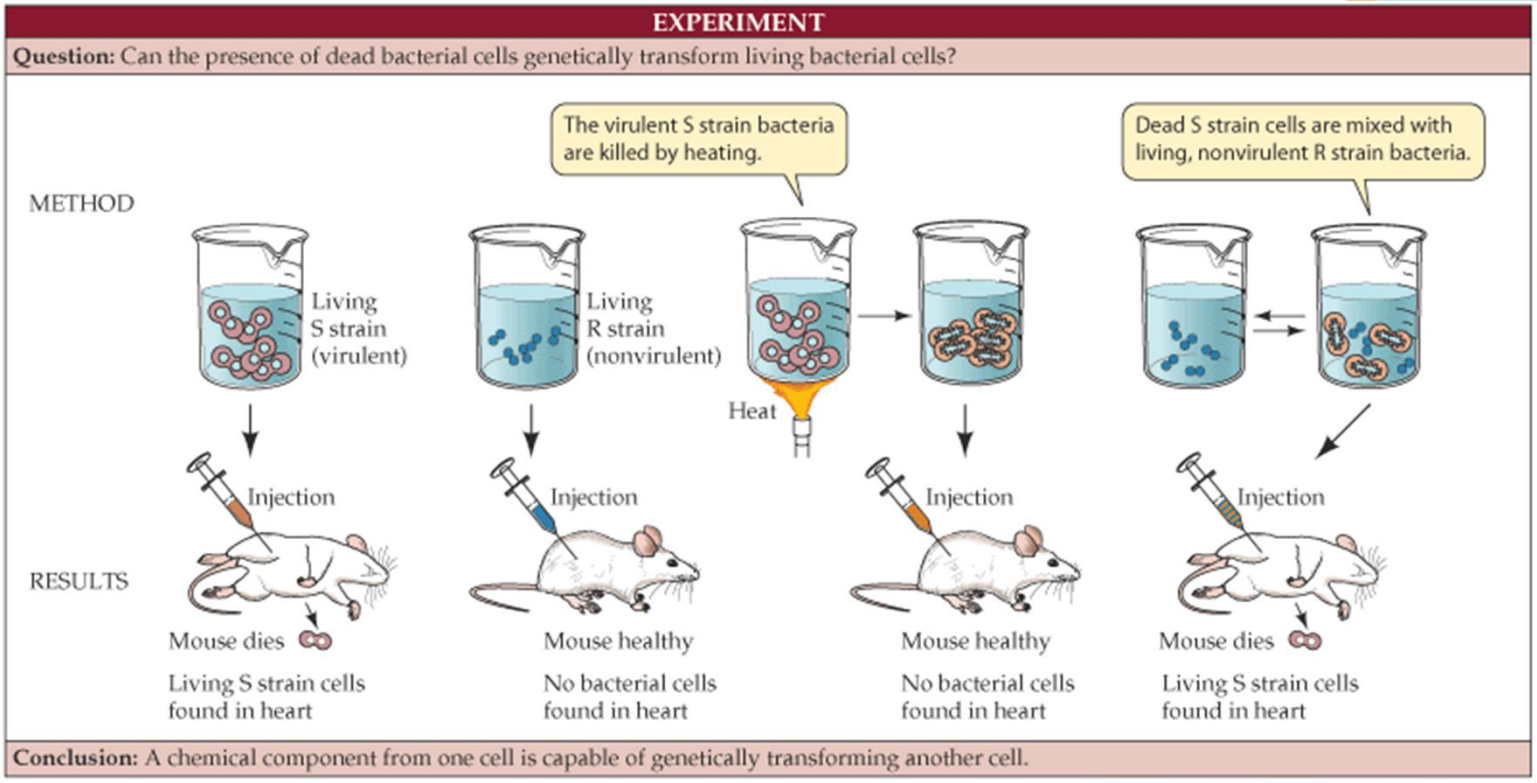

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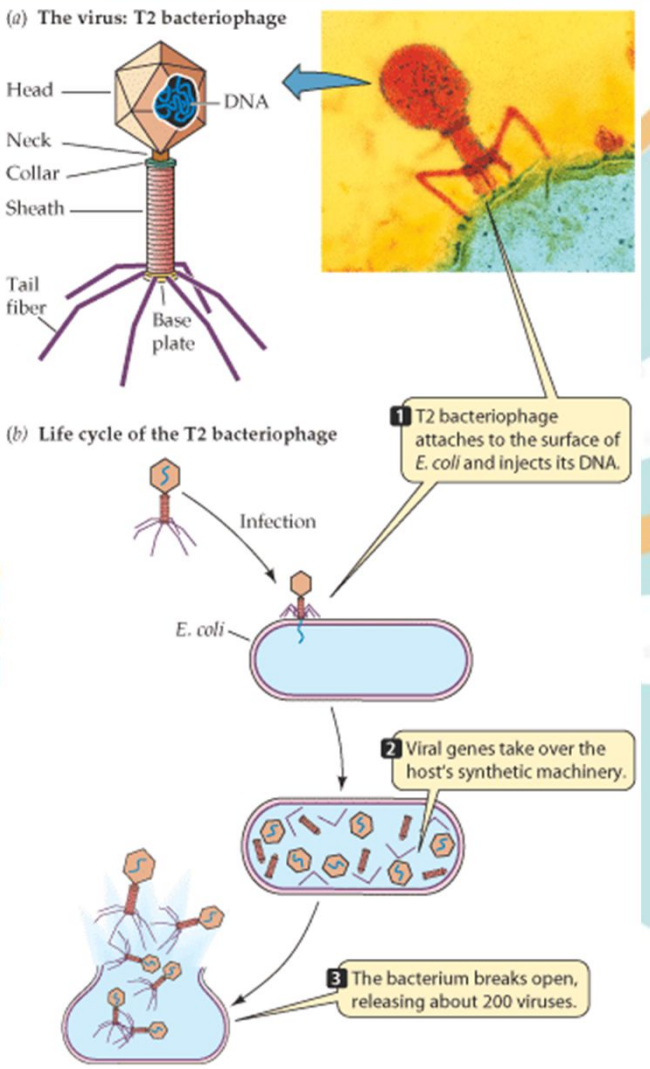

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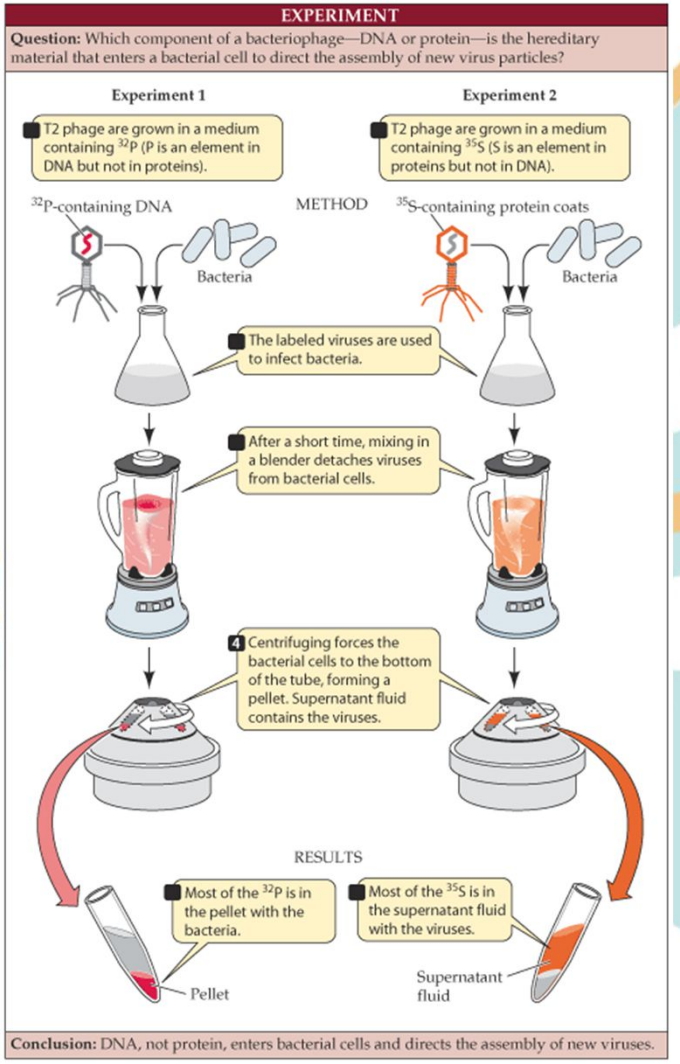

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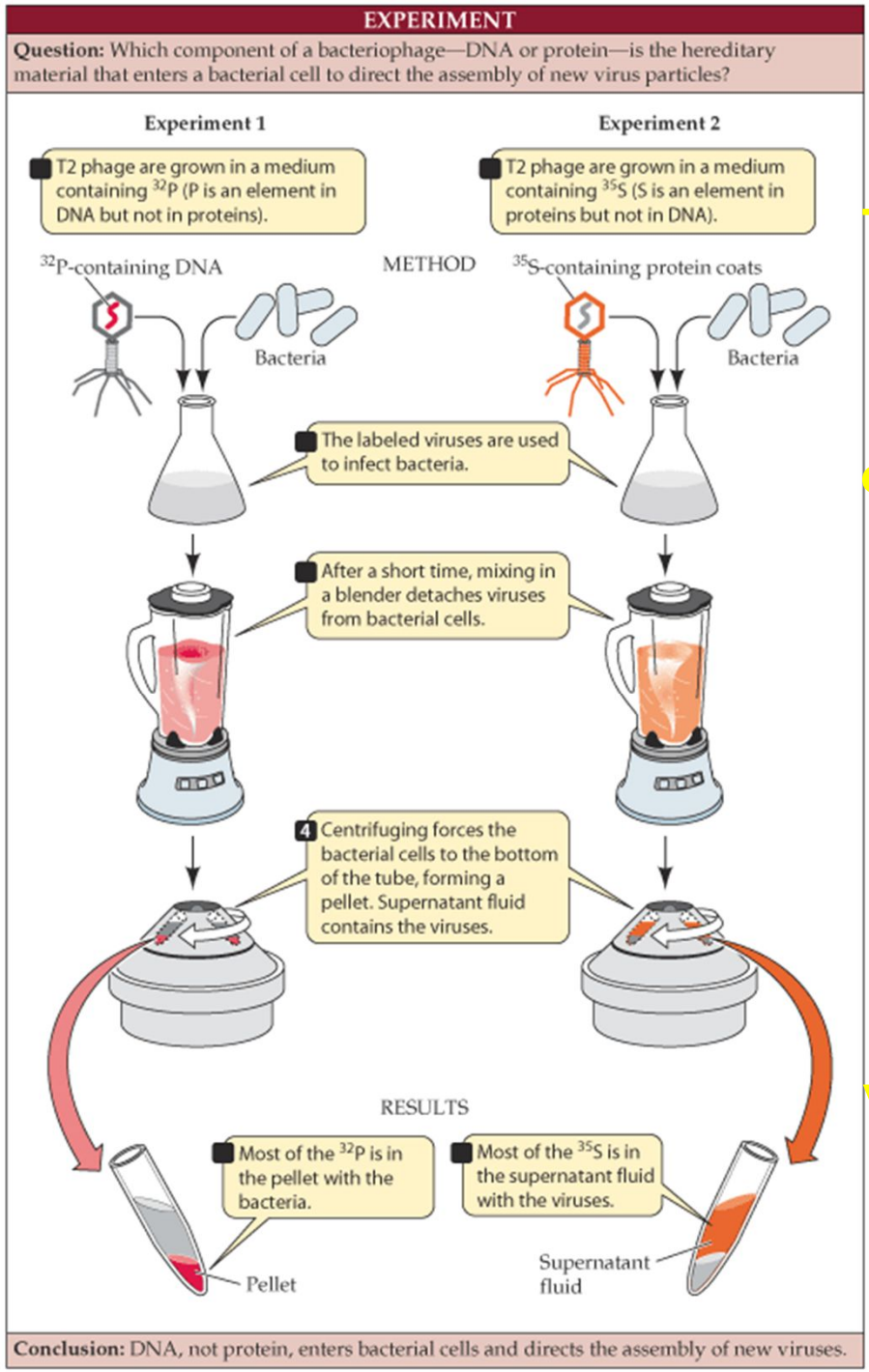

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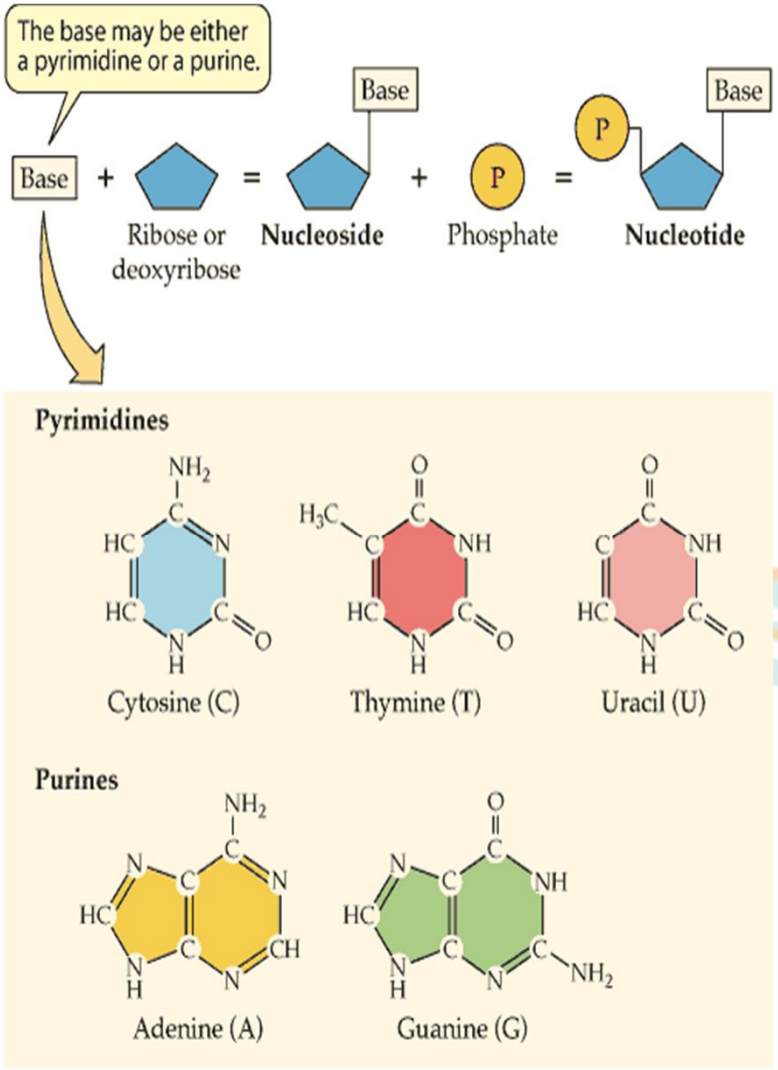

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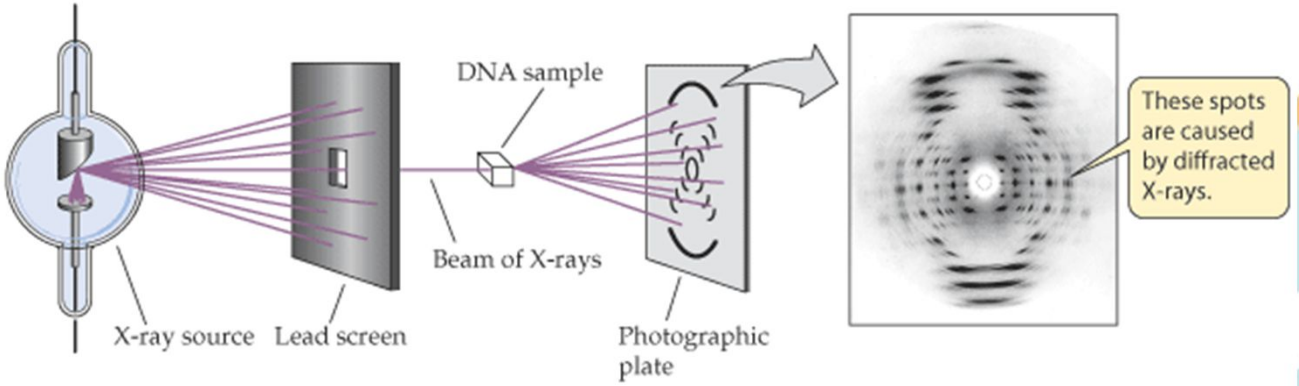

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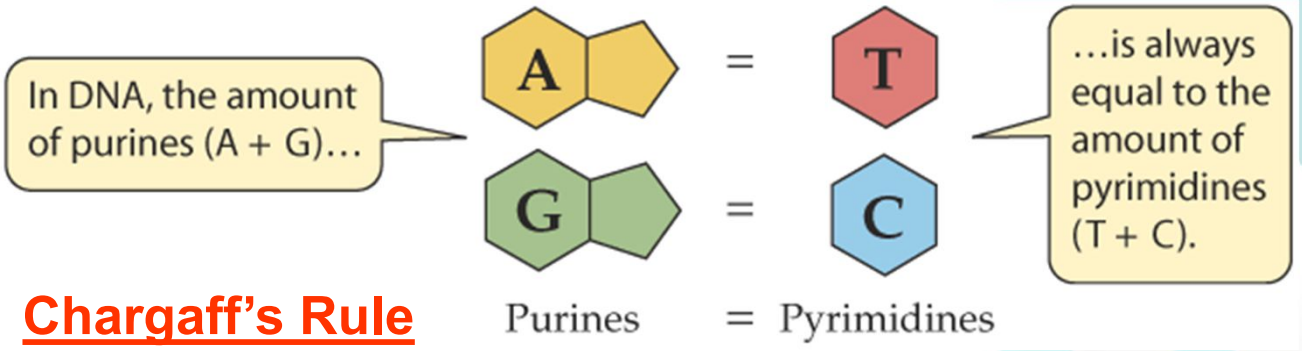

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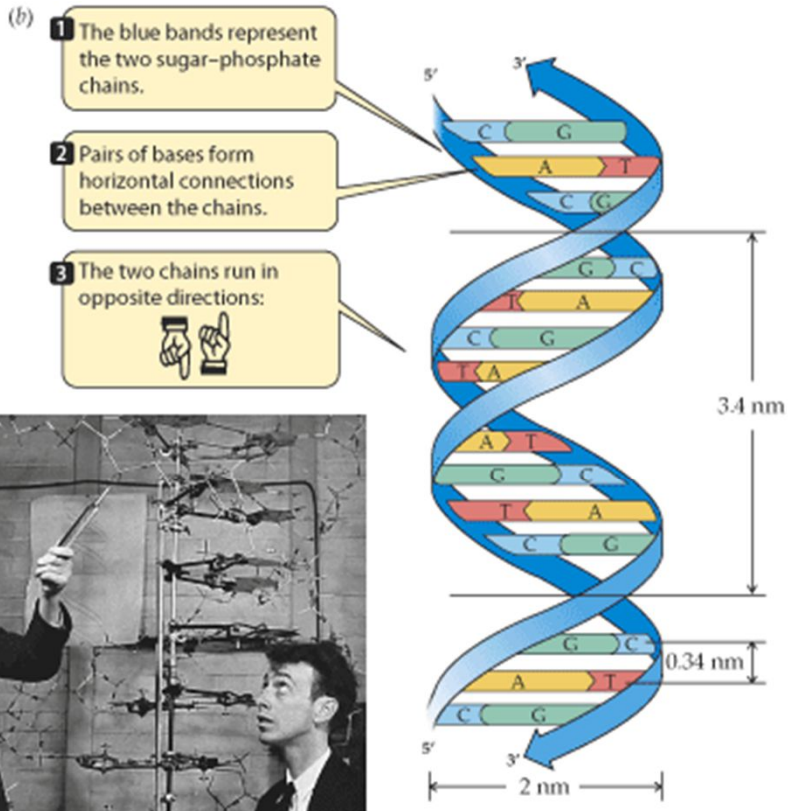
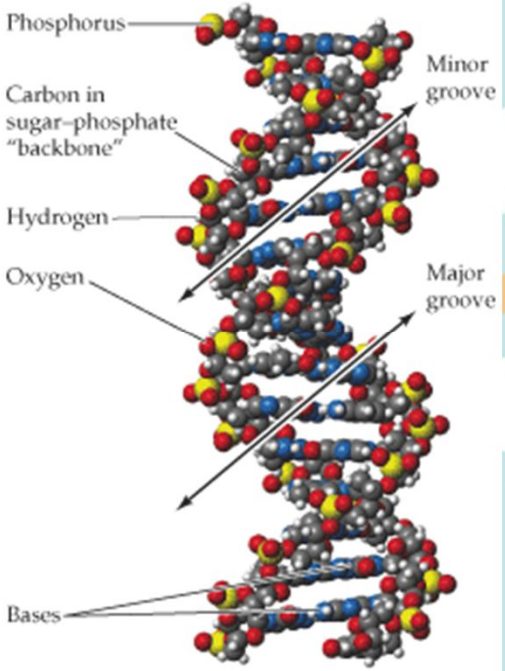
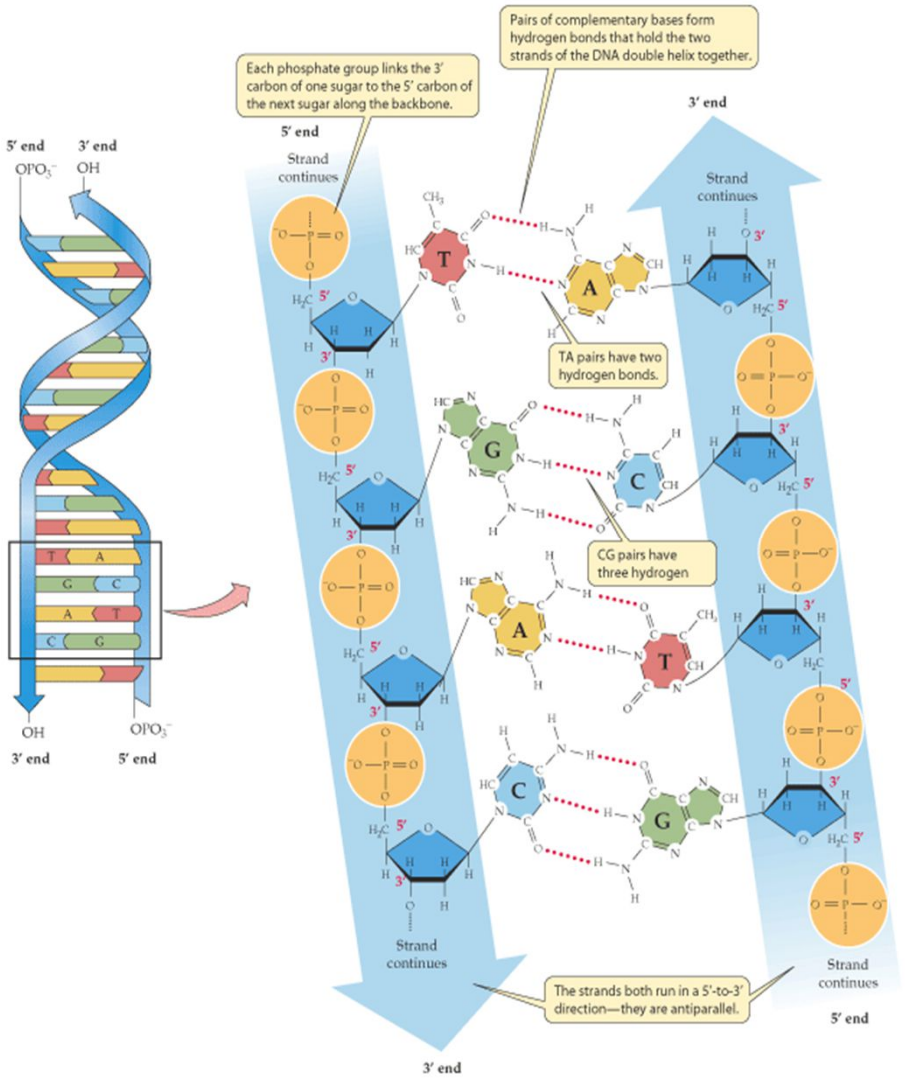




DNA Function From Structure



Double stranded helix

Right handed (threads on a screw)

Antiparellel

Complementary base pairing

A=T

G=C

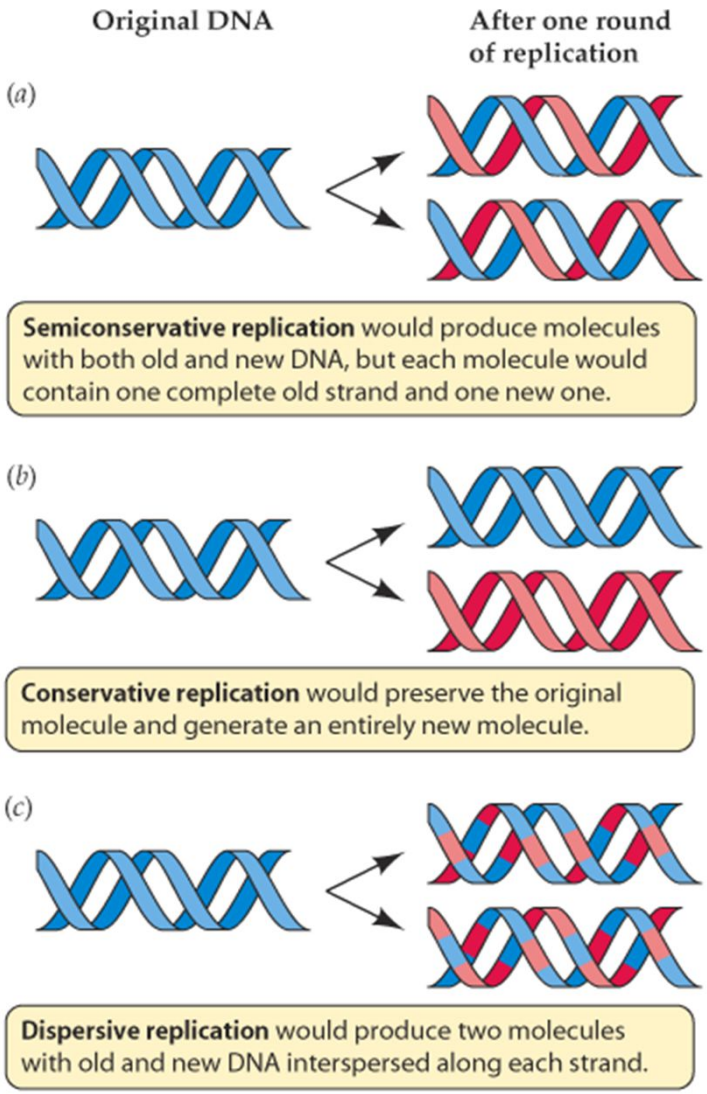
Store an enormous amount of information and could account for species and individual differences

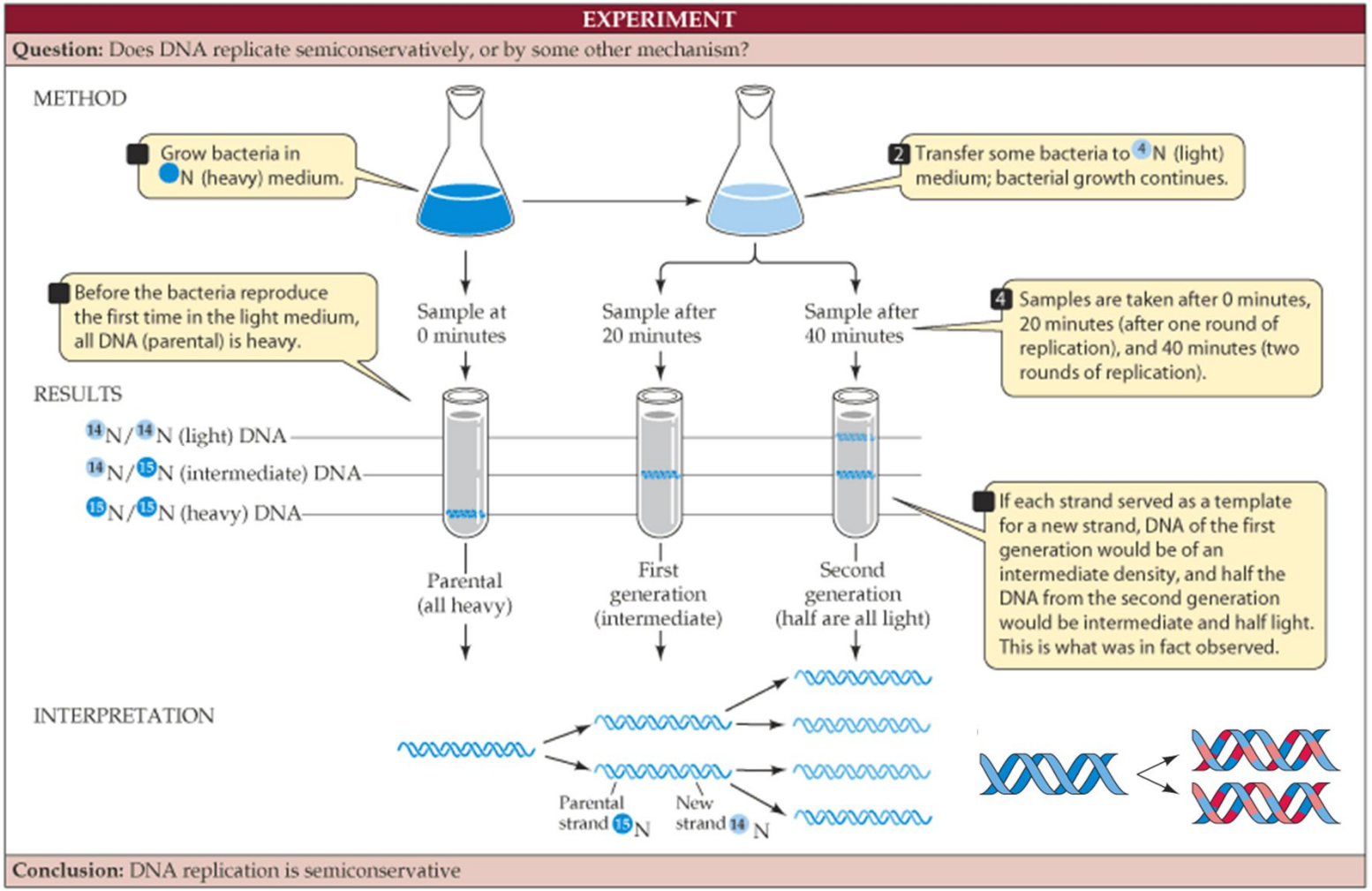
Susceptibility to change (mutate) Change bp

Precise Replication bp is specific

Expression of a phenotype

Replication Models


Original DNA
After one round of replication
(a)
Semiconservative replication would produce molecules with both old and new DNA, but each molecule would contain one complete old strand and one new one.
(b)
Conservative replication would preserve the original molecule and generate an entirely new molecule.
(c)
Dispersive replication would produce two molecules with old and new DNA interspersed along each strand.


EXPERIMENT
Question: Does DNA replicate semiconservatively, or by some other mechanism?
METHOD
1 Grow bacteria in 15N (heavy) medium.
2 Transfer some bacteria to 14N (light) medium; bacterial growth continues.
3 Before the bacteria reproduce the first time in the light medium, all DNA (parental) is heavy.
Sample at 0 minutes
Sample after 20 minutes
Sample after 40 minutes
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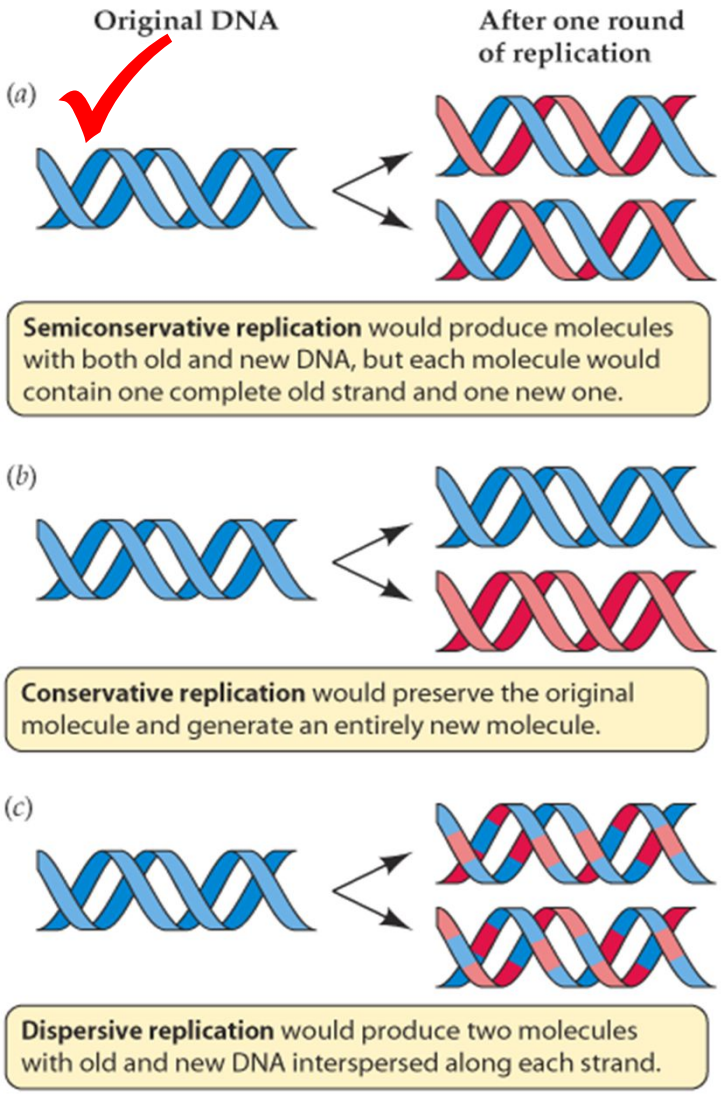
RESULTS
14N/14N (light) DNA
14N/15N (intermediate) DNA
15N/15N (heavy) DNA
Parental (all heavy)
First generation (intermediate)
Second generation (half are all light)
5 If each strand served as a template for a new strand, DNA of the first generation would be of an intermediate density, and half the DNA from the second generation would be intermediate and half light. This is what was in fact observed.
INTERPRETATION
Parental strand 15N
New strand 14N
Conclusion: DNA replication is semiconservative


Original DNA
After one round of replication
(a)
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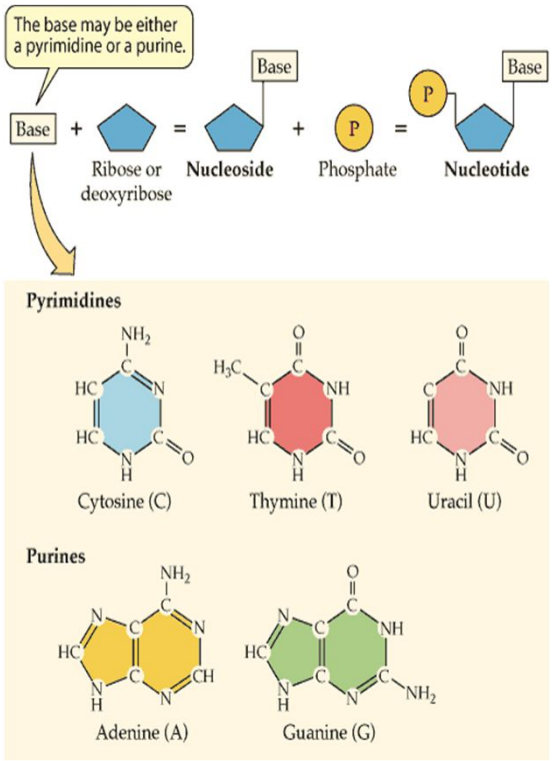
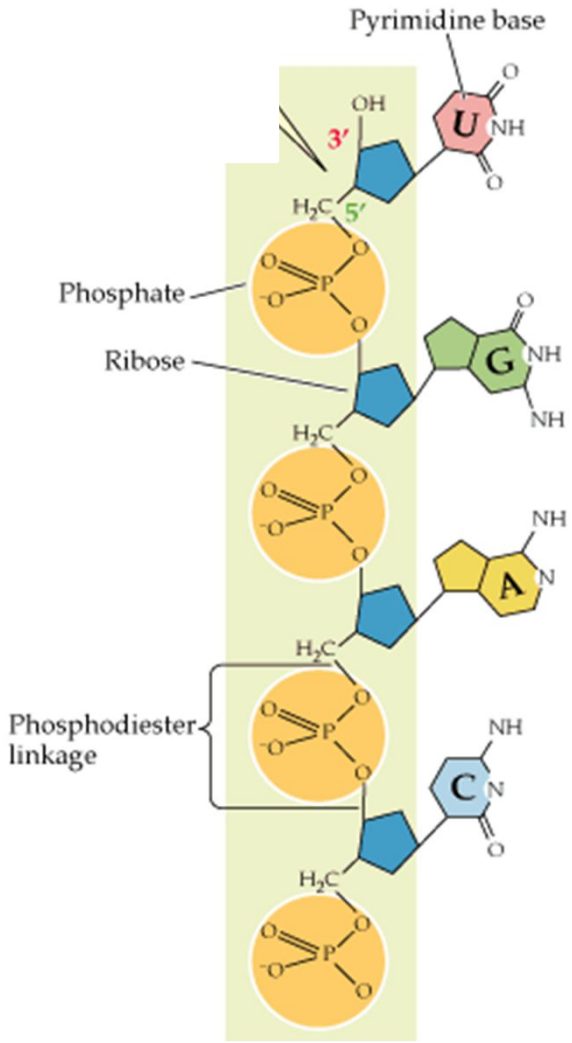
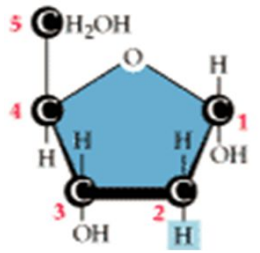
(b)
Conservative replication would preserve the original molecule and generate an entirely new molecule.
(c)
Dispersive replication would produce two molecules with old and new DNA interspersed along each strand.

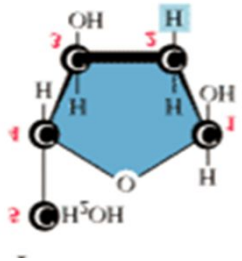
END

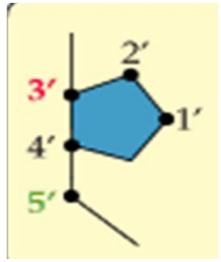
DNA Replication The Molecular View

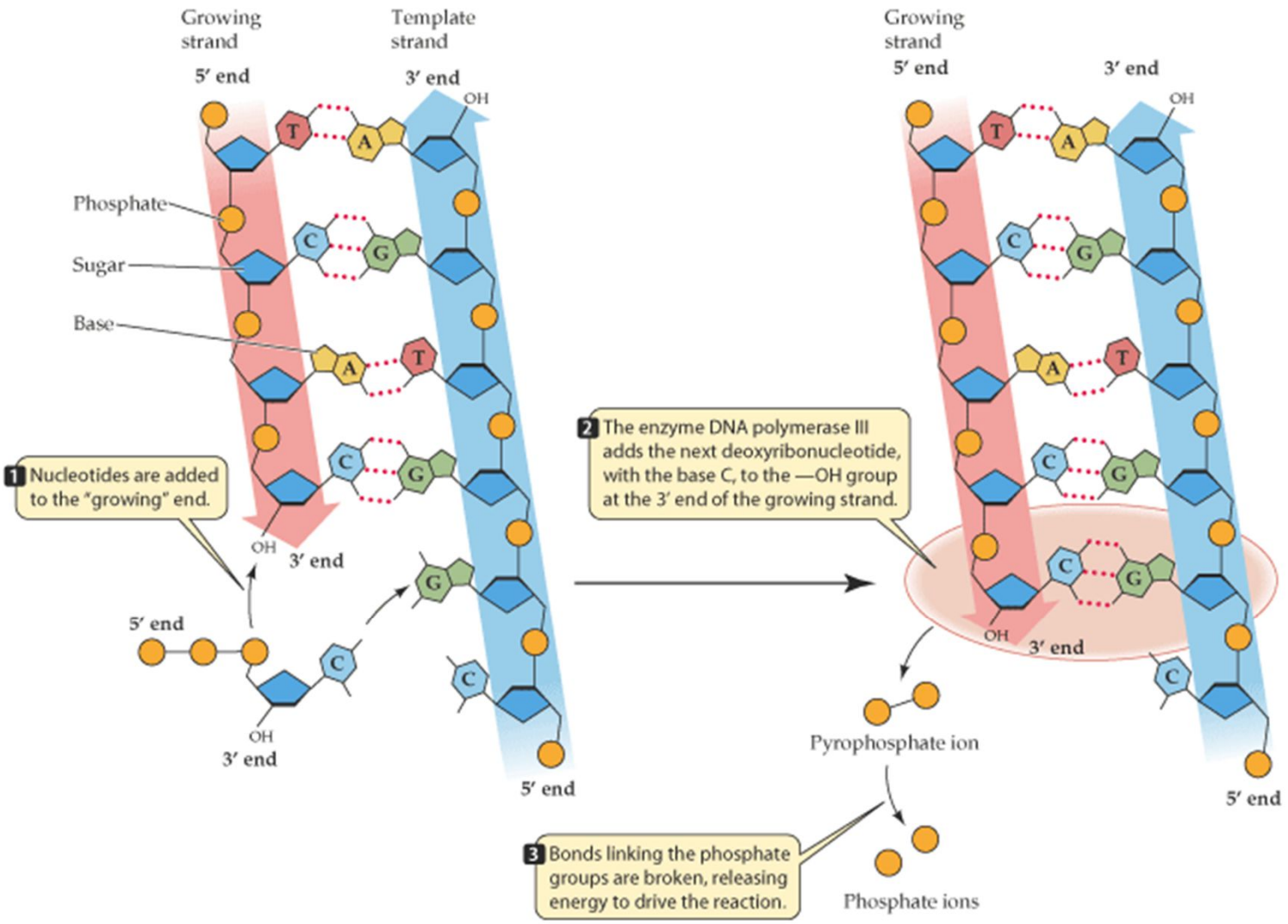

The base may be either a pyrimidine or a purine.
Base
+
Ribose or deoxyribose
=
Base
Nucleoside
+
P
Phosphate
=
P
Base
Nucleotide
Pyrimidines
Cytosine (C)
Thymine (T)
Uracil (U)
Purines
Adenine (A)
Guanine (G)


Pyrimidine base
Phosphate
Ribose
Phosphodiester linkage
3′ end
5′ end
U
G
A
C
OH
3′
5′


5
H2OH
O
H
4
1
H
H
H
OH
3
2
OH
H


OH
H
H
H
H
OH
H
O
H
5OH


2′
3′
1′
4′
5′


Growing strand
Template strand
5′ end
3′ end
OH
Phosphate
Sugar
Base
T
A
C
G
1 Nucleotides are added to the "growing" end.
OH
3′ end
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OH
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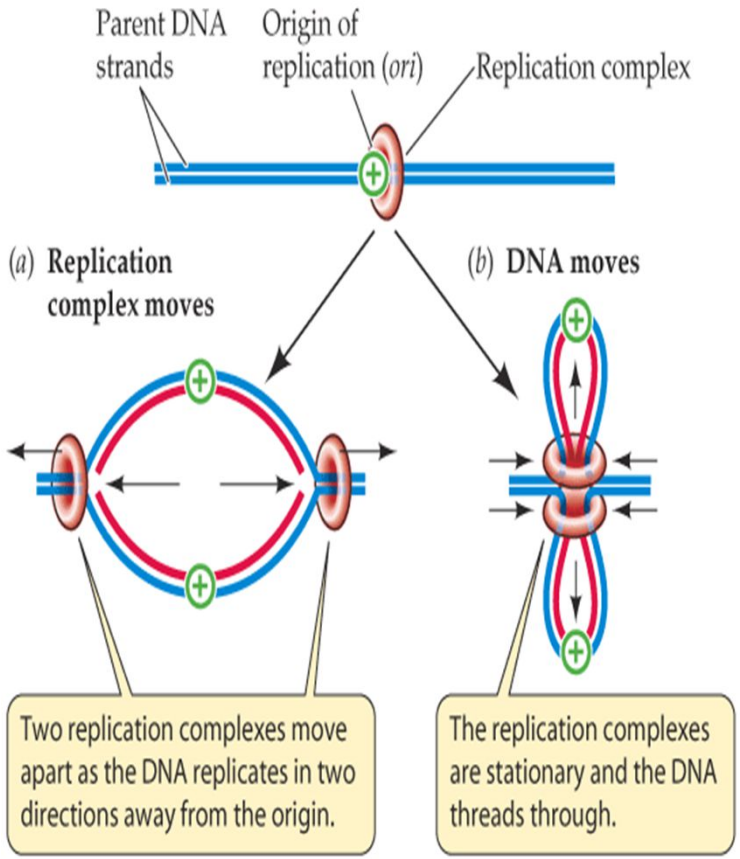
Growing strand
5′ end
3′ end
OH
OH
3′ end
Pyrophosphate ion
5′ end
3 Bonds linking the phosphate groups are broken, releasing energy to drive the reaction.
Phosphate ions


Parent DNA strands
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(a) Replication complex moves
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Two replication complexes move apart as the DNA replicates in two directions away from the origin.
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Requirements

1) The Original

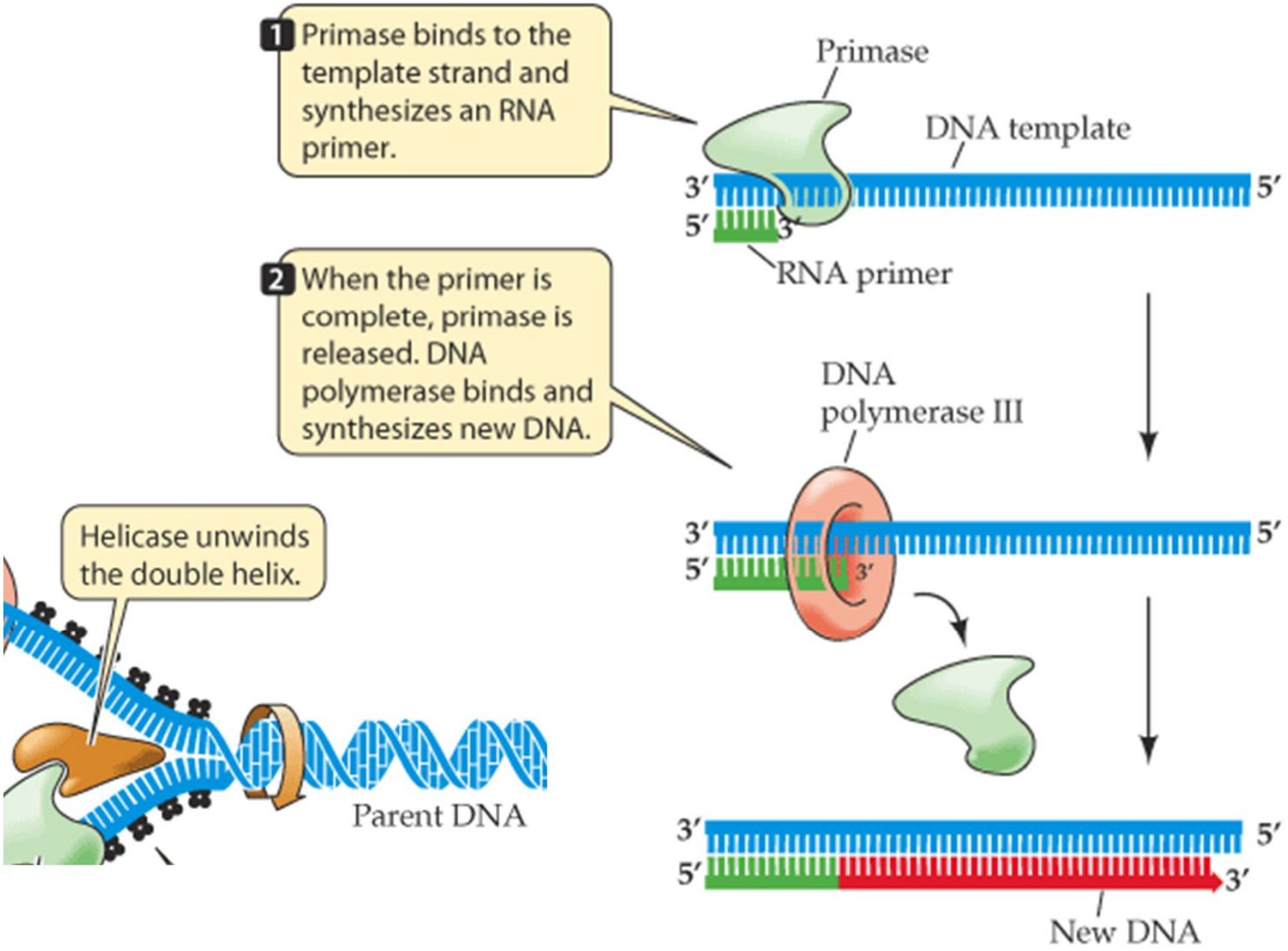
2) The Machinery

3) The Raw Material

5) Energy

4) Initiator

DNA polymerase
useless without primer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Primase
DNA template
3′
5′
5′
3′
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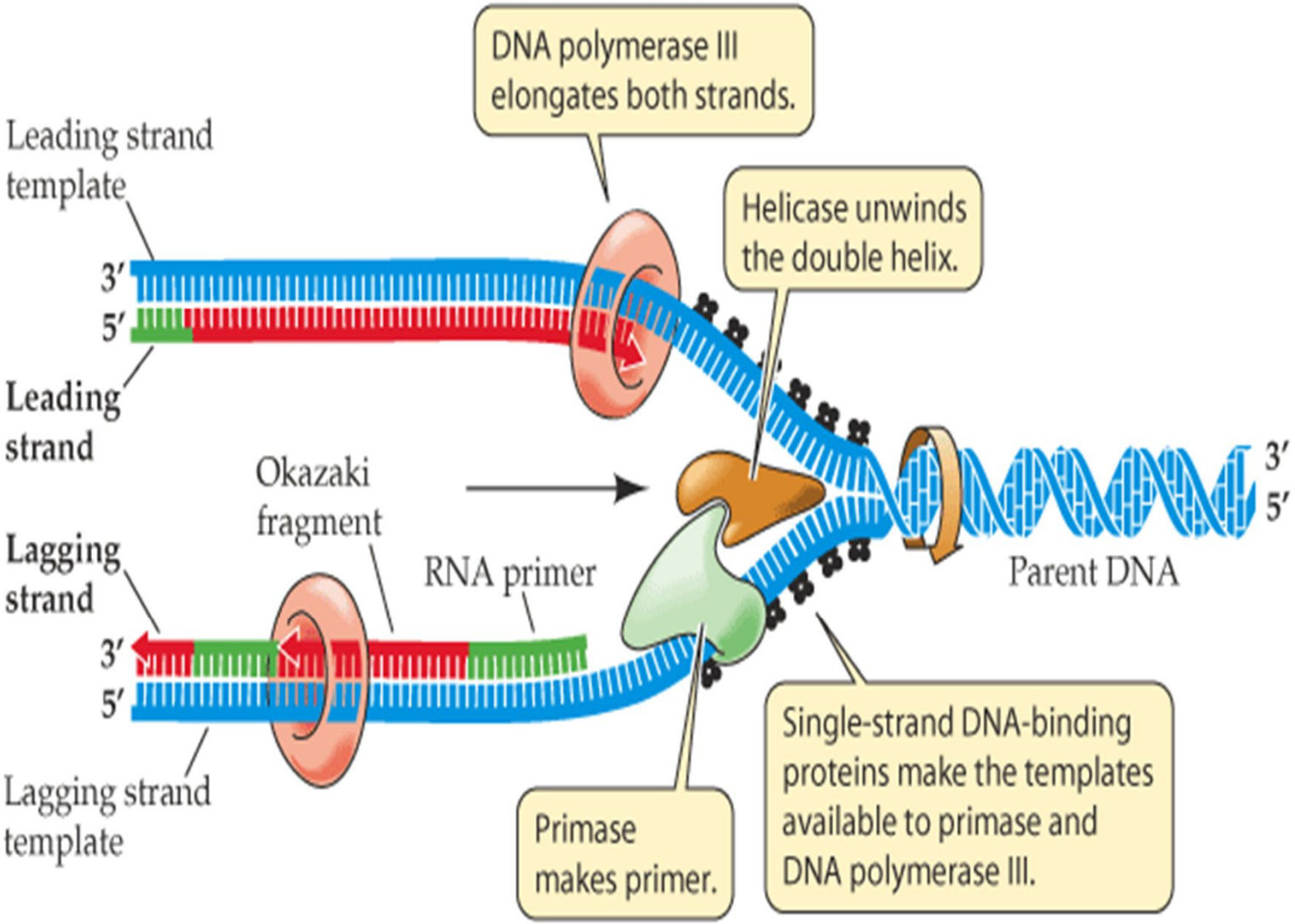
DNA polymerase III
3′
5′
5′
3′
Helicase unwinds the double helix.
Parent DNA
3′
5′
5′
3′
New DNA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Lagging strand
3′
5′
Lagging strand template
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5′
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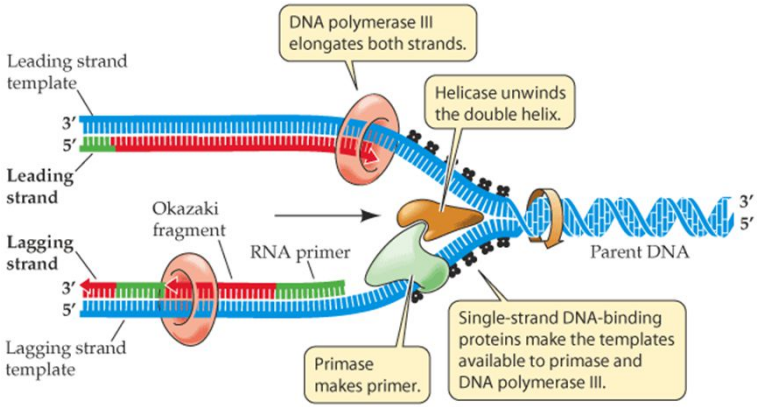
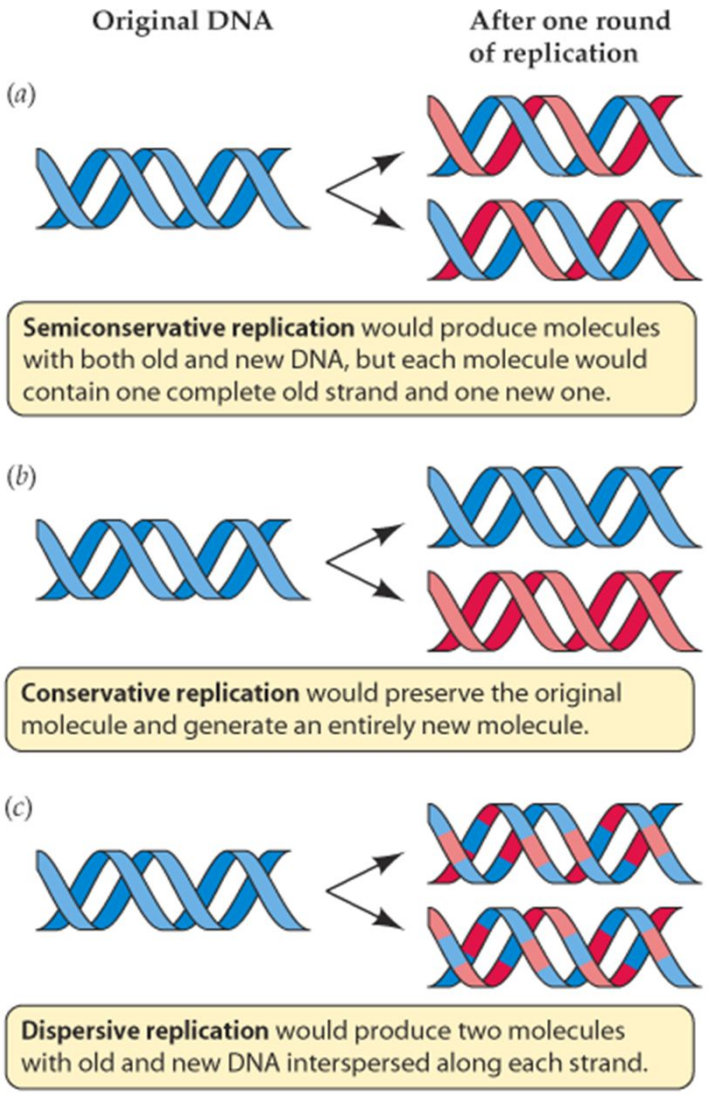
The Three Dilemmas



END

Replication Models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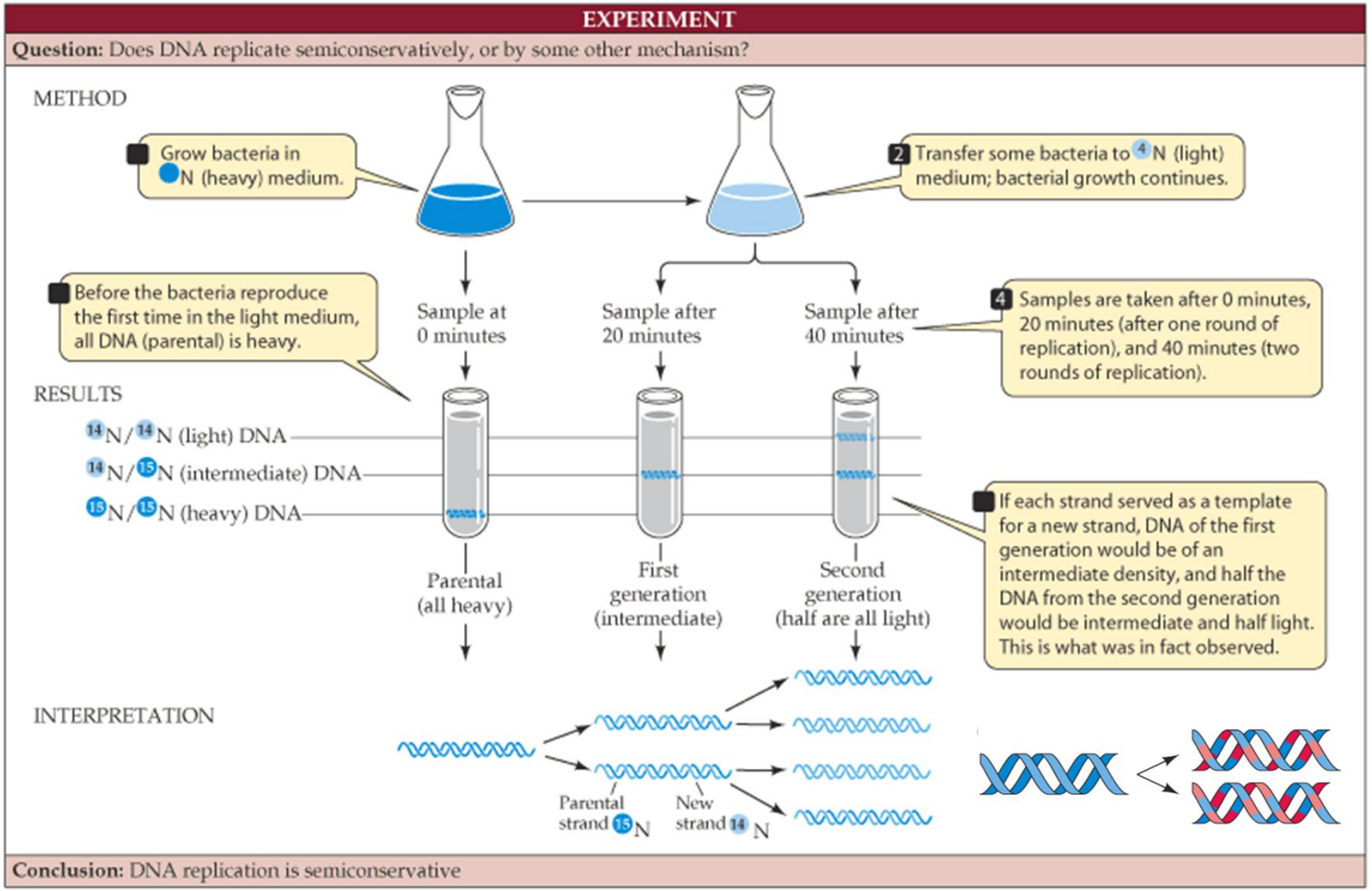
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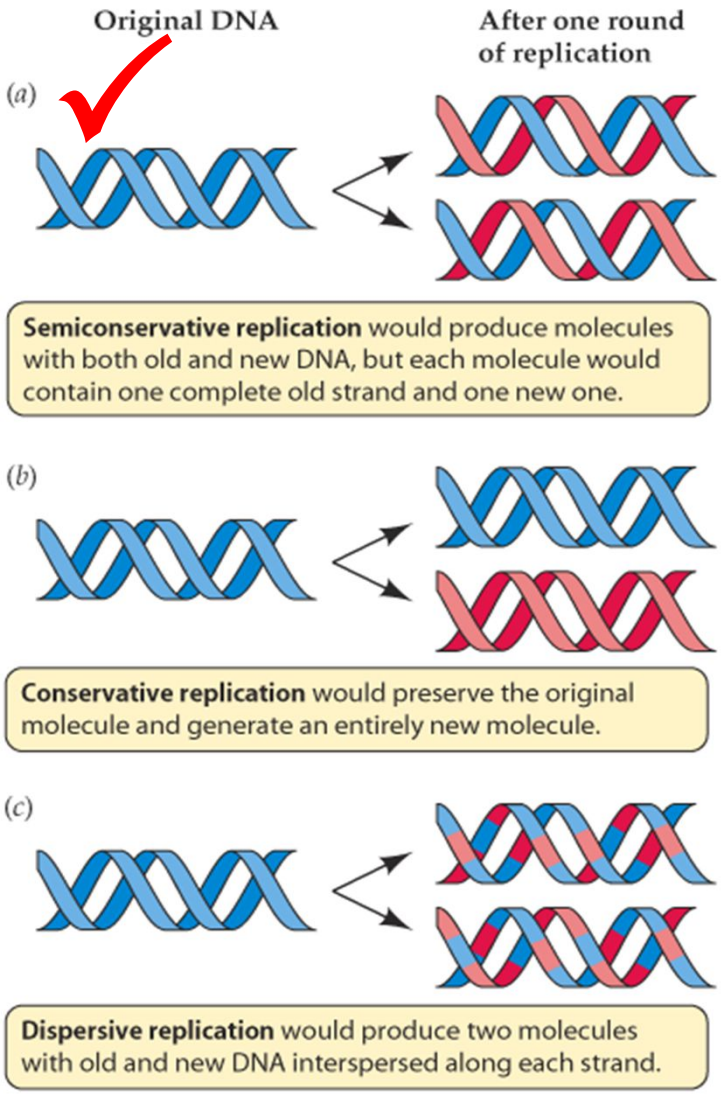
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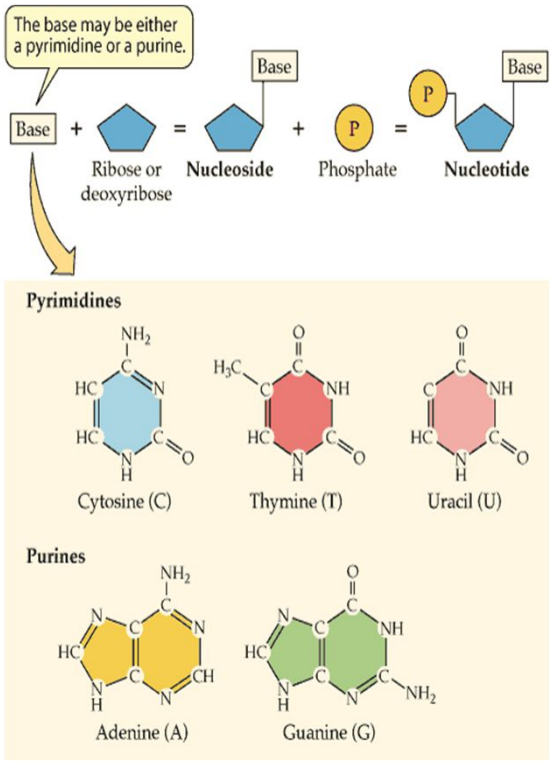
DNA Replication The Molecular View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=
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Nucleoside
+
P
Phosphate
=
P
Base
Nucleotide
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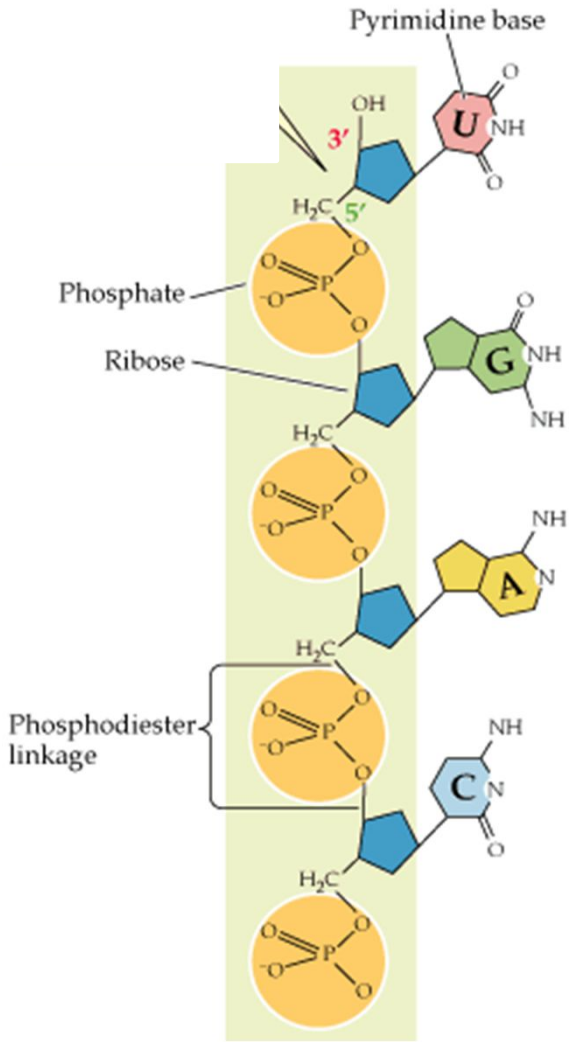
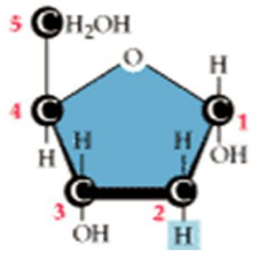
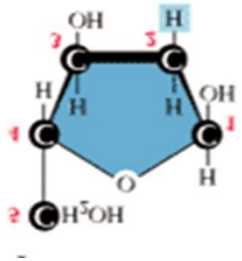
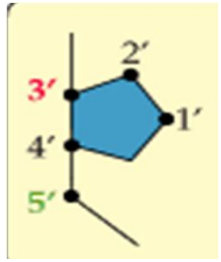
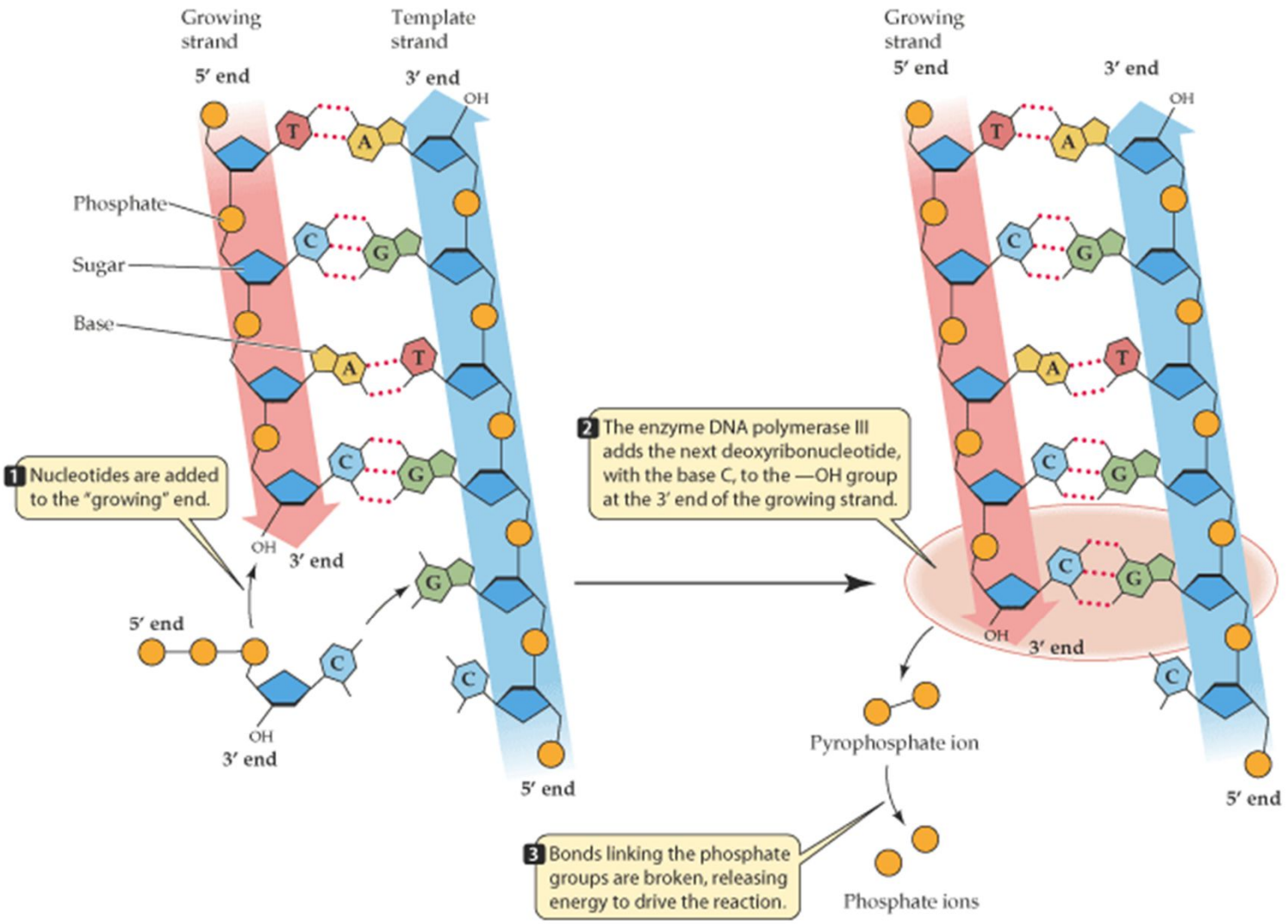
Cytosine (C)
Thymine (T)
Uracil (U)
Purines
Adenine (A)
Guanine (G)


Pyrimidine base
Phosphate
Ribose
Phosphodiester linkage
3′ end
5′ end
U
G
A
C
OH
3′
5′


5
H2OH
O
H
4
1
H
H
H
OH
3
2
OH
H


OH
H
H
H
H
OH
H
O
H
5OH


2′
3′
1′
4′
5′


Growing strand
Template strand
5′ end
3′ end
OH
Phosphate
Sugar
Base
T
A
C
G
1 Nucleotides are added to the "growing" end.
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3′ end
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OH
3′ end
5′ end
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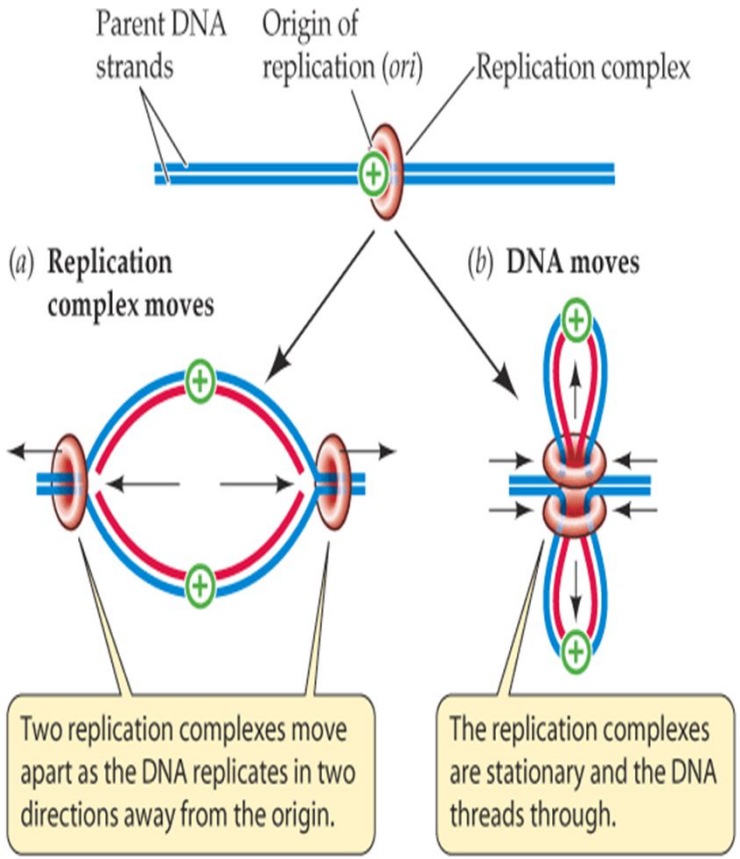
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Replication complex
(a) Replication complex moves
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DNA Replication The Process

Requirements

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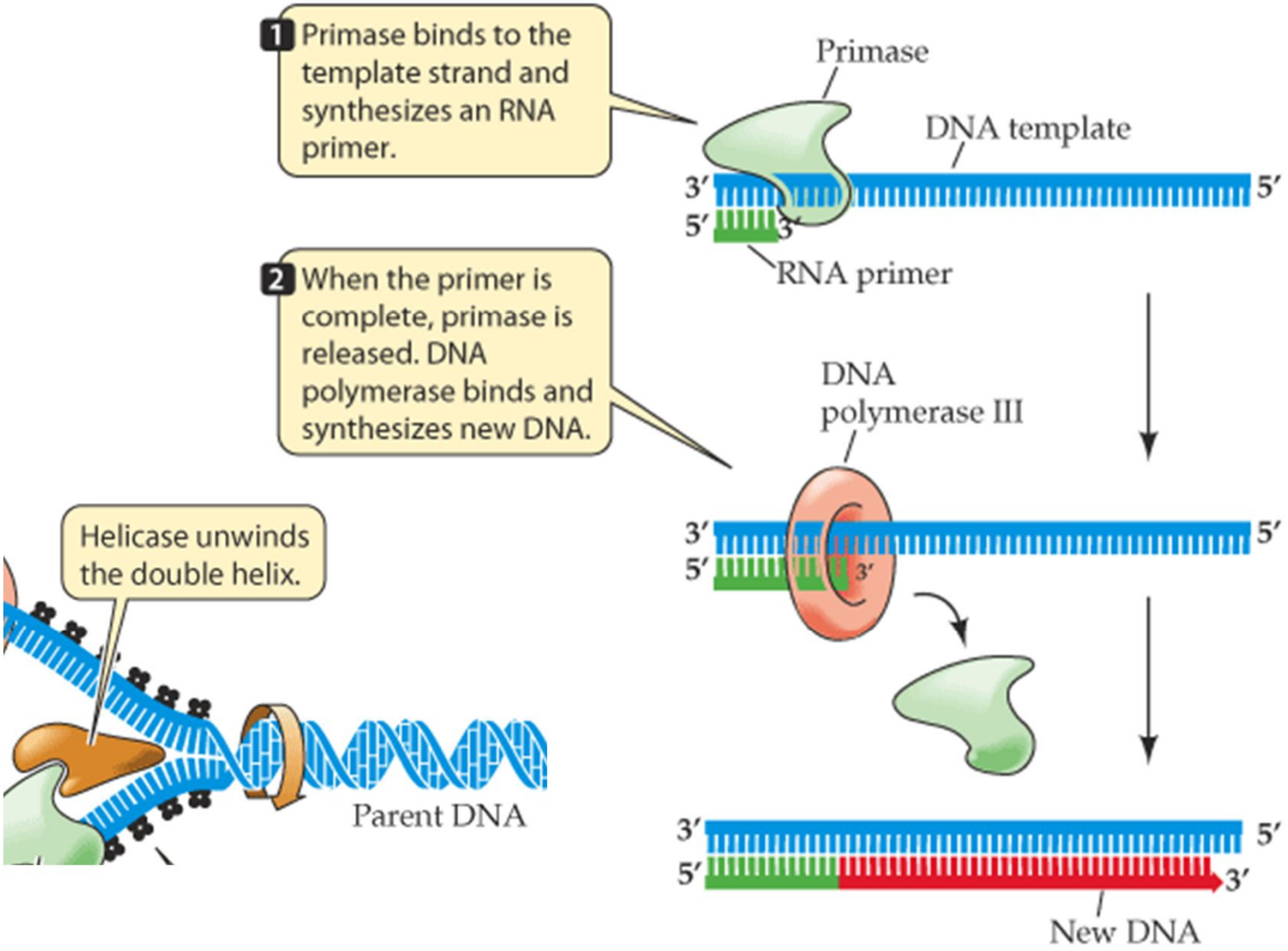
3) The Raw Material

5) Energy

4) Initiator

DNA polymerase
useless without primer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3′
5′
5′
3′
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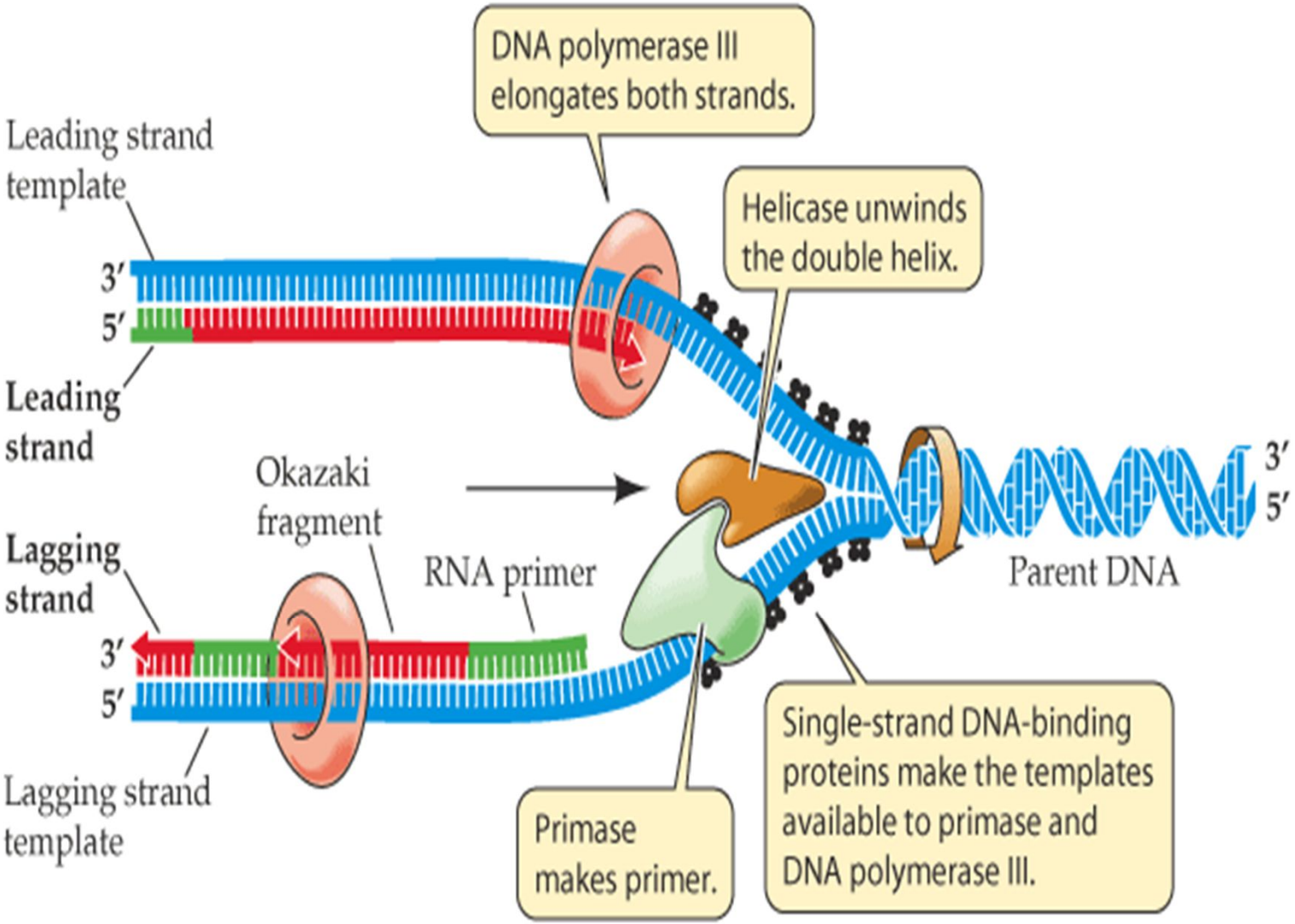
3′
5′
5′
3′
Helicase unwinds the double helix.
Parent DNA
3′
5′
5′
3′
New DNA


DNA polymerase III elongates both strands.
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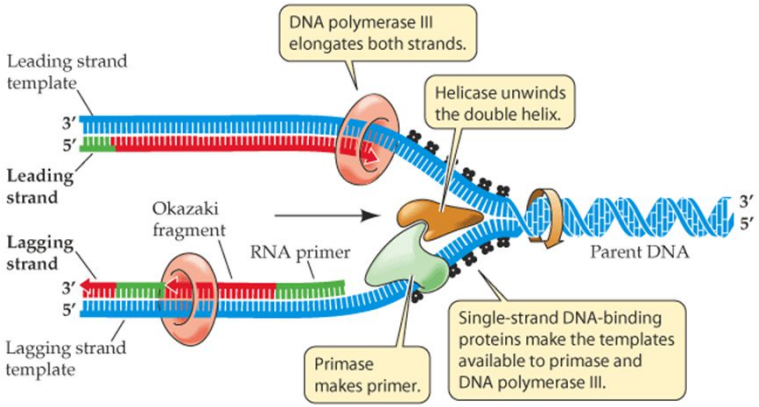
3′
5′
Leading strand
Okazaki fragment
RNA primer
Lagging strand
3′
5′
Lagging strand template
3′
5′
Parent DNA
Primase makes primer.
Single-strand DNA-binding proteins make the templates available to primase and DNA polymerase III.

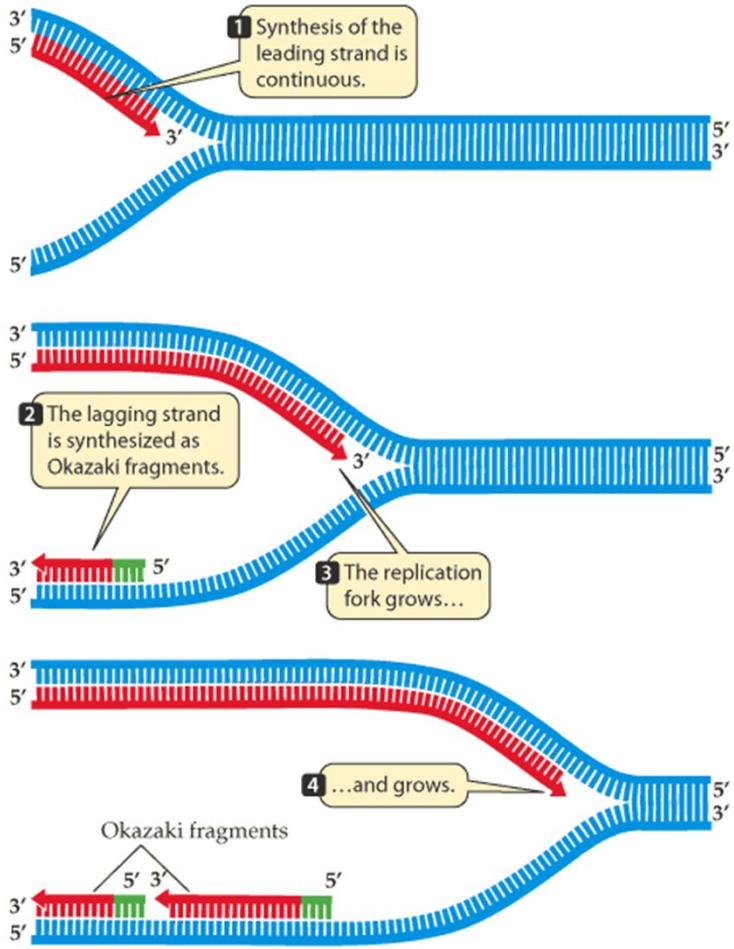
The Three Dilemmas

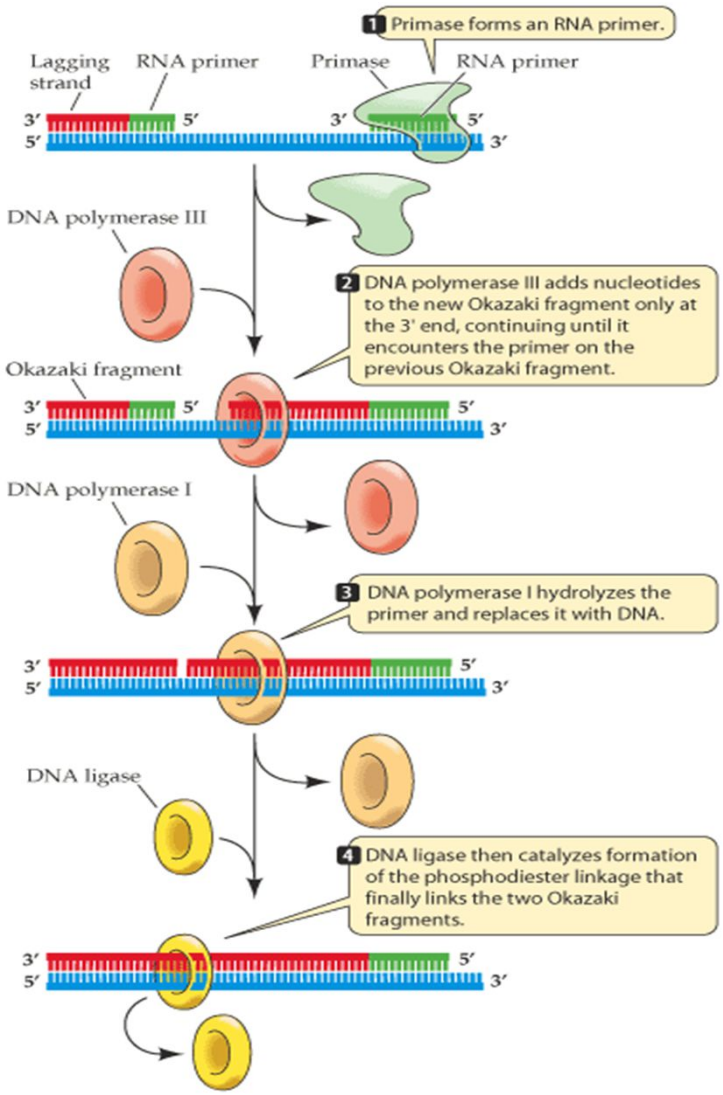


END

The Dilemma 1

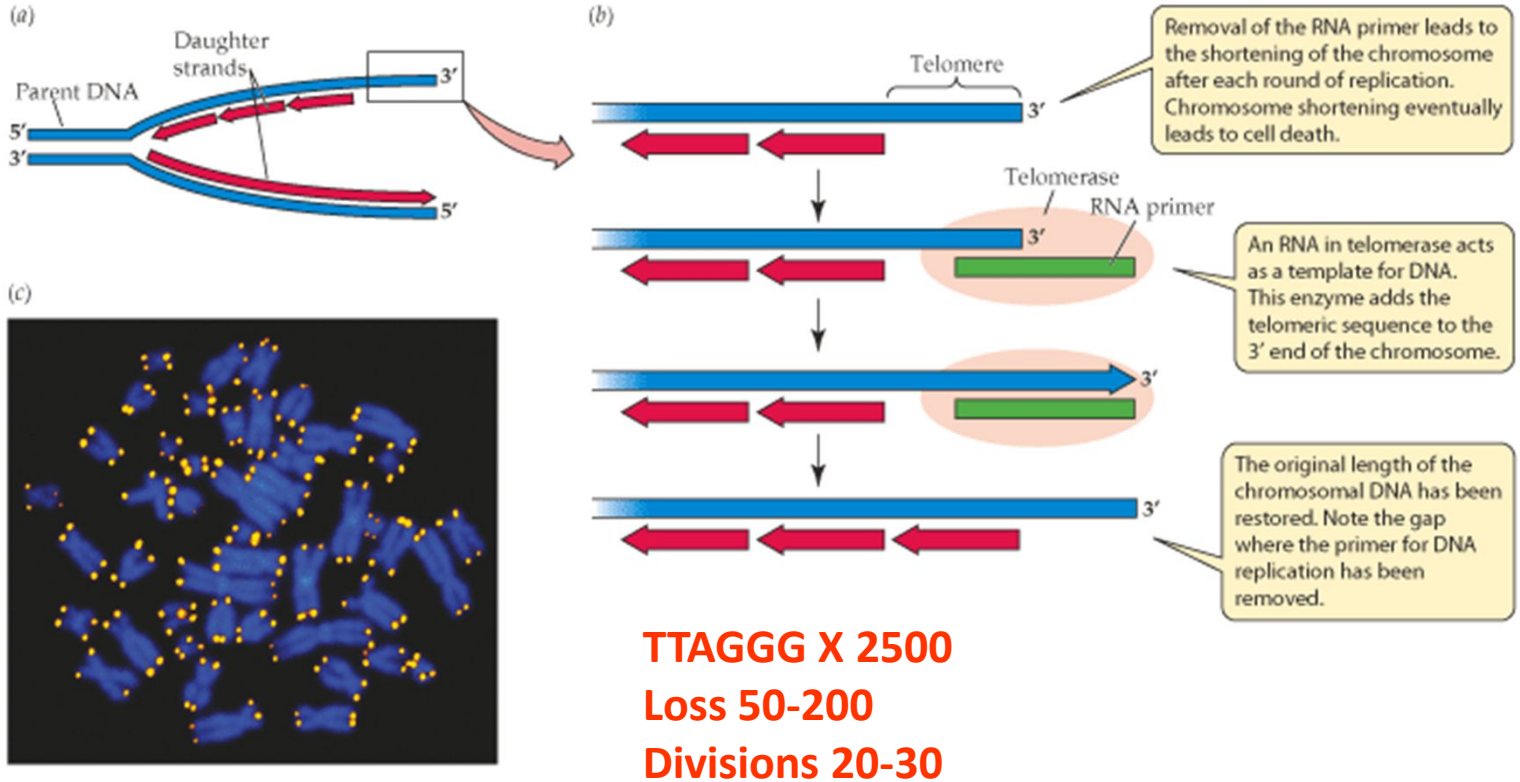
Effort By: AARISH ALI


1 Synthesis of the leading strand is continuous.
2 The lagging strand is synthesized as Okazaki fragments.
3 The replication fork grows…
4 …and grows.
Okazaki fragments
3′
5′


1 Primase forms an RNA primer.
Lagging strand
RNA primer
Primase
RNA primer
3′
5′
DNA polymerase III
2 DNA polymerase III adds nucleotides to the new Okazaki fragment only at the 3′ end, continuing until it encounters the primer on the previous Okazaki fragment.
Okazaki fragment
DNA polymerase I
3 DNA polymerase I hydrolyzes the primer and replaces it with DNA.
DNA ligase
4 DNA ligase then catalyzes formation of the phosphodiester linkage that finally links the two Okazaki fragments.

END

The Dilemma 2 & 3

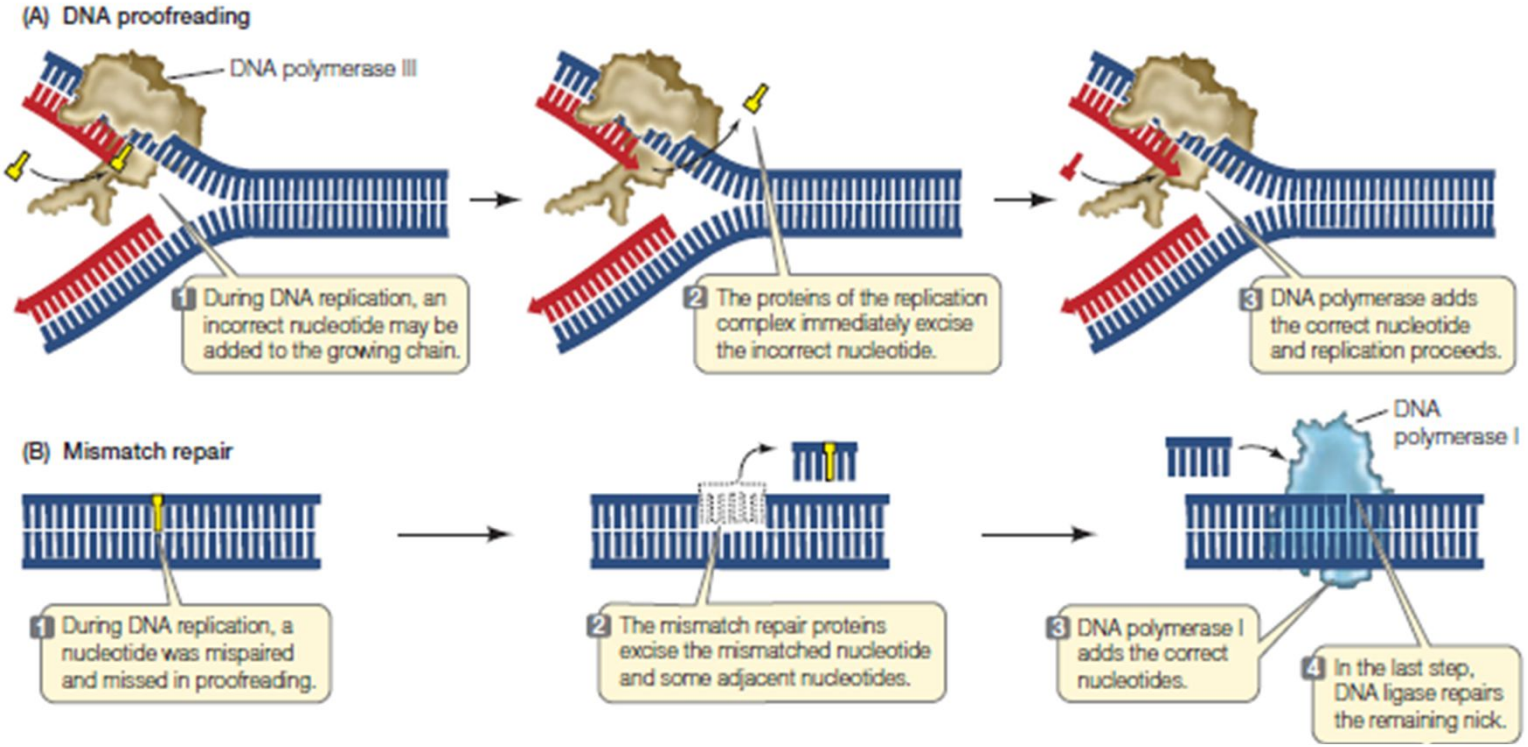


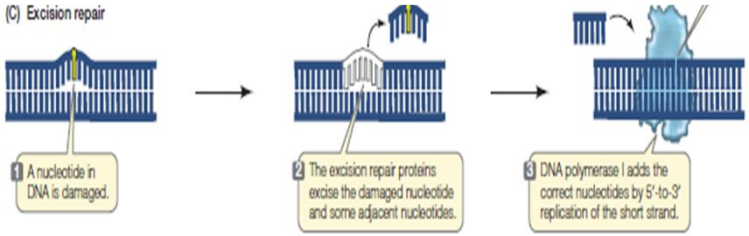
TTAGGG X 2500
Loss 50-200
Divisions 20-30

Error rate for DNA pol 10^6 = 1000 mutations/division

DNA polymerase can recognize mispairing

Efficiency 1 in 10,000 bp Overall error rate = 10^{10}





An excision repair mechanism removes abnormal bases that have formed because of chemical damage and replaces them with functional bases.

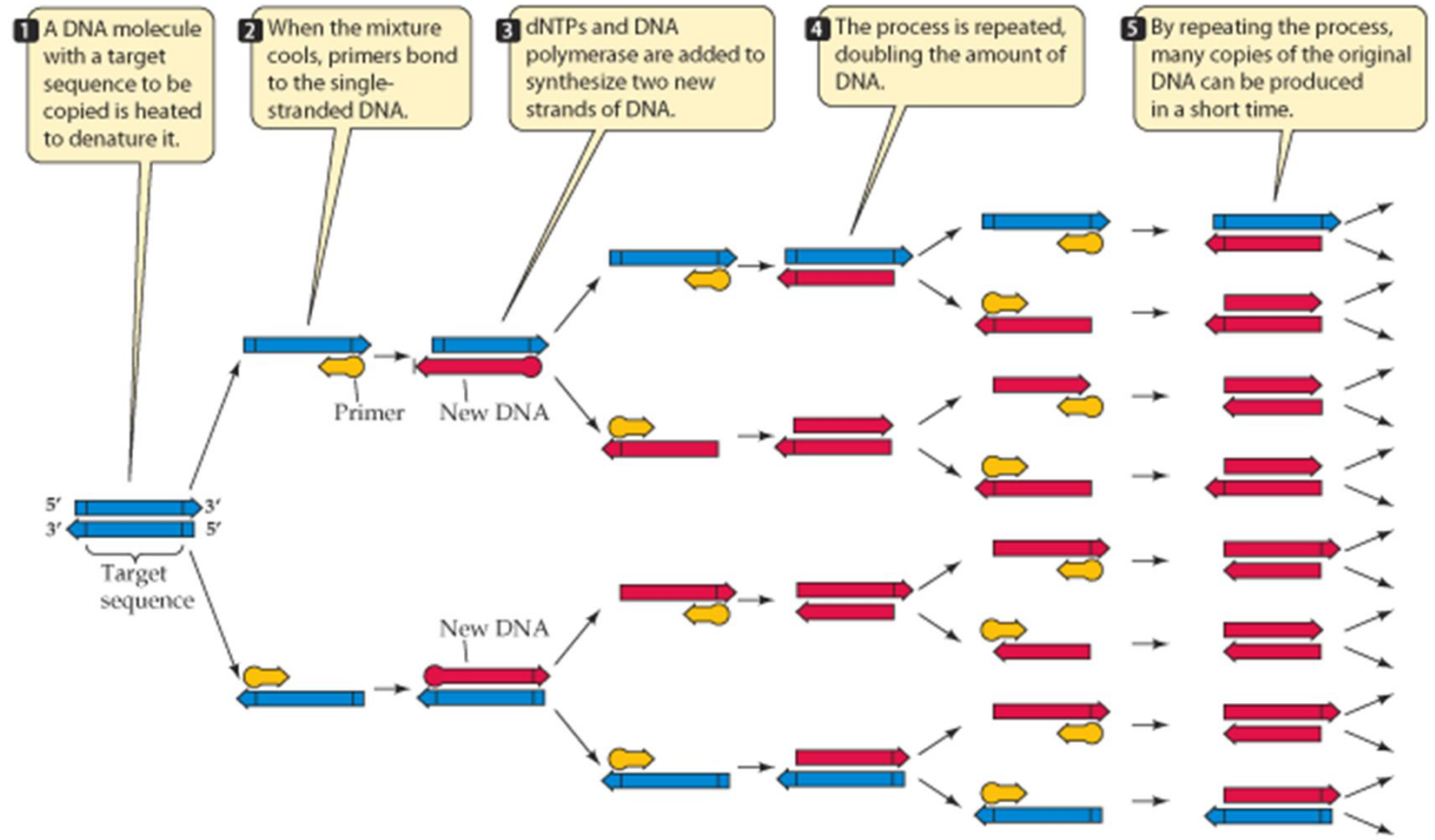
END

The Polymerase Chain Reaction (PCR) Technology

Requirements

1) Target DNA
2) Two oligonucleotide primers
3) DNA polymerase,
4) dNTPs
5) PCR machine

RESEARCH METHOD



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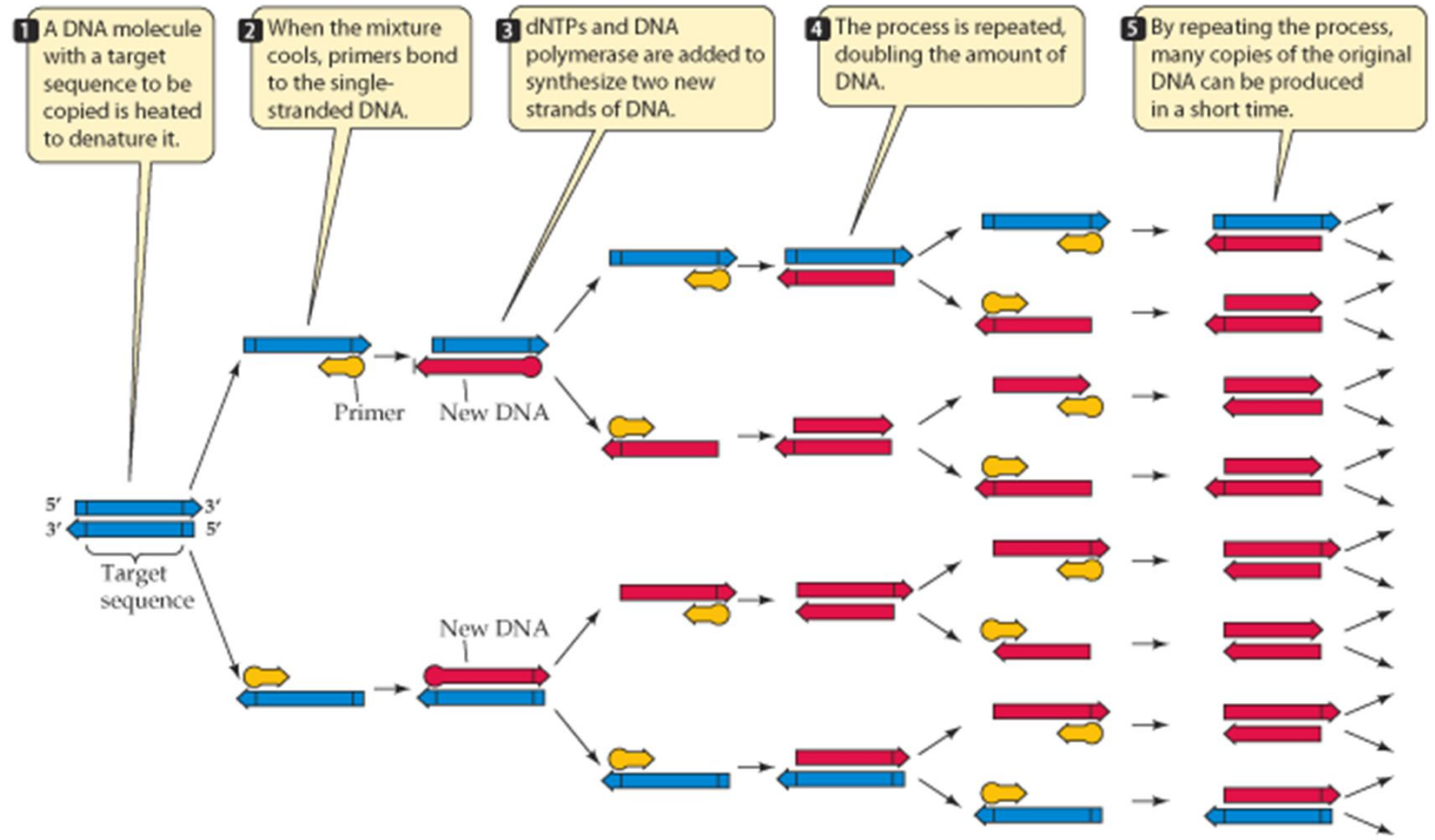

Requirements

1) Target DNA
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END

PCR Animation

PCR Continued

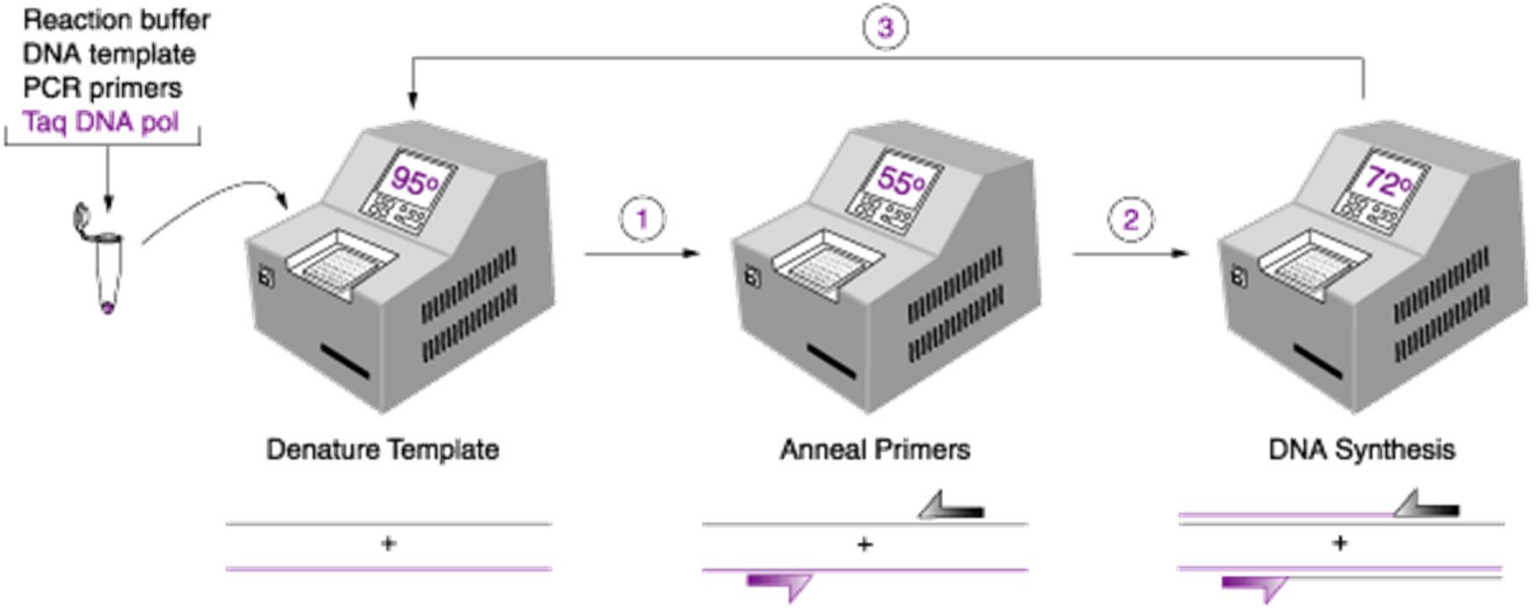
Now

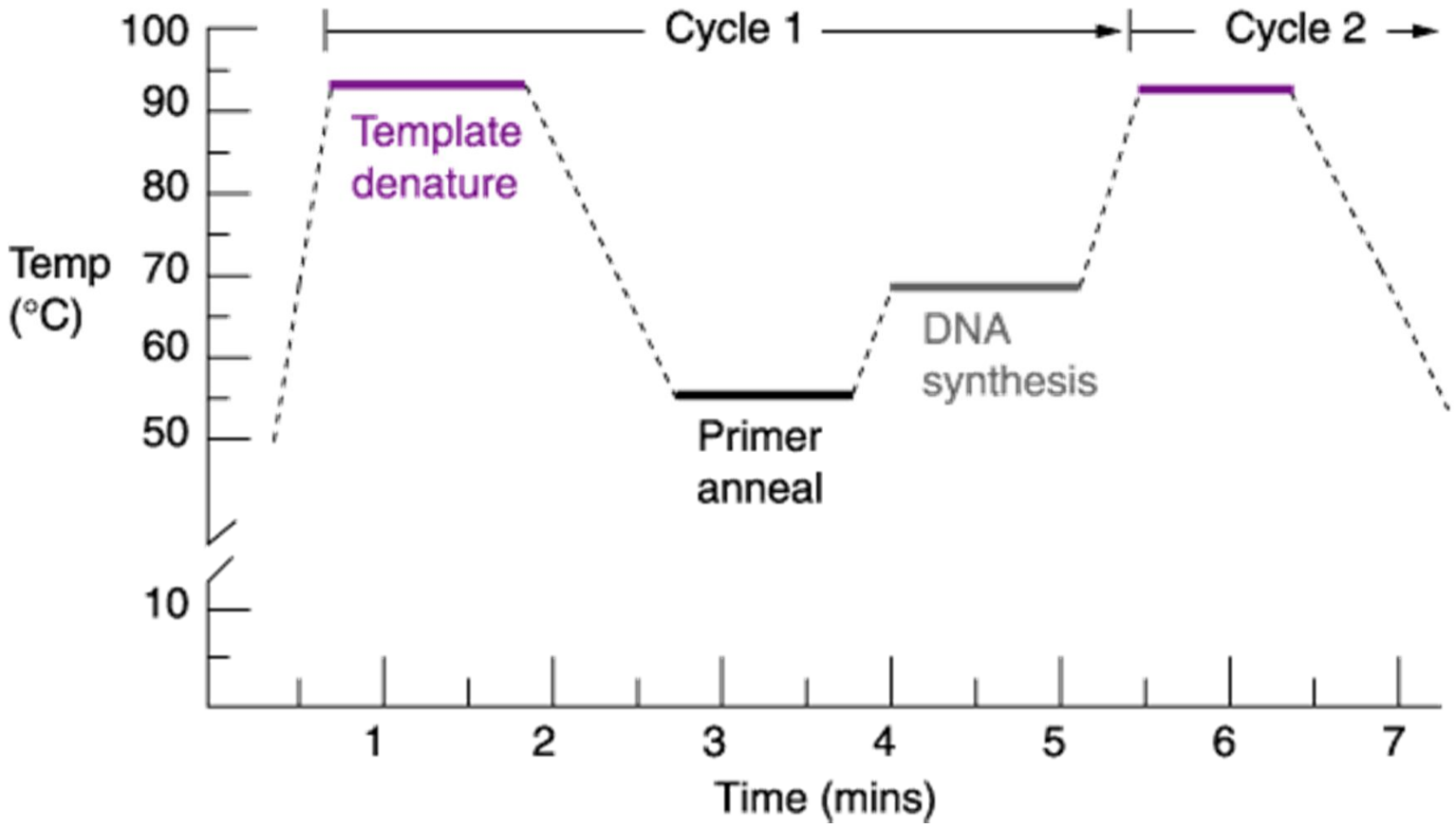
eppendorf 5330 plus
Run
Stop
Caution! Hot Base
eppendorf
Mastercycler 5330
Run
Stop
Prg
Data
Track
Run
Stop
Prg
Data
7 8 9
4 5 6
1 2 3
• 0 Clr

Then

GENE MACHINE
BATH 1
BATH 2
BATH 3
CYCLE CONTROL
INNOVONICS
Please DO NOT leave this machine on overnight.

The Cycles


Reaction buffer
DNA template
PCR primers
Taq DNA pol
95°
55°
72°
1
2
3
Denature Template
Anneal Primers
DNA Synthesis
+
+
+


Cycle 1
Cycle 2
100
90
80
70
60
50
10
Temp (°C)
Template denature
Primer anneal
DNA synthesis
1
2
3
4
5
6
7
Time (mins)

Exponential Amplification

Cycles	Copies
1	2
2	4
4	16
10	1,024
15	32,768
20	1,048,576
25	33,554,432
30	**1,073,741,824**

END

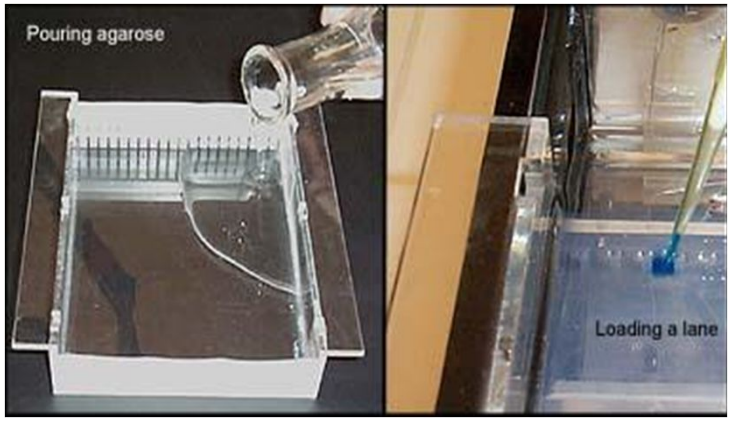
Exponential Amplification

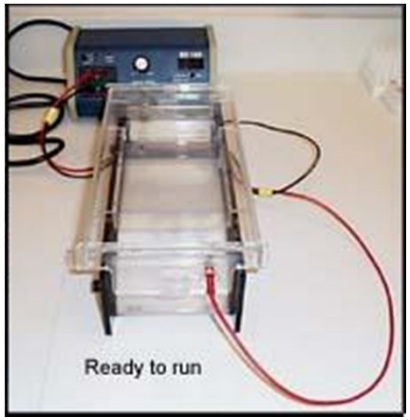
1,073,741,824

END

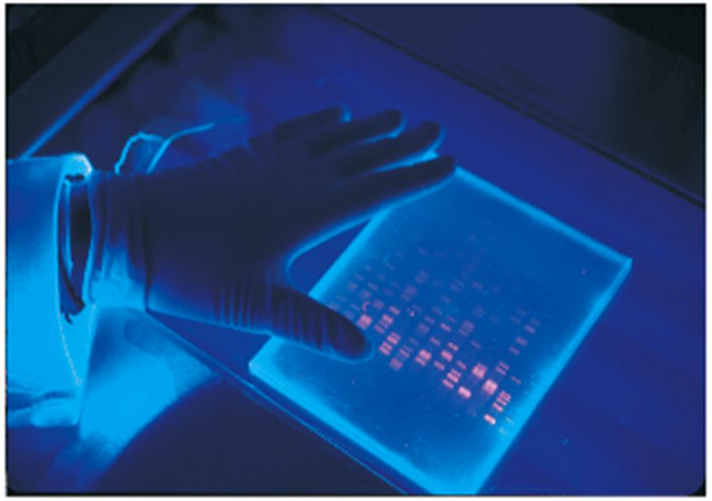
Detecting Amplified DNA

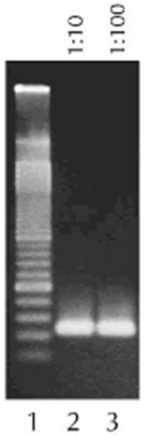
Gel Electrophoresis





UV





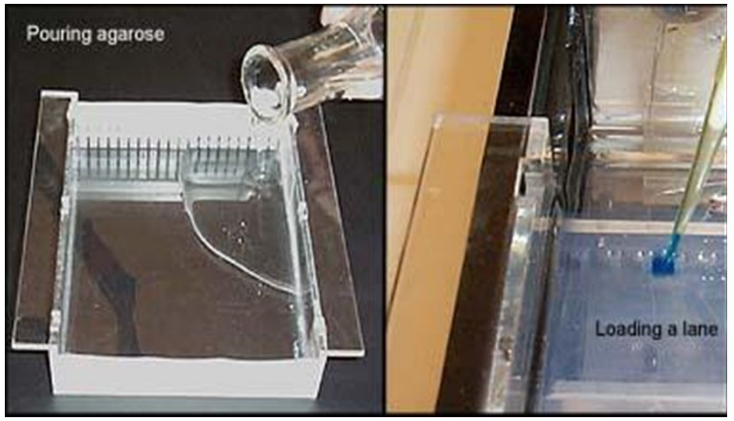
END

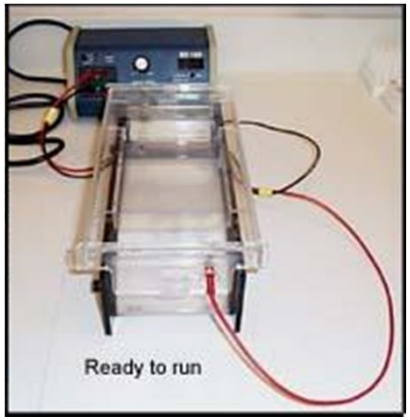
Gel Electrophoresis Animation

Gel Electrophoresis

END

Gel Electrophoresis

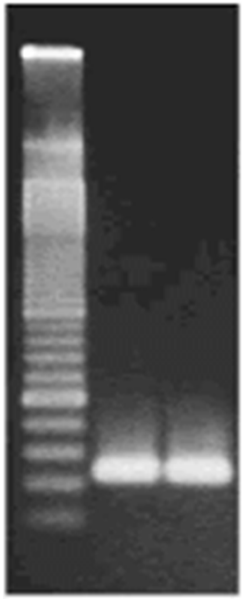




PCR Technology
Applications

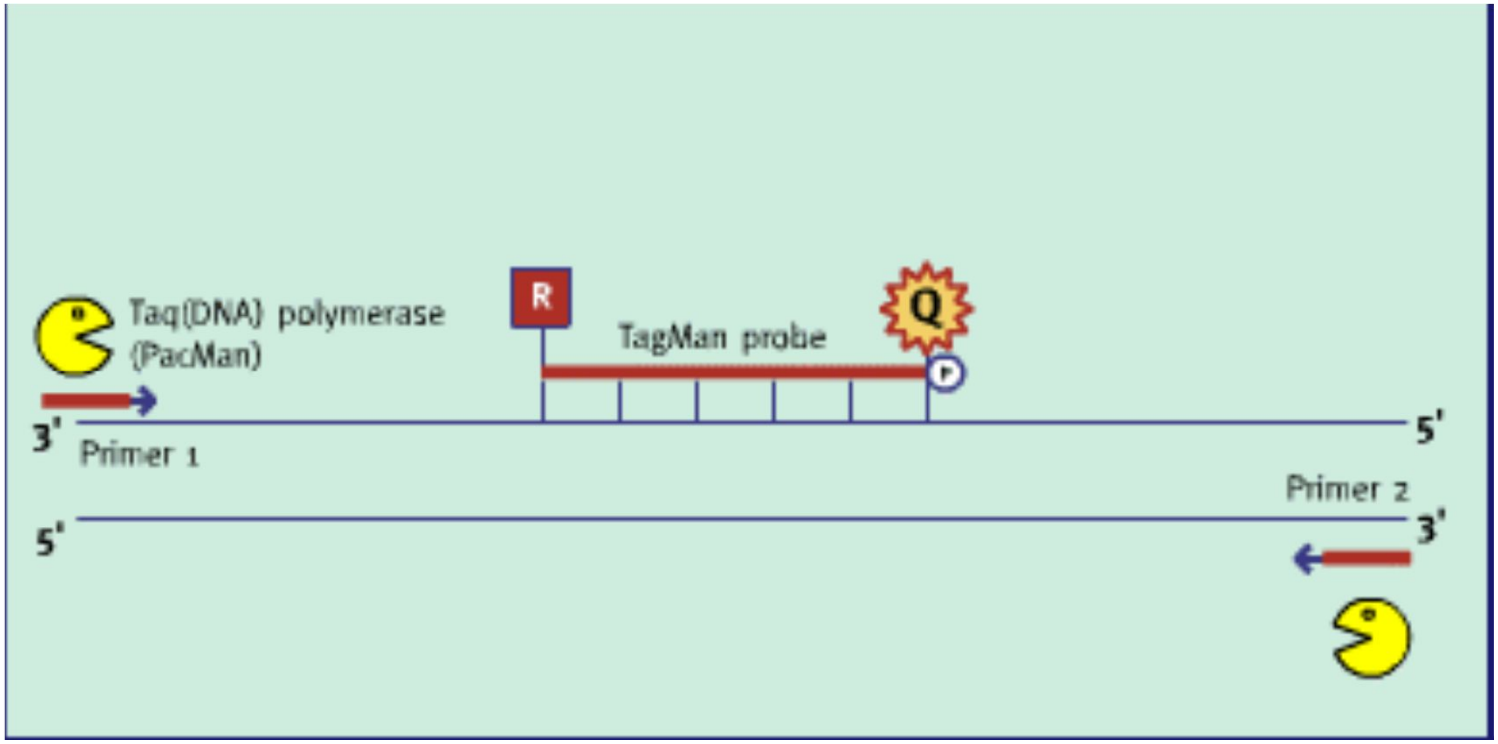
Disease Detection

Many pathogens can be detected from a small sample e.g.

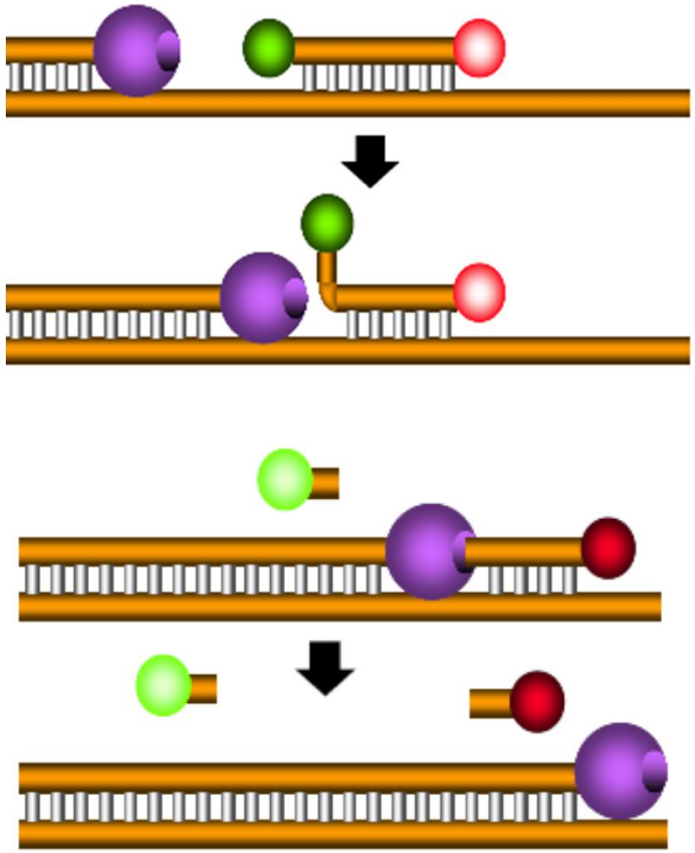


HCV
HBV
HIV
Adenoviruses
Influenza virus
Cytomegalovirus etc

Quantification



Quantification

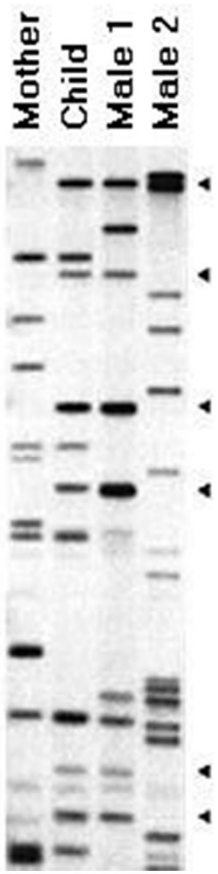


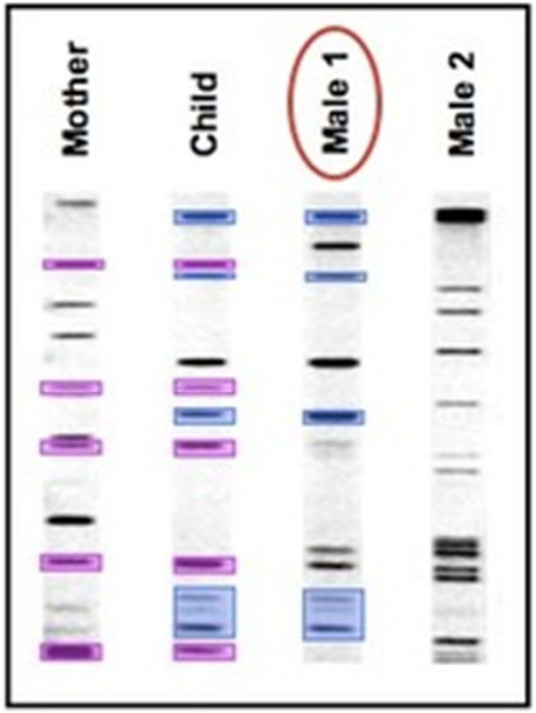
END

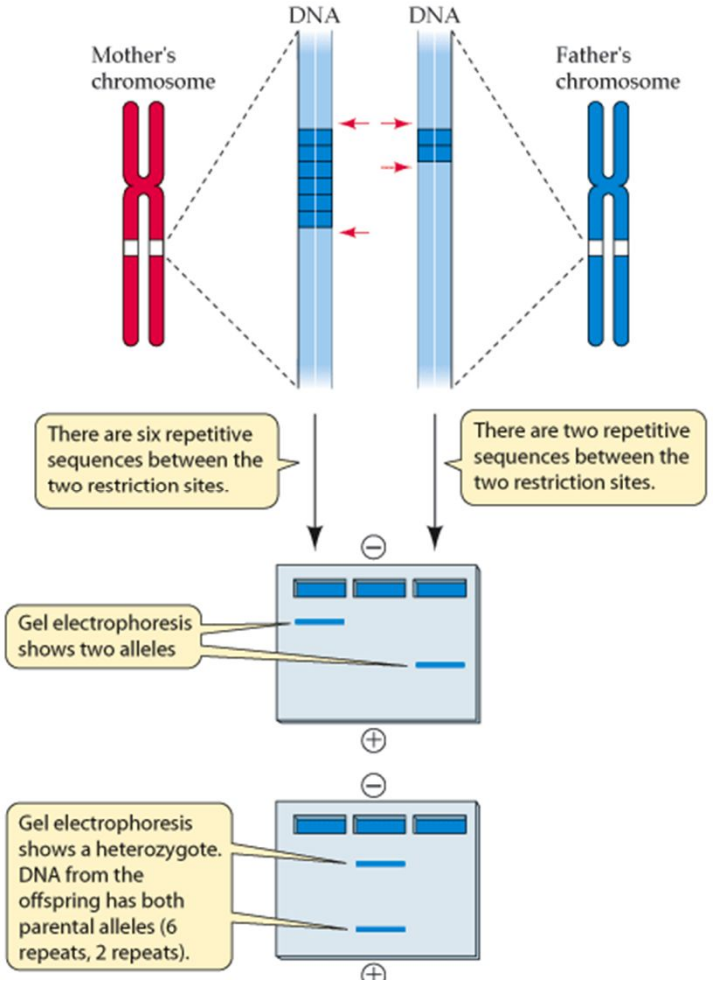
DNA Finger Printing

VNTRs
Variable Number Tandem Repeats

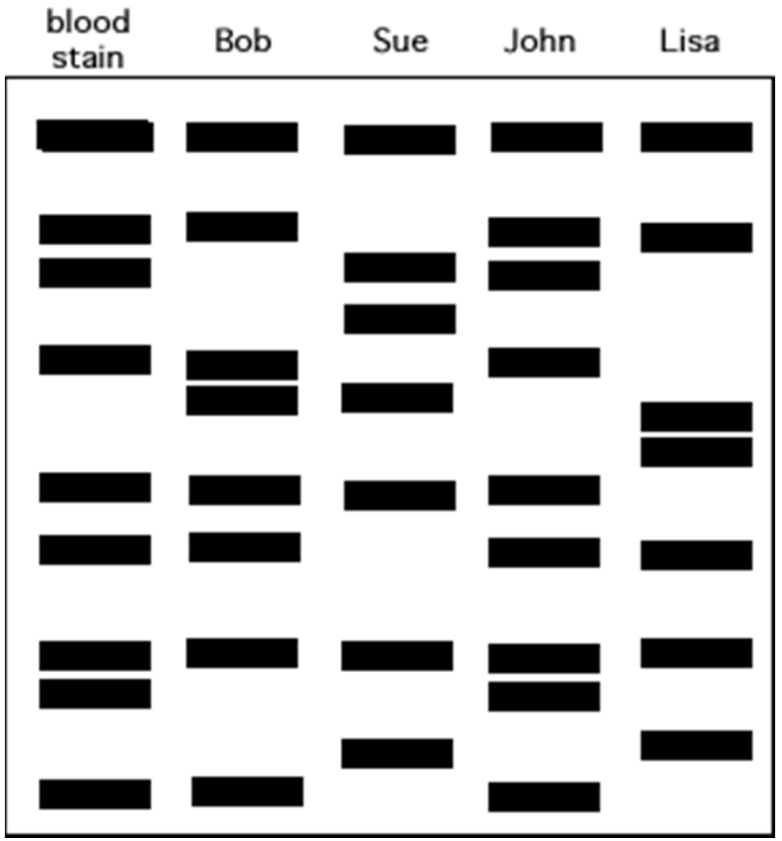
Paternity


Mother
Child
Male 1
Male 2

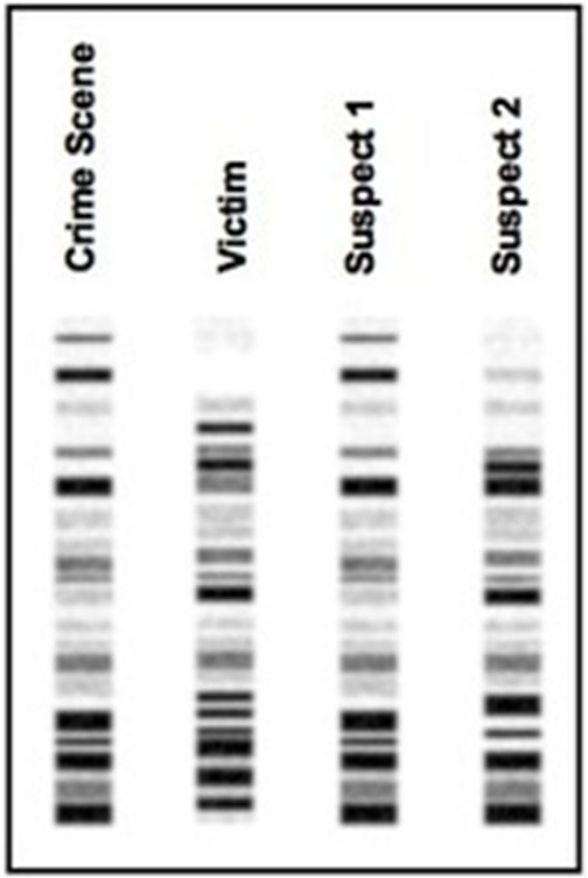

Mother
Child
Male 1
Male 2


DNA
DNA
Mother's chromosome
Father's chromosome
There are six repetitive sequences between the two restriction sites.
There are two repetitive sequences between the two restriction sites.
Gel electrophoresis shows two alleles
Gel electrophoresis shows a heterozygote. DNA from the offspring has both parental alleles (6 repeats, 2 repeats).

Crime Investigation



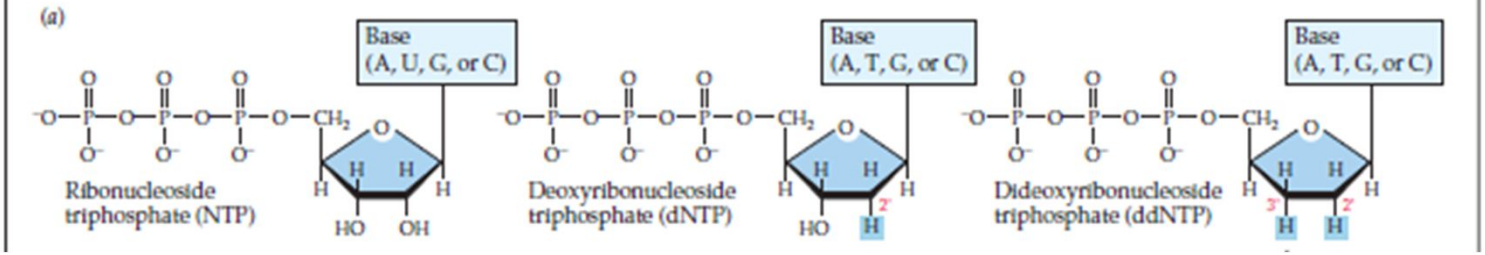
Crime Investigation

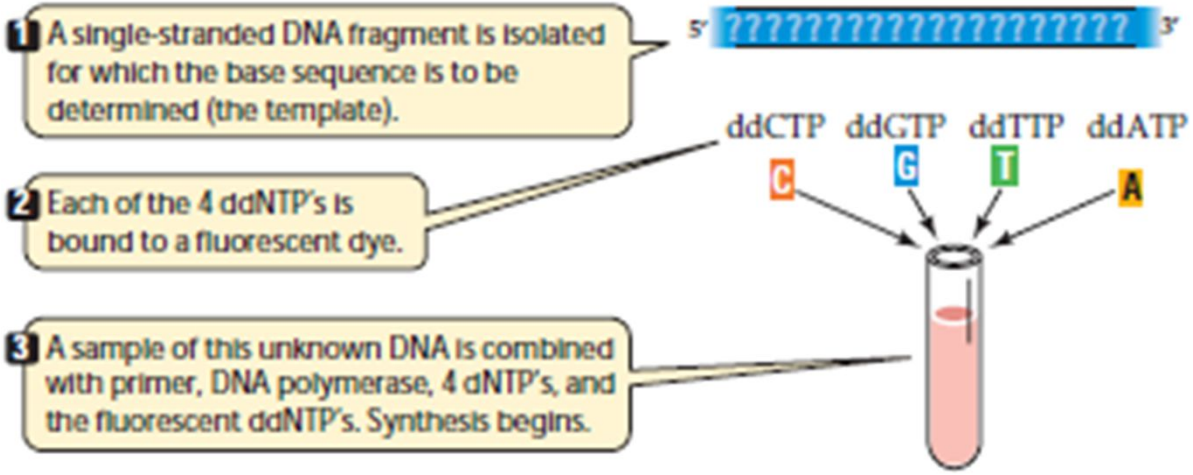


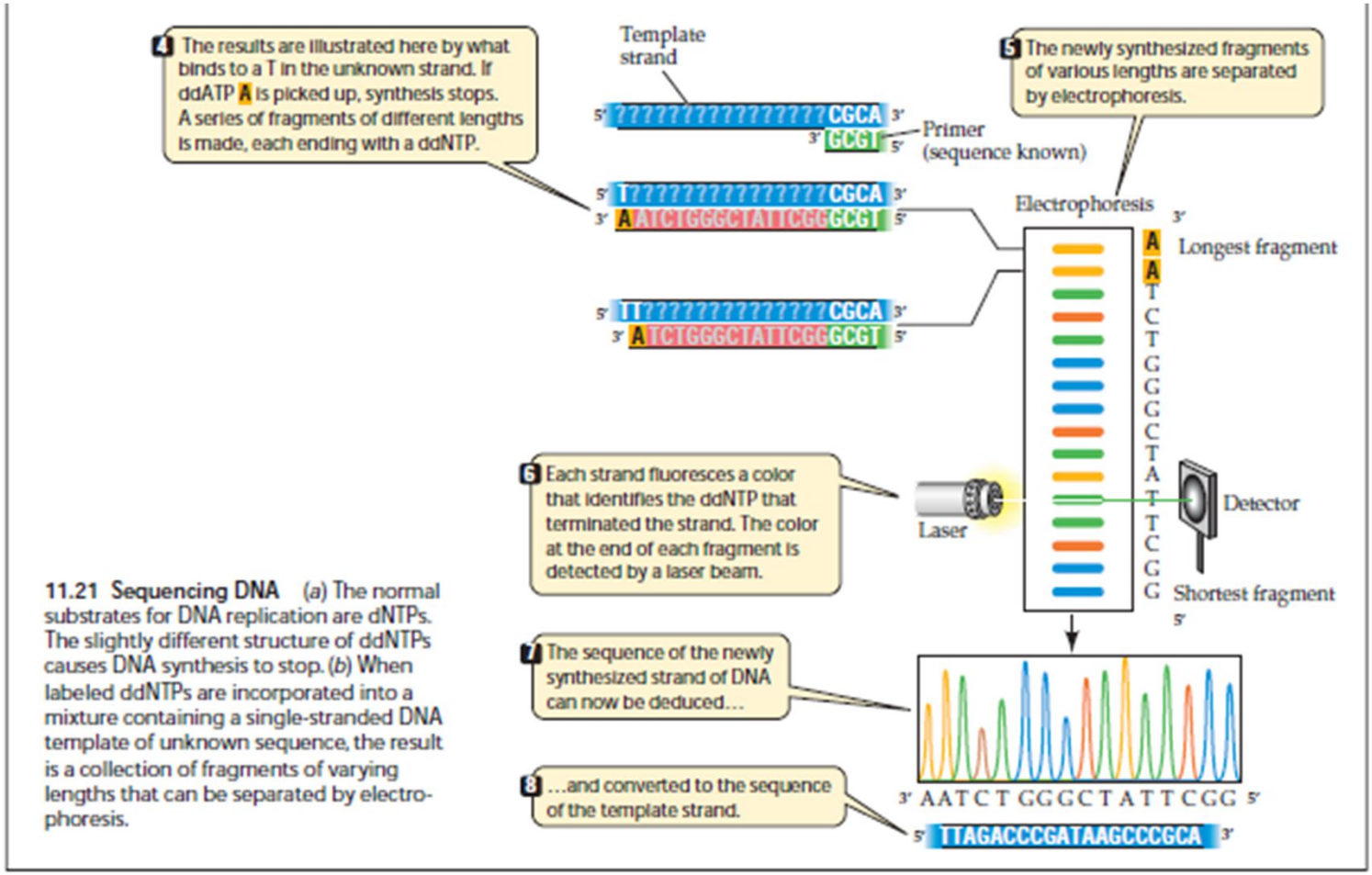
END

DNA Sequencing

RESEARCH METHOD





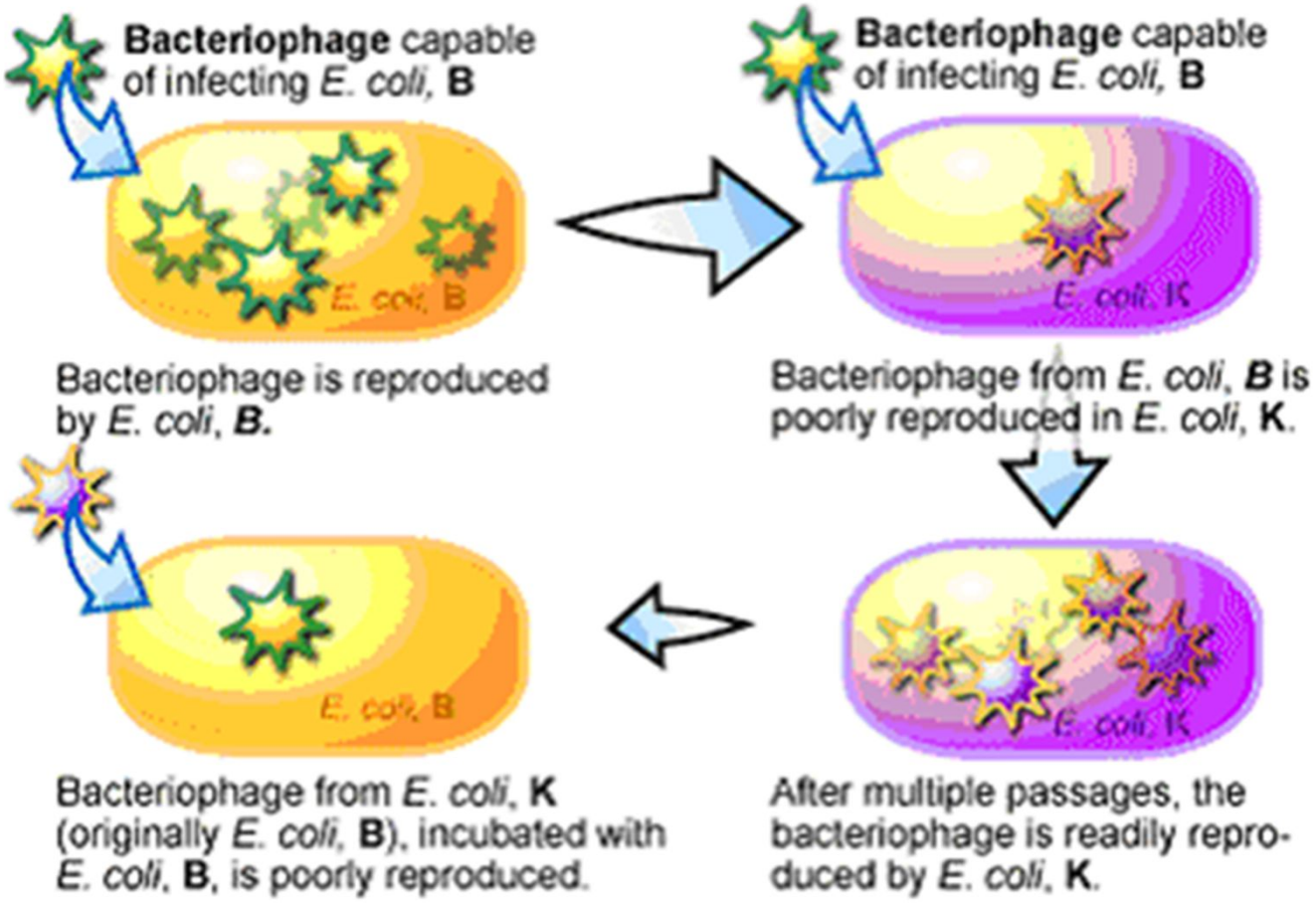


11.21 Sequencing DNA (*a*) The normal substrates for DNA replication are dNTPs. The slightly different structure of ddNTPs causes DNA synthesis to stop. (*b*) When labeled ddNTPs are incorporated into a mixture containing a single-stranded DNA template of unknown sequence, the result is a collection of fragments of varying lengths that can be separated by electrophoresis.

END

Restriction Endonucleases

Arber, Nathans & Smith (1978 Nobel Prize



Special bacterial enzyme can cut double stranded DNA at specific sites in a test tube.

Restriction endonucleases (restriction enzymes) act as a kind of immune system, protecting the cell from the invasion of foreign DNA (virus).

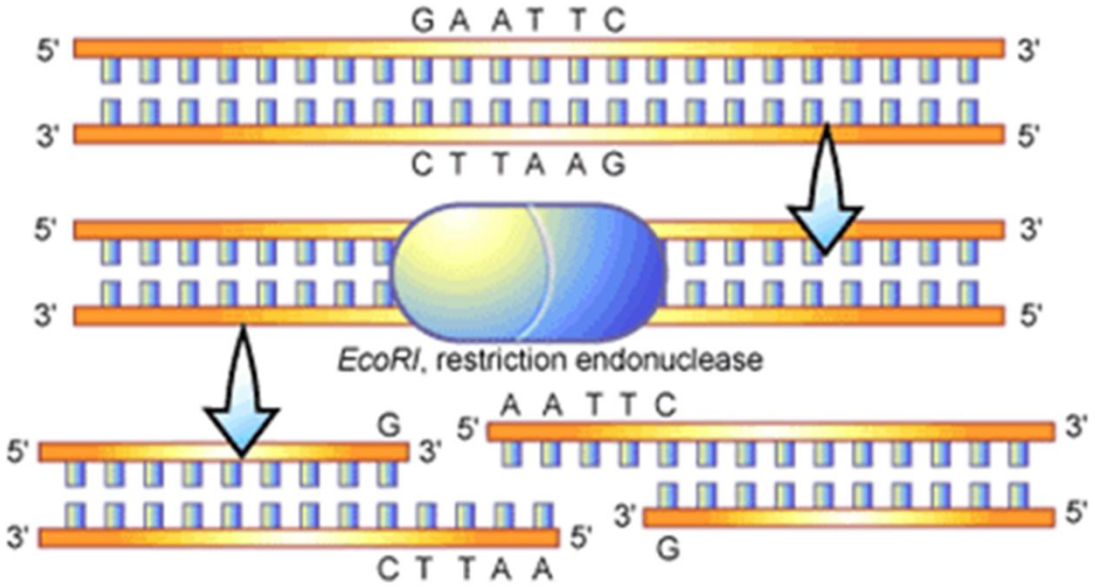
Many recombinant DNA technologies, which the field of biotechnology heavily relies on, are unlikely to have been developed without the discovery of restriction enzymes

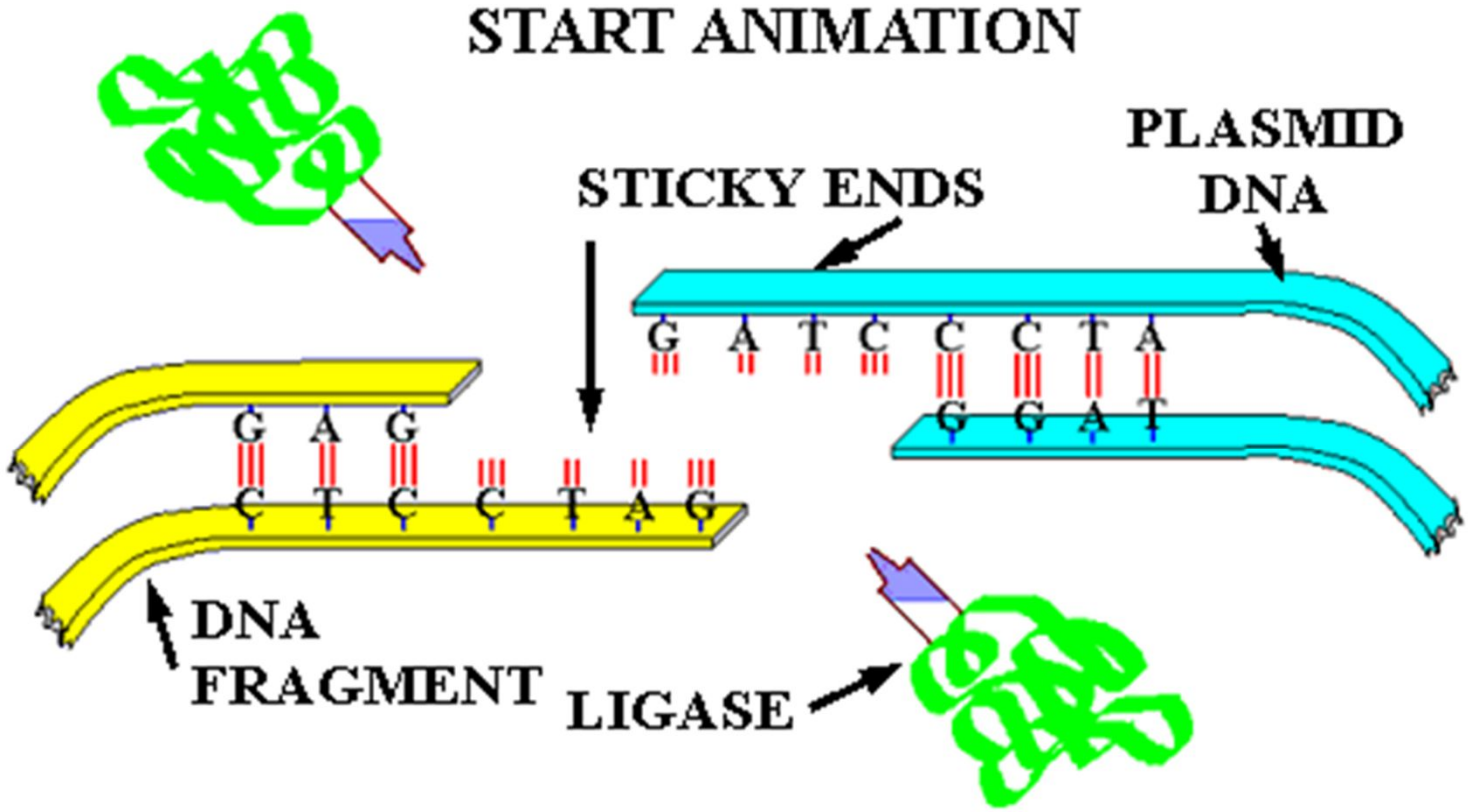
Palindromic Sequences: A palindromic sequence is the same when read in 5′ to 3′ direction on either complementary strand of DNA.

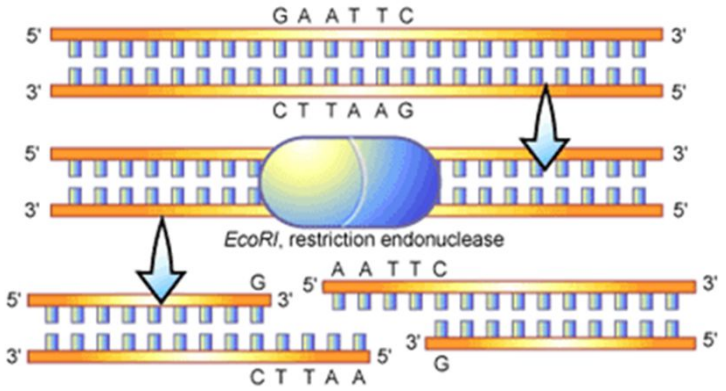
DENNIS SINNED

GTAATG VS GTATAC

.



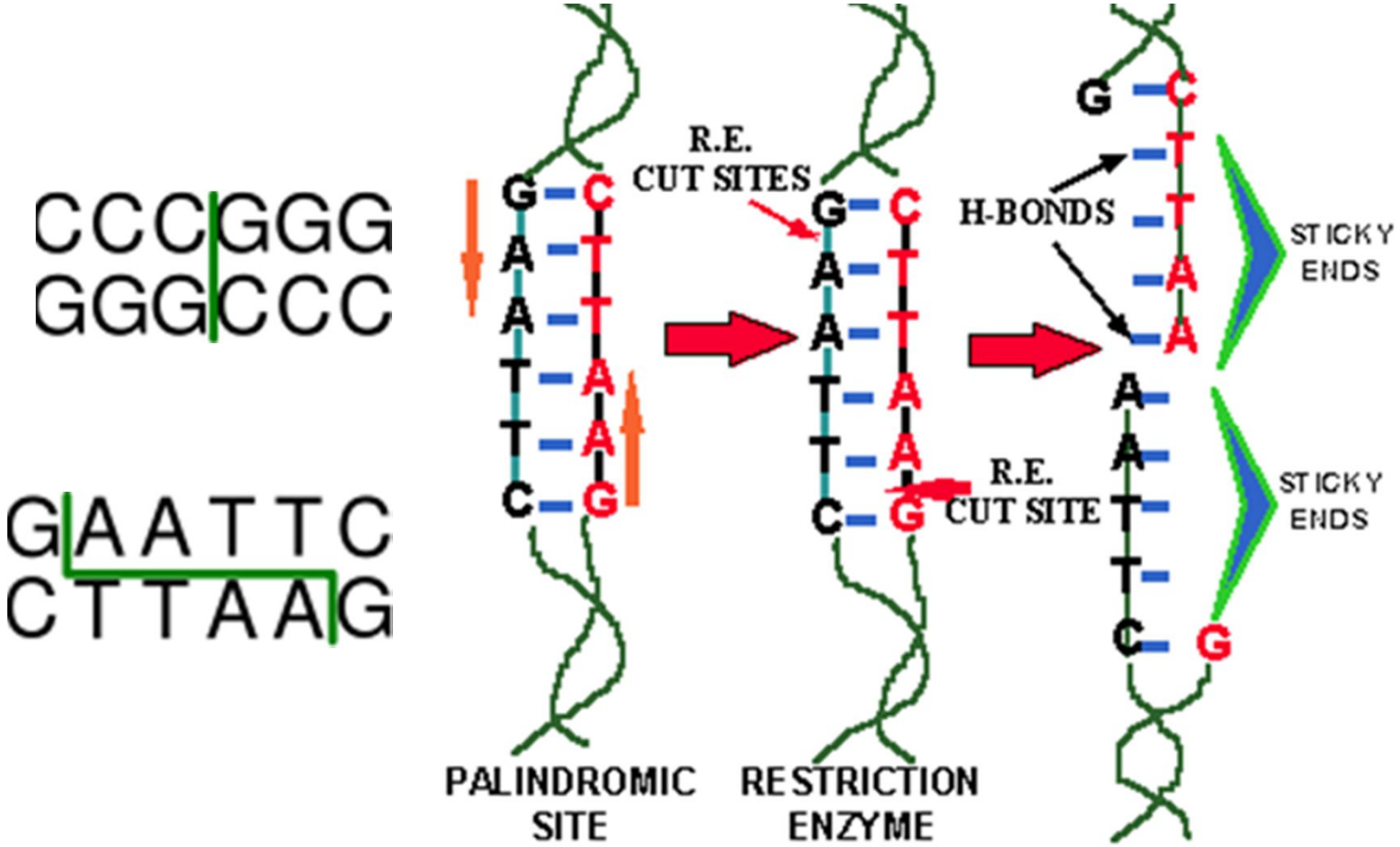

START ANIMATION
PLASMID DNA
STICKY ENDS
G A T C C C T A
G G A T
G A G
C T C C T A G
DNA FRAGMENT
LIGASE

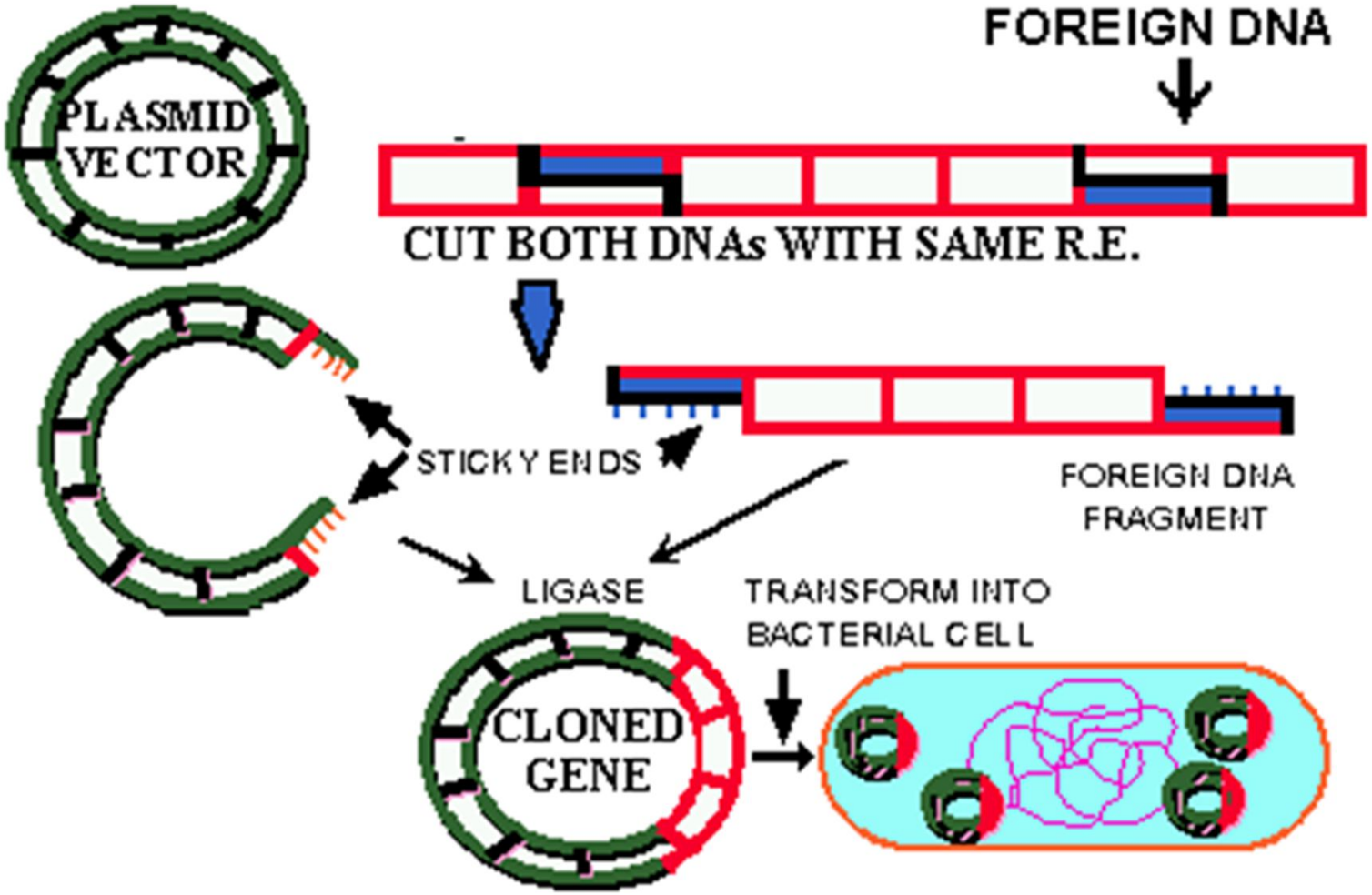

G A A T T C
5'
3'
3'
5'
C T T A A G
5'
3'
3'
5'
5'
3'
EcoRI, restriction endonuclease
G
5'
3'
A A T T C
5'
3'
3'
5'
3'
5'
C T T A A
3'
G

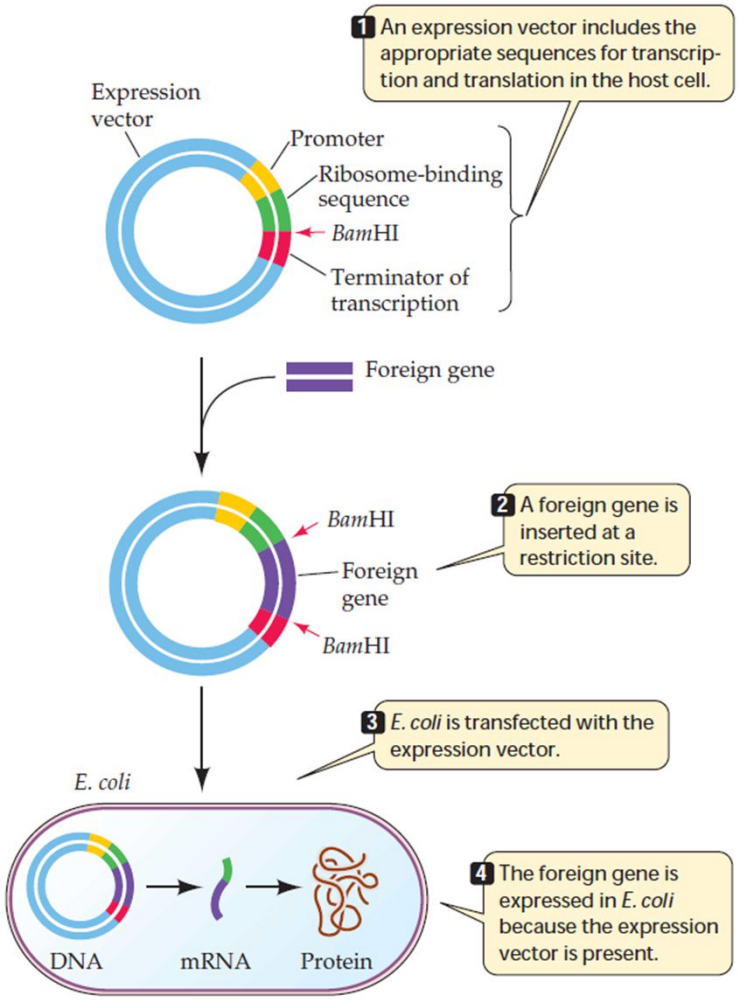
END

Genetic Engineering

Blunt Ends Sticky Ends

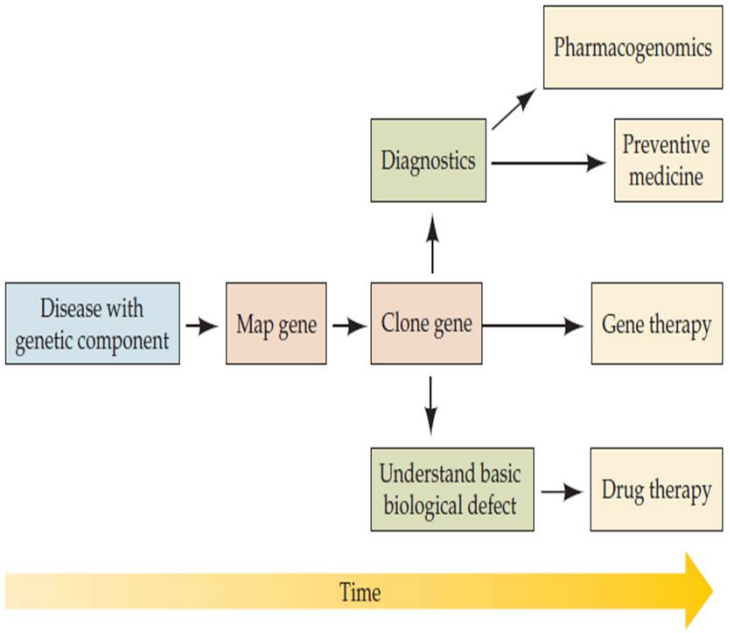



PLASMID VECTOR
FOREIGN DNA
CUT BOTH DNAs WITH SAME R.E.
STICKY ENDS
FOREIGN DNA FRAGMENT
LIGASE
TRANSFORM INTO BACTERIAL CELL
CLONED GENE


1 An expression vector includes the appropriate sequences for transcription and translation in the host cell.
Expression vector
Promoter
Ribosome-binding sequence
BamHI
Terminator of transcription
Foreign gene
BamHI
Foreign gene
BamHI
2 A foreign gene is inserted at a restriction site.
3 E. coli is transfected with the expression vector.
E. coli
DNA
mRNA
Protein
4 The foreign gene is expressed in E. coli because the expression vector is present.

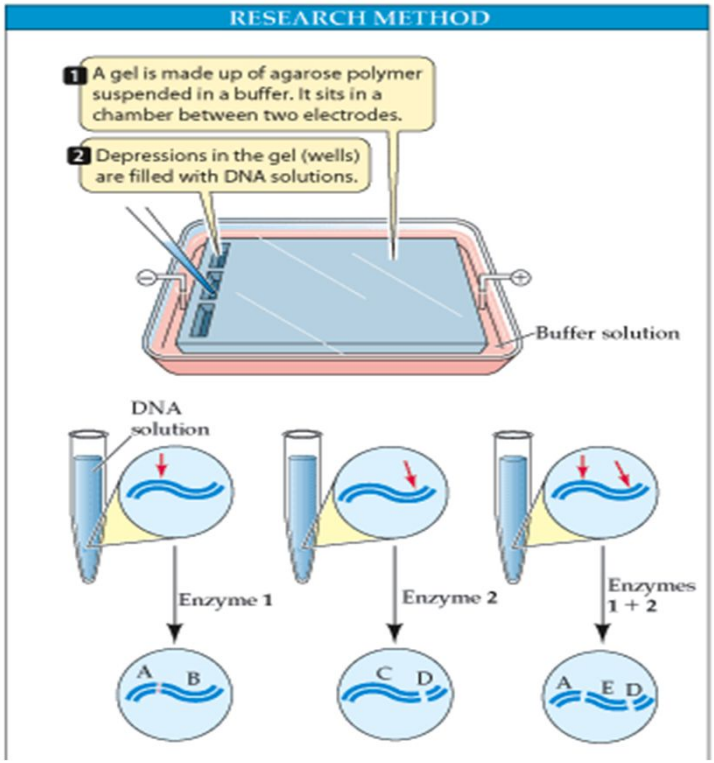
SEQUENCING: Sequencing determines the base pair sequence of a gene. By reading the 3-letter code, sequencing also describes the AMINO ACID SEQUENCE translated from that gene.
MUTATION: It is possible to study the effects of SINGLE AMINO ACID CHANGES on the function of the gene product, which is, after all, the ultimate purpose of the exercise (rev genetics).
MASS PRODUCTION OF PROTEINS: To use the amplified gene to make HUMONGOUS QUANTITIES of the GENE PRODUCT for commercial purposes.

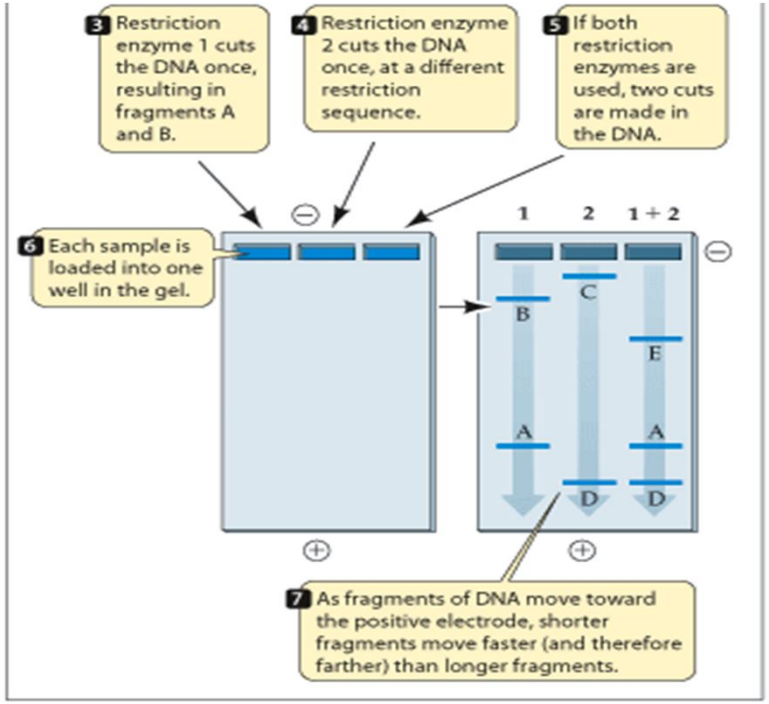
Human Health



END

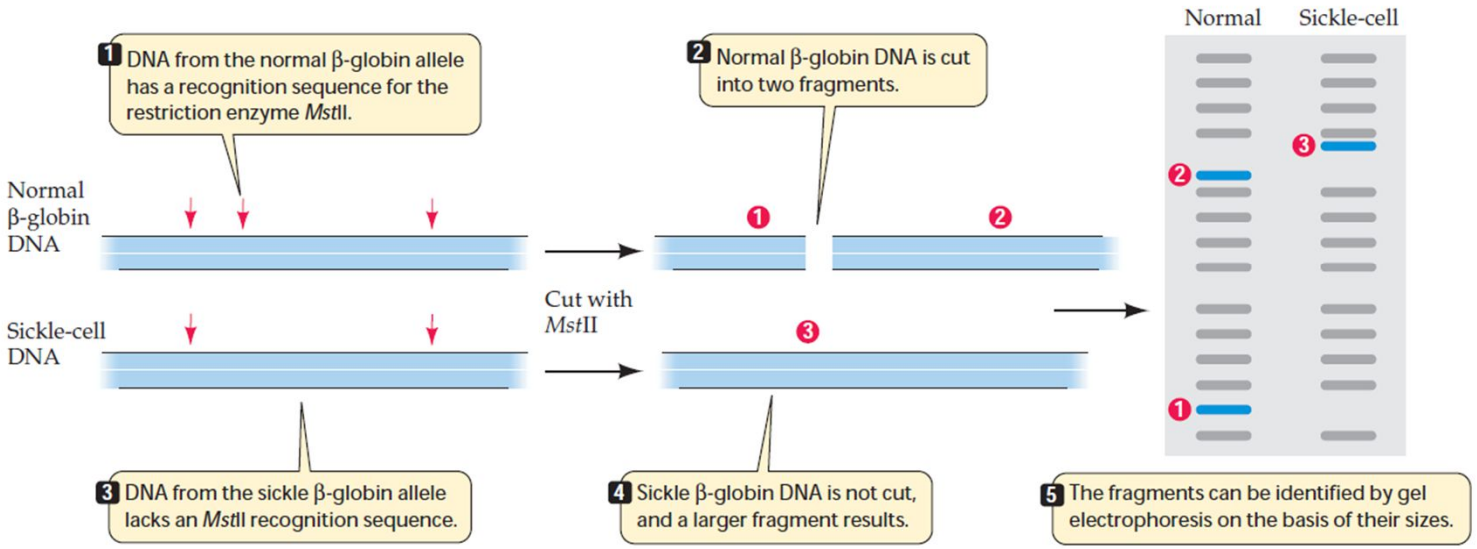
RFLP

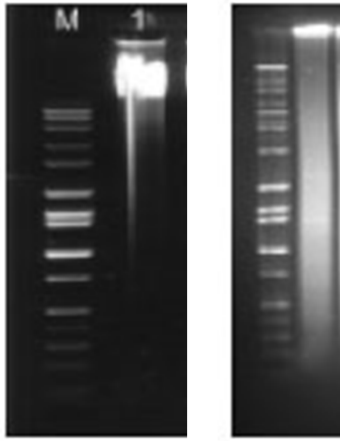




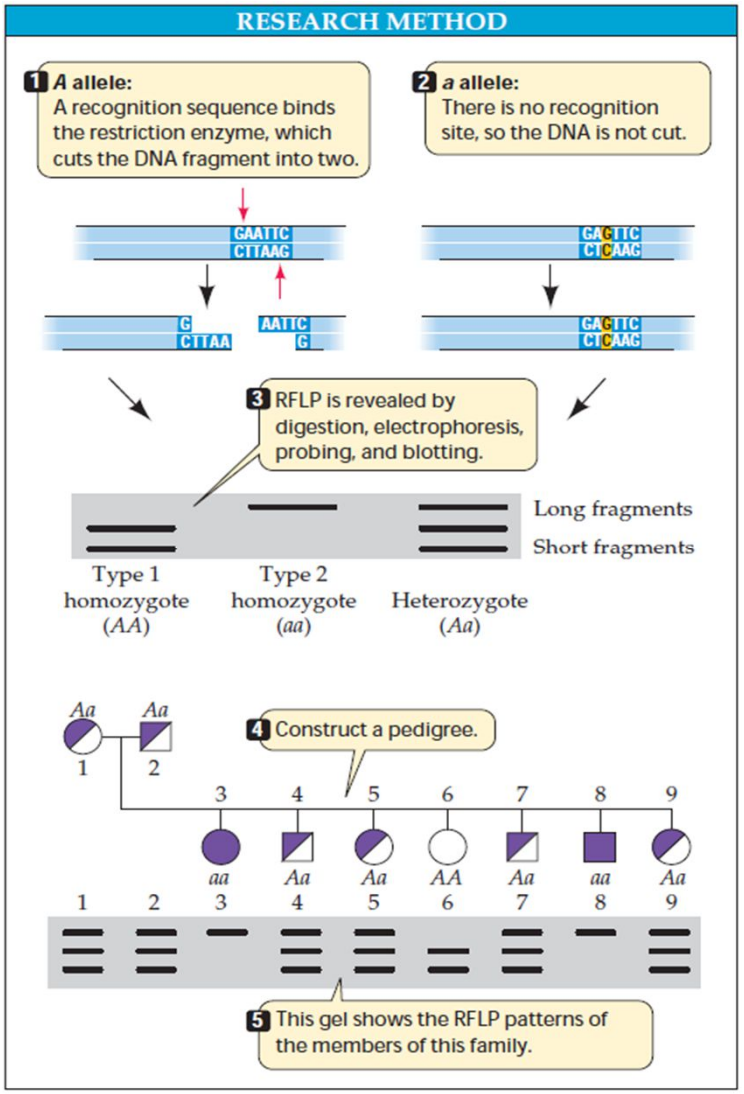
Effort By: AARISH ALI

RESEARCH METHOD

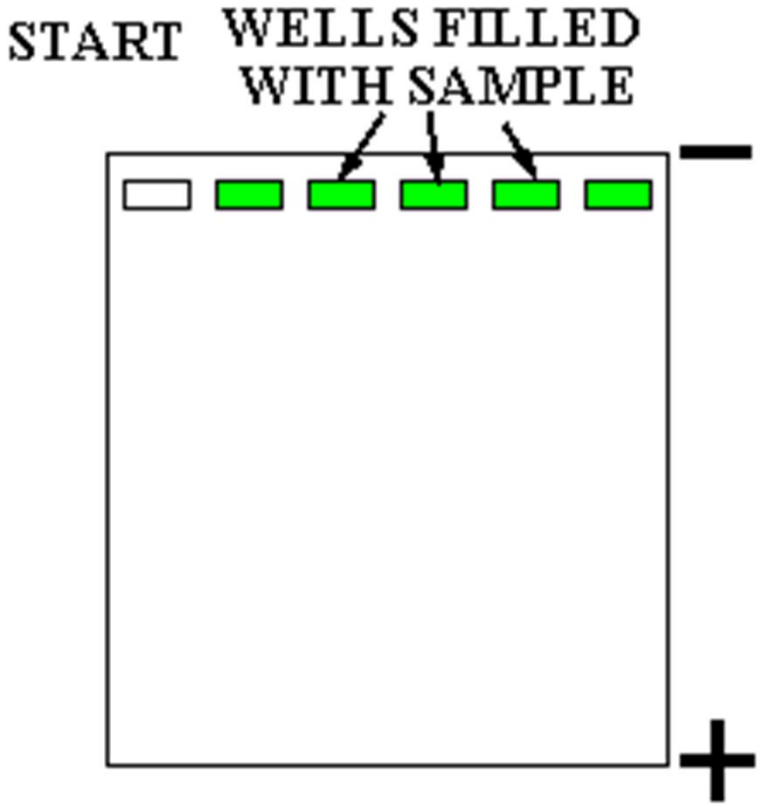




Restriction Fragment
Length polymorphisms (RFLP)
Can reveal mutations in DNA which may indicative of disease

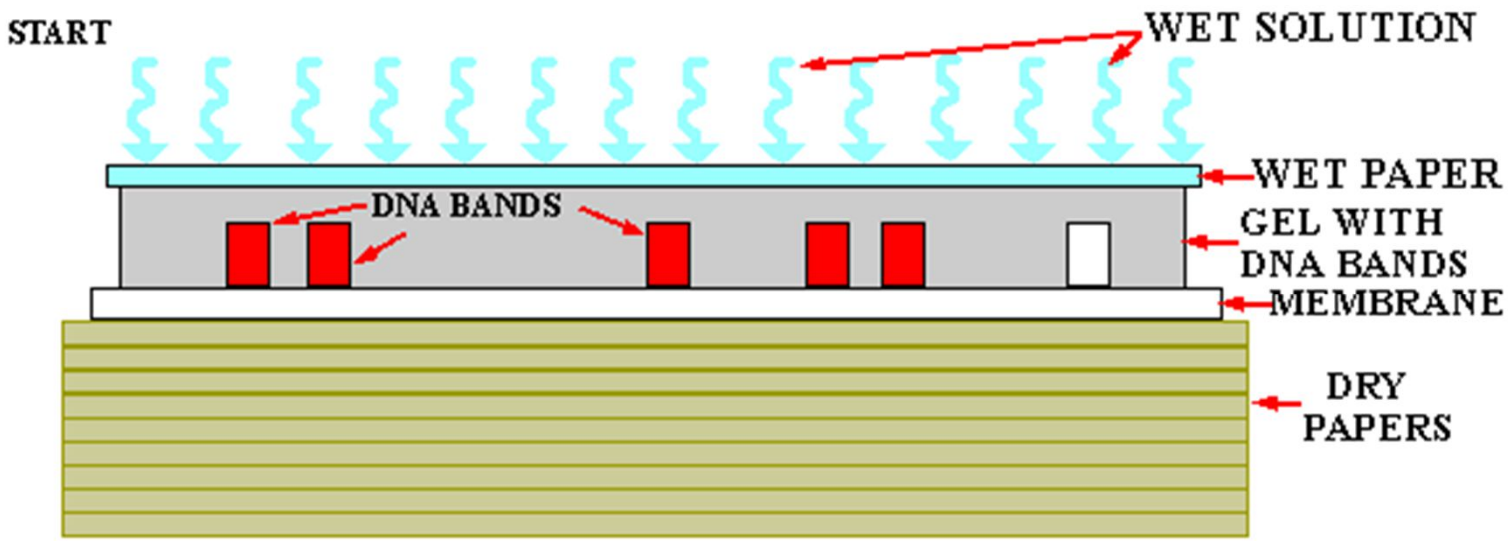


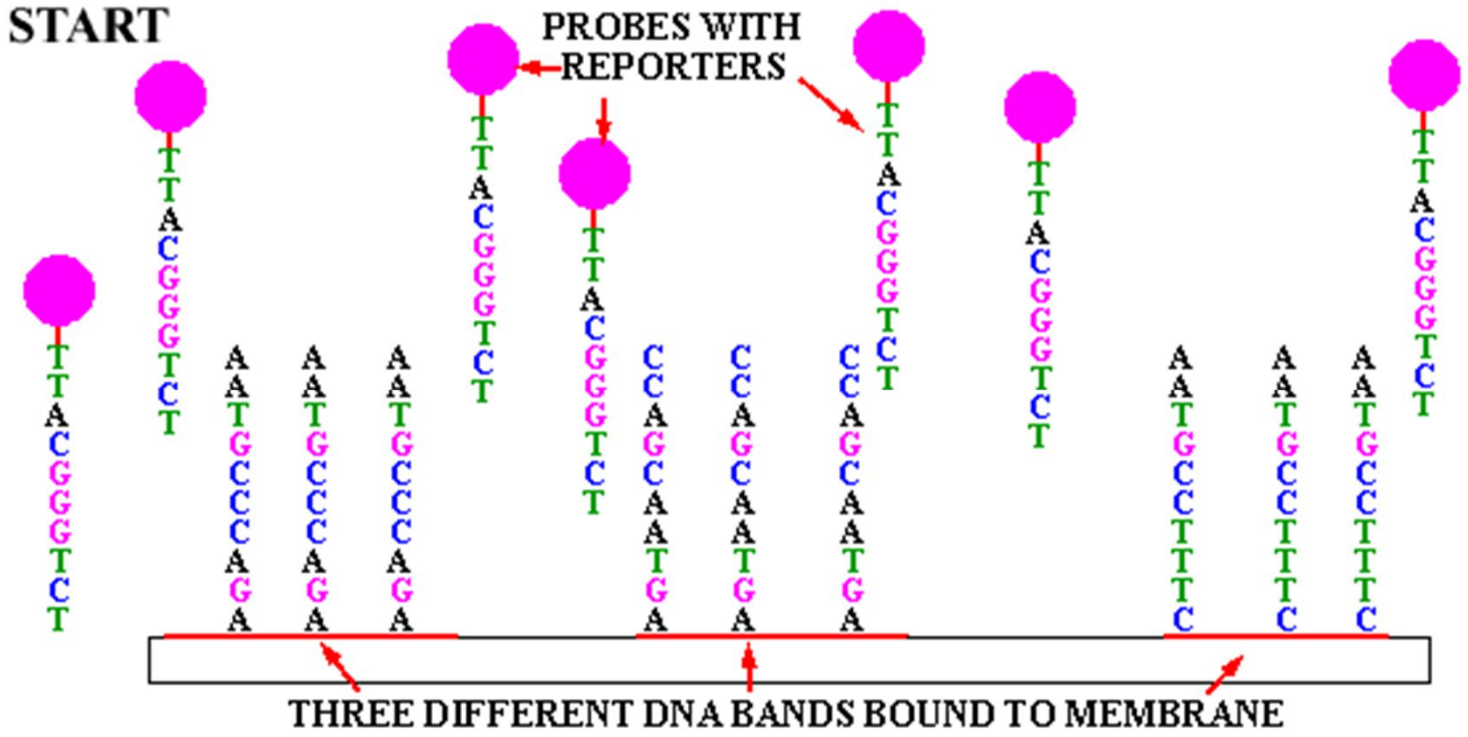
Cut Gel Pic



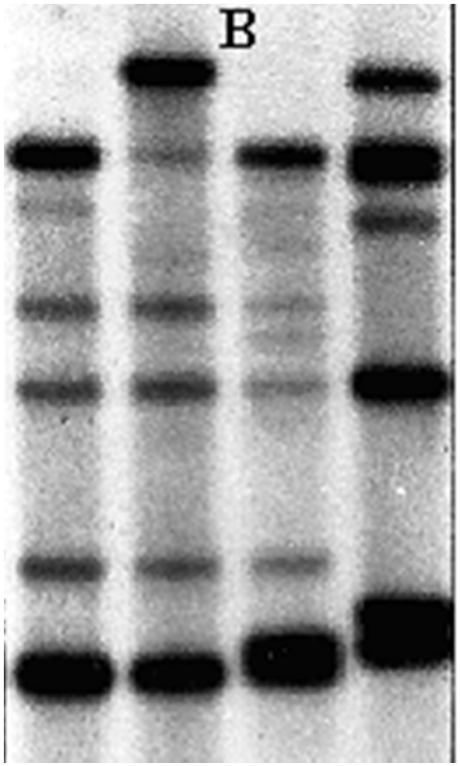
END

DNA Transfer


START
WET SOLUTION
WET PAPER
DNA BANDS
GEL WITH
DNA BANDS
MEMBRANE
DRY
PAPERS


START
PROBES WITH REPORTERS
THREE DIFFERENT DNA BANDS BOUND TO MEMBRANE

DNA Bound to Probe Exposed to X-ray film



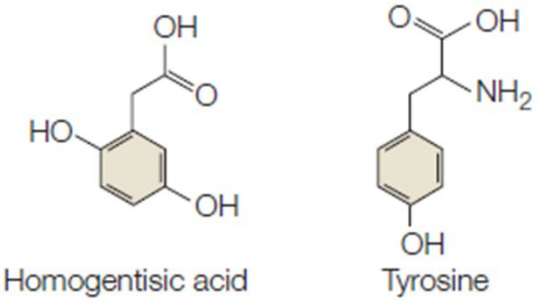
Genetic Information is Converted into Protein

Alkaptonuria ("black urine")

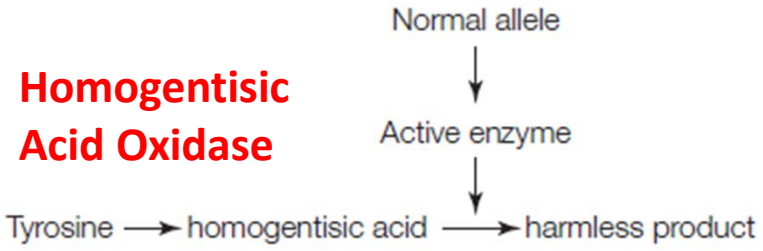
The molecular basis of phenotypes was actually discovered before it was known that DNA was the genetic material.

He linked the biochemical phenotype of the disease to an abnormal gene and a missing enzyme.

Most common in children whose parents were first cousins 12.5%



Homogentisic Acid Oxidase



Neurospora

An altered gene resulted in an altered phenotype, associated with an altered enzyme

Neurospora haploid (n) recessive alleles

X-rays, which act as a mutagen prototrophs (original) converted to auxotrophs (increase)

Some mutant strains could no longer grow on the minimal medium. One group could grow if supplemented with the arginine (*arg* mutants)

Neurospora

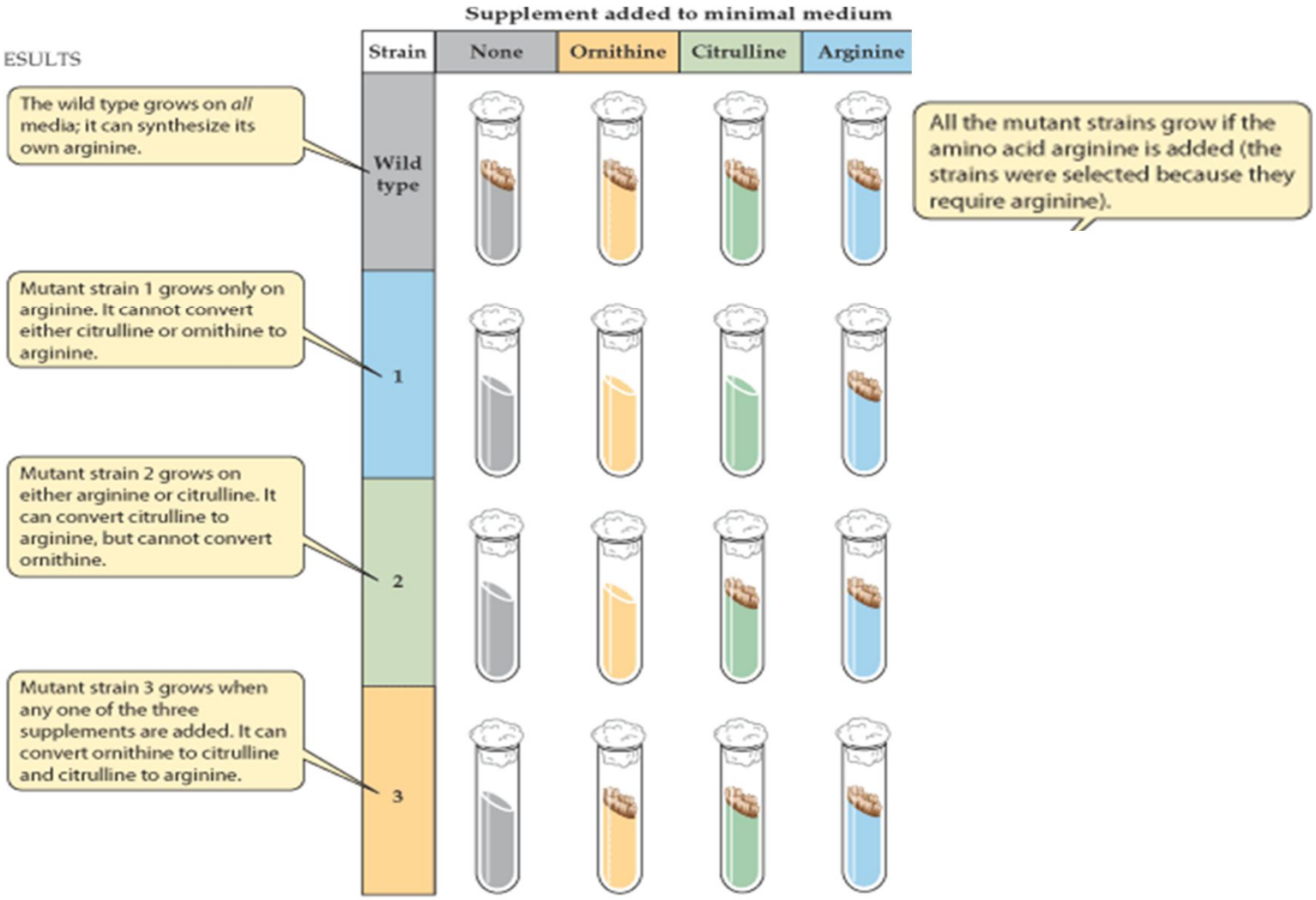
arg mutants were grown in presence of various compounds *suspected intermediates* in the synthetic metabolic pathway for arginine,

B & T classified each mutation as affecting one enzyme or another.

wild-type and mutant cells examined for enzyme activities.

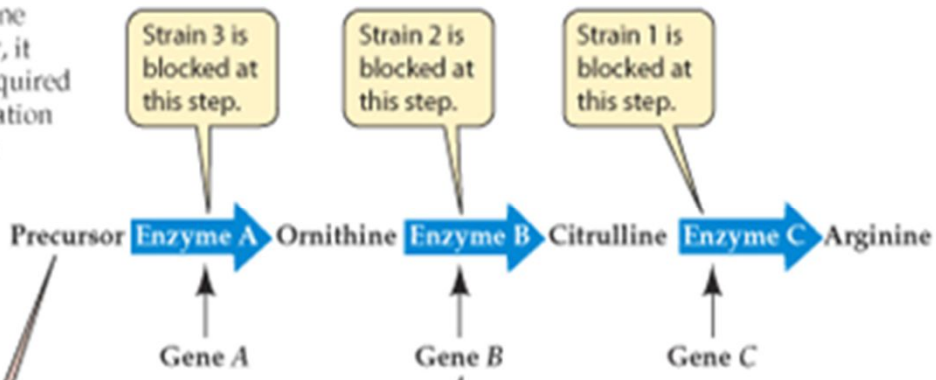
results confirmed: Each mutant strain was indeed missing a single active enzyme in the pathway.

ESULTS

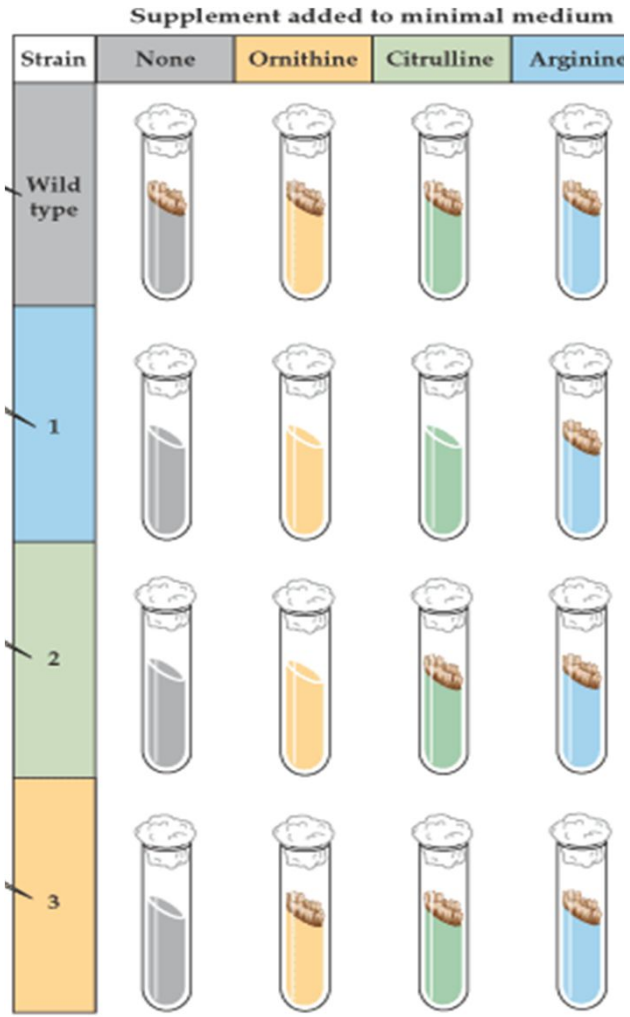

Supplement added to minimal medium
Strain
None
Ornithine
Citrulline
Arginine
Wild type
1
2
3
The wild type grows on *all* media; it can synthesize its own arginine.
Mutant strain 1 grows only on arginine. It cannot convert either citrulline or ornithine to arginine.
Mutant strain 2 grows on either arginine or citrulline. It can convert citrulline to arginine, but cannot convert ornithine.
Mutant strain 3 grows when any one of the three supplements are added. It can convert ornithine to citrulline and citrulline to arginine.
All the mutant strains grow if the amino acid arginine is added (the strains were selected because they require arginine).

INTERPRETATION

If an organism cannot convert one particular compound to another, it presumably lacks an enzyme required for the conversion, and the mutation is in the gene that codes for that enzyme.



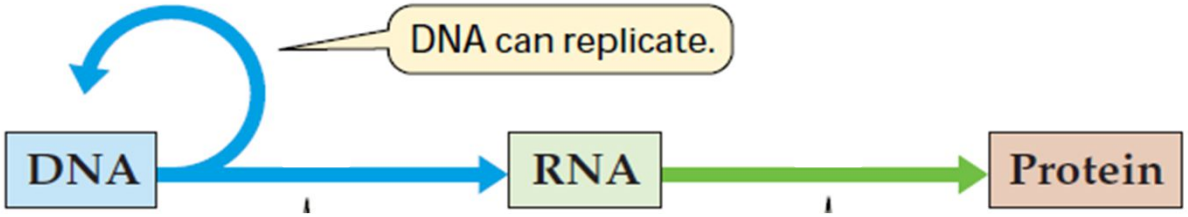
Conclusion: The synthesis of arginine proceeds like this. Each gene specifies a particular enzyme.


Supplement added to minimal medium
Strain
None
Ornithine
Citrulline
Arginine
Wild type
1
2
3

END

The Dogma

Info flows from DNA to RNA to proteins



How does genetic information get from the nucleus to the cytoplasm?

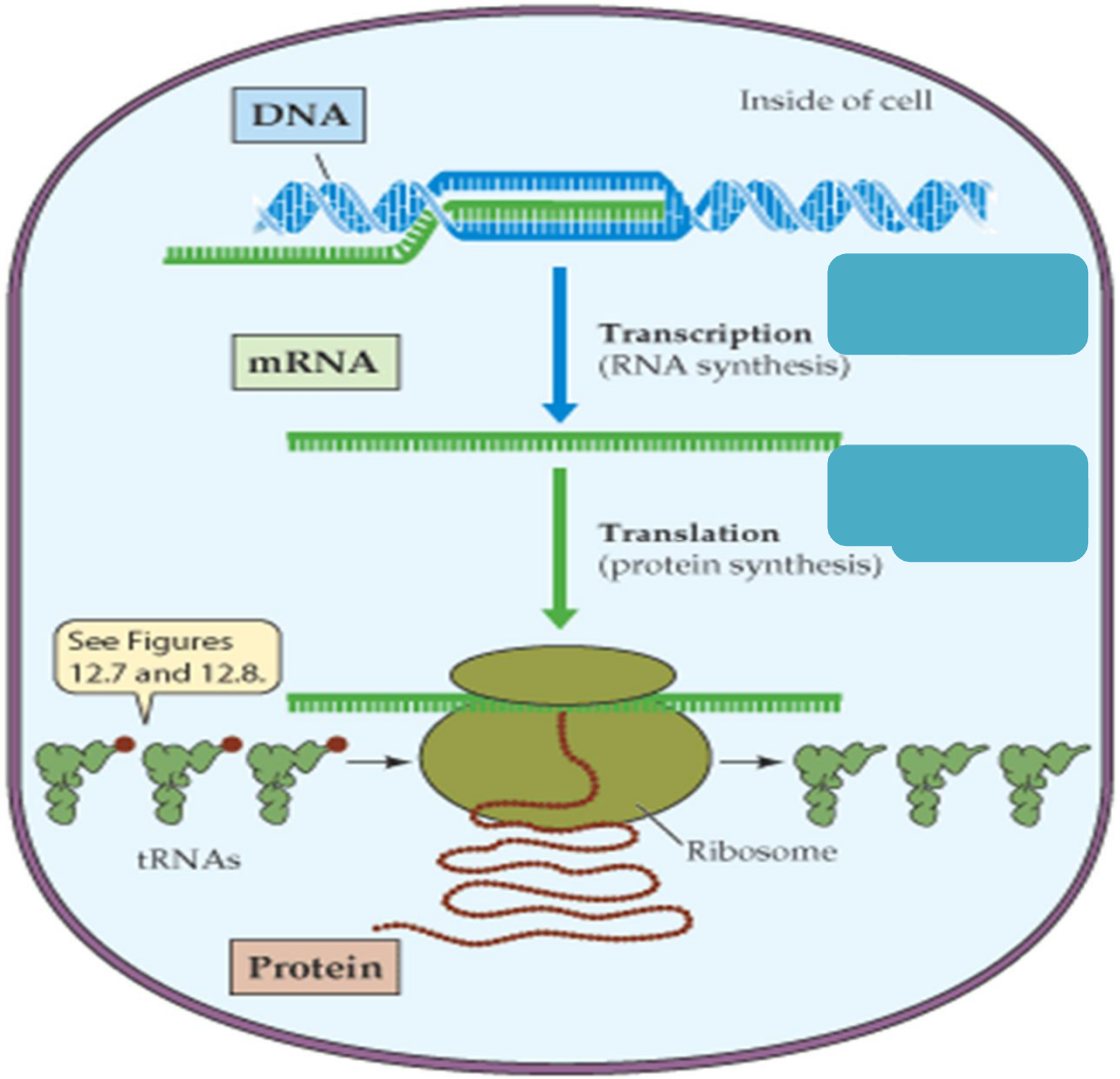
What is the relationship between a specific nucleotide sequence in DNA and a specific amino acid sequence in a protein?)

THE MESSENGER HYPOTHESIS AND TRANSCRIPTION

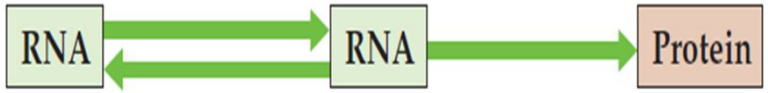
RNA molecule forms complementary copy of one DNA strand, moves to cytoplasm serves as template for protein synthesis.

THE ADAPTER HYPOTHESIS AND TRANSLATION

Adapter molecule binds a specific amino acid with one region and recognize a sequence of nucleotides with another region to translate the language of DNA into the language of proteins

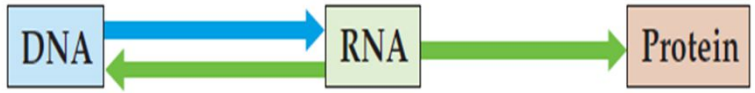

DNA
Inside of cell
mRNA
Transcription
(RNA synthesis)
Translation
(protein synthesis)
See Figures
12.7 and 12.8.
tRNAs
Ribosome
Protein

Influenza virus, & poliovirus



Transcribing from RNA to RNA

HIV

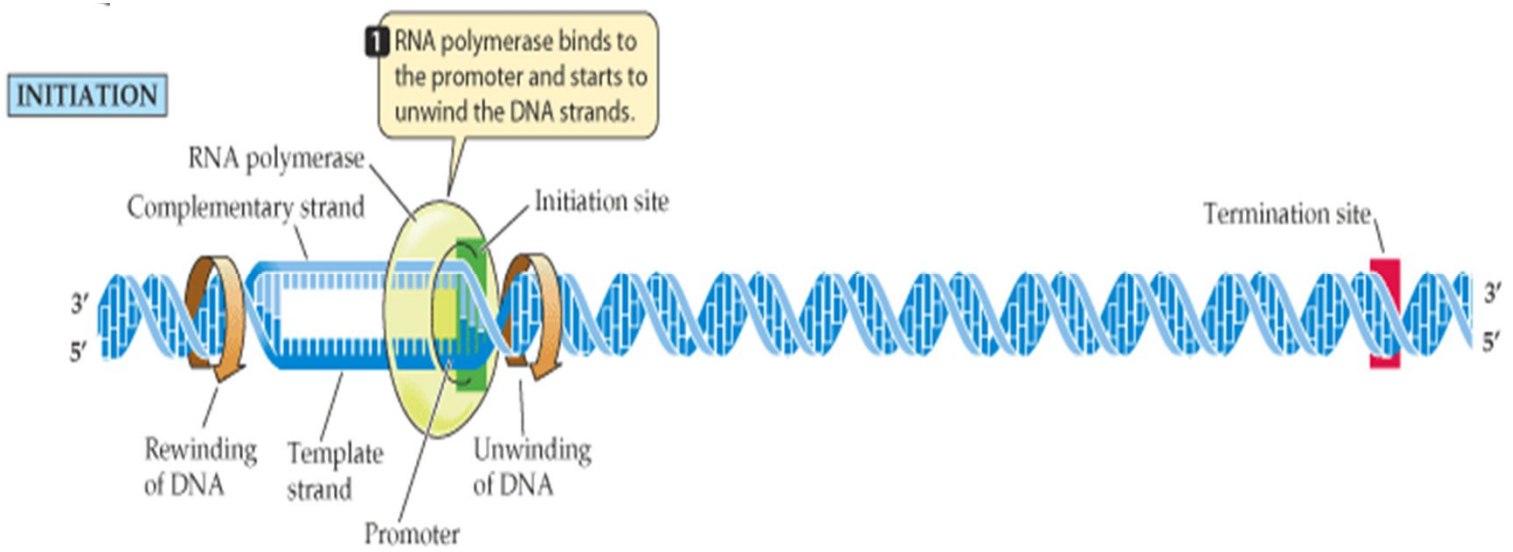


make a DNA copy of their genome

END

Transcription

Initiation begins at promoter -tightly binds RNA Pol

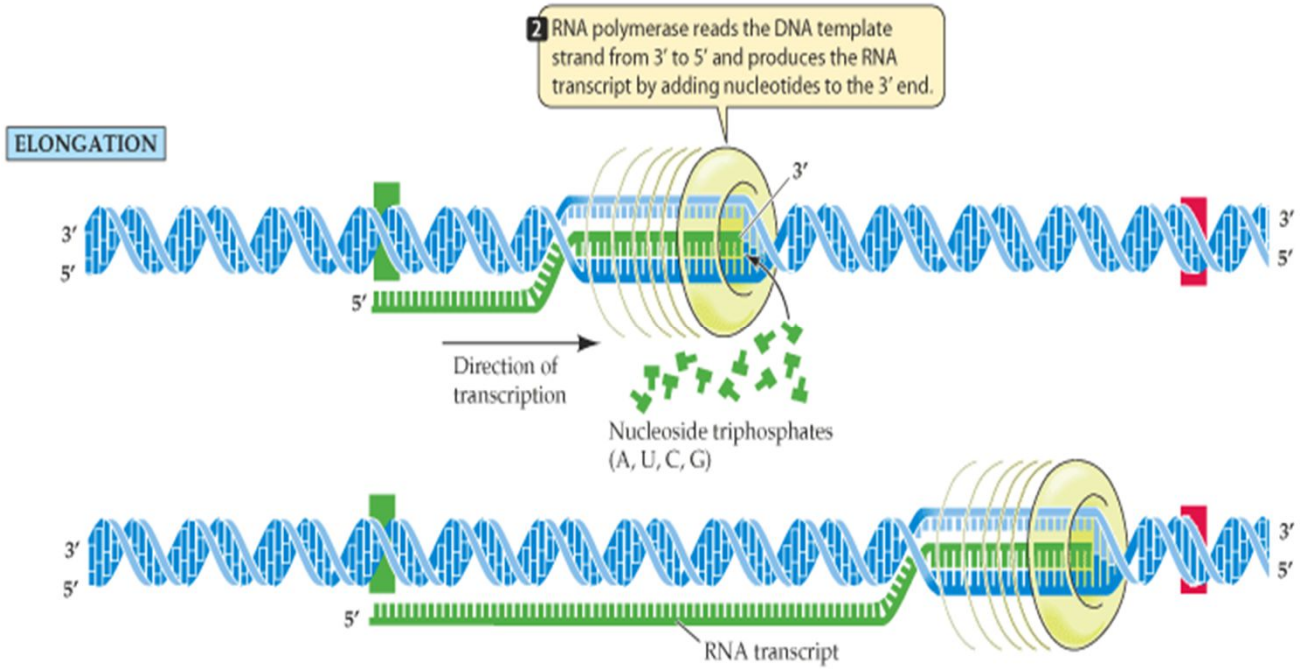


1) Where to start transcription.
2) Which strand to read.
3) The direction to take start from.

Part of promoter is initiation site (transcription start).

RNA Pol moves 3' to 5', synthesis 5' to 3', ~~primer~~

Unwinds 20 bp at a time.

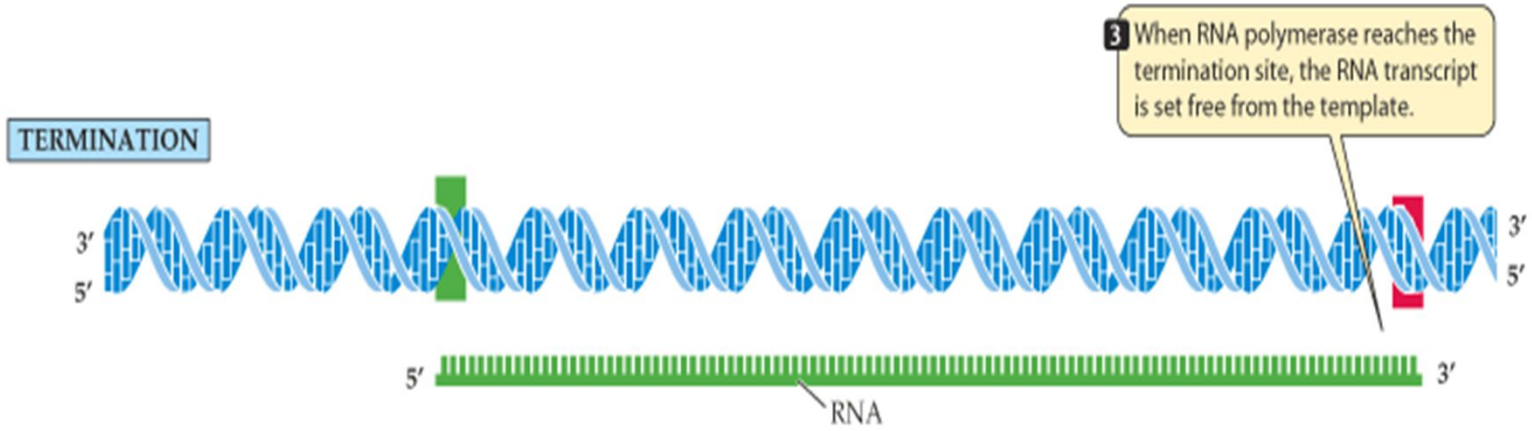


Transcript antiparallel to DNA template strand.

No proof reading error rate 1 in 10,000-100,000

Translation starts before termination in prokaryotes.

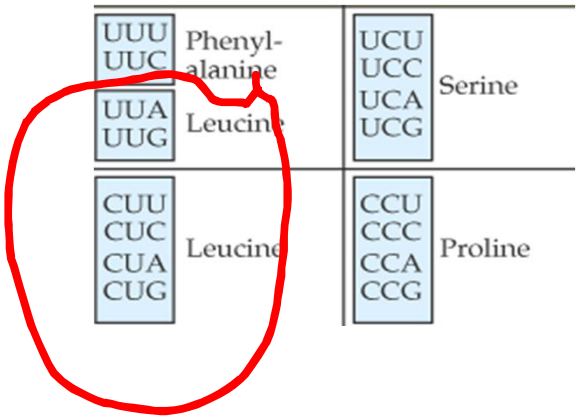
Pre-mRNA 1st product of transcription in eukaryotes.



Promoter
Terminator
5'
3'
3'
5'

Genetic Code

Three mRNA nucleotides form a codon

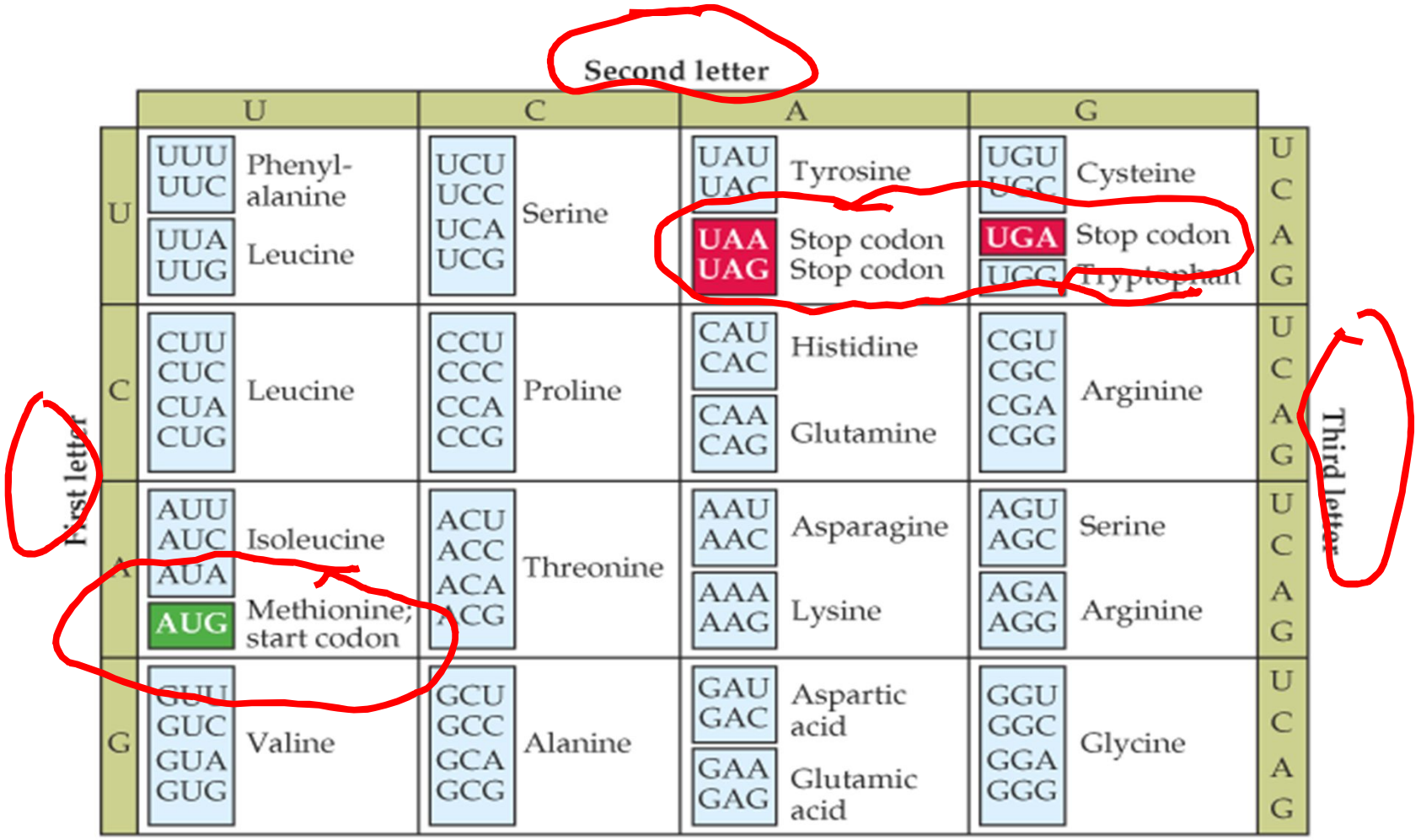


which is complementary & antiparallel to corresponding DNA.

Genetic code relates codons to their specific AA.

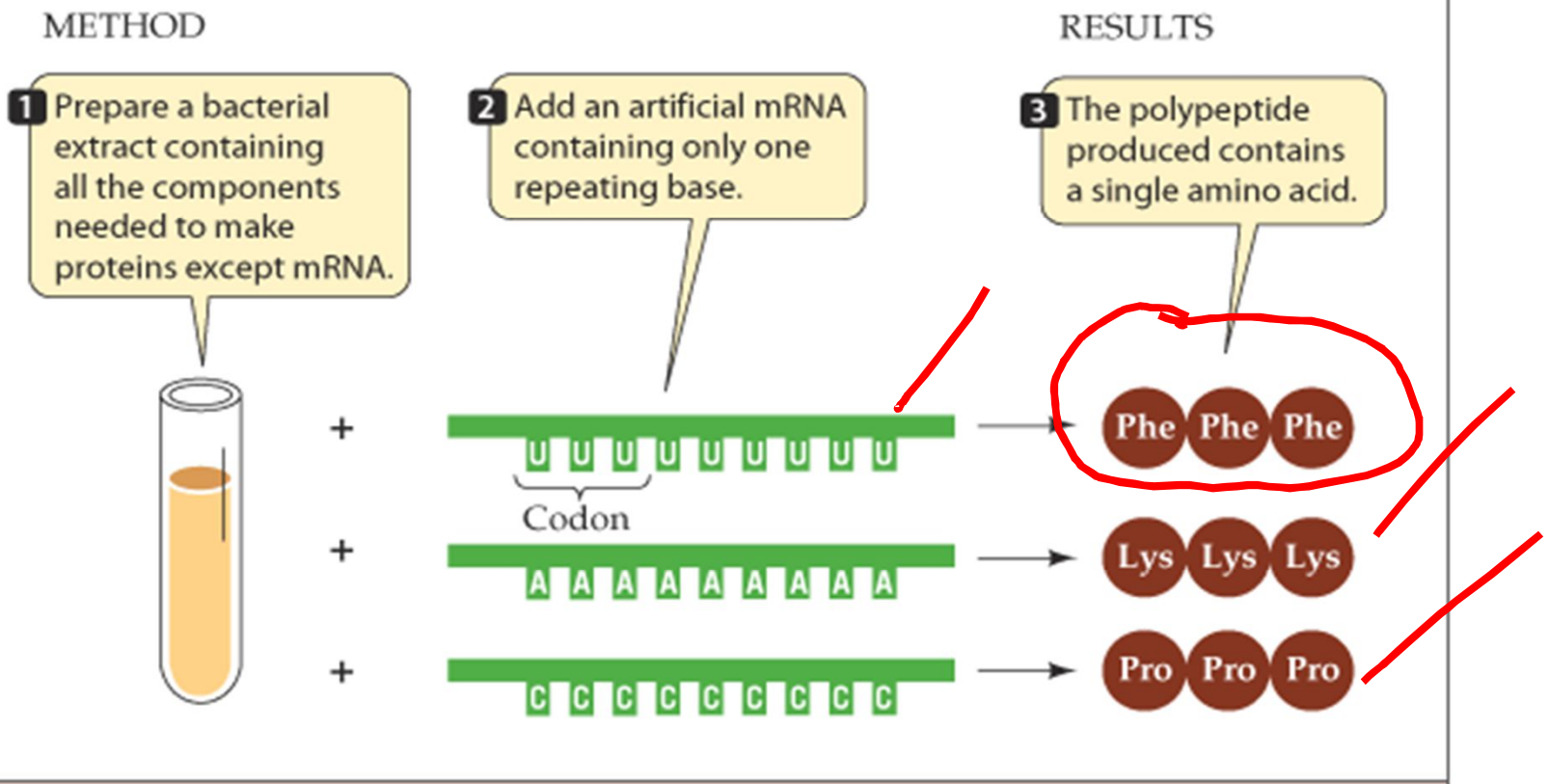
~ Universal & Redundant but not ambiguous some exceptions.

Start & Stop


Second letter
U
C
A
G
First letter
Third letter
U
UUU UUC Phenyl-alanine
UUA UUG Leucine
UCU UCC UCA UCG Serine
UAU UAC Tyrosine
UAA Stop codon
UAG Stop codon
UGU UGC Cysteine
UGA Stop codon
UGG Tryptophan
C
CUU CUC CUA CUG Leucine
CCU CCC CCA CCG Proline
CAU CAC Histidine
CAA CAG Glutamine
CGU CGC CGA CGG Arginine
A
AUU AUC AUA Isoleucine
AUG Methionine; start codon
ACU ACC ACA ACG Threonine
AAU AAC Asparagine
AAA AAG Lysine
AGU AGC Serine
AGA AGG Arginine
G
GUU GUC GUA GUG Valine
GCU GCC GCA GCG Alanine
GAU GAC Aspartic acid
GAA GAG Glutamic acid
GGU GGC GGA GGG Glycine

EXPERIMENT

Question: What are the amino acids specified by the triplet codons UUU, AAA, and CCC?



Conclusion: UUU is an mRNA codon for phenylalanine.
AAA is an mRNA codon for lysine.
CCC is an mRNA codon for proline.

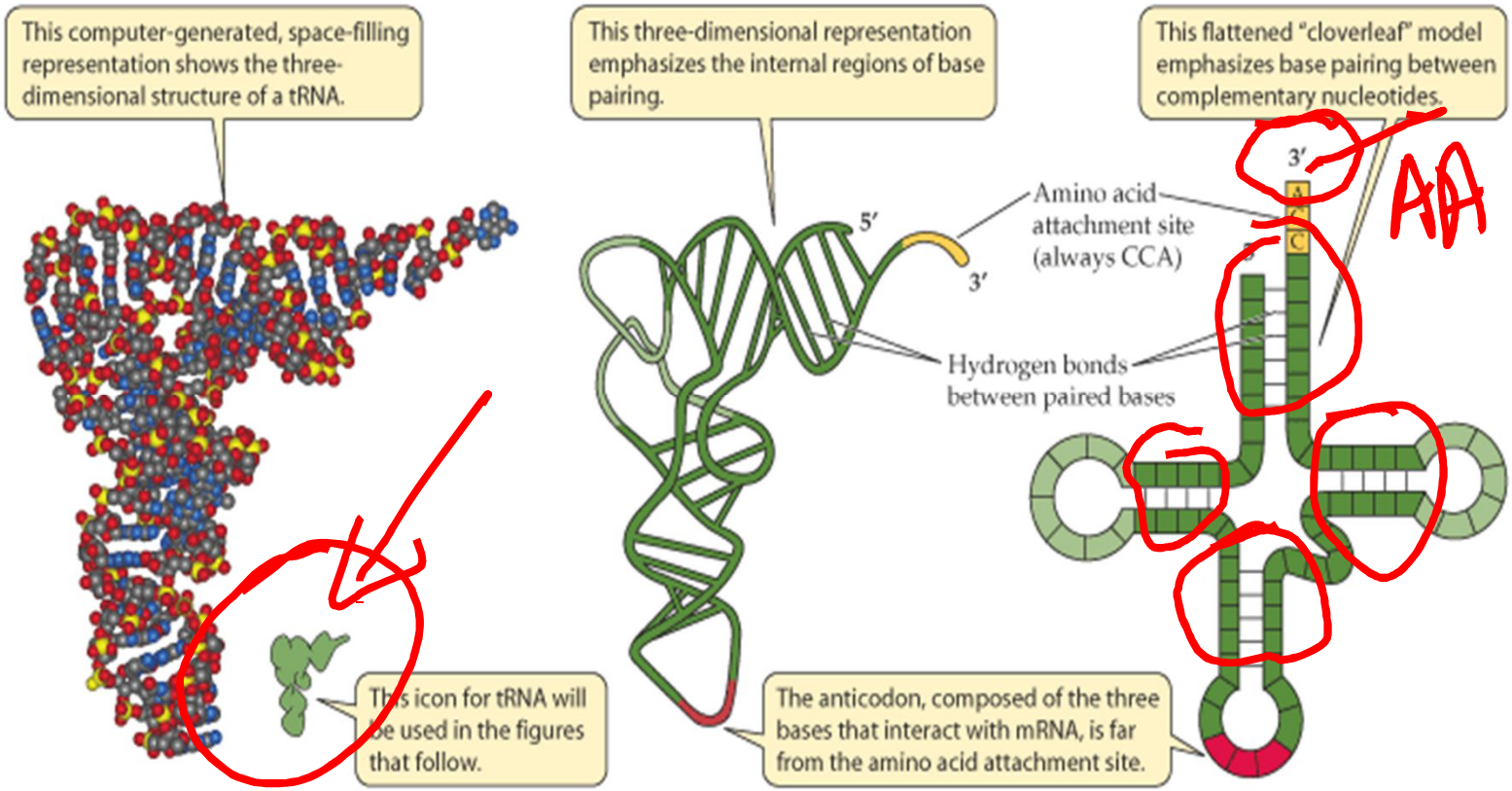
tRNA Translates Nucleotide Language into AA Language

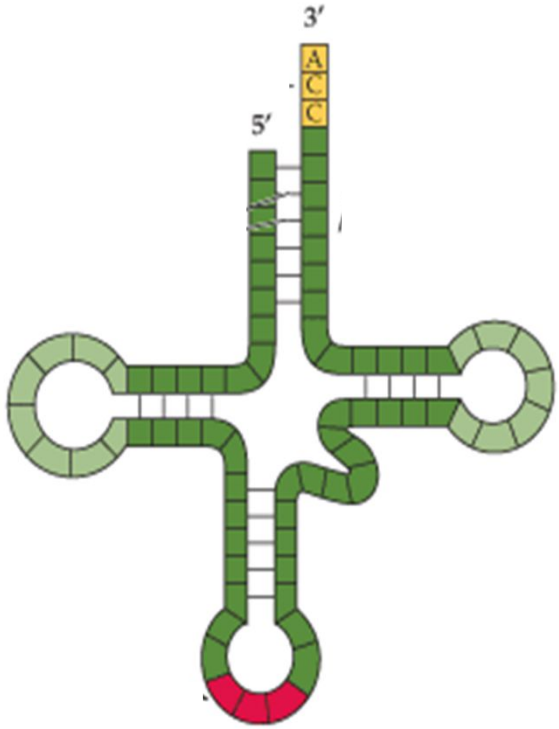
1) Carries AA
2) Associates with mRNA
3) Binds ribosome

tRNA has 75-80 nucleotides with internal H-bonding.

Conformation of tRNA allows it to bind ribosome.

3' end of tRNA (CCA) is attached to AA & mid point has anticodon (antiparallel).


This computer-generated, space-filling representation shows the three-dimensional structure of a tRNA.
This three-dimensional representation emphasizes the internal regions of base pairing.
This flattened "cloverleaf" model emphasizes base pairing between complementary nucleotides.
5′
3′
Amino acid attachment site (always CCA)
3′
A
C
C
Hydrogen bonds between paired bases
This icon for tRNA will be used in the figures that follow.
The anticodon, composed of the three bases that interact with mRNA, is far from the amino acid attachment site.


3′
A
C
C
5′

Charging tRNA

Aminoacyl-tRNA Synthetase

Specific for one AA & its corresponding tRNA.

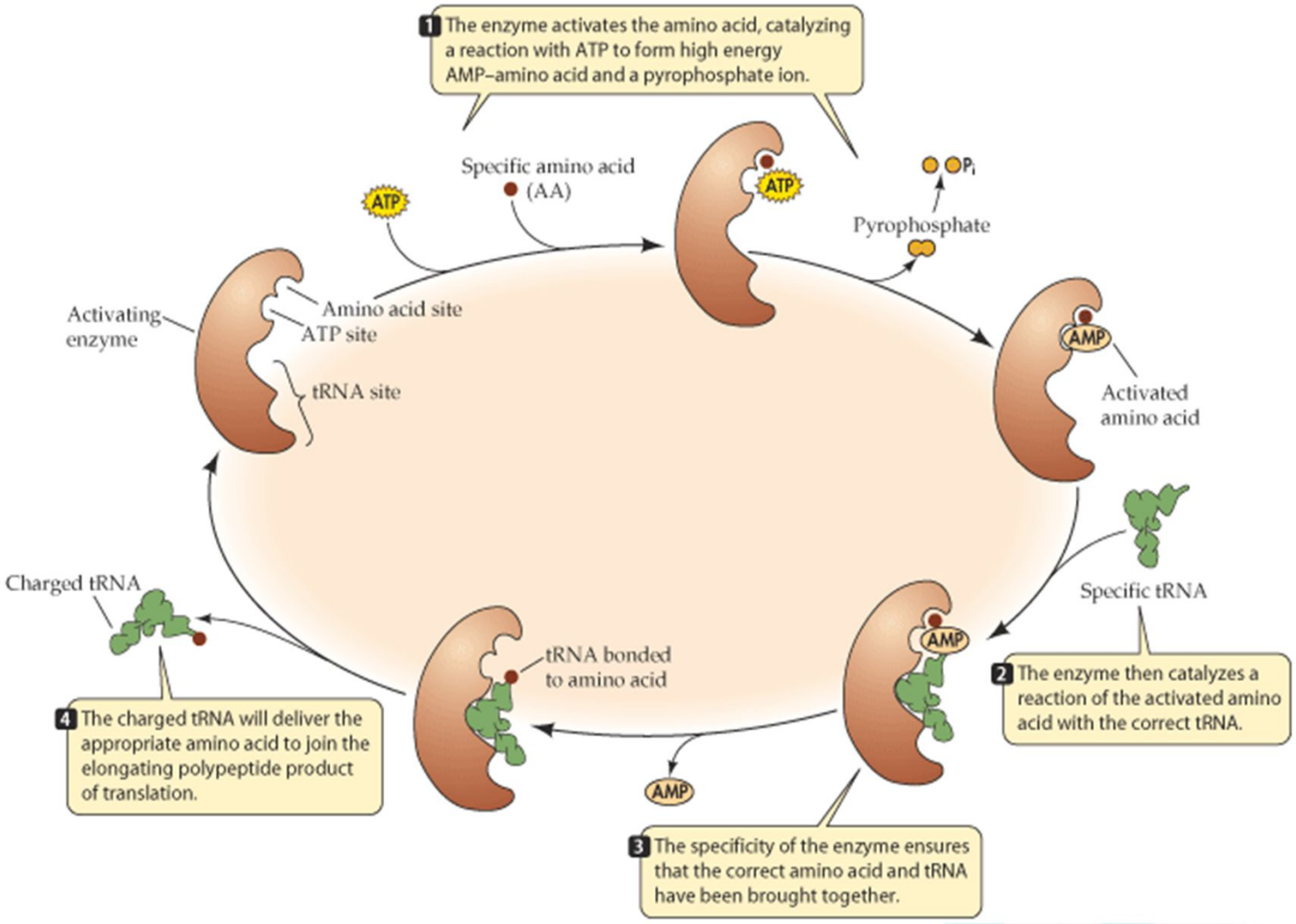
Enzyme has three active sites for three molecules.

amino acid + ATP → aminoacyl-AMP + PPi
aminoacyl-AMP + tRNA → aminoacyl-tRNA + AMP

-COOH of AA is attached to 3' end of tRNA's free OH on ribose.

This high energy bond provides energy for peptide bond synthesis (2nd G code).

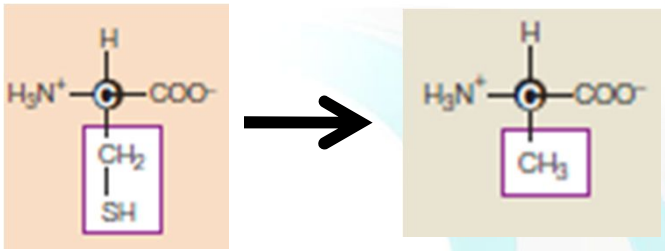
Charging


1 The enzyme activates the amino acid, catalyzing a reaction with ATP to form high energy AMP–amino acid and a pyrophosphate ion.
Specific amino acid (AA)
ATP
Activating enzyme
Amino acid site
ATP site
tRNA site
Pyrophosphate
Pi
AMP
Activated amino acid
Specific tRNA
2 The enzyme then catalyzes a reaction of the activated amino acid with the correct tRNA.
3 The specificity of the enzyme ensures that the correct amino acid and tRNA have been brought together.
tRNA bonded to amino acid
Charged tRNA
4 The charged tRNA will deliver the appropriate amino acid to join the elongating polypeptide product of translation.

Benzer Experiment

Amino acid/tRNA recognized?

1) Took the cys-tRNA molecule and chemically modified the cysteine, converting it into alanine.
2) Hybrid charged tRNA was put into a protein synthesizing system.
3) Everywhere in the synthesized protein where cysteine was supposed to be, alanine appeared instead.
4) Protein synthesis machinery recognizes the anticodon of the charged tRNA, not the amino acid attached to it.



Charging

Charging

Translation Workbench

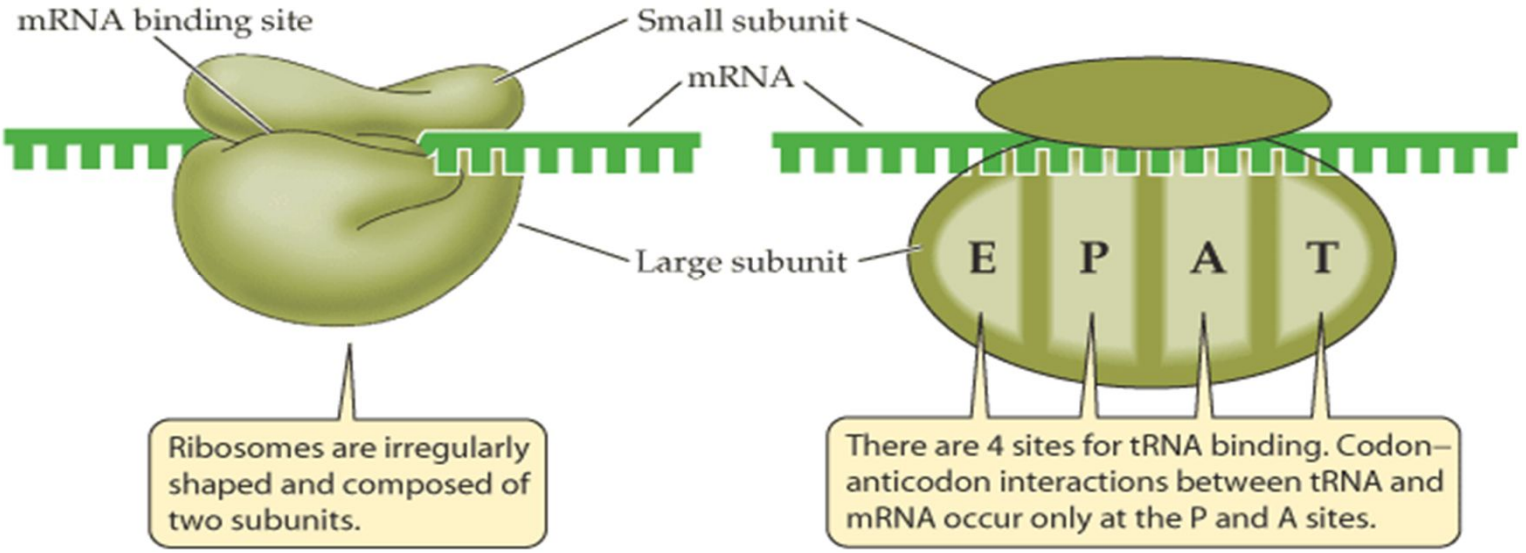
Two Subunits

Large subunit = 3 rRNAs + 45 poly peptides.

Small subunit = 1 rRNA + 33 poly peptides.

Ionic & hydrophobic forces hold complex together.

Complex self assembles



Ribosomes

A ribosome can use any mRNA and all species of charged tRNAs, and thus can be used to make many different polypeptide products.

The ribosomes of prokaryotes Have different ribosomal proteins and RNAs.

Antibiotics that target the prokaryotic ribosome (tetracycline) can be used to kill bacteria without harming a patient's cells.

Mitochondria and chloroplasts also contain ribosomes, some of which are similar to those of prokaryotes

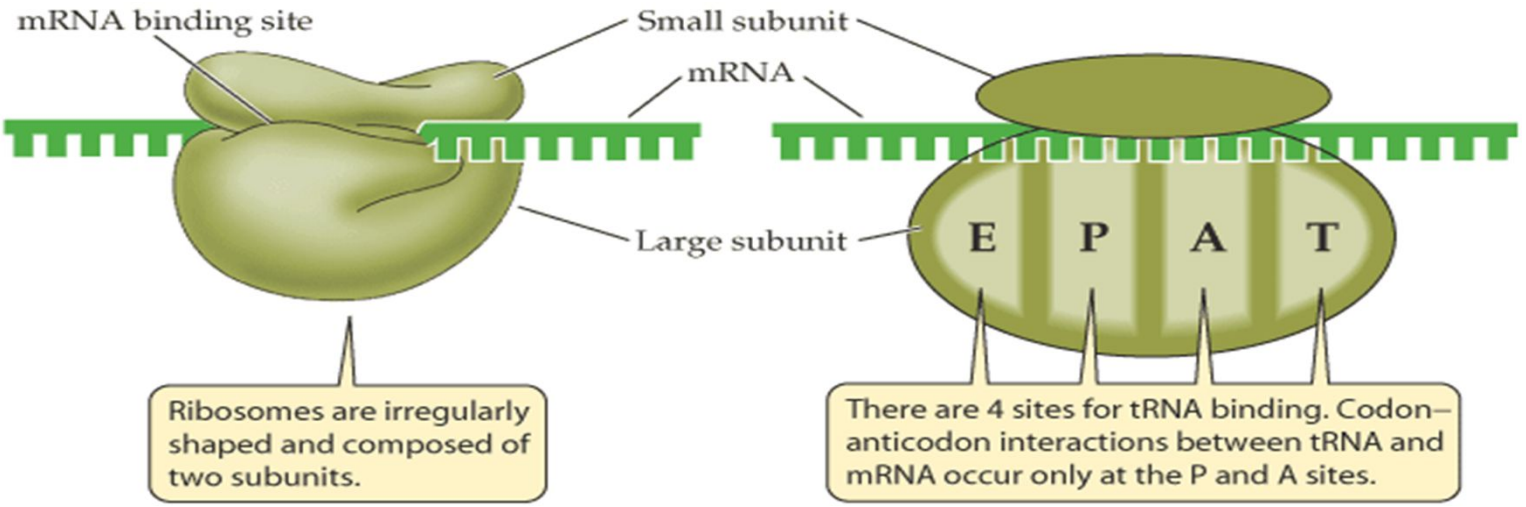
Large Subunit

T (transfer) site: first site where charged tRNA lands with the help of transfer factor

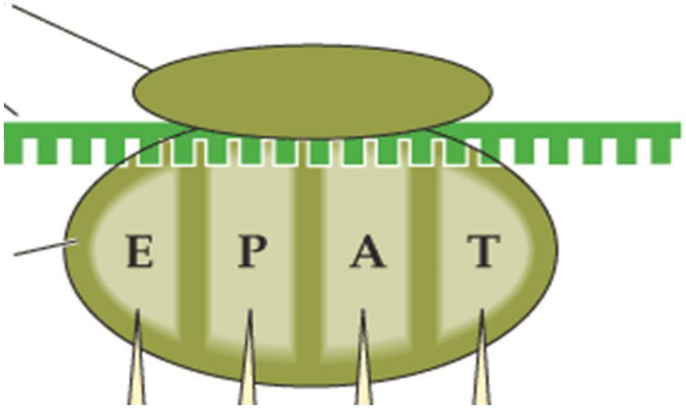
A (AA) site: Anticodon binds to mRNA codon

P (polypeptide) site: tRNA adds AA to polypeptide

E (exit) site: after AA transfer tRNA exits here



Ribosome



END

Midterm Term Syllabus
Lecture 01 to 88

Efforts By

AARISH ALI (+96569002641)

Contact us to join our esteemed group

“All About Biotech”

Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

REMEMBER IN PRAYERS

KEEP SMILING

SAY NO TO CURRUPTION

HELP TO OTHERS

Regards,

AARISH ALI & TEAM

“All About Biotech”

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